



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 15832-001.1
Date: 9/25/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: Odor Control Fans
Reviewed By: Khalil Kairouz, Kellen Gross

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☒	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☐	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. Provide high temp switch on Odor Control Fan Motors.

High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number:
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification		
Check Either (A) or (B):		
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .	
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.		
General Contractor's Reviewer's Signature: 		
Printed Name and Title:		
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.		
Firm:	Signature:	Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
SPECIFICATION SECTION # _____
SPECIFICATION _____
PARAGRAPH _____
DRAWING _____
SUBCONTRACTOR _____
SUPPLIER _____
MANUFACTURER _____

CERTIFIED BY CQCM or Designee : 

SECTION 15832
ODOR CONTROL FANS

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
 - 1. Fiberglass reinforced plastic centrifugal fans for biological odor control systems as specified herein.
 - 2. The fans and associated accessories shall be the responsibility and be provided in the scope of the Odor Control supplier (OCS) as specified in Section 11395D - Headworks Odor Control System.

1.02 REFERENCES

- A. American Bearing Manufacturers Association (ABMA):
 - 1. 9 - Load Ratings and Fatigue Life for Ball Bearings.
 - 2. 11 - Load Ratings and Fatigue Life for Roller Bearings.
- B. Air Movement and Control Association International, Inc. (AMCA):
 - 1. 210 - Laboratory Methods of Testing Fans for Certified Aerodynamic Performance Rating.
 - 2. 211 - Certified Rating Program- Product Rating Manual for Fan Air Performance.
 - 3. 300 - Reverberant Room Method for Sound Testing of Fan.
 - 4. 301 - Methods for Calculating Fan Sound Ratings from Laboratory Test Data.
- C. ASTM International (ASTM):
 - 1. D4167 - Standard Specification for Fiber Reinforced Plastic Fans and Blowers.
 - 2. E84 - Standard Test Method for Surface Burning Characteristics of Building Materials.
- D. National Electrical Code (NEC).
- E. National Electrical Manufacturers Association (NEMA):
 - 1. 250 – Enclosures for Electrical Equipment (1000 V Maximum).
- F. National Fire Protection Association (NFPA):
 - 1. 90A - Standard for Installation of Air Conditioning and Ventilating Systems.
 - 2. 820 - Standard for Fire Protection in Wastewater Treatment and Collection Facilities.
- G. Occupational Safety and Health Administration (OSHA).
- H. Underwriters' Laboratories, Inc. (UL).

1.03 SYSTEM DESCRIPTION

A. Design requirements:

1. Provide fans that have sharply rising pressure characteristics which extend throughout the operating range and continue to rise beyond the efficiency peak.
2. Provide fans that peak as close as possible to the maximum efficiency and whose operating range is within the normal fan selection range.
3. When scheduled, provide guided vibration isolator for fans, so that not more than 10 percent of the vibration amplitude of the fan and motor is transmitted to the supporting structure.
4. Design fan inner scroll and air stream surfaces to maintain smoothness for entire fan service life.
5. Electrical components: UL listed and meeting the design and installation requirements of the NEC.
6. Motors supplied with fans: as specified herein in this Section and in Section 16222 - Low Voltage Motors up to 500 Horsepower, whichever more stringent requirements shall apply. Motor is not to be overloaded at any point on the fan curve including belt losses.
7. Belt drive systems: Where specified, and applicable to the application, provide adjustable drive for minimum within 5 percent speed change, rated for 1.5 times maximum horsepower motor available for the scheduled fan size or model.
8. Insulation and adhesives: Meet NFPA 90A requirements for flame spread and smoke generation.
9. Finishes: When not specified with fan type, coat ferrous metals as specified in Section 09960 - High-Performance Coatings.
10. Accessories: Provide accessories specified and those scheduled and as indicated on the Drawings. Refer to Section 16222 -- Low Voltage Motors up to 500 Horsepower for motor accessory specifics.

B. Performance requirements:

1. Performance requirements are included in the Fan Schedule located at the end of this Section.
2. Fan performance: Rated and licensed to bear the AMCA label in accordance with AMCA 210 and AMCA 211.
3. Total sound power levels in the 8-octave band range as measured in accordance with ASHRAE 68, AMCA 301, or AMCA 300 as appropriate for each fan.
4. Bearings: Rated per ABMA 9 or 11 for a L10 life rating of not less than 200,000 hours; provide greater life when specified with each fan type.
5. Provide fan vibration levels and field-testing method according to AMCA 204 Standard for fan balance and vibration, factory and in situ conditions.

C. Electrical and control system design:

1. Contractor to design and supply necessary electrical power and control systems, components, and wiring to make a complete functioning system. Design to perform the system ventilating functions as specified and as indicated on the Drawings.
2. Provide control panels as specified in the following Sections:
 - a. Section 11395D- Headworks Odor Control System

- b. Section 17101E – Specific Control Strategies – Headworks Odor Control
3. Provide vibration switch for each fan to shut down the fan in high vibration. The vibration switch shall be monitored from both the VLCP and the VMCP and shall provide high vibration alarm upon shutdown. Explosion proof vibration switch to be mounted on the fan and wired to alarm and shutdown fan in the event of excess fan vibration. The switch shall be made by Robertshaw, Murphy; or approved equal. The switch shall have a base mounted case, explosion-proof aluminum alloy housing; meets NEMA 7/IP50 specifications; Class I, Division I, Groups C and D; UL and CSA listed with 120 VAC rating, SPDT switch operation.
 4. Provide an indication of fan belt failure and Zero speed. Alarm on both. These alarms will also be transmitted to the plant area PLC via network communications.

1.04 SUBMITTALS

-  A. Submit as specified in Section 15050 - Common Work Results for Mechanical Equipment and specified herein.
1. Use the equipment tags shown on the Drawings for all submittals.
-  B. Product data:
1. Materials.
 2. Primary and ancillary equipment.
 3. Sound reduction wrap and estimated sound fan reduction in dba at 5 ft from the fan.
 4. Sound Power Level, dB, in each of 8 octave bands and overall dBA or Sones.
 5. Fan system layout, mechanical, electrical power, and control diagrams.
 6. Supports, vibration isolators, and seismic bracing calculations and details.
 7. Factory certified wheel and fan trim balance reports per AMCA 204 for appropriate BV grade and contractor to field balance the fan per AMCA 204 in situ levels for appropriate BV grade.
 8. Bearings and Bearings life calculations.
 9. Fan performance curves showing specified operating condition with multiple speed curves showing the fan will have no negative impacts when operating at a minimum speed producing the design cfm at 3 in w.c. up to the speed producing the design cfm at the design pressure specified in the fan schedule.
-  C. Provide vendor operation and maintenance manual as specified in Section 01782 - Operation and Maintenance Manuals.
1. Furnish bound sets of installation, operation, and maintenance instructions for each type of fan.
- D. Provide Manufacturer's Certificate of Source Testing as specified in Section 01756 - Commissioning.

1.05 SPARE PARTS

- A. Provide the following spare parts:
1. Inboard and outboard bearing set for each fan and motor.

2. Any other spare parts as recommended by manufacturer.
3. Contractor and fan manufacturer must specify during bidding the spare parts provided.

1.06 QUALITY ASSURANCE

- A. Provide fans:
1. Rated in accordance with AMCA.

1.07 DELIVERY, STORAGE, AND HANDLING

- A. Deliver units in 1 piece, factory assembled, internally wired, and lubricated.
- B. Protect equipment from dust and atmospheric exposure as recommended by the unit manufacturer:
1. As a minimum provide temporary closures for equipment openings designed for airflow.

1.08 WARRANTY

- A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 FRP BASE MOUNT CENTRIFUGAL FANS

- A. Manufacturers: One of the following or equal:
- ✓ 1. Hartzell Fan.(Basis of design)
 2. New York Blower Company.
 3. Verantis.
- ✓ B. Characteristics:
1. Fan design/construction: In accordance with ASTM D 4167.
 2. Fan housing: Solid fiberglass reinforced plastic (FRP).
 3. Sound reduction wrap: Fan shall have 1/8-inch-thick flexible fused reinforced loaded vinyl (LVR) sheet wrap with high-density insert filters (10 to 15 dBA sound reduction at 5 feet from fan). Other sound reduction wraps configuration can be acceptable as long as they provide the sound reduction performance criteria and the ease of maintenance for the fan and associated motor and components.
 4. Fan wheel: Backward inclined, non-overloading throughout entire curve at rated speed; solid FRP, totally encapsulated aluminum core insert for secure attachment to shaft.
 5. Shaft: Ground and polished ASTM A 108 grade 1040/1045 steel with Hastelloy sleeve in air stream or Type 316 stainless steel. Shaft to have maximum 1 mil run-out. Hardware in air stream shall be Type 316 stainless steel.
 6. Bearings: Minimum ABMA L-10 life rating of 200,000 hours, grease lubricated with lube tubes extended to outside of housing.
 7. Shaft seal assembly: Teflon for use with 316 stainless steel double lip.

8. Duct Connections: Provide flanged connections per manufacturer standard drill pattern.
9. Drain: Provide 1.5-inch diameter FRP pipe bonded to a low point in the housing; provide and install PVC ball valve and threaded plug where applicable. The drain connection to the inside of the housing should be smooth with no ridges or rough surfaces.
10. Fan inspection port: Provide minimum 8-inch round or square fan housing port to allow examination of interior.
11. Vibration switch: provide as specified in Section 11395D - Headworks Odor Control System.
12. Motors shall be severe and inverter duty TEFC, premium high-efficiency, 1.15 service factor, 3 phase/60hz /460V rated for Class 1 Division 2. Motors shall be stainless steel washdown/inverter duty induction type motors with sealed, lubricated bearings mounted on a corrosion resistant stainless-steel motor pedestal. No painted carbon steel bases shall be allowed. Motors shall meet the requirements of Section 16222 - Low Voltage Motors up to 500 Horsepower.
13. Bearings shall be heavy duty, self-aligning grease lubricated ball or roller type with a minimum 200,000-hour B-10 life. OSHA approved shaft guard shall be provided when fan wheel is not direct coupled to the motor.
14. The pressure loss of the combined odor control system and duct work shall not exceed the maximum pressure available from the fan at the specified air flow rate operating at non-overload conditions.
15. Electrical grounding: Coat air stream surfaces with graphite impregnated resin with grounding straps to motor frames or fan base to control static electricity.
16. Base and bearing support structures: Rigid Structural aluminum, or Type 316 stainless steel.
17. Duct connection orientation: As indicated on the Drawings. For fans at the Odor Control Facility, coordinate with the scrubber manufacturer.
18. Variable Frequency Drive (VFD):
 - a. Fan shall be VFD driven. Motor shall be inverter duty rated in accordance with NEMA MG-1 Part 31.
 - b. All variable frequency drives shall comply with 16265 – Reduced Harmonic Variable Frequency Drives.
19. FRP requirements:
 - a. Suitable for continuous operation when exposed to vapors and gases expected to include methane, hydrogen sulfide, chlorine gas, sulfur dioxide, gasoline vapors, ammonia, and water saturated air. Air stream may contain droplets of dilute sulfuric acid, greases, and particulate matter and may vary from 30 degrees Fahrenheit to 130 degrees Fahrenheit.
 - b. Housing resin:
 - 1) Type: Premium polyester resin.
 - 2) Resin shall achieve Class I flame spread rating below 25 when tested in accordance with ASTM E 84 without the use of additives.
 - 3) Manufacturer: One of the following or equal:
 - a) Ashland Chemical Company.
 - b) Derakane or Hetron.
 - c. Wheel resin:
 - 1) Type: Premium Vinyl ester resin.

- 2) Resin shall achieve Class II flame spread rating below 30 when tested in accordance with ASTM E 84.
 - 3) Manufacturer: One of the following or equal:
 - a) Dow Chemical Company.
 - b) Derakane 510A40 or 570A40; Ashland Chemical Company.
 - d. Protect fiberglass surfaces with ultraviolet resistant resin.
 - e. Provide Veil-Nexus corrosion resistant surfacing veil and additional coat of fiberglass reinforced plastic resin on surfaces exposed to the air stream for corrosion resistance and chopped strand fiberglass for structural core strength.
20. Balancing: Factory balance the fan wheel to G2.5 per AMCA 204 and the final fan/motor assembly to BV-4 per AMCA 204 and provide certified reports for approval prior to shipment. Field balance requirements are the responsibility of the contractor.
21. Fan Arrangement: Arrangement 1, belt driven acceptable to the Engineer per AMCA Standards to be installed as shown on the Drawings and recommended by fan manufacturer. Fan installation and arrangement shall not exceed the vibration limits specified in Section 15958 - Mechanical Equipment Testing when operating over the entire range of the speed.
22. Other accessories:
 - a. Inlet box:
 - 1) Provide FRP inlet box specifically sized for the fan selected inlet opening. The box shall be designed and provided by the fan manufacturer (not by the FRP or OCS/manufacturer) to optimize fan inlet conditions.
 - 2) Maximum pressure loss through inlet box shall be 1.0 inches water column at specified design flow rate.
 - 3) Inlet box shall be mounted and supported from integral fan/motor/inlet box base with rigid structural type 316 stainless steel supports. Contractor or Odor Control Supplier shall provide the inlet box supports, whether shown or not on the drawings.
 - b. Fan Mounting: Provide fans that are designed to be installed in a rigid mount. The fans shall be installed anchored to the concrete pad without the need for vibration isolators. The fan shall be properly leveled, grouted into place using epoxy grout as recommended by fan manufacturer and in compliance with AMCA 204 factory and vibration criteria for the appropriate BV grade.
 - c. Provide cover access doors: FRP, neoprene gasket, non-hinged, stainless-steel bolts.
 - d. Fasteners: Type 316 stainless steel and shall be encapsulated in FRP when exposed to air stream.
23. Master Vendor Control Panel (MVCP):
 - a. The fan shall be controlled through the MVCP that controls the associated facility Odor Control System as specified in Section 11395D - Headworks Odor Control System.

2.02 SOURCE QUALITY CONTROL

- ✓ A. Factory test fans listed on the Fan Schedule for proper operation and performance.
 - 1. Test witnessing: Witnessed.
 - a. Refer to Section 01756 - Commissioning for witnessed testing requirements.

PART 3 PROVIDE MANUFACTURER'S CERTIFICATE OF SOURCE TESTING AND FACTORY BALANCING REPORTSEXECUTION

3.01 EXAMINATION

- A. Examine and verify that Work is in condition to receive installation specified in this Section.
- B. Take measurements and verify dimensions to ascertain fit of installation.
- C. Ascertain support and openings are correctly located.

3.02 PREPARATION

- A. Before installation, remove dust and debris from equipment and ducts.
- B. During installation and until equipment is operated, protect equipment and ducts from dust and debris by covering openings with tape or plastic.

3.03 INSTALLATION

- A. Observe applicable installation requirements as specified in Section 15050 - Common Work Results for Mechanical Equipment.
- B. Anchoring and support:
 - 1. Provide anchoring and support for fans and appurtenances.
 - 2. Provide anchoring to sustain seismic and wind forces as specified in Section 01850 - Design Criteria.
- C. Adjust alignment of ducts where necessary to resolve conflicts with architectural features or to resolve conflicts with the work of other trades.
- D. Install and wire unit fans and controls in accordance with manufacturer's recommendations.
- E. Install flexible connections to fans.
- F. For fan housings with threaded water trap drain, provide drain piped from fan housing to the nearest drain channel, floor drain, or sump.

3.04 FIELD QUALITY CONTROL

- A. Testing, Training, and Start-Up: As specified in Section 01756 - Commissioning and as described below.

- B. Test equipment and installation to verify tightness, operation, and unit vibration is within AMCA 204 BV-4 limits and within manufacturer's submitted maximum, whichever is more stringent throughout the operating speed range. Submit vibration test reports for the fan speeds corresponding to the design point, minimum operating condition, and 3 other fan speeds in between.
- C. Test equipment performance and balance equipment as specified in Section 15954 - Testing, Adjusting, and Balancing for HVAC.

3.05 COMMISSIONING AND PROCESS START-UP REQUIREMENTS

- A. As specified in Section 01756 - Commissioning and this Section.
-  B. Require manufacturer's representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 5 trips:
 - 1. Installation assistance: 1 trip; 2 workdays:
 - a. Advise/observe the Contractor on the installation of the equipment.
 - 2. Installation inspection: 1 trip; 2 workdays.
 - 3. Start-up/performance testing assistance: 1 trip; 3 workdays:
 - a. Provide Certificate of Installation and Functionality Compliance per Section 01756 - Commissioning.
 - 4. Training: As defined in Section 01782 – Operation and Maintenance Manuals, Section 01756 - Commissioning and this Section:
 - a. Operations: 2-hour class, 2 sessions.
 - b. Mechanical Maintenance: 2-hour class, 1 session.
 - c. Combined Electrical and I&C Maintenance: 2-hour class, 1 session.
 - 5. Final acceptance checkout: 2 workdays (trip may be combined with training).
- C. Post start-up field visit: 1 trip; 1 workday.

3.06 SCHEDULES

- A. Fan Schedule:

FAN SCHEDULE

Equip. Tag	Equip. Name	Location	Fan						Motor			Additional Requirements (See listed Notes)
			Drive	Min. CFM	Min. ESP"	Max. RPM	Min Fan Diam	Radiated Sound Pressure	Max HP	V / Ph	Max. RPM	
✓ 07-FAN-1110	Headworks Odor Control Fan 1	Headworks Odor Control Area	Belt	23,000	9	1,400	Per Mfr	81 dba at 5 ft	60	460/3	1,800	8, 19
✓ 07-FAN-1120	Headworks Odor Control Fan 2	Headworks Odor Control Area	Belt	23,000	9	1,400	Per Mfr	81 dba at 5 ft	60	460/3	1,800	8, 19
Notes: 1. Provide bird screen. 2. Provide insect screen. 3. Provide backdraft damper, counterbalanced for minimum pressure loss. 4. Provide exterior weather hood. 5. Provide adjustable belt sheaves. 6. Provide vibration isolators. 7. Provide manufacturer std. motor enclosure. 8. Provide TEFC motor enclosure and wiring suitable for Class I, Division 2 locations. 9. Provide explosion proof motor and wiring suitable for Class I, Division 1 locations. 10. Provide variable frequency drive speed controller as specified in this Section. 11. Provide SCR speed controller as specified in this Section. 12. Provide exterior disconnect switch at fan, NEMA Type 3R. 13. Provide NEMA Type 1 disconnect switch at fan inside housing.						14. Interlock fan with motorized louver dampers, other fans, or equipment as indicated on the Drawings. 15. Provide replaceable filters. 16. Provide fiberglass canopy hood, size as indicated on the Drawings. 17. Fan selection based on existing opening; field verify opening size and select maximum sized, slowest rpm fan to fit opening and meet performance conditions. 18. Provide fire/smoke control system for all fans greater than 2,000 cfm. 19. Provide Inverter duty motor, VFD driven.						

END OF SECTION

HARTZELL SUBMITTAL

910 S. Downing St., Piqua, OHIO 45356
PH: (937) 773-7411 | Fax: (937) 773-8994



Hartzell Order:	2521509	Date:	Aug 25, 2025	Submittal Revision :	3
Customer:	Nan Inc.			Customer P.O.:	24077-00083
Transmitted To:	Hartzell Contact Information:				
03ES	- Technical Contact: Dakota Iddings diddings@hartzell.com ext3024				
Paul Scott	- All Other Inquires: customerservice@hartzell.com 937.773.7411				

THE FOLLOWING DOCUMENTS ARE TRANSMITTED FOR:

☐	Record Only including Hartzell Air Movement, Inc. Terms & Conditions of Sale – No Response Required
☐	For Approval including Hartzell Air Movement, Inc. Terms & Conditions of Sale – Written Authorization Required Prior to Production
☒	Revised for Approval – Written Authorization Required Prior to Production
☐	Final Copies

TYPES OF DRAWING(S) INCLUDED:

☐	Dimensional Drawings – standard General Arrangement drawing which may not include factory quoted items or all accessories
☐	As-Built Drawings – includes factory quoted items & all accessories
☒	Certified Drawings – As-Built drawings with registered PE signature, valid in State of Ohio

FOR APPROVAL

☐	APPROVED AS SUBMITTED
☐	NOT APPROVED – RESUBMIT

SIGNED BY: _____

TITLE: _____

COMPANY: _____

DATE: _____

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.03	<u>Line: 03 A41-0-271FA100FGI8MT - Fiberglass Backward Curved Centrifugal Fan</u>
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SUBMITTAL REVISION HISTORY



REV	Description of Change	Date
1	Original Submission	05/21/2025
2	Added Seismic Calculation and Isolator Package	06/24/2025
3	Added engineering comments to section 2 of submittal. Added bearing and inlet box calculations to each line item. Updated line 3 fan drawing.	08/25/2025

SECTION 02 HARTZELL EXCEPTIONS



Hartzell reserves the right to take exception to specifications delivered, if section is blank a specification has not been received and will incur a Change Order fee to review specification after initial submittal.

Markup Legend (if applicable):



Applicable to Fan(s) Submitted upon



Exception taken for Item Listed



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 15832-001.0
Date: 8/18/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: ODOR CONTROL FANS
Reviewed By: Khalil Kairouz

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☒	Make Corrections Noted - Confirm
Not Approved	☐	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. The 50 HP motor provided is acceptable in lieu of the 60 HP motor specified. However, any modifications to the electrical design, if any, should be at the cost of the contractor, otherwise provide a 60 HP motor as specified.
2. Provide calculations showing the L10 bearing life meet or exceed the specified 200,000 hours per section 2.01 B 13.
3. Provide calculations showing the maximum pressure loss does not exceed the 1" w.c. as specified in section 2.01 B. 22.a.2
4. Verify per section 3.05 the manufacturer representative services.
5. Verify that fan manufacturer testing will be performed per section 2.02. The factory testing will be witnessed by the engineer. Provide 3 weeks advance notice prior to factory testing.

HARTZELL SUBMITTAL

RESPONSE TO INQUIRIES

Hartzell Sales Order: 2521509



Sales Order: 2521509 Date: 8/20/2025
Re: Nan Inc. Review Comments Response

To Whom It May Concern,

In response to inquiries made on this submittal, Hartzell's response is as follows:

- 1) The 50 HP motor provided is acceptable in lieu of the 60 HP motor specified. However, any modifications to the electrical design, if any, should be at the cost of the contractor, otherwise provide a 60 HP motor as specified. There are no modifications to the electrical design. The 50 HP motor will be provided.
 - a) There are no modifications to the electrical design. The 50 HP motor will be provided.
- 2) Provide calculations showing the L10 bearing life meet or exceed the specified 200,000 hours per section 2.01 B 13.
 - a) The L10 bearing calculations have been added to the revised submittal under each associated line.
- 3) Provide calculations showing the maximum pressure loss does not exceed the 1" w.c. as specified in section 2.01B. 22.a.2
 - a) The inlet box maximum pressure loss calculations have been added to the revised submittal under each associated line.
- 4) Verify per section 3.05 the manufacturer representative services.
 - a) Verification of the manufacturer's services is provided on page 148 of the submittal package.
- 5) Verify that fan manufacturer testing will be performed per section 2.02. The factory testing will be witnessed by the engineer. Provide 3 weeks advance notice to factory testing.
 - a) Verification of the manufacturer's testing is provided on page 147 of the submittal package.
- 6) Line item 3 (fan tag 02-EF-1390 & 02-EF-1396): The drawing was originally submitted with an inlet box. After reviewing the specifications, it was confirmed that an inlet box is not needed for the fans on line 3. Additionally, there was no inlet box included on the order for line 3. This was an error by Hartzell and has been corrected. There is a small deduct to remove the inlet box drain that is listed on line 3. This was a typo and was added by mistake to the line. The updated drawing has been added to the revised submittal.

Appreciatively,

Dakota Iddings

SECTION 03.01 LINE ITEM TECHNICAL INFORMATION



Line No.: 2521509-01

Description: A41-1-402FA100FGFQT3 - Fiberglass Backward
Curved Centrifugal Fan

Tag: 07-FAN-1110/1120



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 1 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date:	5/19/2025 8:45:57 AM	Ship Method:	PPD CC
Salesman Code:	99WE	FOB:	Piqua
Partial Ship:	False	AR Terms:	Per Invoice

Ship To : WILL ADVISE
WILL ADVISE
WILL ADVISE

Contact Name:

Ship To Phone:

Ship To Fax:

Line Nbr	Item ID	Req Date	Req Qty	Unit Price	Extended Price
	Item Name			Unit of Measure	

1 HFA/PBA-FRP
HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE

12/31/2025

2.00

Each

A41-1-402FA100FGFQT3

41 - Fiberglass Backward Curved Centrifugal Fan

Tag: 07-FAN-1110/1120

Performance

Volume Flow Rate: 23000.0 cfm

SP: 9.0 in. w.g. / TP: 9.37 in. w.g.

RPM: 1230.0

Density: 0.075 lbs/ft³

Operating Temp: 70 °F

OPwr: 41.375 hp / SPwr: 41.375 hp

Motor

Operating Voltage: 460

HP: 50, RPM: 1750, Volts: 460, Hertz: 60, Phase: 3

Enclosure: TEFC IEEE-841, Frame Size: 326T, 1 Speed / 1 Winding

Drill and Tap: No

Special Features: MOTOR SPEC. 16222, INCLUDES 120V SPACE HEATERS. INTERNAL AEGIS RING w/RR#9, MEETS STD. 841, INVERTER READY NAMEPLATED, ---NOTE: NO RTD'S REQUIRED.

Vendor: BALDOR/ABB, Vendor P/N: ECP84115T-4-M21A-M39B-1, Vendor Lead Time: 2-4 weeks

Extended Electrical Leads: None, Extended Lube Lines:Yes

Fan

Arrangement: 01

Rotation and Discharge: Bottom Horizontal (Clockwise)

Mounting Location: Floor

Motor Position: W

Material: Fiberglass

Construction: ASTM D4167 97 (Includes Single Veil & Electric Grounding)

Fan Coating: Vinyl Ester

Prop/Wheel: FA - Backward Airfoil Centrifugal Wheel - Fiberglass

Prop/Wheel Coating: Electric Grounding

Standard Accessories (Included)

Water Slinger

Inlet Support, Steel - Inorganic Zinc Coating w/ Epoxy Topcoat

WHERE IS TEMP
SWITCH?



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 2 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 99WE
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:

Ship To Phone:

Ship To Fax:

Line Nbr	Item ID	Req Date	Req Qty	Unit Price	Extended Price
	Item Name			Unit of Measure	

Additional Accessories
Bolted Access Door, Fiberglass
Drain, CPVC w/ CPVC Plug (Factory Installed)
Extended Lube Lines to Fan Bearings, 316SS
316SS Hardware to Complete Fan
316SS Shaft
Teflon Double Lip Seal w/ purge option in a 316SS Casing
Drilled Inlet Flange, PS15-69
Drilled Outlet Flange, PS15-69
Inlet Box, Fiberglass w/ 316SS Support Legs ; 12:00 Orientation
Arrangement #1 Sub Base, 316SS

1.03, B, 4 – HARTZELL – L-10 life rating of 200,000 bearings.

Vitech C1D1 Vibration Switch 
Description and Dimensions: PART# 6124-083
HARTZELL – Quote neoprene gasketing-ADDER , (For bolted access door.)

316 SST base and bearing supports and inlet support.
Material of Construction: 316 Stainless Steel,

LVR Sound Reduction Wrap

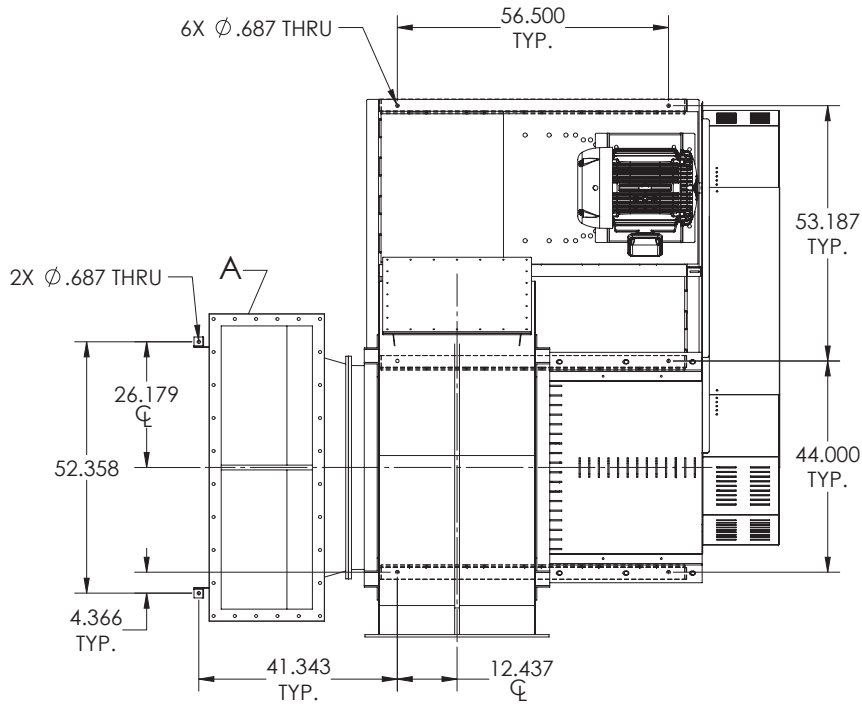
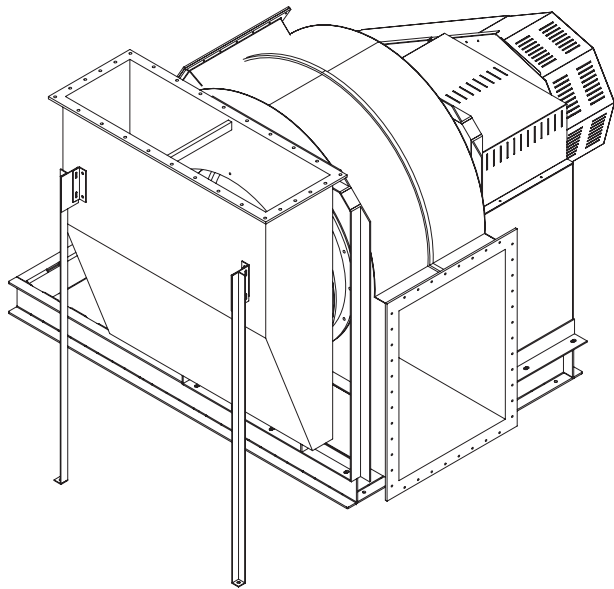
Shaft & Belt Guard
Material of Construction: 316 Stainless Steel,

Spare set of bearings

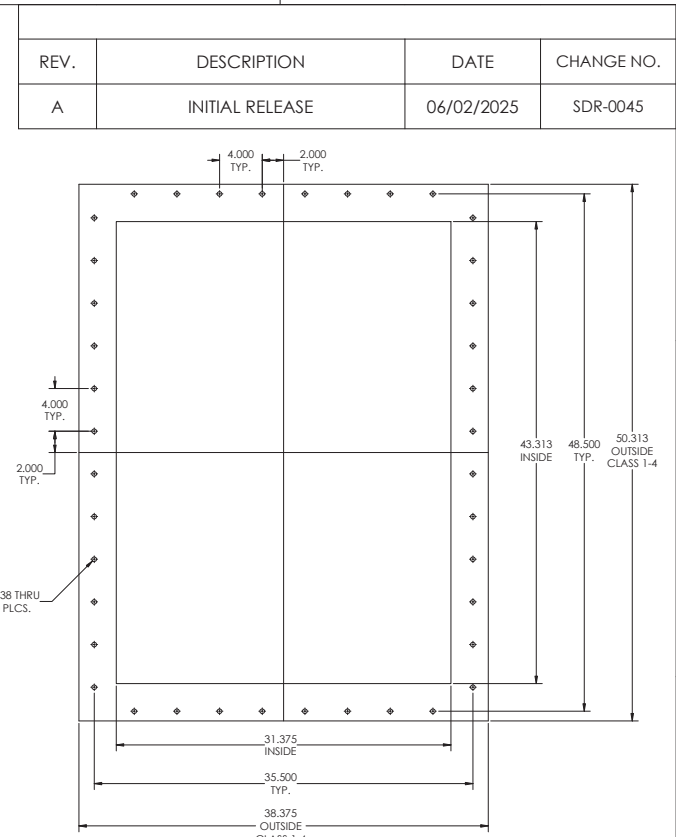
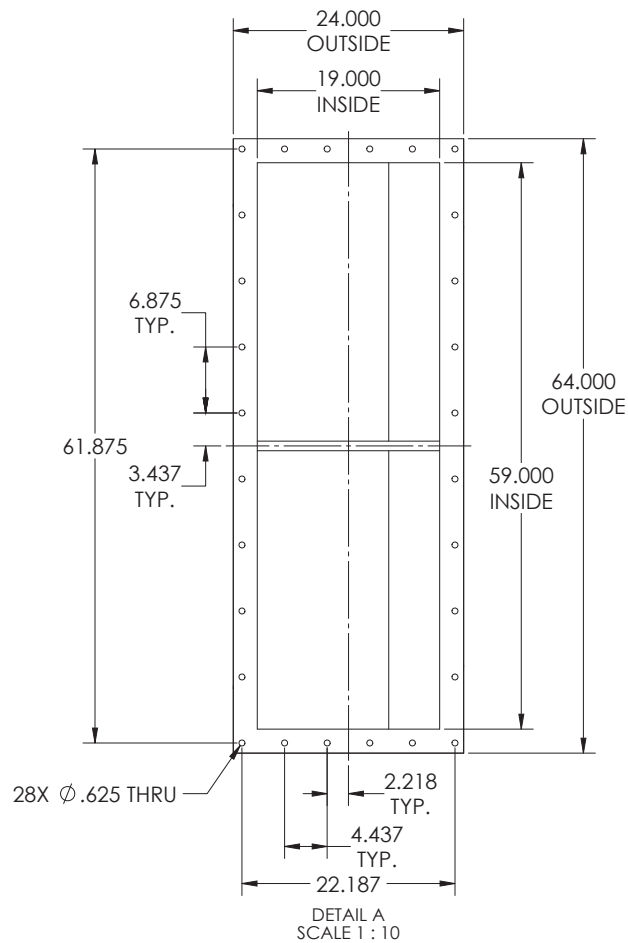
Total Weight (approx.): 3276 lbs. (Each) (quoted items are not included in the weight.)

Blanket Qty: 0.00

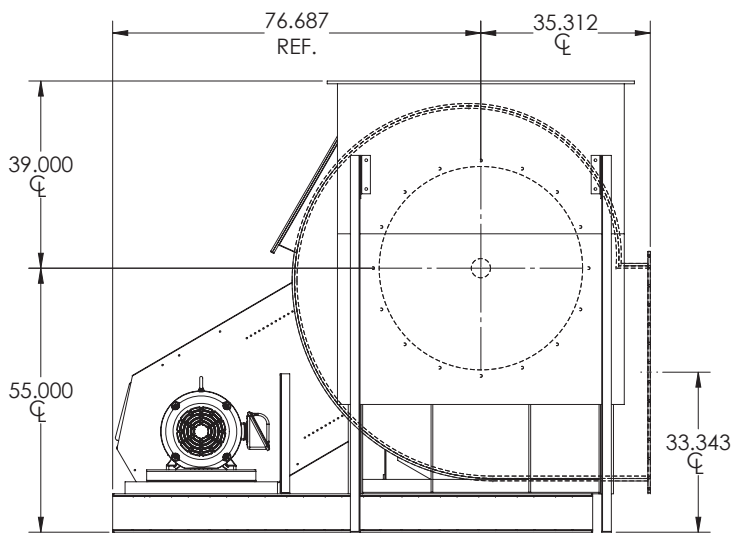
Blanket Exp Date:



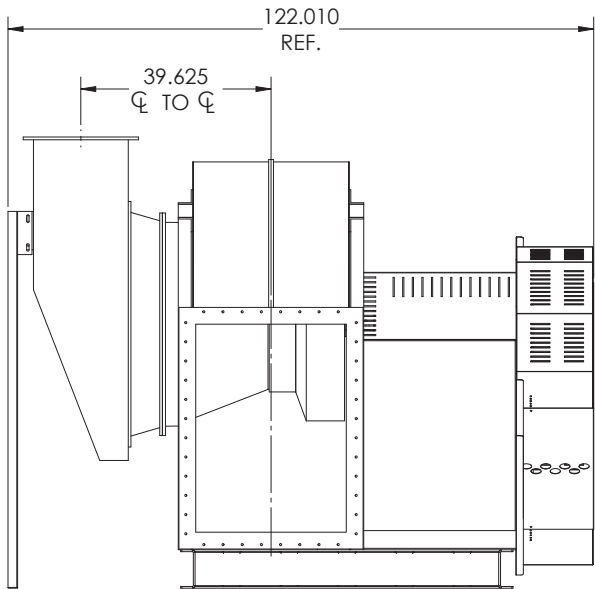
TOP VIEW



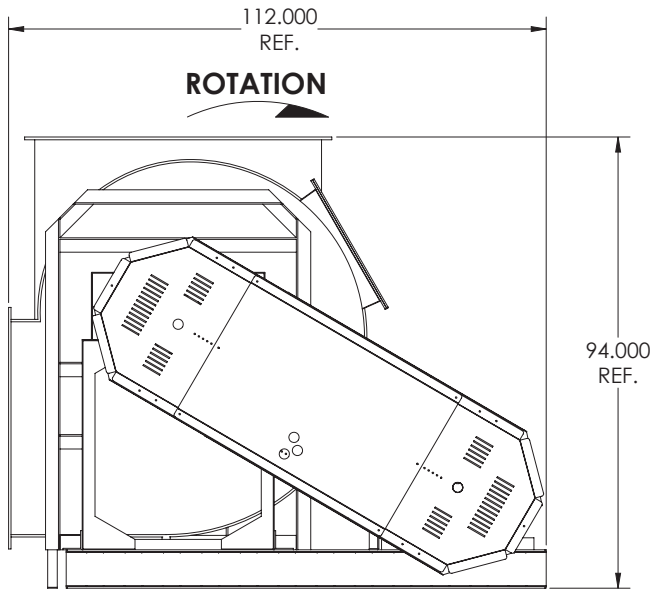
OUTLET FLANGE



INLET SIDE VIEW



FRONT VIEW



DRIVE SIDE VIEW

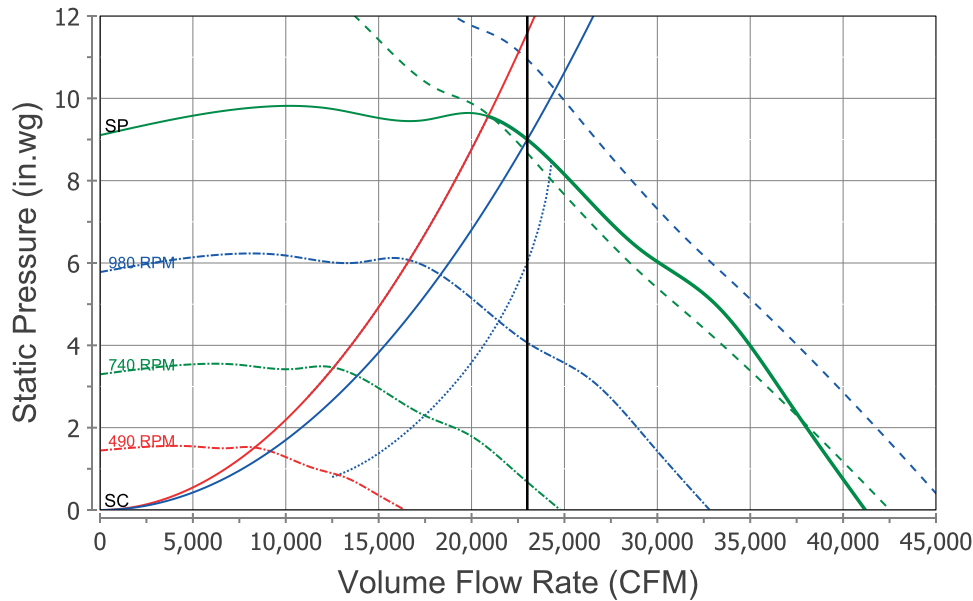
CERTIFIED PRINT
BY John Harty DATE: 06/02/2025

SALES ORDER # 2521509
CUSTOMER PO # 24077-00083
FAN TAG # 07-FAN-1110/1120
APPROX. WEIGHT (lb): 3596

TYPE	PRODUCT SERIES	ARRANGEMENT / DUTY / COLOR			SIZE	# OF BLADES / CLASS		BLADE / WHEEL		BLADE ANGLE / WHEEL WIDTH		MATERIAL OF CONSTRUCTION	MOTOR ENCLOSURE	MOTOR HP/RPM		ADDITIONAL INFORMATION:			
		ARRANGEMENT	DUTY	COLOR		# OF BLADES	CLASS	BLADE	WHEEL	BLADE ANGLE	WHEEL WIDTH			HP	RPM	MOTOR FRAME	BLADE OD	ROTATION	DISCHARGE
A	41-	1			40		2		FA		100	FG	FQ	50	1750	326T		CW	BH

Hartzell AIR MOVEMENT PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF HARTZELL AIR MOVEMENT. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF HARTZELL AIR MOVEMENT IS PROHIBITED. REL LOCATION: C:\Vault\Custom\ORDERS\2520000\2521509\	MATERIAL INFORMATION MATERIAL: SEE BOM MATERIAL THICKNESS WEIGHT: 3596 SURFACE AREA: 155861 PERIMETER LENGTH	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES DIMENSIONAL TOLERANCE = ±1/8" ANGULAR = MACH. ± 5° BEND ± 1° DIMENSIONS MARKED AS REF. ARE NOT SUBJECT TO TOLERANCE.		Hartzell Air Movement Piqua, Ohio 45356	
		DRAWN BY: NDW	DATE: 6/2/2025	TITLE: 41-40" 100% ARR. 1 FIBERGLASS BACKWARD CURVED CENTRIFUGAL FAN	
		CHECKED: APPROVED:		SIZE: DWG. NO. D 41-2521509-01	
				REV A	

Hartzell-FLOW™ v1.0.18 / February 2019
A41-1-402FA100FGFQT3



Fan Tag#: 07-FAN-1110/1120

Vol Flow Rate	23000
Pressure	9
Density (lbs/ft³)	0.075
Oper. Temp. (°F)	70
Fan RPM	1230
Max Safe RPM	1376
Operating Power	41.375
Standard Power	41.375
Static Efficiency	0.787
Outlet Velocity (fpm)	2439
Fan Energy Index (FEI)	1.31
Fan Efficiency Grade (FEG)	FEG85

Discharge Sound Power Levels referred to 10 ⁻¹² watts							
1	2	3	4	5	6	7	8
100	97	99	96	94	95	85	78

Radiated Sound Power Levels referred to 10 ⁻¹² watts							
1	2	3	4	5	6	7	8
96	90	89	85	82	82	72	65
Radiated Sound is not AMCA Licensed							

Hartzell Air Movement certifies that the model shown is licensed to bear the AMCA Seal. The ratings shown are based on tests and procedures performed in accordance with AMCA publication 211 and AMCA publication 311 and comply with the requirements of the AMCA Certified Ratings Program. FEI values are calculated in accordance with ANSI/AMCA Standard 208 and are based on four-pole TEFC motors of the size shown. Power rating BHP excludes transmission losses. Performance certified is for Installation Type D: Ducted Inlet, Ducted Outlet. AMCA Licensed for Sound and Air Performance. Performance ratings do not include the effects of appurtenances (Accessories). Sound ratings are based on sound level data obtained in accordance with AMCA Standard 300. The sound power level ratings shown are in decibels, referred to 10⁻¹² watts, calculated per AMCA Standard 301. Fan Outlet Sound Testing. Values shown are for outlet Lwo sound power levels for: Installation Type D: Ducted Inlet, Ducted Outlet. Ratings include the effects of duct end correction.

Discharge Sound Pressure = 89 dBA @ 5ft

Radiated Sound Pressure = 77 dBA @ 5ft

Discharge Sound Power = 100 LwA

Radiated Sound Power = 89 LwA

The A-weighted sound pressure level (dBA) is based on Hartzell Laboratory sound power tests, and is calculated in accordance with AMCA standard 303.

The FEG, dBA, LwA and radiated values are not AMCA International Licensed.

The calculation assumes a free field condition with a directivity factor for hemi-spherical radiation (Q=2).

The installed sound pressure levels are influenced by the installation and acoustic environment, and cannot be guaranteed. Use of this estimate level along for field acceptability test is not recommended.

Although the calculation can be done for any stated distance, the free field does not start until 20 to 50 ft from the equipment in most installations.

Contact Hartzell Air Movement for more information concerning dBA values.



Fan Shaft Critical Speed and Bearing Life Analysis

Model: 412-40FA100

Reference: 2521509-01

Actual Fan Diameter (in.): 40.25

Tip Speed: 12,961

Motor Horsepower: 50

Fan Sheave Dia. (in.): 10.9

Sheave Wt. (lbs.): 30

Shaft Speed (rpm): 1230

Shaft Diameter (in.): 2.9375

Wheel Wt. (lbs.): 242.3

Shaft Material: 316 SS

Tensile Yield Strength: 30000

Hub Material: 1020 HR

Tensile Yield Strength: 30000

Arrangement No.: 1

Fan Air Temp (deg. F): 70

Ambient Temp (deg. F): 70

Belt Tension Factor: 1.75

Belt Angle from 6 O'Clock (deg.): 45

Shaft Modulus of Elasticity (lbs/in^2): 2.92E+07

Wheel WR2 (lb-ft^2): 422.7

WR2 Factor: 0.5

Distance From Wheel to Drive Bearing (in.): 41.321

Distance Between Bearings (in.): 28.6875

Distance From Drive Bearing to Sheave (in.): 7.063

Wheel Overhang: 12.6335

	w/o Belt Pull	w/Belt Pull
Critical Speed	4453	3000
Safety Factor	3.620	2.439

G Shock Load: 10

Minimum Shaft Dia.: 2.260

Hub O.D.: 4.750

Hub Wall: 0.90625

Static Pressure (neg. for downblast): 9

Expeller Vanes?: No

Minimum Bearing L10 Life: 200,000

Shaft Orientation: Horizontal

Belt Pull: 823

Belt Pull: 823

Key Size: 0.75

Key Length: 0.310

Load on Bearing A (Wheel): 300

Load on Bearing B (Drive): 1007

Total Axial Thrust: 163

Bearing A: Fixed

Bearing B: Fixed

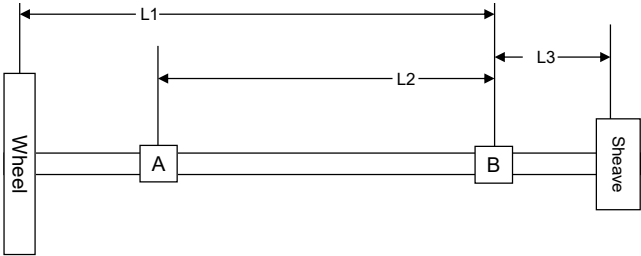
Base Load

Base Load

Series	Position	Dynamic Rating	Status	Equivalent Load	Fixed or Expansion	L10 Life
Link-Belt B22400	A	41,800	Good Selection	495	Fixed	>1 million
	B	41,800	Good Selection	1203	Fixed	>1 million

V04155

5/21/2025



L1

L2

L3

Thrust Multiplier

0.67

Today is: Wednesday, May 21, 2025



Document:

ES 7.5.1

Rev

1Title: **Inlet Boxes for Centrifugal Fans – Series 03, 05, and 41**Owner: **Engineering**Document Type: **Engineering Standard**Creation Date: **5/2/1989**

Inlet Boxes for Centrifugal Fans – Series 03, 05, and 41

1.0 Scope

This standard defines the dimensional and performance requirements for inlet boxes used on centrifugal fans

2.0 Products Affected

- 2.1 Series 03 Backward Curved Centrifugal Fan, Single Width
- 2.2 Series 05 Industrial Exhausters
- 2.3 Series 41 Fiberglass Backward Curved Centrifugal Fans

3.0 Dimensions

Inlet dimensions are based on the fan wheel diameters as shown below. For Series 03 and Series 41 fans, they are approximately 1.5 wheel diameters wide by 0.5 wheel diameters deep. For Series 05 fans, they are approximately 1.0 wheel diameters wide by 0.5 wheel diameters deep

Series 03 / 41			Series 05		
Fan Size	Inlet Width (in.)	Inlet Depth (in.)	Fan Size	Inlet Width (in.)	Inlet Depth (in.)
12	18	6	12	12	6
15	24	8	15	16	8
18	28	10	19	20	10
22	34	12	22	22	12
24	36	12	26	26	14
27	42	14	29	30	14
30	46	16	33	34	16
33	50	16	36	36	18
36	54	18	40	40	20
40	69	20	45	46	22
44	66	22	50	50	26
49	74	24	57	58	28
54	86	28			
60	96	30			
66	105	34.25			
73	116	38			
81	129	41.5			
89	141.75	46			
98	156.5	51			

Actual Inlet Vel. - 2760 fpm
Actual Press. Drop - 0.61 in w.c.

4.0 Performance

The pressure drop through an inlet box is based on tests and is shown below. Use the inlet areas listed above to calculate the inlet velocity. These values include all system effects

Inlet Velocity (fpm)	1300	1500	2000	2500	3000	3500	4000	4500	5000
Pressure Drop (in W.G.)	0.10	0.16	0.31	0.50	0.72	0.99	1.30	1.64	1.97

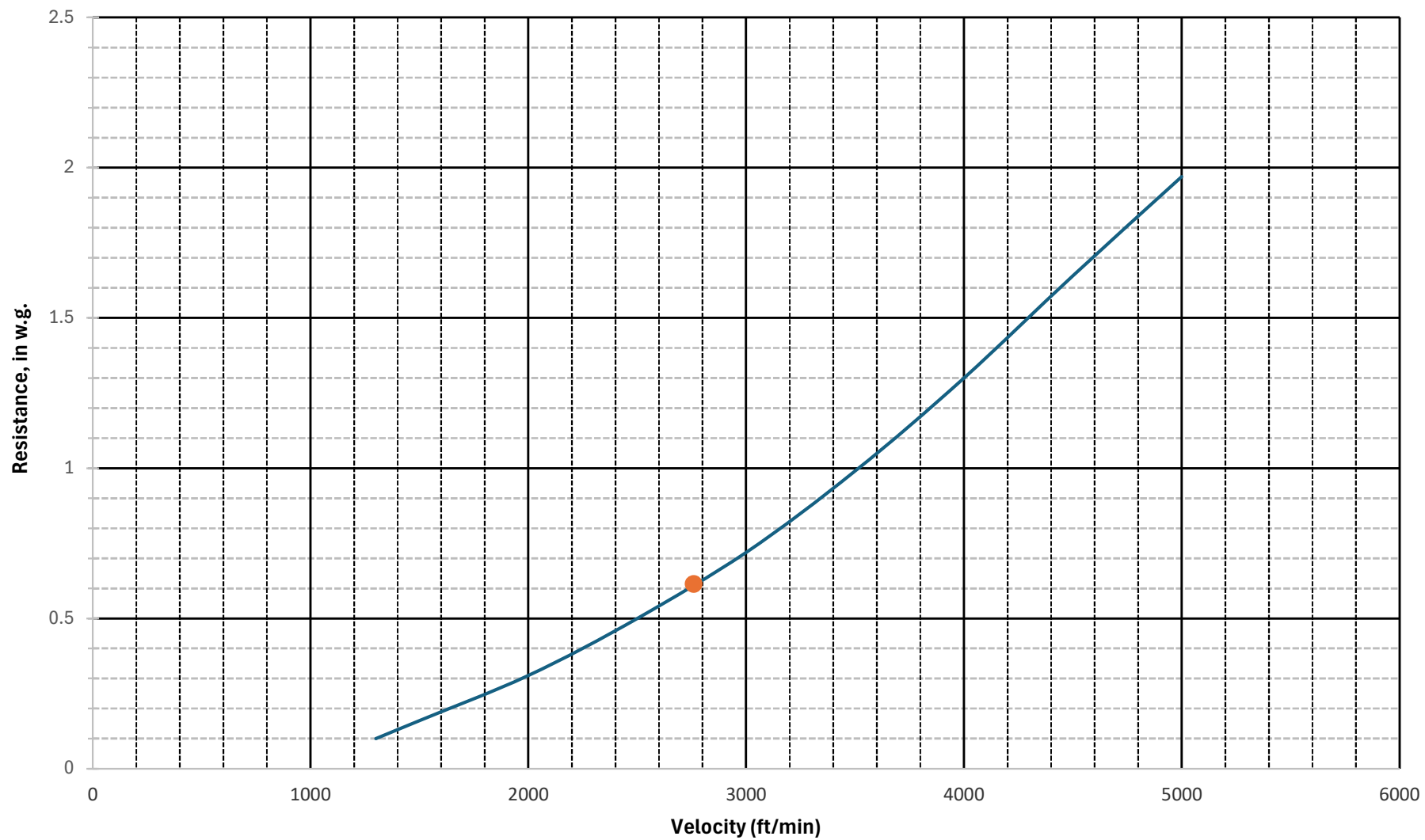
5.0 References

- 5.1 N/A

6.0 Revision History

Rev	Description	Date	Changed By
1	Remove from Drive System and Release in QMS	10/30/2024	J.Felver

Inlet Box Resistance





Customer information packet

ECP84115T-4

50HP, 1770RPM, 3PH, 60HZ, 326T, TEFC, F1

Class - CLI GP A,B,C,D

Division - Division II

Specifications

Enclosure	TEFC
Frame	326T
Frame Material	Iron
Frequency	60.00 Hz
Haz Area Class and Group	CL I GP A,B,C,D
Haz Area Division	Division II
Motor Letter Type	Three Phase
Output @ Frequency	50.000 HP @ 60 HZ
Phase	3
Synchronous Speed @ Frequency	1800 RPM @ 60 HZ
Voltage @ Frequency	460.0 V @ 60 HZ
Agency Approvals	CCSA US CSA EEV NEMA PREMIUM NEMA_PREMIUM UR
Ambient Temperature	40 °C
Auxillary Box	No Auxillary Box
Auxillary Box Lead Termination	None
Base Indicator	Rigid
Bearing Grease Type	Polyrex EM (-20F +300F)
Blower	None
Constant Torque Speed Range	1
Current @ Voltage	57.000 A @ 460.0 V
Design Code	B
Drip Cover	No Drip Cover
Duty Rating	CONT
Efficiency @ 100% Load	94.5 %
Electrically Isolated Bearing	Not Electrically Isolated
Enclosure Modification	841
Feedback Device	NO FEEDBACK
Haz Area Temp Code	T3

Part detail

Revision	M
Type	AC
Mech. spec.	
Base	
Status	PRD/A
Elec. spec.	12WGX731
Layout	12LYG135
Eff. date	03-21-2025
CD Diagram	CD0006
Poles	04
Leads	3#8
Proprietary	False
Created date	12-12-2018

Heater Indicator	No Heater
High Voltage Full Load Amps	57.0 a
Insulation Class	F
Inverter Code	Inverter Duty
IP Rating	IP56
KVA Code	G
Lifting Lugs	Standard Lifting Lugs
Locked Bearing Indicator	Locked Bearing
Max Speed	2700 rpm
Motor Lead Termination	Ring Terminals
Motor Standards	NEMA
Motor Type	1272M
Mounting Arrangement	F1
Number of Poles	4
Overall Length	30.74 IN
Power Factor	87
Product Family	Chem Process S/P 32-8 IEEE 841
Pulley Face Code	Standard
Rodent Screen	None
Service Factor	1.15
Shaft Diameter	2.125 IN
Shaft Ground Indicator	No Shaft Grounding
Shaft Rotation	Reversible
Speed	1770 rpm
Speed Code	Single Speed
Starting Method	Direct on line
Thermal Device - Bearing	None
Thermal Device - Winding	None
Vibration Sensor Indicator	No Vibration Sensor
Winding Thermal 1	None
Winding Thermal 2	None

Nameplate

NP4332

STK.NO.	ECP84115T-4			P/N					ENCLOSURE		TEFC					
SPEC.	12-0000-0367			CC	010A	FRAME		326T	S/N							
 HP	50			CLASS		F	HZ	60	ODE BRG		6312	DE BRG		6312		
RPM	1770	RPM MAX	2700	PH	3	DES		B	D.E. BRG.		60BC03X30X					
VOLT	460			KVA-CODE		G		O.D.E. BRG.		60BC03X30X						
AMP	57			MOTOR WEIGHT				679	GREASE	POLYREX EM						
RATING	40C AMB-CONT			MAX. KVAR				9.4	BLANK							
NEMA-NOM-EFF	94.5	# OF ROTOR BARS			40	SER.F.		1.15	IP	56						
G.MIN.EFF	93.6	# OF STATOR SLOTS			48	PF	87	INV.TYPE		PWM						
TEMP CODE	T3	INVERTER-TEMP-CODE				200										
TEMP =	200	C HP FR	60	C HP TO	90											
CT HZ FROM	1	CT HZ TO	60													
HTR-VOLTS	HTR-AMPS			HTR-WATTS				MAX. SPACE HEATER TEMP.					VT HZ FROM	O	VT HZ TO	60

AC Induction Motor Performance Data

Record # 93262

Typical performance - not guaranteed values

Winding: 12WGX731-R055			Type: 1272M	Enclosure: TEFC	
Nameplate Data			460 V, 60 Hz: Single Voltage Motor		
Rated Output (HP)		✓ 50	Full Load Torque	148 LB-FT	
Volts		460	Start Configuration	direct on line	
Full Load Amps		✓ 57	Breakdown Torque	446 LB-FT	
R.P.M.		1770	Pull-up Torque	183 LB-FT	
Hz	60	Phase	3	Locked-rotor Torque	254 LB-FT
NEMA Design Code	B	KVA Code	G	Starting Current	362.5 A
Service Factor (S.F.)		1.15	No-load Current	18.3 A	
NEMA Nom. Eff.	94.5	Power Factor	87	Line-line Res. @ 25°C	0.118 Ω
Rating - Duty		40C AMB-CONT	Temp. Rise @ Rated Load	61°C	
S.F. Amps			Temp. Rise @ S.F. Load	75°C	
			Locked-rotor Power Factor	24.4	
			Rotor inertia	9.59 lb-ft²	

Load Characteristics 460 V, 60 Hz, 50 HP

% of Rated Load	25	50	75	100	125	150	S.F.
Power Factor	58	78	85	87	87	87	88
Efficiency	92.1	94.7	94.9	94.5	93.9	92.9	94.1
Speed	1791	1785	1779	1772	1765	1756	1768
Line amperes	22.3	31.9	43.5	56.9	71.3	87.1	65.5

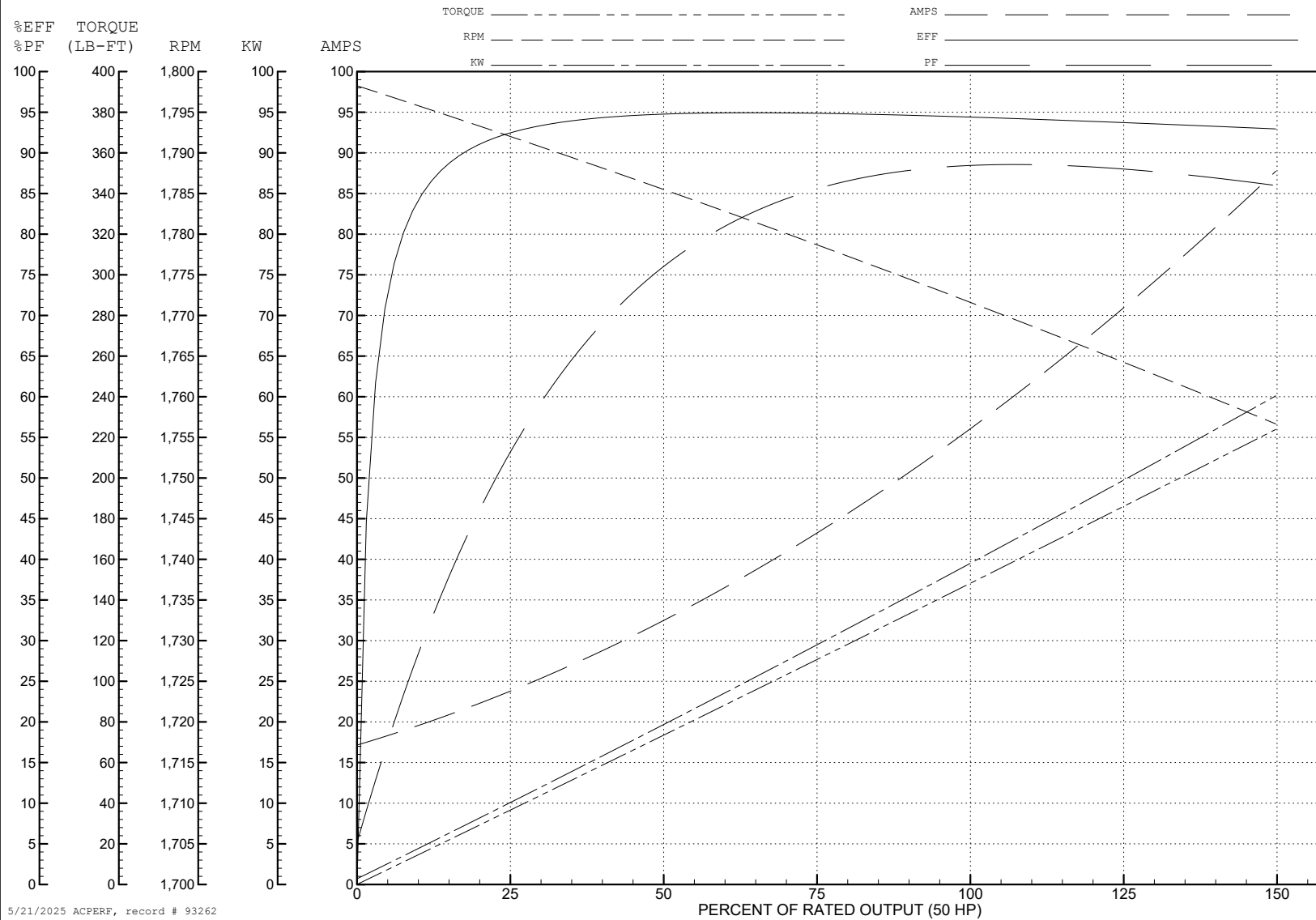
ABB Motors and Mechanical Inc.

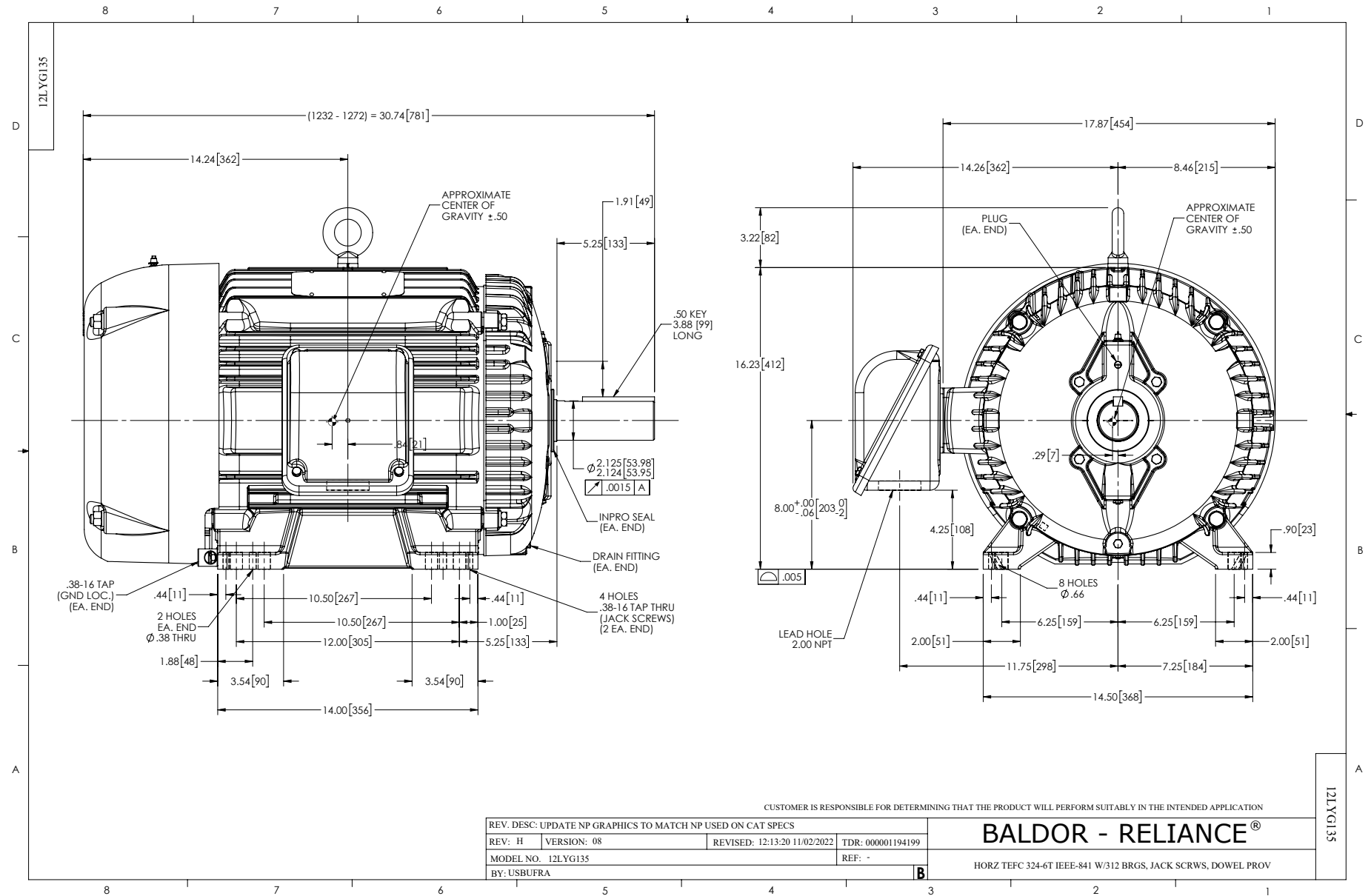
WINDING # 12WGX731

Typical performance - not guaranteed values.

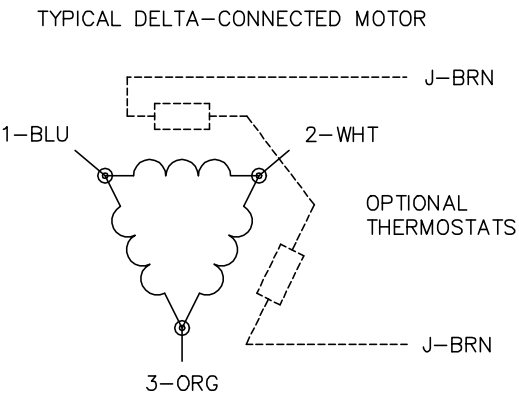
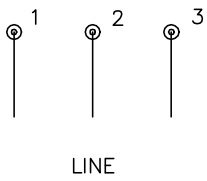
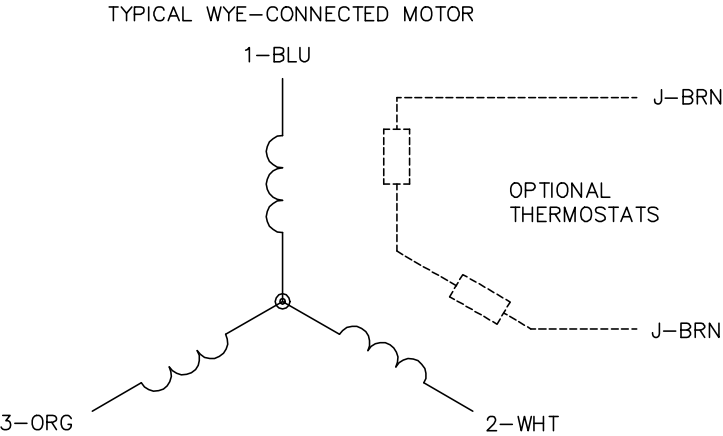
50 HP 3 PH 60 HZ 1770 RPM 460 V 1272M

TORQUES (LB-FT): PO=446 PU=183 LR=254 LRA=362.5





CD0006



- NOTES:
- 1. THREE LEAD MOTOR MAY BE EITHER WYE CONNECTED OR DELTA CONNECTED.
 - 2. INTERCHANGE ANY TWO LINE LEADS TO REVERSE ROTATION.
 - 3. OPTIONAL THERMOSTATS ARE PROVIDED WHEN SPECIFIED.
 - 4. ACTUAL NUMBER OF INTERNAL PARALLEL CIRCUITS MAY VARY.
 - 5. LEAD COLORS ARE OPTIONAL. LEADS MUST BE NUMBERED AS SHOWN.

CD0006

REV. DESC: ADD CLASS CONN00000007		
REV. LTR: E	VERSION: 01	TDR: 000001099922
FILE: \AAA\00005\141	REVISED: 10: 24: 49 02/19/2019	BY: ENBRIRO
MTL: —		

BALDOR - RELIANCE®

3PH, SV, 3 LEADS, WYE OR DELTA CONNECTED

SH 1 of 1

Motor Mod Express®

Modification number	Motor Mod Express®
Balance	
M1A	Dynamic balance of rotors
Bearings	
M2A	Sealed
M2B	Ball to roller 1800 RPM or below
M2C	RPM AC ball to roller
M2F	Add isolated bearings NEMA 56 through 449T (SKF Insocoat or equivalent)
Brake motors	
M3D	Change stearns brake voltage
Conduit box	
M4A	Tap or provide additional lead hole
M4B	Rotate conduit box lead opening
M4F	Replace die cast aluminum conduit box
M4G	Add conduit box for thermostat or heater leads
Cord	
M5A	Install line cord in conduit box or terminal panel
M5B	Install 6 foot, 3 conductor, type SO cord with plug
Drains	
M7A	Add condensation drain holes
M7C	Close standard condensation drain holes
M7D	Add breather drains to cast iron explosion-proof motor
Dripcovers	
M8A	Install dripcover on TEFC or ODP
Leads	
M10A	Install terminal lugs
M10B	Reconnect motor from nine leads to three
M10C	Reconnect motor to high or low voltage in conduit box
M10D	Longer leads, additional leads added
Lubrication	
M11A	Install low temperature lubricant in bearings
M11B	Add high temperature grease
Seals	
M12A	install oil seals
M12B	Install Inpro/Seal® drive end only
M12C	Install V-ring seals to motor

Motor Mod Express®

Modification number	Motor Mod Express®
Mountings	
M13A	Add rigid base
M13B	Install NEMA C-face – steel band motor construction
M13C-1	Install NEMA C-face – cast iron construction
M13C-2	Install NEMA C-face – IEEE 841 motors
M13D	Add D-flange
M13E	Remove rigid base from foot mounted steel band motors
M13F-2	Convert mounting configuration for cast iron motor
M13G	Install C-Face to explosion-proof motors
M13I	Labor only to mount grinder or buffer onto a pedestal and package for shipment
Mounting drawings	
M44A	RPM AC mounting conversions
Export packaging	
M14A	Open crate packaging
M14B	Totally enclosed crate packaging
M14C	Motor and nameplate photograph
Nameplate	
M15A	Provide original nameplate for remote mounting
M15B	Replace nameplate
M15C	Laser engrave nameplate into stainless steel motor
M15E	Mail nameplates to customer
Blowers	
M16A	Add constant velocity blower unit
M16B	Convert from a 3 phase to a single phase blower
Paint	
M17A	Paint motor Baldor-Reliance® stock paint color
M17B	Paint motor or parts customer specified color
Lifting lugs	
M18A	Add lifting lugs up to a 326T frame size
Hardware	
M20A	Exchange plated hardware with stainless steel
Space heaters	
M21A	Add space heaters to TEFC or open motors
Thermal protectors	
M23A	Thermostats
Winding thermocouples	
M24A	Iron constantan (3 per motor)
Winding thermistors	
M25A	Thermistors

Motor Mod Express®

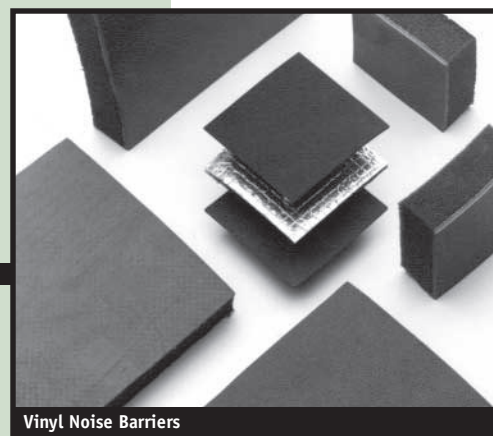
Modification number	Motor Mod Express®
Tropicalization	
M26A	Protection of windings and mechanical parts
Weatherproofing	
M27A	Treatment of windings and mechanical parts
Shaft modifications	
M29A	Drill and tap one hole in end of shaft or mill flat on shaft extension
M29B	Convert from TEFC to TEAO
M29C	Threading
M29D	Convert "T" frame to "U" frame shaft dimension or smaller (TS)
M29F	Shorten shafts without reducing the overall diameter
Tach and encoder mounting kits	
M31D	Add encoder kit to inverter duty V*S Master or RPM AC motor
M31E	Additional mounted encoder options for IDRPM motors
Testing	
M32A	CSA C390 method 1 equivalent to IEEE 112 method B performance testing
M33A	Short motor test
M33B	Routine motor test
M33C	Complete performance test
M33D	Calibration test
M33E	Sound measurement test
Terminal blocks	
M35A	Install IEC terminal blocks in conduit box and connect leads
Starters	
	Add magnetic or manual starter to grinder or buffer
M36A	Starter – single phase
M36B	Starter – three phase
M36C	Magnetic starter – three phase
M36D	Overload protection for dust control unit
Shaft grounding	
M39A	Shaft grounding systems
M39B-1	Shaft grounding ring - bearing protection
M39C	Baldor-Reliance® shaft grounding brush assembly
M39D	Add shaft grounding ring's colloidal silver shaft coating to shaft
Check Total Indicated Runout (TIR)	
M40A	Check Total Indicated Runout (TIR)
ABB Ability™	
M43A	Install ABB Ability™ Smart Sensor for motors

SOUNDSCREEN™ Vinyl Noise Barriers and Mats

a McGill AirSilence™ Product

Product Bulletin 4241

United McGill® products



SOUNDSCREEN Vinyl Noise Barriers Effectively Block Airborne Noise

SOUNDSCREEN Vinyl Noise Barrier is flexible fused vinyl sheet material loaded with high density inert fillers. This combination of materials provides the limpness and mass necessary for the effective blocking of airborne noise. Easy to cut and install, the material may be laid on floors, draped over uneven surfaces or suspended as a curtain wall.

Standard Products

Reinforced Loaded Vinyl (LVR): The basic 1/8-inch-thick, 1 lb/sq ft density loaded vinyl is reinforced in the center with a polyester scrim. Use in all hanging or other applications where superior integral material strength is needed.

Clear Vinyl (LVC): Used as a transparent curtain or as a window in a curtain enclosure, SOUNDSCREEN standard LVC is 0.08 inches thick to 0.5 lb/sq ft density. Where required, 0.75 and 1 lb/sq ft densities can be furnished.

Plain Black Reinforced (LV): Our lowest-cost LV bonds to sheet metal or plywood to make machinery covers or walls. Lay it under rugs or over ceilings to cut down on noise transmitted into plant offices.

Floor Mat (FVM-1 or FVM-1S): Unreinforced black loaded vinyl with a 1/4-inch backing of decoupling foam fused to LV. It is used on floors, bulkheads, and side panels to cut down road, engine, and drive train noise. Plain surface model (FVM-1) is coated with a scuff-resistant vinyl to improve wear resistance (FVM-1S).

Installation

The 1/8-inch-thick loaded vinyl comes in rolls and can be easily cut with a knife, shears, or steel rule die. When suspended from an existing wall or partition, the vinyl should be pinched between bolted support strips and suspended from a suitable structural support.

For curtain applications, a grommeting kit (SOUNDSCREEN GK-1) containing a tool and grommets is available for economical self-installation. Grommets should be spaced on maximum 12-inch centers, and adjacent

sections can be overlapped 6 inches to reduce acoustical leakage.

A variety of methods can be used to acoustically seal overlapping seams. For flashing existing partitions up to a ceiling, the adjoining vinyl sheets can be riveted or bolted together. For custom-designed curtain enclosures and barriers, Velcro® strips can be sewn onto the vinyl edges to provide a positive sealing mechanism.

Loaded vinyl contains a plasticizer ingredient that can create adhesion problems. For applications where an adhesive seal is appropriate, using a two-part urethane is recommended.

Our vinyl noise barriers are easy to install. Materials may be laid on floors, draped over uneven surfaces or suspended as a curtain wall.

Products depicted in this specification sheet were current at the time of publication. As a quality-conscious manufacturer, McGill AirSilence is continually seeking ways to improve its products to better serve its customers. Therefore, all designs, specifications, and product features are subject to change without notice.

United McGill® is a registered trademark, and McGill AirSilence™ and SOUNDSCREEN™ are trademarks of United McGill Corporation.

Velcro® is a registered trademark of Velcro Industries B.V.

Acoustical Properties

Material	Typical Sound Loss in dB						STC
	125	250	500	1K	2K	4K	
1 lb/sq ft LV and LVR	15	19	21	26	33	37	27
0.5 lb/sq ft LVC	8	19	16	21	25	30	20

Physical Properties

Product	Thickness and Weight	Temperature Range (°F)	Color	Resistance (all products)
LVR	1/8", 1 lb/sq ft	0-180	black	water: excellent
LV	1/8", 1 lb/sq ft	0-180	black	petroleum: excellent
LVC	0.08", 0.5 lb/sq ft	0-120	clear	alkalis: good
				flame: meets MVSS 302

Product Availability

Model	Roll Size	Vinyl Description
LVR	54" x 60'	gray, reinforced vinyl
LVC	48" x 60'	clear vinyl
LV	54" x 60'	black, unreinforced vinyl
FVM-1	54" x 60'	black, vinyl floor mat
FVM-1S	54" x 60'	black, vinyl floor mat with scuff coating

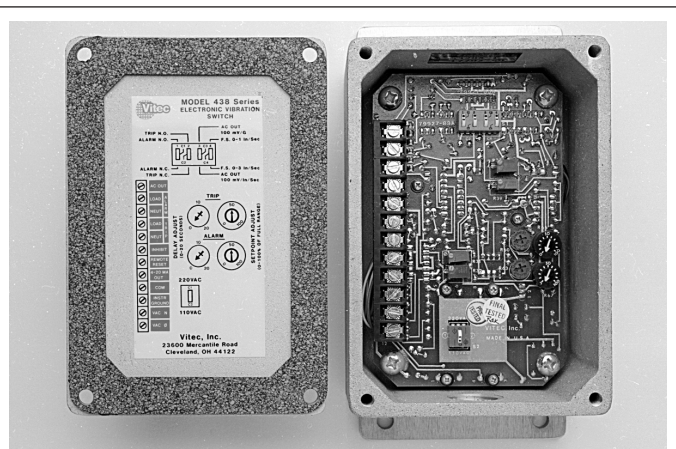
McGill AirSilence LLC

An enterprise of United McGill Corporation — Founded in 1951

2400 Fairwood Avenue
Columbus, Ohio 43207-2700
614/443-5520, Fax: 614/542-2620
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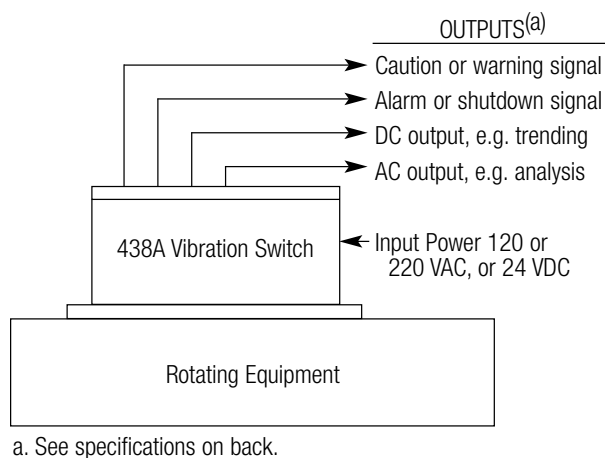


438A Vibration Switch



Applications

- Cooling Towers, Centrifuges, Hammer Mills, Compressors, Generators, Rolling Mills, Ball Mills
- Fans, Pumps, Motors, Coilers, Grinders, Conveyors, Mixers
- General Purpose Rotating Machinery



438A Lowers Operating Costs

The 438A helps lower your operating costs by giving advance notice of equipment problems and allowing you to plan your repairs in advance. The ability to schedule repairs reduces the cost of parts, overtime and, most significantly, lost revenues due to production downtime.

User Benefits

Helps Eliminate Catastrophic Equipment Failures

Should your equipment vibration level increase, the 438A gives you notification, and can also shut down the equipment, before major damage occurs.

Allows You to Plan Repairs in Advance

Trending the vibration level enables you to detect changes in vibration which allows you to schedule repairs to coincide with your normal maintenance schedule.

Compatible with Predictive Maintenance Programs

The 438A is compatible with most data collectors, analyzers and online analysis systems. Analyzing the vibration signatures allows you to find specific causes of high vibration such as imbalance, misalignment and bent shafts as well as faulty belts, bearings and gears.

Eliminates False Alarms

The 438A replaces mechanical vibration switches that can give false alarms, or trips, at machine startup, or when transient vibrations from external sources occur.

Easy Selection

The 438A is designed to be configured by the Customer in the field to fit a wide variety of applications. When ordering the 438A you only need to specify the type of enclosure and the power input required.

Corresponding Features

Dual Vibration Level Setpoints

Alarm and shutdown, or trip, contact closures are supplied, that activate when Customer specified vibration levels are reached or exceeded.

Analog Output

The 438A has a 4-20 mA DC analog output proportional to the vibration level, typically connected to a PLC, DCS or chart recorder.

AC Signal Output

An AC signal output, field selectable for 100 mV/g or 100 mV/in/sec, is standard. This output is available on a terminal block for connection to a Customer supplied remote connector.

Time Delays and Inhibit Functions

Each relay has a Customer adjustable time delay ranging from 0 to 20 seconds. An inhibit function is included, which eliminates relay operation during machine maintenance.

Multiple Configurations

The 438A comes in three different enclosure types: NEMA 4, NEMA 4X or explosion proof. It also has three different input powers: 120 or 220 VAC, or 24 VDC.

SPECIFICATIONS

Full-Scale Range:

0.0 – 1.0 in/sec or 0.0 – 3.0 in/sec (field selectable)

Frequency Response:

3 – 1,000 Hz

Dual Set Points:

Two Customer selectable, field adjustable from 5 to 100% of full scale

Time Delay:

Field adjustable from 0 – 20 seconds

Relays (solid-state triacs):

Two solid-state relays rated at 10 amp for either 120 or 220 VAC unit, and 1 amp for the 24 VDC unit.

Field selectable, normally open or normally closed below setpoints. Field selectable, latching or non-latching. Remote reset capability. Relay inhibit capability.

Input Power:

A 120 or 220 VAC, Customer selectable, or 24 VDC.

DC Analog Output:

4 – 20 mA DC, proportional to overall vibration

AC Signal Output:

Field selectable, velocity, 100 mV/in/sec, or acceleration, 100 mV/g, for analysis purposes

Temperature Range:

–25 to +160 F

Measurement Mode:

Velocity

Housing:

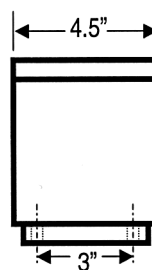
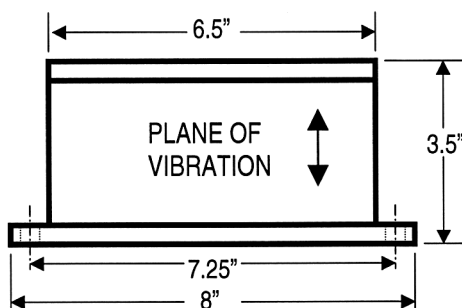
NEMA 4, cast aluminum
NEMA 4X, epoxy coated cast aluminum with a stainless steel mounting plate
Explosion proof, cast aluminum

Weight:

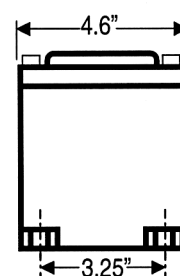
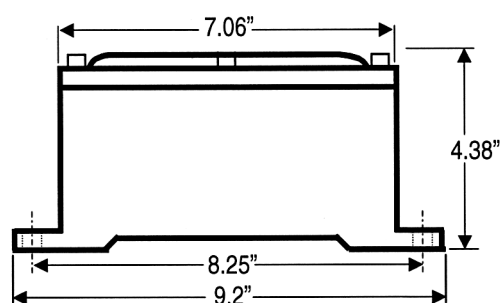
5.5 Pounds, NEMA 4 and 4X
7.5 Pounds, Explosion Proof

DIMENSIONS:

NEMA 4 and NEMA 4X Enclosures



Explosion Proof Enclosure



- The NEMA 4 and 4X 438A should be mounted using four 1/4 – 20 UNC x 1/2 inch long bolts, washers and lock-washers.
- The explosion proof 438A should be mounted using four 5/16 – 18 UNC x 3/4 inch long bolts, washers and lock-washers.

Environmental Rating:

NEMA 4 – Weatherproof
NEMA 4X – Weatherproof and corrosion resistant

Hazardous Rating:

Explosion proof –
Class I, Divisions 1 and 2, Groups C and D, and
Class II, Divisions 1 and 2, Groups E, F and G.

Vitec's Vibration Switch Categories:

438A Plate Mounting

- 120 and 220 VAC
- 24 VDC

438D

- Consists of transducer, cable assembly and electronics
- 120 and 220 VAC
- Low-pass, Bandpass, High-pass filter
- Multiple transducers
 - Acceleration
 - Velocity
- Multiple cable assemblies

480 Stud Mounting

- 120 and 220 VAC
- 24 VDC

438R

- Consists of transducer, cable assembly and electronics
- 120 and 220 VAC
- 24 VDC
- Multiple transducers
 - Acceleration
 - Velocity
- Multiple cable assemblies



Protecting the Machines of Production
for More than a Quarter-Century.



OPERATOR'S MANUAL

438A VIBRATION SWITCH: 24 VDC

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Vitec, Inc.
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Phone: 216-464-4670 Fax: 216-464-5324



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LIST OF ACRONYMS AND ABBREVIATIONS

AC	Alternating Current
C1	Circuit 1
C2	Circuit 2
C3	Circuit 3
C4	Circuit 4
F	Fahrenheit
FS	Full Scale
g	Vibration Units of Acceleration
Hz	Hertz, (Cycles per Second)
In/Sec	Inches/Second Vibration Units of Velocity
mA	Milliamp
Mils	Vibration Units of Displacement
mV	Millivolts
mV/g	Millivolt/g Force
PLC	Programmable Logic Controller
R60	Potentiometer No. 60 to adjust shutdown time delay
R63	Potentiometer No. 63 to adjust alarm time delay
R66	Potentiometer No. 66 to adjust shutdown setpoint
R67	Potentiometer No. 67 to adjust alarm setpoint
S1	Switch to field adjust the state of the relays, full scale range and AC out
S2	Switch to field adjust the voltage of the AC power in
sec	Second



List of Acronyms and Abbreviations - continued

TB	Terminal Block
TB1-1	Terminal Block 1, Location 1
TB1-7	Terminal Block 1, Location 7
TB1-8	Terminal Block 1, Location 8
TB1-9	Terminal Block 1, Location 9
UNC	Unified National Course
VAC	Volts Alternating Current
VDC	Volts Direct Current

LIST OF ENGINEERING SYMBOLS AND ABBREVIATIONS

A	Acceleration: g's peak
ALM	Alarm
AMP	Ampere
ANA	Analog
COM	Common
CPM	Cycles Per Minute
D	Displacement: Inches peak to peak
d	Displacement: Mils peak to peak
DC	Direct Current
g	Gravity
GND	Ground
Hz	Cycles per Second
INSTR	Instrument
In/Sec	Inches per Second
MAX	Maximum
NORM	Normally
Ω	Ohm
N	Neutral
$\emptyset$	Phase
REM	Remote
RPM	Revolutions Per Minute
Shut	Shutdown
Stpt	Setpoint
VDC	Voltage, Direct Current
V	Velocity: Inches per second peak
W	Watt



1.0 INTRODUCTION

The Vitec 438A^(a) Vibration Switches are an economical approach to accurate and reliable vibration protection for rotating machinery. The 438A is a sophisticated vibration monitoring system and, therefore, requires some care during installation.

Installation and operation of this unit will be simple and easy if the instructions are followed.

READ THIS MANUAL THOROUGHLY TO AVOID ANY INSTALLATION-RELATED PROBLEMS

2.0 PRODUCT DESCRIPTION

The 438A measures the vibration level of the machine to which it is mounted. It provides two solid state relay (transistor) closures when preset vibration levels are exceeded, thereby offering protection from excessive vibration. One provides an alarm or warning, the second provides for shutdown or "trip".

The 438A incorporates a built-in vibration sensing element and electronics to measure the vibration in terms of the velocity of vibration. Those familiar with setpoints given in displacement (mils) can convert to velocity (in/sec) via the Setpoint Conversion Figure and Formulas Chart shown in Figure 1 on page 5.

The 438A also includes two electrical outputs that are very useful. A 4 - 20 mA signal proportional to the overall vibration level will allow the unit to be connected to a remote device for purposes of reading or recording the vibration level. Common applications include connections to meters, data loggers, recorders or Programmable Logic Controllers (PLC's).

An AC signal, switch selectable for either acceleration or velocity, is also provided. This signal is proportional to the actual vibration being measured by the transducer. Common applications for this signal include connection to an oscilloscope to view the raw transducer signal or connection to a real-time analyzer for analysis of the vibration being measured.

3.0 ELECTRICAL INSTALLATION

Proper electrical installation is essential. A little care here will assure a trouble-free installation.

Follow the wiring diagram shown in Figure 3 on page 7. Make special note of the following:

a. In this Manual, the Vitec 438A means the 24 VDC version.



1. The system requires a good instrument earth ground. Do not use the machine itself as a ground, as it normally will not provide proper grounding.

Do not use a ground that is common to other large pieces of electrical machinery. Electrical noise or surges from these machines can feed back into the 438A and cause false tripping.

2. The alarm and shutdown outputs are completely isolated from the monitor electronics. One or two separate sources of DC power can be used because of the isolation. The maximum load current is 1.0 amp with a 24-volt DC supply. There is no minimum current required for switch operation.
3. Make sure that the voltage input lines and the relay output lines are fused as noted on the wiring diagram in Figure 3 on page 7.
4. If the AC output signal is utilized, use a good grade of shielded twisted cable. Be sure to keep the cable separated from other AC or power cables.

Reread this section to ensure the electrical installation conforms to the Figure 3 wiring diagram. Any damage to the 438A due to incorrect wiring is not covered under warranty.

4.0 MECHANICAL INSTALLATION

Figures 4 and 5 on pages 8 and 9 show mounting dimensions and mounting procedures for the 438A NEMA 4 and Explosion Proof versions, respectively.

The 438A Vibration Switch is sensitive to vibration in one direction only. Make sure the unit is oriented properly to measure the type of vibration being monitored.

If the required mounting location is not known, the vibration switch should generally be mounted to measure the RADIAL vibration, meaning the axis of sensitivity is perpendicular to the rotating shaft. See Figure 6 on page 10 for typical mounting locations. Note that the mounting locations shown are only general in nature. The manufacturer of the equipment being monitored is the best source of mounting location information.

Mounting tips:

1. Be sure the vibration switch is mounted FIRMLY to the machine. A switch that is not mounted firmly to the machine will measure its own vibration, in addition to the vibration of the machine (see Figure 6).



2. Be sure to replace the cover AND gasket after installation. Water or moisture in the electronics will cause FAILURE of the unit! Water entry through the 3/4 inch cable entry hole due to condensation or other reasons can be just as catastrophic. Use proper procedures to seal the cable entry hole. Failures caused by improper sealing are NOT covered under warranty.
3. Try to orient the unit so that the alarm setpoint adjustments are accessible after the unit is mounted. Someone may want to change the setpoints in the future.
4. Make sure that the electrical connections are not putting excessive force on the terminal block or printed circuit card. Don't try to stuff excess wire into the box; trim the cable to proper length.

5.0 SETPOINT ADJUSTMENT

The ALARM and SHUTDOWN setpoints are adjusted via two potentiometers located on the printed circuit card (see Figure 2 on page 6, items marked R66 and R67). For alarm setpoint adjust R67; for shutdown setpoint adjust R66. Do not attempt to adjust the small square potentiometers that are also located on the printed circuit card; these potentiometers are used for calibration of the unit.

The setpoint adjustments are calibrated for 0 to 100% of full scale selected, meaning of the selected range, 0.0 to 1.0 or 0.0 to 3.0. Simply turn the potentiometer until the arrow lines up with the required setpoint level. Do not use excessive force. Do not try to turn the potentiometers beyond their stops. Both are single-turn potentiometers.

A 0 to 20 second adjustable time delay is built into the alarm and shutdown circuits to help avoid false trips. The unit is shipped with a 0 second time delay, but can be adjusted in the field for up to 20 seconds. Turn the potentiometer clockwise to increase the time delay. Adjust R63 for alarm time delay and R60 for shutdown time delay (see Figure 2 on page 6, items marked R63 and R60).

6.0 RELAYS

Both the alarm and shutdown solid-state relays (transistors) are set at the factory to be normally open below the setpoint. Think of them as single pole light switches that are normally off, or open, but turn on, or close, when the vibration level gets too high. To operate properly, the switched load must operate at 24 VDC, with a maximum current of 1 A.

The relays can be changed to normally closed below setpoint (open when the vibration setpoints are exceeded) using switch S1. When switch S1 is in positions 1 and 2, alarm and shutdown relays are normally open. Switching S1 to the opposite position makes the relays normally closed, see Figure 2, on page 6, items identified as S1.



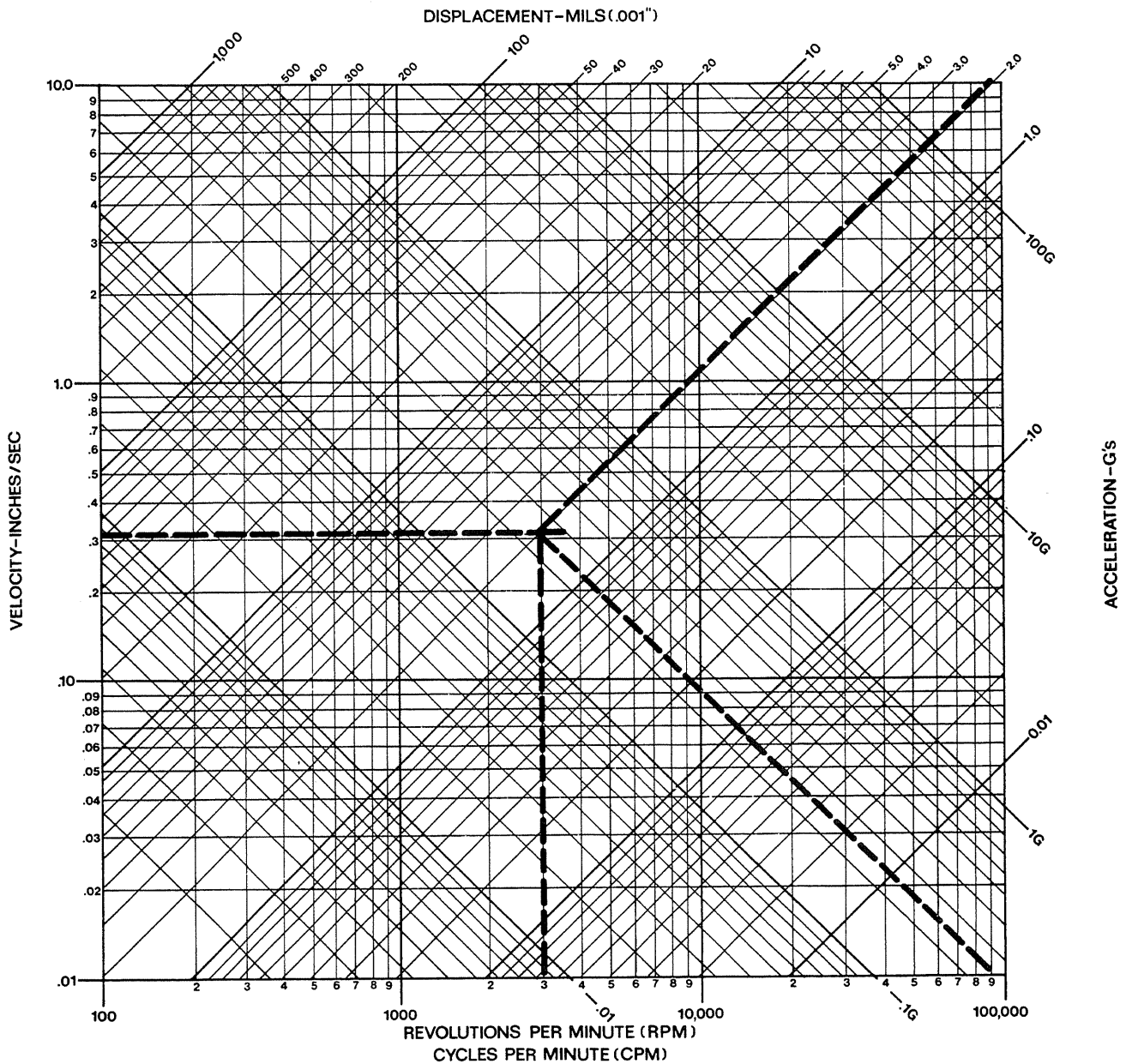
When shipped from the factory, the relays are LATCHING unless wired to be NON-LATCHING at the request of the Customer. Latching means that after the vibration returns to a normal level, the relays will NOT reset themselves. They must be manually reset via a remote reset switch which you, the Customer, must provide, since it does not come with the 438A.

The 438A can be changed to a NON-LATCHING configuration in the field by installing a jumper between terminal block points TB1-7 and TB1-9. With this jumper installed, the relays will automatically reset themselves when the vibration returns to a level less than the setpoint.

If desired, the relay action can be inhibited by connecting terminal points TB1-6 and TB1-9. This feature is normally only used during machine tests or repair procedures. BE AWARE that when you use this feature it totally eliminates operation of the relays. No vibration protection is available while this feature is in use.

7.0 SPECIFICATIONS

The 438A Vibration Switch Specifications are shown in Figure 7 on page 11.



CONVERSION FORMULAS

Symbols:

D = Displacement: Inches peak to peak
d = Displacement: Mils peak to peak
V = Velocity: Inches per second peak
A = Acceleration: G's peak
Hz = Cycles per second
CPM = Cycles per minute

$$D = 0.318 \frac{V}{\text{Hz}}$$

$$D = 19.607 \frac{A}{(\text{Hz})^2}$$

$$V = \pi(\text{Hz}) (D)$$

$$V = 61.440 \frac{A}{\text{Hz}}$$

$$A = 0.051 (\text{Hz})^2 (D)$$

$$A = 0.016 (V) (\text{Hz})$$

$$d = (1.910) (10^4) \frac{V}{\text{CPM}}$$

$$d = (7.059) (10^7) \frac{A}{(\text{CPM})^2}$$

$$V = (5.236) (10^{-5}) (\text{CPM}) (d)$$

$$V = (3.696) (10^3) \frac{A}{\text{CPM}}$$

$$A = (1.417) (10^{-8}) (\text{CPM})^2 (d)$$

$$A = (2.704) (10^{-4}) (\text{CPM}) (V)$$

EXAMPLE: 2.00 Mils at 3000 RPM

By Chart V = 0.3 inches per second

A = 0.25 G's

By Formula $V = (5.236) (10^{-5}) (3000) (2.00) = 0.31$ inches per second

$A = (1.417) (10^{-8}) (3000)^2 (2.00) = 0.26$ G's

FIGURE 1 SETPOINT CONVERSION FIGURE AND FORMULAS

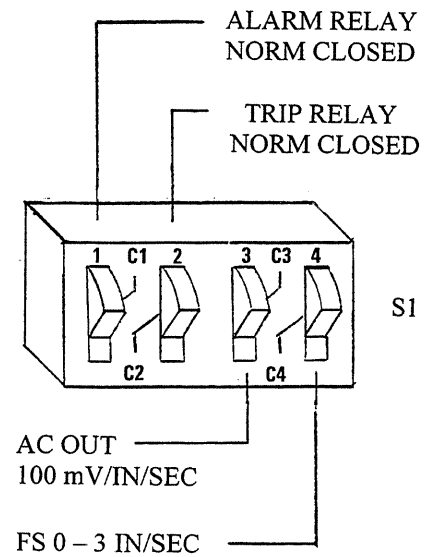
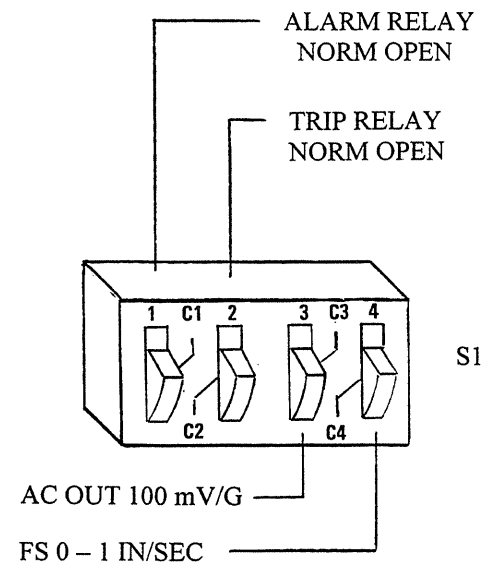
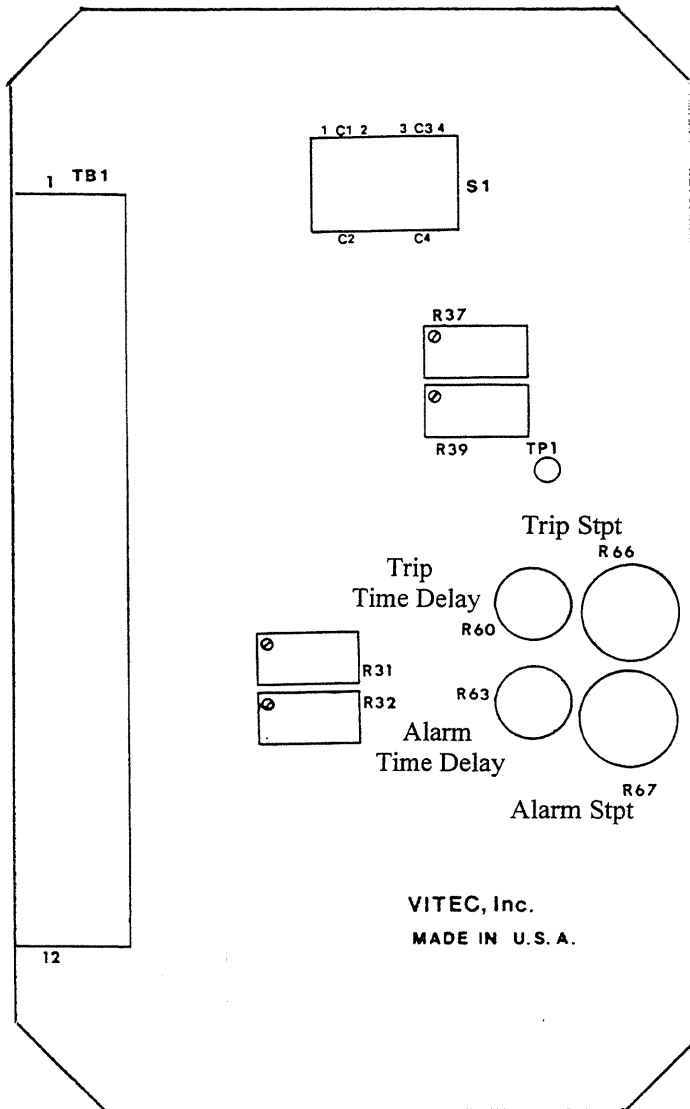
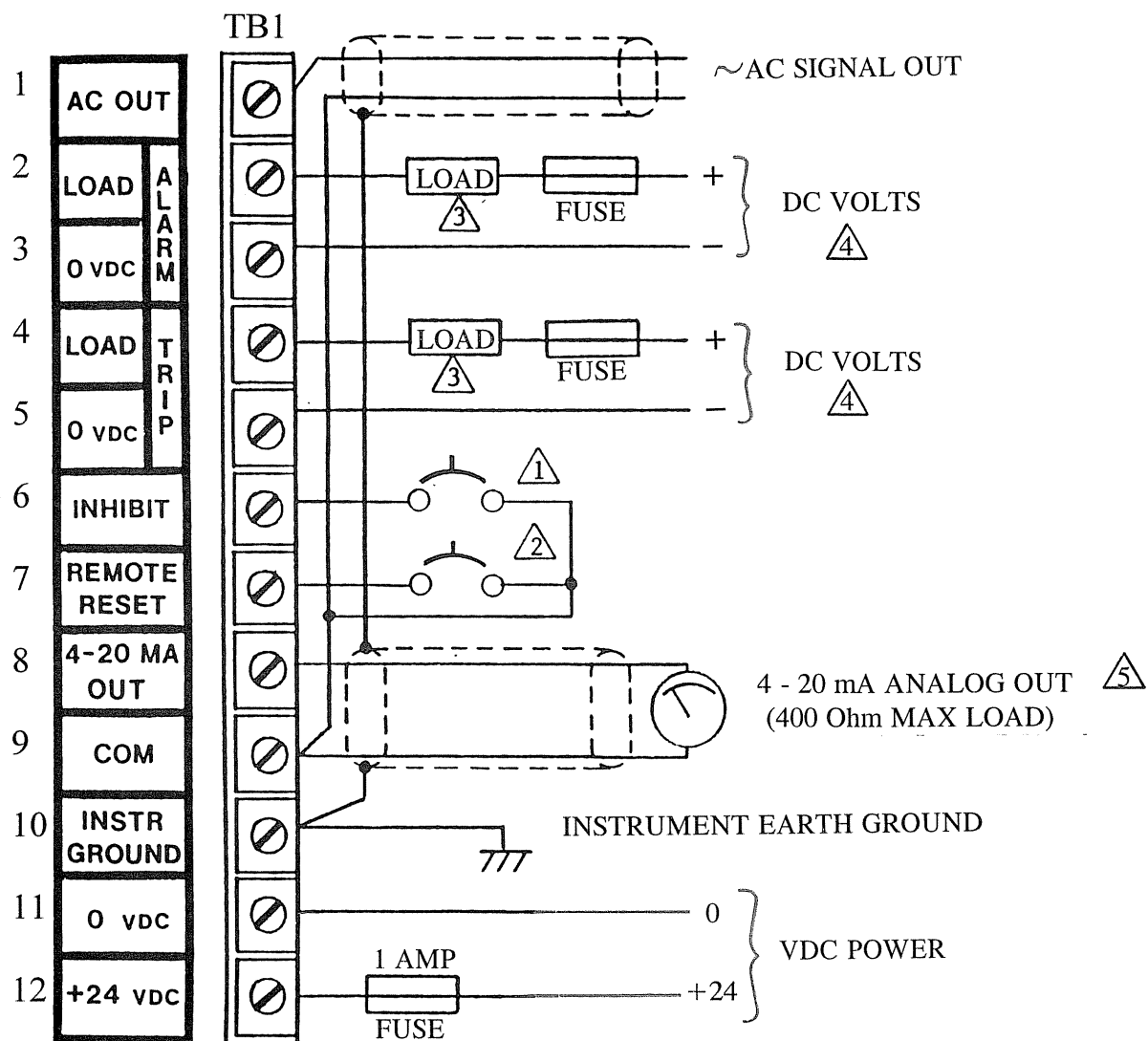


FIGURE 2 COMPONENT LOCATION AND IDENTIFICATION OVERVIEW



Notes:

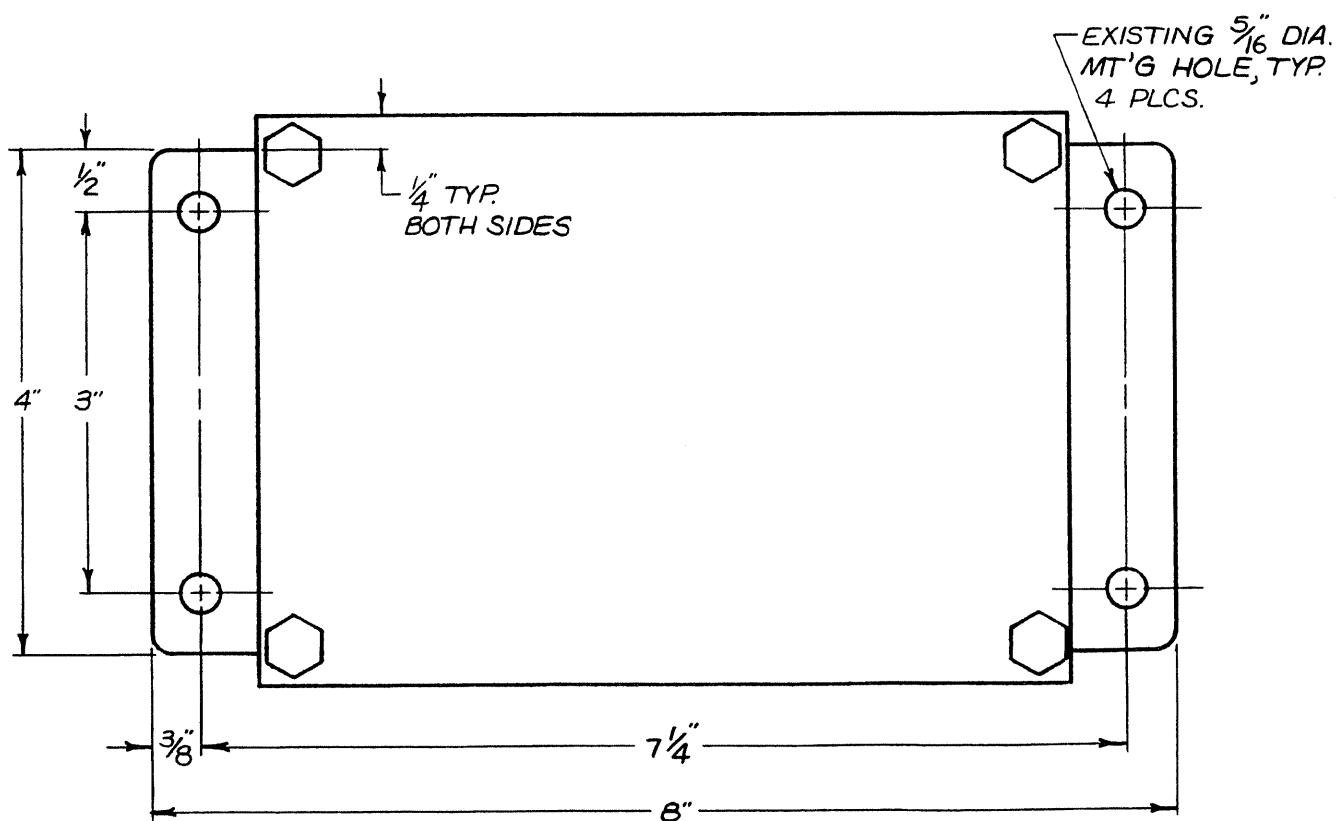
- ① Relays inhibited while momentary switch is activated.
- ② Latching relays reset when momentary switch is activated. For non-latching relays, Customer to install jumper between TB1-7 and TB1-9.
- ③ Maximum load current is 1 amp.
- ④ Maximum voltage is 24 VDC.
- ⑤ If 4 - 20 mA analog output is not used, install a 100 Ω ¼ W resistor across TB1-8 and TB1-9.

FIGURE 3 WIRING DIAGRAM FOR 24 VDC 438A VIBRATION SWITCHES



RECOMMENDED MOUNTING

1. Locate an easily accessible mounting surface for the switch. Points of contact with the base should be reasonably flat and smooth.
2. Drill No. 6 (.2040) approximately 9/16 inches deep (7/16 inches deep for bottoming tap) and tap 1/4 - 20 approximately 5/16 inches deep, typical four places.
3. Securely fasten switch to mounting surface with four each of 1/4-20 x 9/16 inch HHCS, 1/4 inch flat washer, 1/4 inch split lockwasher.
4. Make all necessary electrical connections.



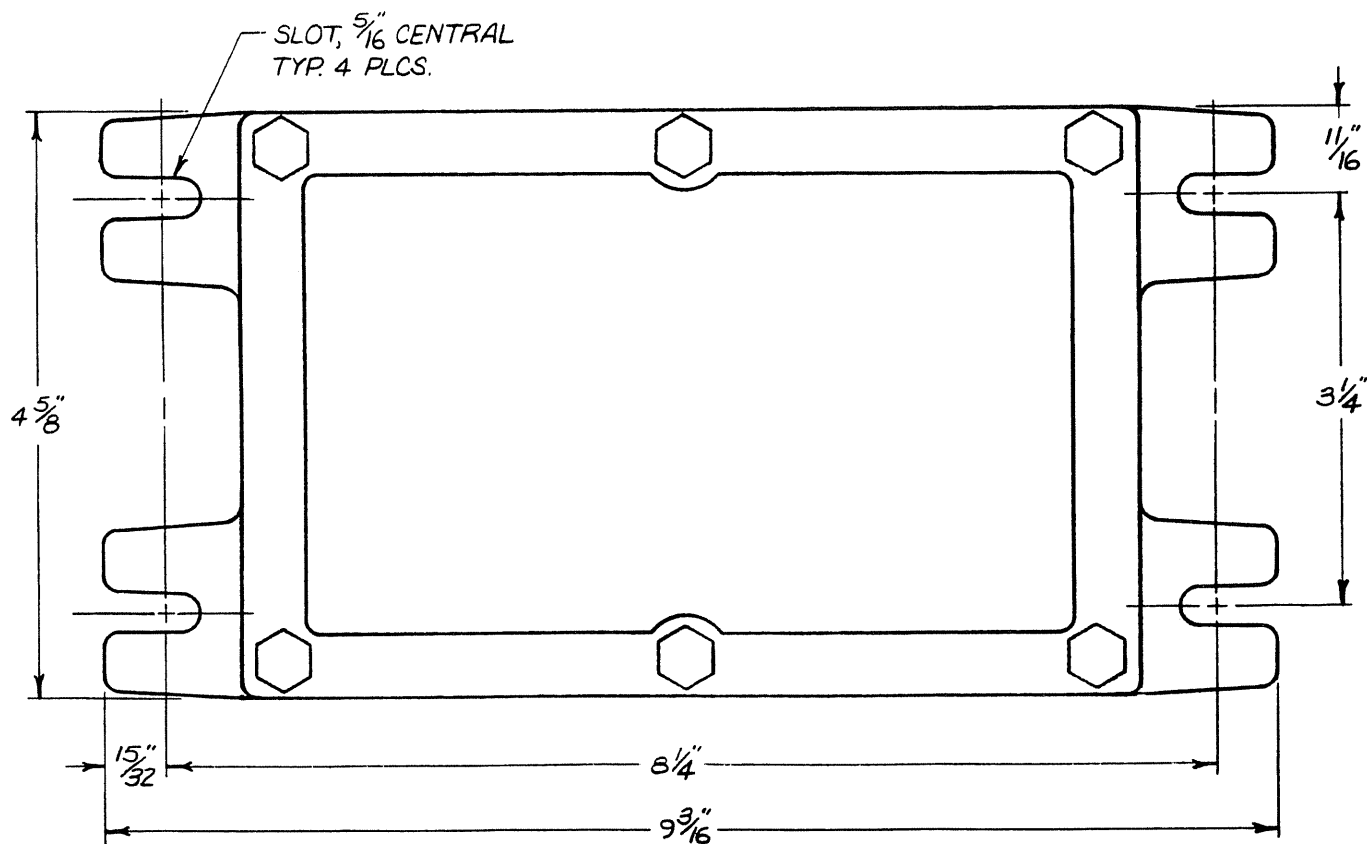
Overall Dimensions: 8" L x 4-1/2" W x 3-5/16" H

FIGURE 4 MOUNTING PROCEDURE FOR THE 438A NEMA 4 VIBRATION SWITCH



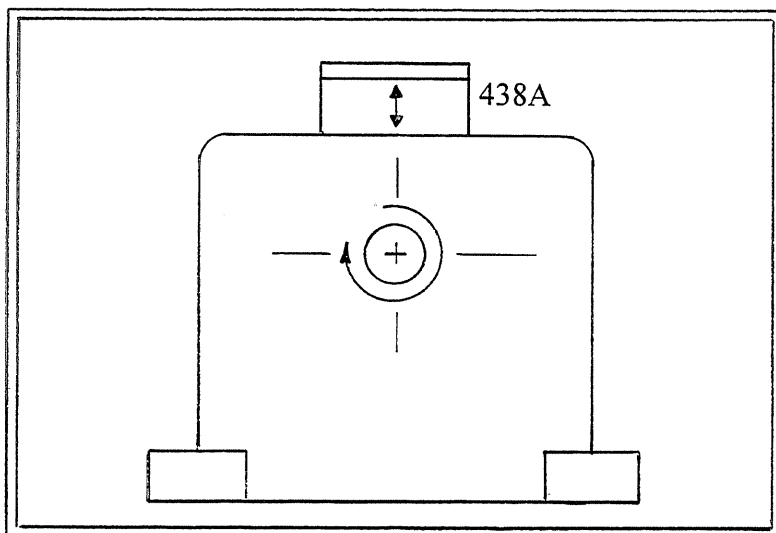
RECOMMENDED MOUNTING

1. Locate an easily accessible mounting surface for the switch. Points of contact with the base should be reasonably flat and smooth.
2. Drill No. 6 (.2040) approximately $\frac{9}{16}$ inches deep ($\frac{7}{16}$ inches deep for bottoming tap) and tap $\frac{1}{4}$ -20 approximately $\frac{5}{16}$ inches deep, typical four places
3. Securely fasten switch to mounting surface with four each of $\frac{1}{4}$ -20 x $\frac{9}{16}$ inch HHCS, $\frac{1}{4}$ inch flat washer, $\frac{1}{4}$ inch split lockwasher.
4. Make all necessary electrical connections.

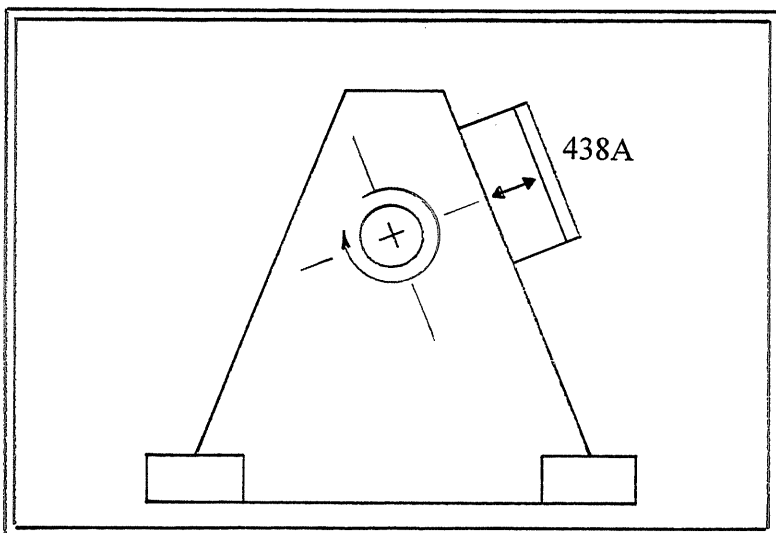


Overall Dimensions: $9 \frac{3}{16}$ \"

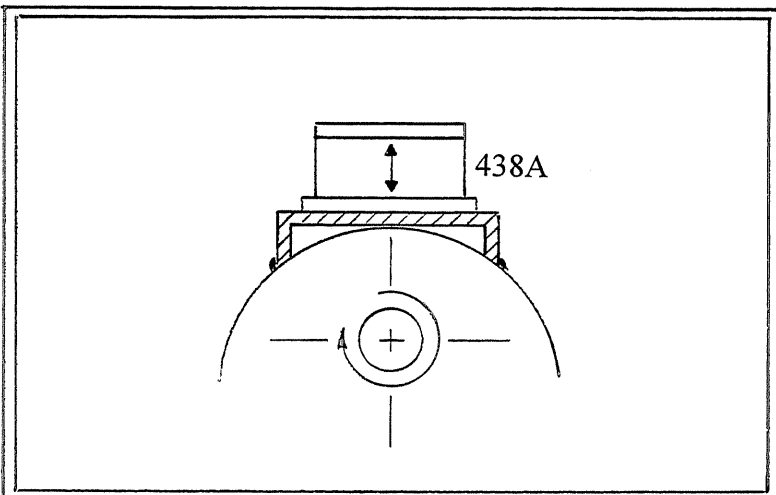
FIGURE 5 MOUNTING PROCEDURE FOR THE 438A EXPLOSION PROOF VIBRATION SWITCH



Flat Mounting,
Vertical or
Horizontal



Angled
Surface
Mounting



Small or Curved
Surface.
Mounting plate
to be welded or
bolted solidly to
curved surface.

FIGURE 6 TYPICAL MECHANICAL MOUNTING LOCATIONS



438A Vibration Switches are shipped from the factory as follows:

Full Scale Range ^(a)	0-1 in/sec (S1 in position 4)
Input Power	24 VDC
Frequency Range	3 to 1,000 Hz
Setpoint Adjustability	5% to 100% of full scale
Time Delay Adjustability	0 to 20 seconds
Setpoint Accuracy	± 5% of full scale
Setpoint Repeatability	± 1% of full scale
Operating Temperature Range	-25 F to +160 F
Analog Output	4 - 20 mA proportional to full scale
AC Signal Output ^(a)	100 mV/g (S1 in position 3)
Solid State Relays ^(a)	Transistors designated as Alarm and Shutdown 1 Amp maximum load current at 24 VDC Normally open below setpoint Closes and latches above setpoint Alarm: S1 in Position 1 Shutdown: S1 in Position 2

a. Field Selectable Options:

1. Full Scale Range	0-3.0 in/sec	S1 in position opposite 4
2. AC Signal Output	100 mV/in/sec	S1 in position opposite 3
3. Solid -State Relays	Alarm normally closed	S1 in position opposite 1
	Shutdown normally closed	S1 in position opposite 2
	Non-latching relays	Install jumper between TB1-7 and TB1-9

FIGURE 7 438A VIBRATION SWITCH SPECIFICATIONS



Transmittal of Drawings

Kinetics Noise Control Inc.
6300 Irelan Place
Dublin, Ohio 43017
PHONE: (614) 889-0480
FAX: (614) 889-0540

Send Via:

- | | |
|---|-----------------------------------|
| ☐ Email | ☐ Next Day |
| ☐ First Class Mail | ☐ Other |
| ☐ UPS | |

To: Hartzell Air Movement

Date	06/19/2025
KNC #	217055
Submittal #	1.0
PO #	2521509
RE	Nan Inc Hilo WWTP Ph 1

Attn Dakota Iddings

For:

- | | |
|--|---------------------------------|
| ☒ Approval | ☐ Record |
|--|---------------------------------|

Please Return:

- | | |
|---|---|
| ☒ Completed Submittal Review/Approval Sheet | ☐ No Return Required |
|---|---|

Order:

- | | |
|---|--|
| ☒ Released After Approval | ☐ Released to Manufacturing |
|---|--|

Comments:

NOTES:

- 1.ASSUMED ATTACHED TO STEEL.
2. BUILDING HEIGHT NOT PROVIDED.



THIS SHEET MUST BE RETURNED TO KINETICS NOISE CONTROL PRIOR TO BEGINNING MANUFACTURE OF THE REFERENCED ITEMS IN THIS SUBMITTAL

Job Title:

Nan Inc Hilo WWTP Ph 1

Customer:

Hartzell Air Movement

Customer P.O.:

2521509

Date Submitted:

6/19/2025

KNC Order No.:

217055

Submittal No.:

1.0

KNC Rep Name:

Dakota Iddings

- ☐ Drawings Approved As Shown.
- ☐ Revise Drawings As Shown. Resubmittal Required.
- ☐ Revise Drawings As Shown. Resubmittal Not Required.
- ☐ KNC Is To Release All To Manufacture.
- ☐ KNC Is To Release Only The Tags Noted Below.
- ☐ Preliminary approval pending PE Review

Schedule of Drawing Changes:		
Drawing No.	Change Notes	Reason for Change

Note:

ALL VERBAL COMMENTS OR MODIFICATIONS TO SUBMITTAL PACKAGE MUST BE DOCUMENTED IN WRITING TO KINETICS NOISE CONTROL PRIOR TO RELEASE TO MANUFACTURING. K.N.C. ASSUMES NO LIABILITY FOR NON-CONFIRMED CHANGES OR FOR ANY CHANGES OTHER THAN THOSE LISTED.

Revised By:

Name:

Company:

Date:

Disclaimer:
By approving this submittal you are confirming that the information provided to Kinetics is correct and complete and that all pertinent dimensions and data included herein have been checked and found to be in conformance with the requirements.

NOTES ON TAGS

Sub Num	Tag	Tag Notes
1.0	2521509-02	USED ONLY 4 ISOLATORS. USED FHS-2s TO GET ATLEAST MINIMUM 1" DEFLECTION AS REQUESTED BY CUSTOMER.
1.0	2521509-03	USED ONLY 4 ISOLATORS. USED FHS-2s TO GET ATLEAST 1" MINIMUM DEFLECTION AS REQUESTED BY CUSTOMER.
1.0	2521509-01	ASSUMED 10% FROM CENTER FOR C.G.

P.O.: 2521509

Date: 6/19/2025

Project: Nan Inc Hilo WWTP Ph 1

DWG No.	Description
-	Transmittal of Drawings
-	Review/Approval
-	Tag Notes
-	Submittal Index
-	MATERIAL REQUIRED LIST
-	SEISMIC CERTIFICATION
GSS-1	Seismic and Wind Certification Disclaimer (2 pgs)
2521509-01	Seismic Calculations
2521509-02	Seismic Calculations
2521509-03	Seismic Calculations
-	GENERAL ISOLATION LOCATIONS
S-01-35-21	FHS-2-100/1975
SS-20111000-L	Isolator Weld Data
INS-01-35-0A.PDF	MODEL FHS INSTALLATION INSTRUCTIONS
-	PE Appendix

Material Required List:

Thank you for choosing KINETICS NOISE CONTROL, the world leader in innovative products and solutions for acoustics, noise control, vibration isolation and seismic restraint systems since 1959.

WARNING

IMPORTANT NOTE REGARDING CERTIFICATIONS PROVIDED BY KINETICS NOISE CONTROL – PLEASE READ CAREFULLY
 Certifications provided by Kinetics Noise Control (KNC) pertain ONLY and SOLELY to the components provided by KNC with the following exception: The use of industry standard hardware to connect the equipment and the structure to components provided by KNC is permitted ONLY and SOLELY if said hardware is specified in the KNC Certification.
 For equipment directly connected to the structure (with no KNC provided component in between) the use of non-KNC provided hardware will invalidate the Certification.

Tag/Drawing No.	Part #	Material Required Description	Qty Each	Total Qty	Reference Drawing
2521509-01	5100410	5/8-11 GR 5 HARDWARE KIT, 3.5" FULL THREAD NUT WASHERS	6	6	
2521509-02	013510003	FHS-2-500	2	2	S-01-35-21
	013510019	FHS-2-225	2	2	S-01-35-21
	5100410	5/8-11 GR 5 HARDWARE KIT, 3.5" FULL THREAD NUT WASHERS	8	8	
2521509-03	013510003	FHS-2-500	2	2	S-01-35-21
	013510019	FHS-2-225	2	2	S-01-35-21
	5100410	5/8-11 GR 5 HARDWARE KIT, 3.5" FULL THREAD NUT WASHERS	8	8	

KINETICS RESTRAINT COMPONENT CERTIFICATION

Date: 06/19/2025
Project: Nan Inc Hilo WWTP Ph 1
Location: HONULULU, Hawaii 96819
KNC Number: 217055
Tags: 2521509-01, 2521509-02, 2521509-03

Reviewed By: James Gerloff

Registration: Hawaii

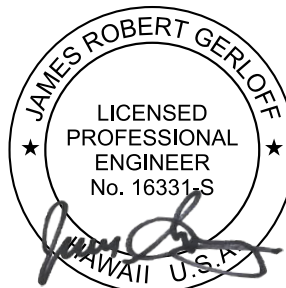
License Number: PE-16331

Expiration Date: 4/30/2026

This packet contains certifications of Kinetics Noise Control restraint products tailored to the specific load conditions indicated on the individual certification documents. Unless indicated otherwise on the attached documents, the basis of the certification is code minimum requirements. Unidentified Equipment Importance factors are assumed to be 1.5 for components in critical and life safety structures and 1.0 in other structures. Load and equipment data has been drawn from information provided to Kinetics Noise Control by our client. Where required input data was not provided, Kinetics Noise Control will attempt to extract data from publicly available mapped data, however it remains the responsibility of those with full access to the project data to verify that the loads used, as well as the equipment and installation conditions are consistent with the system as certified. Any significant deviations from the conditions identified in these documents including equipment substitutions, relocations, mounting arrangements, Importance Factors, special load conditions, etc. will void this certification.

It is critical to note that this certification covers only those hardware items which have been provided by Kinetics Noise Control. It does not extend to global or localized stresses in the building/supporting structure, the equipment, or where relevant, piping/ductwork and their suspension components.

This document and all documents referenced above have been sealed in the name of Kinetics Noise Control, Inc. by the listed Professional engineer. The seal represents only that components have been selected to meet the requirements of ASCE 7 and IBC as indicated in the project requirements. Any specialized Professional Engineering services desired for this project are beyond the scope of Kinetics Noise Control. In conjunction with these certifications, Kinetics Noise Control, Inc. guarantees only the use of that degree of care and skill ordinarily exercised under similar conditions by reputable members of our profession. No other warranty, expressed or implied, is made or intended.



06/22/2025

THIS WORK WAS PREPARED BY
ME OR UNDER MY SUPERVISION. 06/22/2025
Signature4/30/2026
Expiration Date of the License

Certification Agency:

SEISMIC AND WIND CERTIFICATION – GENERAL ASSUMPTIONS AND DISCLAIMER

All Seismic and Wind Certifications performed by Kinetics Noise Control, Inc and/or its associates, unless clearly stated otherwise in the body of that certification document, will be performed in accordance with the assumptions and disclaimers identified here in.

Loads Considered

The loads considered in this certification are limited to those forces described in the seismic portion of the code and/or that portion of the specification for the project provided to Kinetics Noise Control. If Kinetics Noise Control is not otherwise informed, the most recent version of the appropriate code will be used. Because equipment can be indoors, or shielded, wind will not be considered during the analysis unless it is specified to be included in the seismic certification request. In the absence of wind velocity and the appropriate factors, Kinetics Noise Control will use the equivalent of ~45 psf (2,155 Pa) as a wind load requirement. If this is not adequate, it is the responsibility of the Design Professional of Record to notify Kinetics Noise Control.

Extent of the Certification

This certification applies only to those components provided by Kinetics Noise Control. It does not extend to the building structure, equipment, or pipe/duct and their supports.

Equipment Data

The equipment weight, geometry and CG data used to perform the certification have been provided to Kinetics Noise Control by others, no attempt has been made by Kinetics Noise Control to verify its accuracy and it is up those providing the information to do so. Where CG data is not provided, associates of Kinetics Noise Control will attempt to make reasonable, yet conservative estimates as to the magnitude of any imbalance although it must be recognized that the direction of the imbalance is often unknown. Unless the equipment orientation is obvious from the diagram in the certification document, it should be assumed that the orientation is not known. Under these conditions, the worst case restraint, attachment and/or anchorage selection indicated for any particular location must be used for all locations.

Equipment Durability

Kinetics Noise Control and its associates make no representations as to equipment durability and its ability to survive a seismic event and remain functional.

Installation

Where detailed installation procedures are not addressed in documentation provided by Kinetics Noise Control, all seismic hardware and components must be installed in conformance with FEMA 412, 413, and 414, Installing Seismic Restraints for Mechanical Equipment, Installing Seismic Restraints for Electrical Equipment, and Installing Seismic Restraints for Pipe and Duct respectively. Free copies available from FEMA (1-800-480-2520) or through Kinetics Noise Control

SEISMIC & WIND CERTIFICATION – ASSUMPTIONS & DISCLAIMER

PAGE 1 of 2

GSS-1

RELEASED ON: 08/12/2009



Dublin, Ohio, USA • Cambridge, Ontario, Canada

Toll Free (USA Only): 800-959-1229
International: 614-889-0480
FAX: 614-889-0540
World Wide Web: www.kineticsnoise.com
E-mail: sales@kineticsnoise.com



Member

Equipment, Restraint, and Component attachment holes

For seismic restraint, it is necessary that any attachment bolts positioned in the path between the equipment to be restrained and the building structure, be a tight fit with their mating holes (The hole is to be not more than 1/8" (3 mm) in diameter larger than the attachment bolt). In the case of Kinetics Noise Control supplied restraint components, attachment safety factors are based on hardware sized per the above. In the case of directly attached equipment, the hardware and components provided by Kinetics Noise Control are the minimum required to withstand the seismic loading. If attachment holes in the equipment exceed the recommendation above, the attachment hole is to be sleeved or grouted to bring its effective diameter down to not more than 1/8" (3 mm) larger than the attachment hardware used.

Anchor Capacity and Edge Distances

All anchor allowable loads are based on ICC – ES test data and assume full anchor embedment in 3,000 psi (20,684 kPa) concrete and a minimum spacing between the anchor centerline and the edge of the slab into which it is sunk in accordance with the included anchorage data. The anchor data used is appropriate for the anchors provided by Kinetics Noise Control, unless otherwise noted. Under some conditions as noted in the calculations, "Undercut" type anchors may be required

Stamps

Stamped documents are intended to support the Engineer of Record on the project. If the project is located in an area for which Kinetics does not have a valid PE license, the documents will be signed and sealed by a Licensed Professional Engineer in the State of Ohio. This practice is intended solely to indicate that a competent individual has reviewed the document. It is not intended to imply that the licensee is legally empowered to practice in the jurisdiction of the project.

General

Kinetics Noise Control, Inc. and its associates guarantee that we will use that degree of care and skill ordinarily exercised under similar conditions by reputable members of our profession to determine restraint and/or attachment safety factors based on customer supplied input data. No other warranty, expressed or implied is made or intended.

SEISMIC & WIND CERTIFICATION – ASSUMPTIONS & DISCLAIMER

PAGE 2 of 2

GSS-1

RELEASED ON: 08/12/2009



Toll Free (USA Only): 800-959-1229
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FAX: 614-889-0540
World Wide Web: www.kineticsnoise.com
E-mail: sales@kineticsnoise.com



Dublin, Ohio, USA • Cambridge, Ontario, Canada

Member



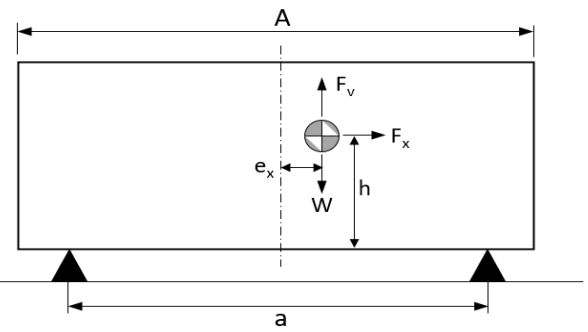
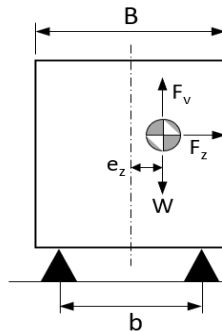
Project:	Nan Inc Hilo WWTP Ph 1	KNC#: 217055
Tag:	2521509-01	Designer: FJ
Date:	6/19/2025	Rev. 0.0

Seismic Design Basis

Building Code:	2018 IBC	Site Class:	C
Risk Category:	III	Spectral Acceleration, S_{DS} :	1.20
Component Importance Factor, I_p :	1.5	Overstrength Factor, Ω_0 :	2.0
Component Response Factor, R_p :	6	Minimum Lateral Acceleration (g's):	NA
Component Amplification Factor, a_p :	2.5	Minimum Vertical Acceleration (g's):	NA
Mounting Height, z (ft)	0	Roof Height, h (ft)	N/A

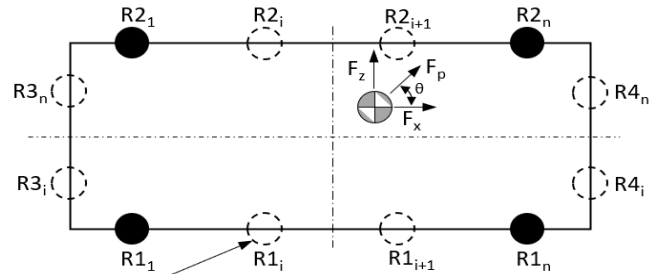
Geometry Description

Isolated System =	No
Eq Length, A (in) =	122
Eq Width, B (in) =	112
Eq Height, H (in) =	94.00
Eq Weight, W (lbs.) =	3596.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	97.84
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	52.358002
Angle of Maximum Effect, θ (deg) =	90
x-dir CG offset, e_x (in) =	12.20
z-dir CG offset, e_z (in) =	11.20
y-dir CG offset, h (in) =	55.00



$$F_x = F_p \cos(\theta)$$

$$F_z = F_p \sin(\theta)$$



Additional restraints when required

Seismic Force Calculation

$$F_p = \frac{0.4 a_p S_{DS} W}{(R_p/I_p)} \left(1 + 2 \frac{z}{h}\right)$$

$$0.3 S_{DS} I_p W \leq F_p \leq 1.6 S_{DS} I_p W$$

$$F_v = 0.2 S_{DS} W$$

$$F_p, (\text{lbs.}) = 1941.8$$

$$F_v, (\text{lbs.}) = 863.0$$

For Anchoring to Concrete

$$F'_p = \Omega_0 F_p$$

Otherwise

$$F'_p = F_p$$

Governing Load Combinations (LRFD)

$$0.9W - F_v + F'_p$$

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	6	15.09
Level 2					
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

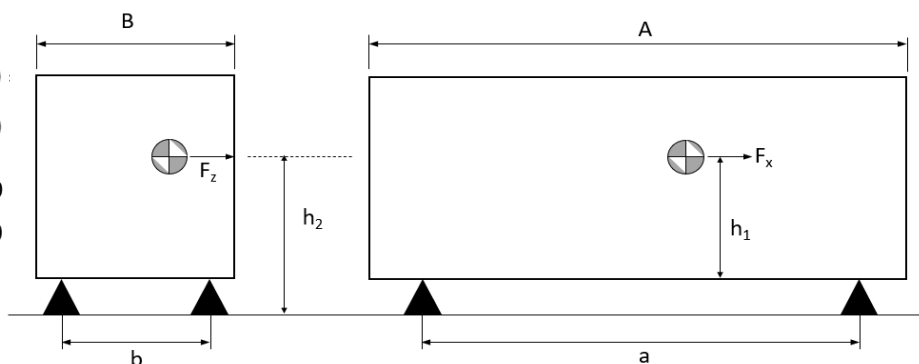
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

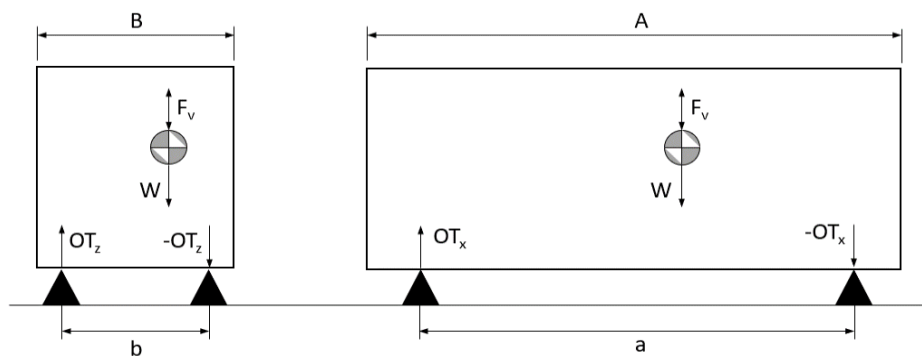
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



Note: The direction of F_v is dependent on which load combination is being applied.

$$(F_v)_{comb} = \pm F_v + OT_x + OT_z - W^*$$

Level 1 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

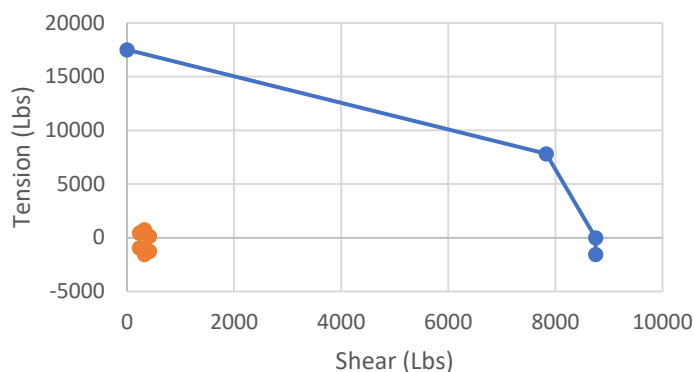
Notes:

- For non-sustained loading conditions.
- Based on maximum shear stress theory.

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$

Capacity Curve - LRFD



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).

Product Type: Gr5 Bolts

Product Size: 5/8 Dia

Product Qty: 6

Orientation: Vertical

Worst Case Tension Load
(lbs.): 760

Worst Case Shear Load
(lbs.): 413

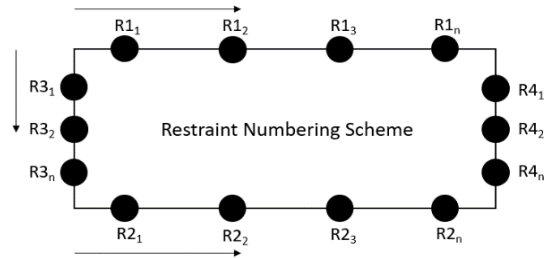
Worst Case Factor of
Safety: 15.09



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-01	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	97.843	52.358	55	3596
2							
3							
4							



Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
3		0		1		0		0		90	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
1	1	Yes	-110	-99	0	234	27	0	527	234	454
1	2	Yes	-343	-309	0	324	82	0	986	324	760
1	3	Yes	-575	-518	0	413	138	0	527	413	147
2	1	Yes	-623	-561	0	234	150	0	-527	234	-938
2	2	Yes	-856	-770	0	324	205	0	-986	324	-1551
2	3	Yes	-1088	-979	0	413	261	0	-527	413	-1245



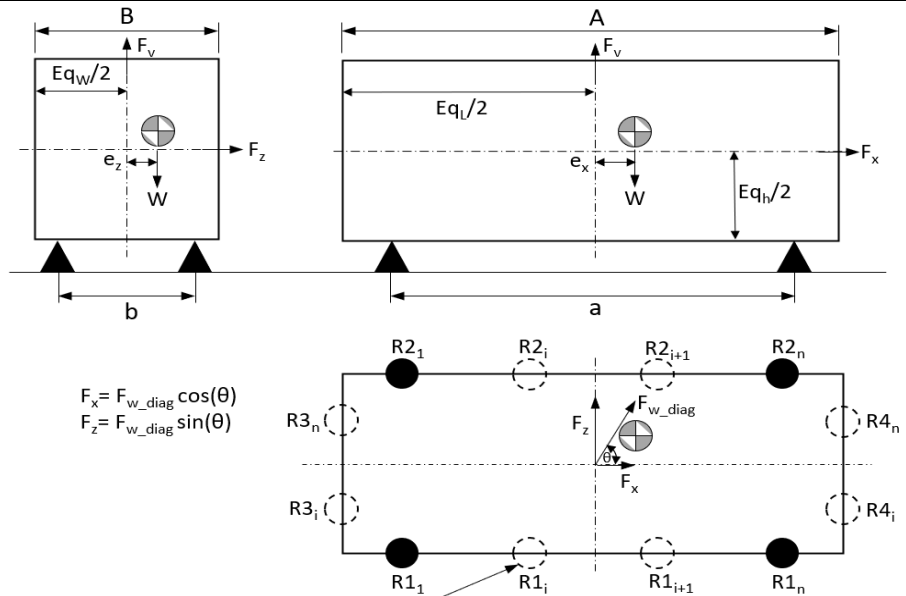
Project:	Nan Inc Hilo WWTP Ph 1	KNC: 217055
Tag:	2521509-01	Designer: FJ
Date:	6/19/2025	Rev: 0.0

Wind Design Basis

Building Code:	2018 IBC	Velocity Press, $P = 0.00256 K_d K_{zt} K_z V^2$ (psf)	49.9
Risk Category:	III	Gust Factor, G	0.85
Basic Wind Speed, V (mph):	145	Lateral Force Coefficient, C_{fh}	1.09
Wind Directionality Factor, K_d :	0.90	Vertical Force Coefficient, C_{fv}	1.09
Exposure Category:	D	GC_{fh}	0.93
Topographic Factor, K_{zt} :	1	GC_{fv}	0.93
Velocity Pressure Coefficient, K_z :	1.03	Mean Roof Height, h (ft)	N/A
Mounting height, z (ft)	0		

Geometry Description

Isolated System =	No
Eq Length, A (in) =	122.0
Eq Width, B (in) =	112.0
Eq Height, H (in) =	94.0
Eq Weight, W (lbs.) =	3596.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	97.8
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	52.4
Angle of Diagonal Wind Force, θ (deg) =	47.4
x-dir CG offset, e_x (in) =	12.2
z-dir CG offset, e_z (in) =	11.2



Additional restraints when required

Wind Force Calculation

Front Area, (Eq Length * Eq Height)/144 (ft ²) =	79.6	x-Dir Wind Force (lbs.) =	3383.8
Side Area, (Eq Width * Eq Height)/144 (ft ²) =	73.1	z-Dir Wind Force (lbs.) =	3686.0
Top Area, (Eq Length * Eq Width)/144 (ft ²) =	94.9	Diagonal Wind Force (lbs.) =	5003.7
Lateral Wind Pressure, $P * GC_{fh}$ (psf) =	46.3	Vertical Wind Force (lbs.) =	4391.8
Vertical Wind Pressure, $P * GC_{fv}$ (psf) =	46.3		

Governing Load Combinations (LRFD)

0.9W + Fw,lat - Fw,vert

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	6	4.10
Level 2					
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

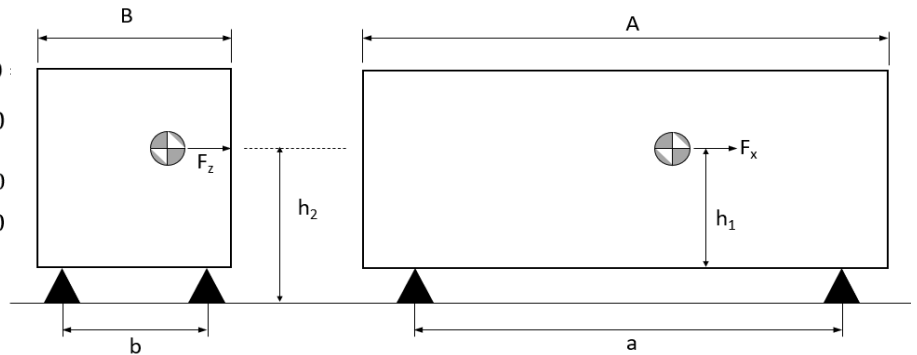
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

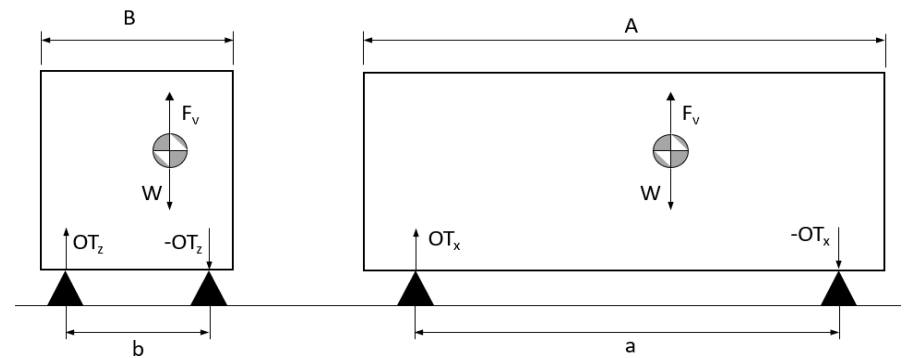
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



$$(F_v)_{comb} = F_v + OT_x + OT_z - W^*$$

Level 1 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

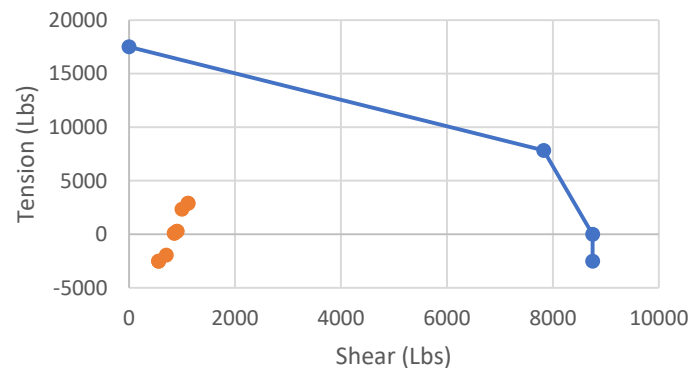
Notes:

- For non-sustained loading conditions.
- Based on maximum shear stress theory.

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$

Capacity Curve - LRFD



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).

Product Type: Gr5 Bolts

Product Size: 5/8 Dia

Product Qty: 6

Orientation: Vertical

Worst Case Tension Load
(lbs.): 2897

Worst Case Shear Load
(lbs.): 1113

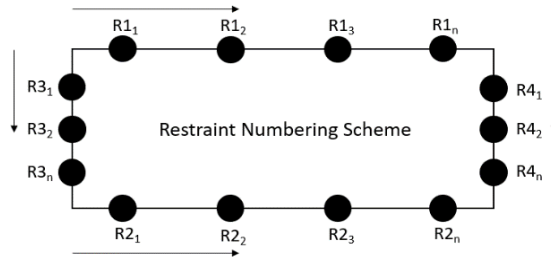
Worst Case Factor of
Safety: 4.10



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-01	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	97.843	52.358	55.0	3596
2							
3							
4							



Level 1 Reaction Loads (lbs)

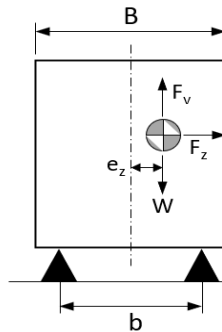
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
3		0		1		0		0		47	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
1	1	Yes	-110	-99	789	785	1329	813	855	1113	2897
1	2	Yes	-343	-309	789	614	1045	0	1599	1000	2336
1	3	Yes	-575	-518	789	444	761	-813	855	905	285
2	1	Yes	-623	-561	339	785	703	813	-855	855	100
2	2	Yes	-856	-770	339	614	419	0	-1599	702	-1951
2	3	Yes	-1088	-979	339	444	135	-813	-855	559	-2512

Seismic Design Basis

Building Code:	2018 IBC	Site Class:	C
Risk Category:	III	Spectral Acceleration, S_{DS} :	1.20
Component Importance Factor, I_p :	1.5	Overstrength Factor, Ω_0 :	2.0
Component Response Factor, R_p :	2	Minimum Lateral Acceleration (g's):	NA
Component Amplification Factor, a_p :	2.5	Minimum Vertical Acceleration (g's):	NA
Mounting Height, z (ft)	0	Roof Height, h (ft)	N/A

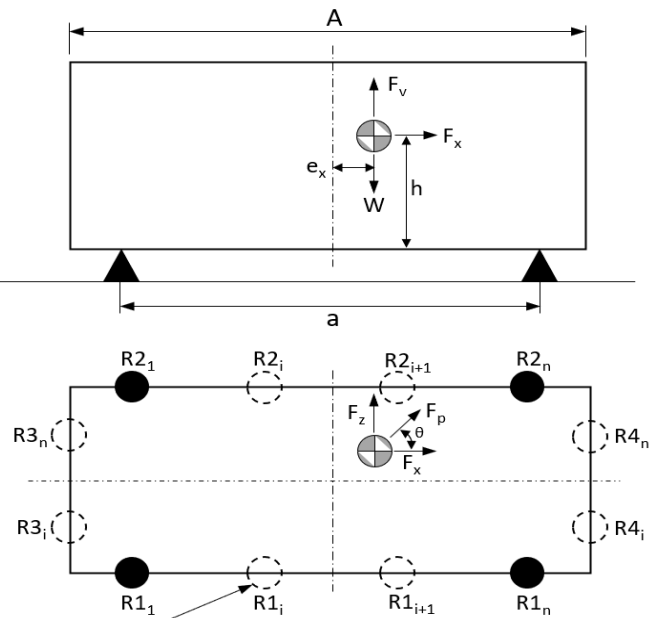
Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	85.263
Eq Width, B (in) =	58.700001
Eq Height, H (in) =	64.63
Eq Weight, W (lbs.) =	1078.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	72.93
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	22
Angle of Maximum Effect, θ (deg) =	290
x-dir CG offset, e_x (in) =	-12.49
z-dir CG offset, e_z (in) =	-1.65
y-dir CG offset, h (in) =	27.64



$$F_x = F_p \cos(\theta)$$

$$F_z = F_p \sin(\theta)$$



Additional restraints when required

Seismic Force Calculation

$$F_p = \frac{0.4 a_p S_{DS} W}{(R_p/I_p)} \left(1 + 2 \frac{z}{h}\right)$$

$$0.3 S_{DS} I_p W \leq F_p \leq 1.6 S_{DS} I_p W$$

$$F_v = 0.2 S_{DS} W$$

$$F_p, (\text{lbs.}) = 970.2$$

$$F_v, (\text{lbs.}) = 258.7$$

For Anchoring to Concrete

$$F'_p = \Omega_0 F_p$$

Otherwise

$$F'_p = F_p$$

Governing Load Combinations (LRFD)

$$0.9W - F_v + F'_p$$

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	3.93
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	26.09
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

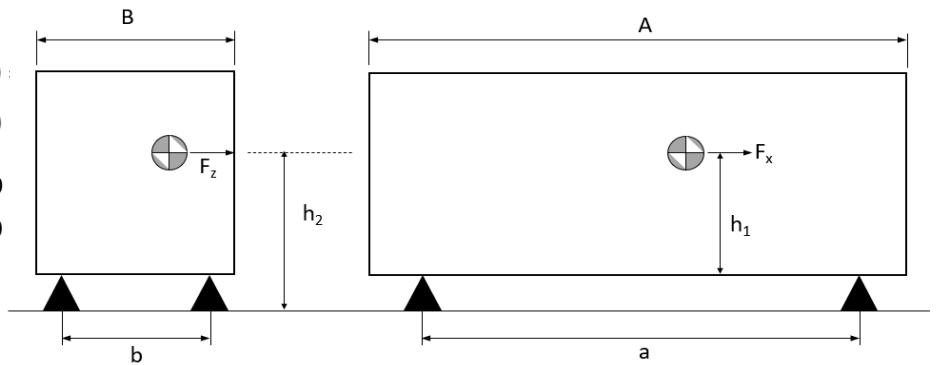
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

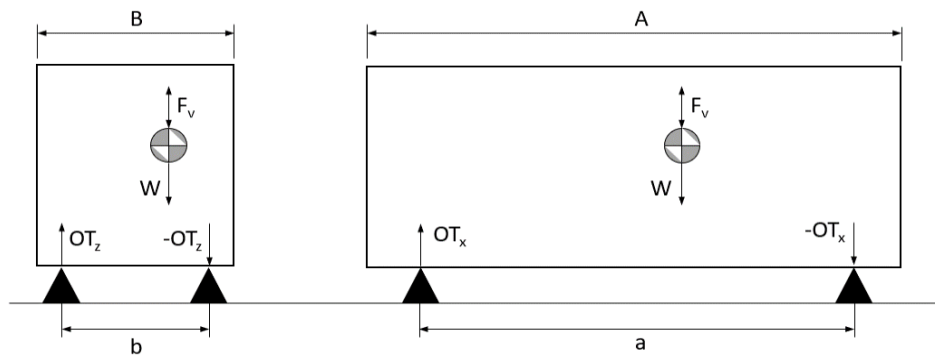
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



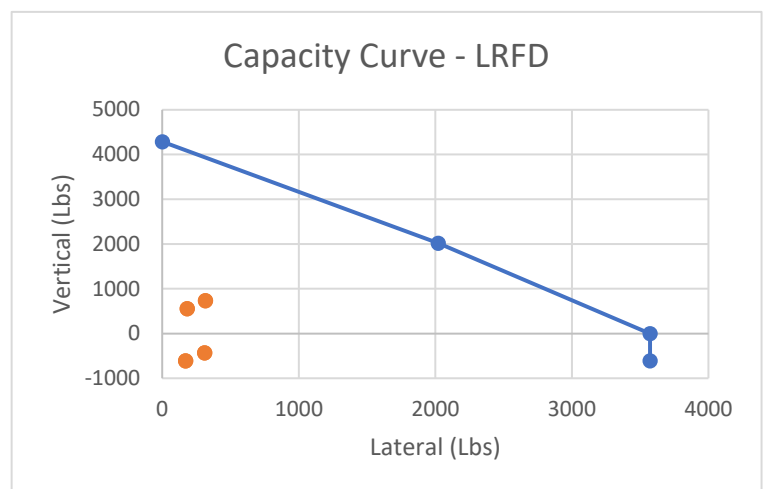
Note: The direction of F_v is dependent on which load combination is being applied.

$$(F_v)_{comb} = \pm F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	738
Worst Case Shear Load (lbs.):	315
Worst Case Factor of Safety:	3.93



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-02

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

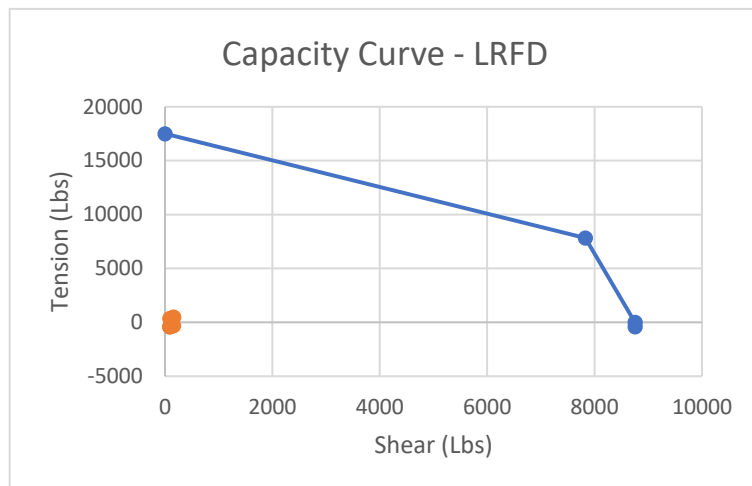
²Shear Capacity = 0.5* Tensile Capacity

Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	476
Worst Case Shear Load (lbs.):	158
Worst Case Factor of Safety:	26.09

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



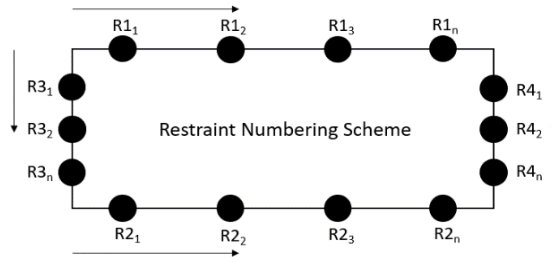
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-02	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	72.929	22	27.644	1078
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	72.929	22	36.954	1078
3							
4							



Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		1		0		0		290	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
	1	No	-426	-383	95	-301	102	63	573	315	738
	1	No	-194	-175	95	-155	47	-63	573	182	557
	2	No	-345	-310	71	-301	83	63	-573	309	-427
2	2	No	-113	-102	71	-155	27	-63	-573	171	-608

Level 2 Reaction Loads (lbs)

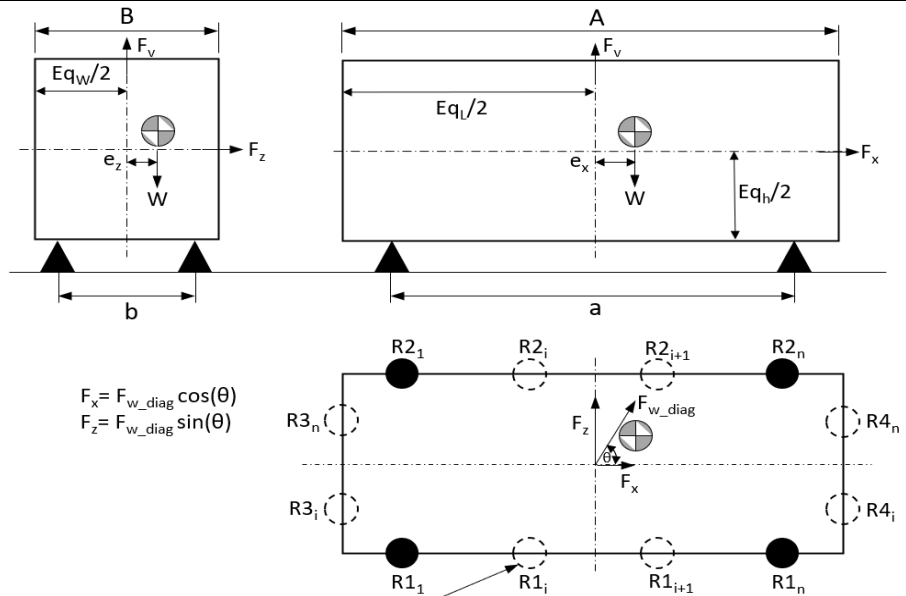
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		2		0		0		290	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-426	-383	95	-301	102	84	766	315	952
1	2	No	-194	-175	95	-155	47	-84	766	182	728
2	1	No	-345	-310	71	-301	83	84	-766	309	-599
2	2	No	-113	-102	71	-155	27	-84	-766	171	-823

Wind Design Basis

Building Code:	2018 IBC	Velocity Press, $P = 0.00256 K_d K_{zt} K_z V^2$ (psf)	49.9
Risk Category:	III	Gust Factor, G	0.85
Basic Wind Speed, V (mph):	145	Lateral Force Coefficient, C_{fh}	1.30
Wind Directionality Factor, K_d :	0.90	Vertical Force Coefficient, C_{fv}	1.30
Exposure Category:	D	GC_{fh}	1.11
Topographic Factor, K_{zt} :	1	GC_{fv}	1.11
Velocity Pressure Coefficient, K_z :	1.03	Mean Roof Height, h (ft)	N/A
Mounting height, z (ft)	0		

Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	85.3
Eq Width, B (in) =	58.7
Eq Height, H (in) =	64.6
Eq Weight, W (lbs.) =	1078.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	72.9
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	22.0
Angle of Diagonal Wind Force, θ (deg) =	55.5
x-dir CG offset, e_x (in) =	-12.5
z-dir CG offset, e_z (in) =	-1.6



Additional restraints when required

Wind Force Calculation

Front Area, (Eq Length * Eq Height)/144 (ft ²) =	38.3	x-Dir Wind Force (lbs.) =	1454.6
Side Area, (Eq Width * Eq Height)/144 (ft ²) =	26.3	z-Dir Wind Force (lbs.) =	2112.9
Top Area, (Eq Length * Eq Width)/144 (ft ²) =	34.8	Diagonal Wind Force (lbs.) =	2565.2
Lateral Wind Pressure, $P * GC_{fh}$ (psf) =	55.2	Vertical Wind Force (lbs.) =	1919.2
Vertical Wind Pressure, $P * GC_{fv}$ (psf) =	55.2		

Governing Load Combinations (LRFD)

0.9W + Fw,lat - Fw,vert

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	1.35
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	7.69
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

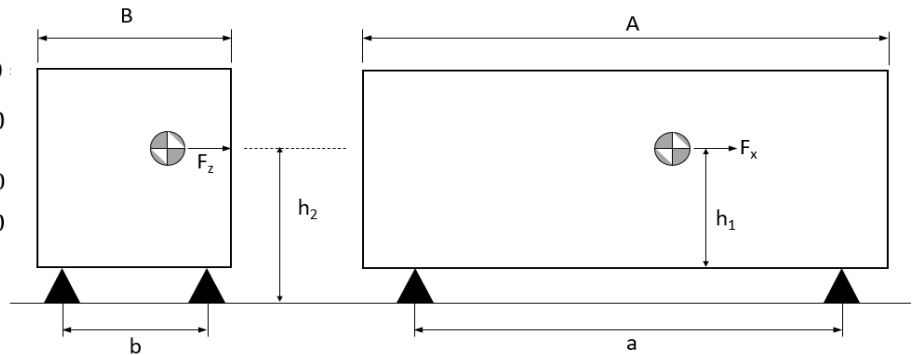
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

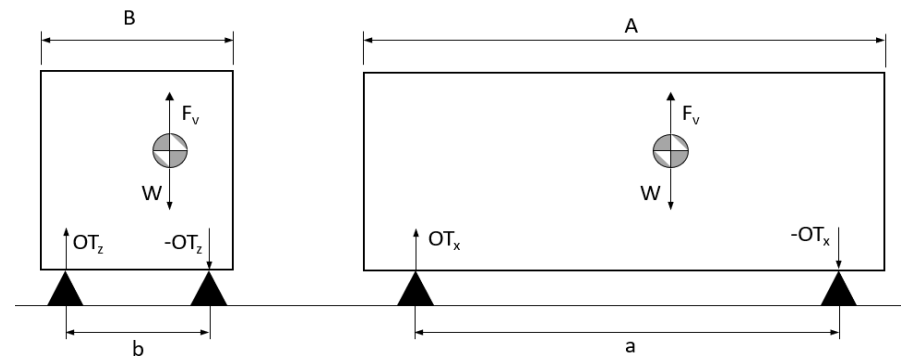
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.

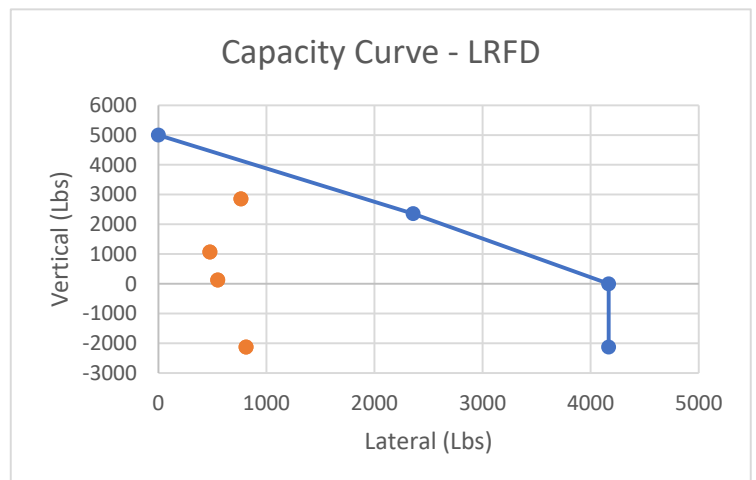


$$(F_v)_{comb} = F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	2852
Worst Case Shear Load (lbs.):	811
Worst Case Factor of Safety:	1.35



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-02

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

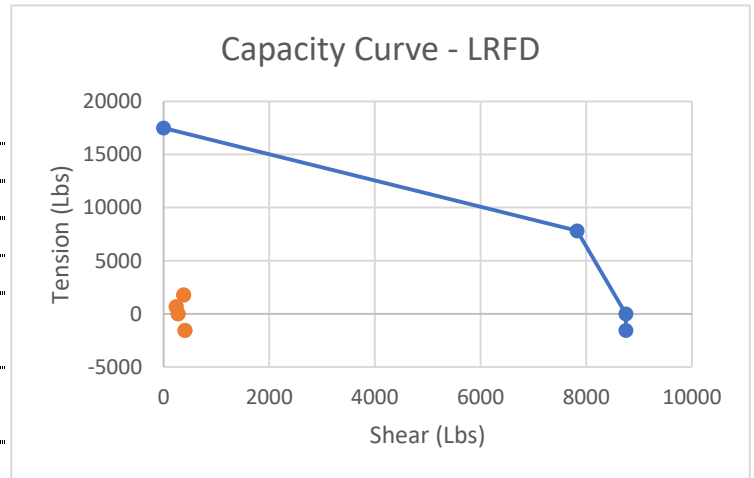
Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	1803
Worst Case Shear Load (lbs.):	405
Worst Case Factor of Safety:	7.69

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



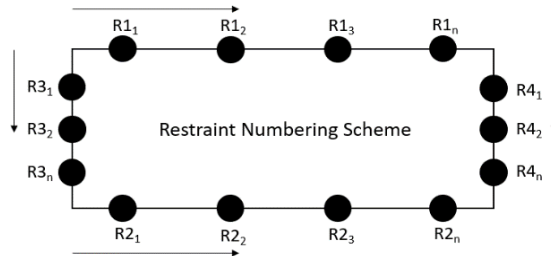
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-02	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	72.929	22	27.6	1078
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	72.929	22	37.0	1078
3							
4							



Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind	
2		0		1		0		0		55	Pass	
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined	
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load	
	1	No	-426	-383	313	360	202	322	543	477	1067	
	1	2	No	-194	-175	313	697	614	-322	2560	764	2852
	2	1	No	-345	-310	415	360	346	322	-543	549	125
2	2	No	-113	-102	415	697	758	-322	-2560	811	-2125	

Level 2 Reaction Loads (lbs)

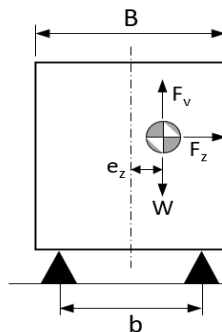
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		2		0		0		55	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-426	-383	313	360	202	431	726	477	1359
1	2	No	-194	-175	313	697	614	-431	3422	764	3605
2	1	No	-345	-310	415	360	346	431	-726	549	51
2	2	No	-113	-102	415	697	758	-431	-3422	811	-3096

Seismic Design Basis

Building Code:	2018 IBC	Site Class:	C
Risk Category:	III	Spectral Acceleration, S_{DS} :	1.20
Component Importance Factor, I_p :	1.5	Overstrength Factor, Ω_0 :	2.0
Component Response Factor, R_p :	2	Minimum Lateral Acceleration (g's):	NA
Component Amplification Factor, a_p :	2.5	Minimum Vertical Acceleration (g's):	NA
Mounting Height, z (ft)	0	Roof Height, h (ft)	N/A

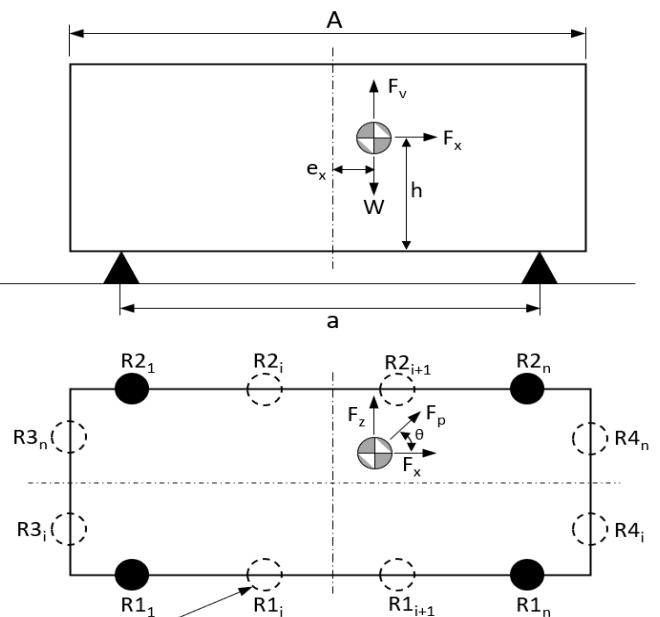
Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	87.763
Eq Width, B (in) =	58.700001
Eq Height, H (in) =	64.63
Eq Weight, W (lbs.) =	1163.00
x Distance Between Outer Restraint and Closest Over	75.43
Turning Point, a (in) =	
z Distance Between Outer Restraint and Closest Over	22
Turning Point, b (in) =	
Angle of Maximum Effect, θ (deg) =	290
x-dir CG offset, e_x (in) =	-13.81
z-dir CG offset, e_z (in) =	-1.64
y-dir CG offset, h (in) =	26.78



$$F_x = F_p \cos(\theta)$$

$$F_z = F_p \sin(\theta)$$



Additional restraints when required

Seismic Force Calculation

$$F_p = \frac{0.4 a_p S_{DS} W}{(R_p/I_p)} \left(1 + 2 \frac{z}{h}\right)$$

$$F_p \text{ (lbs.)} = 1046.7$$

$$0.3 S_{DS} I_p W \leq F_p \leq 1.6 S_{DS} I_p W$$

$$F_v \text{ (lbs.)} = 279.1$$

$$F_v = 0.2 S_{DS} W$$

For Anchoring to Concrete

$$F'_p = \Omega_0 F_p$$

Otherwise

$$F'_p = F_p$$

Governing Load Combinations (LRFD)

$$0.9W - F_v + F'_p$$

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	3.69
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	24.45
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

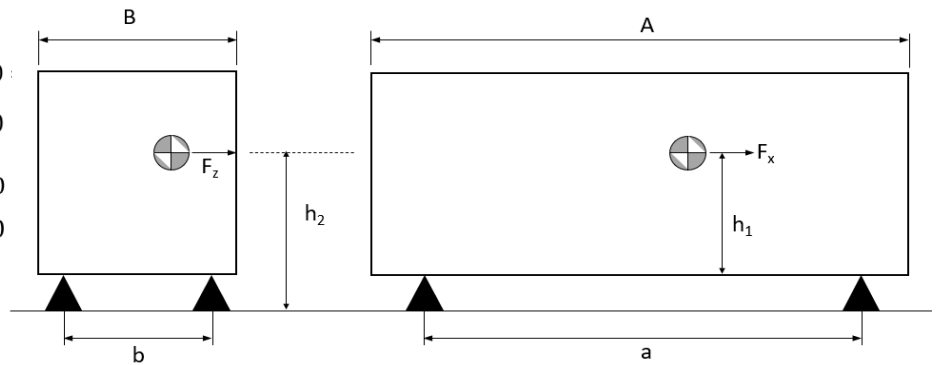
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

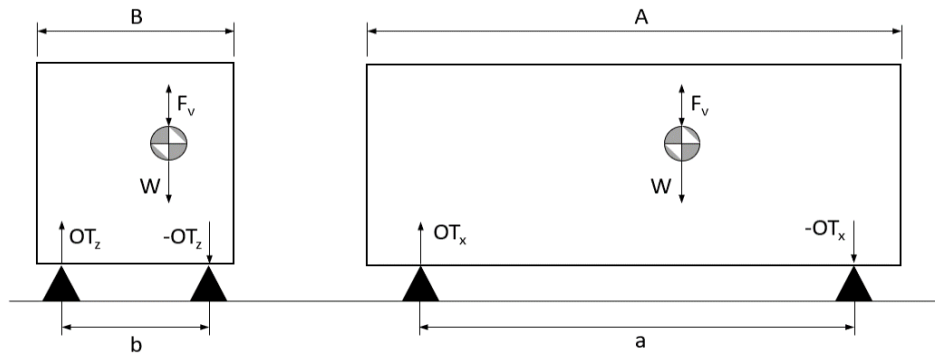
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



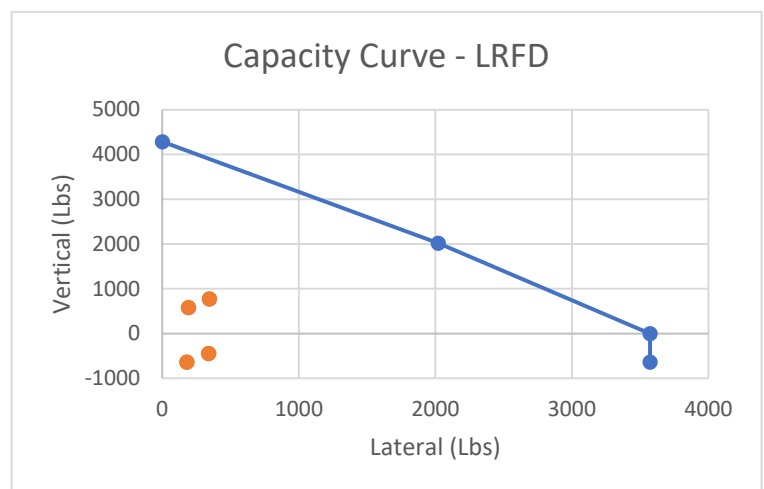
Note: The direction of F_v is dependent on which load combination is being applied.

$$(F_v)_{comb} = \pm F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	774
Worst Case Shear Load (lbs.):	345
Worst Case Factor of Safety:	3.69



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-03

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

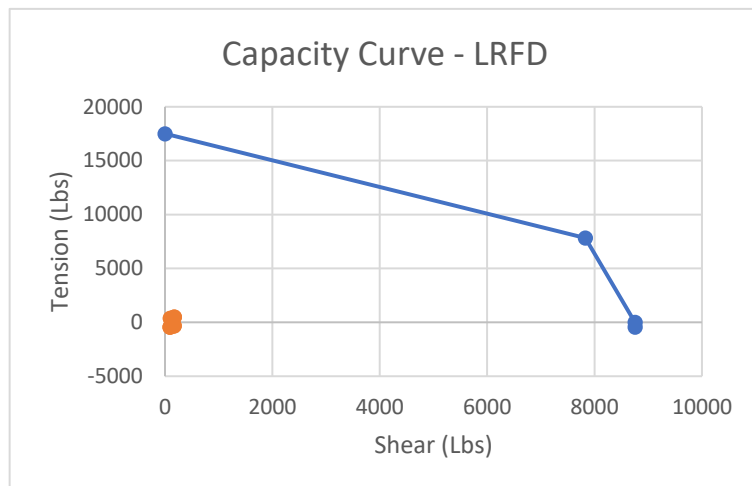
²Shear Capacity = 0.5* Tensile Capacity

Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	502
Worst Case Shear Load (lbs.):	173
Worst Case Factor of Safety:	24.45

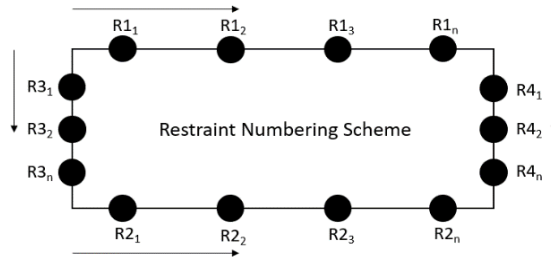
$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).

	Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
	Tag:	2521509-03	Designer:	FJ
	Date:	6/19/2025	Rev:	0.0

Level Inputs							
Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	75.429	22	26.778	1163
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	75.429	22	36.088	1163
3							
4							



Level 1 Reaction Loads (lbs)											
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		1		0		0		290	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat -x Load	Lat -z Load	Vertical Load	OT-x Load	OT-z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-467	-421	102	-330	112	64	599	345	774
1	2	No	-201	-181	102	-162	48	-64	599	191	583
2	1	No	-381	-343	77	-330	91	64	-599	339	-444
2	2	No	-114	-103	77	-162	27	-64	-599	179	-635

Level 2 Reaction Loads (lbs)											
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		2		0		0		290	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-467	-421	102	-330	112	86	807	345	1005
1	2	No	-201	-181	102	-162	48	-86	807	191	769
2	1	No	-381	-343	77	-330	91	86	-807	339	-630
2	2	No	-114	-103	77	-162	27	-86	-807	179	-865



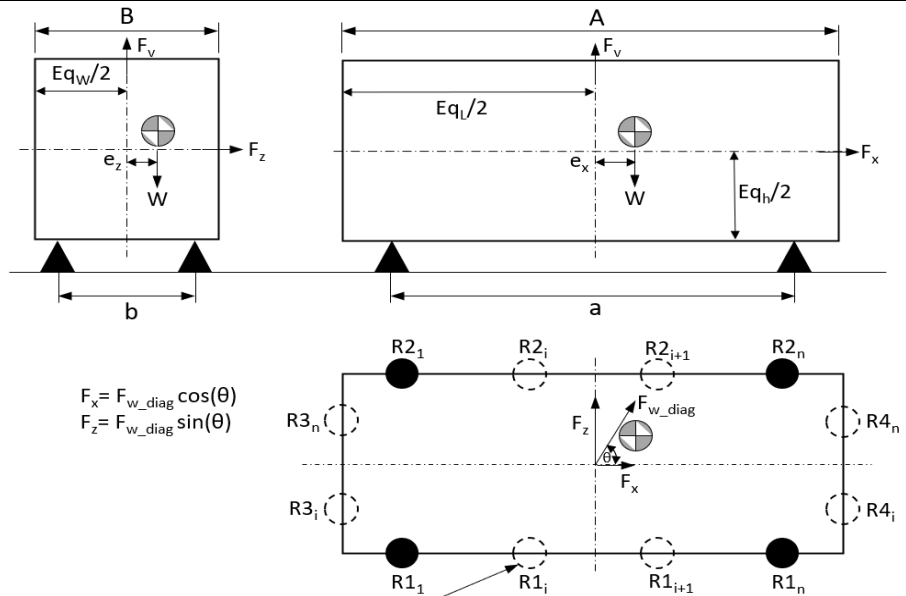
Project:	Nan Inc Hilo WWTP Ph 1	KNC: 217055
Tag:	2521509-03	Designer: FJ
Date:	6/19/2025	Rev: 0.0

Wind Design Basis

Building Code:	2018 IBC	Velocity Press, $P = .00256 K_d K_{zt} K_z V^2$ (psf)	49.9
Risk Category:	III	Gust Factor, G	0.85
Basic Wind Speed, V (mph):	145	Lateral Force Coefficient, C_{fh}	1.30
Wind Directionality Factor, K_d :	0.90	Vertical Force Coefficient, C_{fv}	1.30
Exposure Category:	D	GC_{fh}	1.11
Topographic Factor, K_{zt} :	1	GC_{fv}	1.11
Velocity Pressure Coefficient, K_z :	1.03	Mean Roof Height, h (ft)	N/A
Mounting height, z (ft)	0		

Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	87.8
Eq Width, B (in) =	58.7
Eq Height, H (in) =	64.6
Eq Weight, W (lbs.) =	1163.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	75.4
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	22.0
Angle of Diagonal Wind Force, θ (deg) =	56.2
x-dir CG offset, e_x (in) =	-13.8
z-dir CG offset, e_z (in) =	-1.6



Additional restraints when required

Wind Force Calculation

Front Area, (Eq Length * Eq Height)/144 (ft ²) =	39.4	x-Dir Wind Force (lbs.) =	1454.6
Side Area, (Eq Width * Eq Height)/144 (ft ²) =	26.3	z-Dir Wind Force (lbs.) =	2174.8
Top Area, (Eq Length * Eq Width)/144 (ft ²) =	35.8	Diagonal Wind Force (lbs.) =	2616.5
Lateral Wind Pressure, $P * GC_{fh}$ (psf) =	55.2	Vertical Wind Force (lbs.) =	1975.4
Vertical Wind Pressure, $P * GC_{fv}$ (psf) =	55.2		

Governing Load Combinations (LRFD)

0.9W + Fw,lat - Fw,vert

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	1.30
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	7.35
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

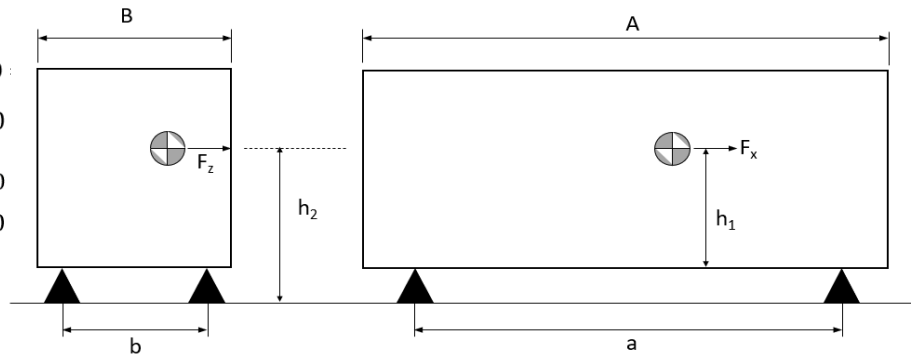
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

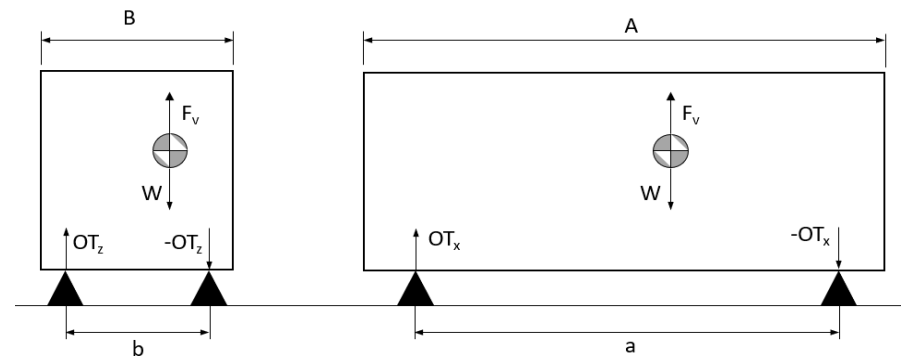
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.

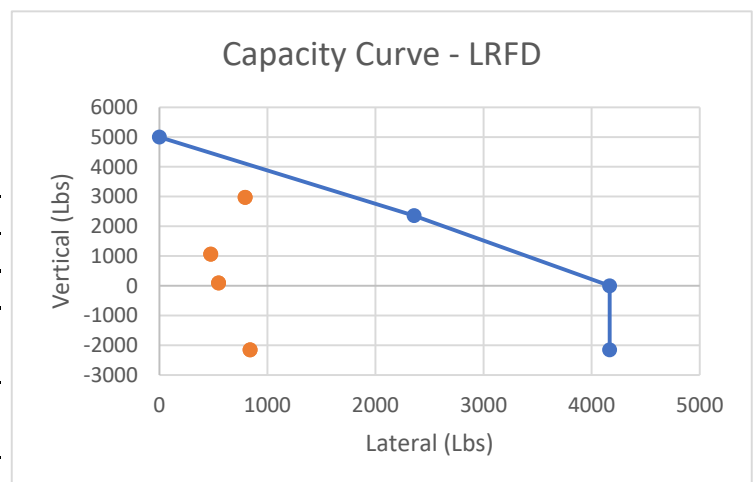


$$(F_v)_{comb} = F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	2970
Worst Case Shear Load (lbs.):	839
Worst Case Factor of Safety:	1.30



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-03

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

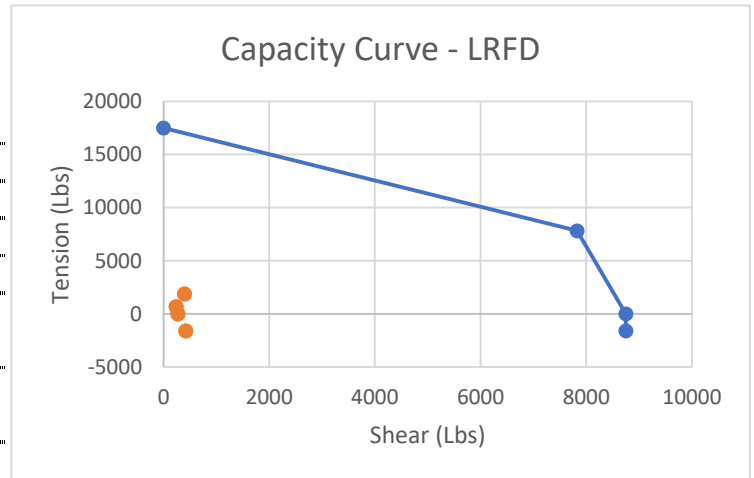
Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	1889
Worst Case Shear Load (lbs.):	419
Worst Case Factor of Safety:	7.35

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



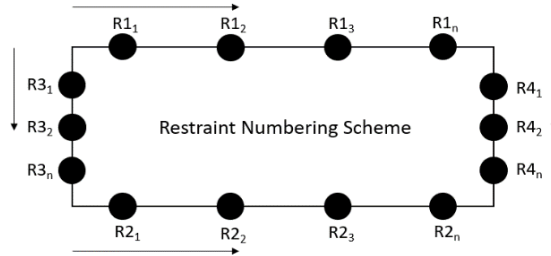
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-03	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	75.429	22	26.8	1163
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	75.429	22	36.1	1163
3							
4							

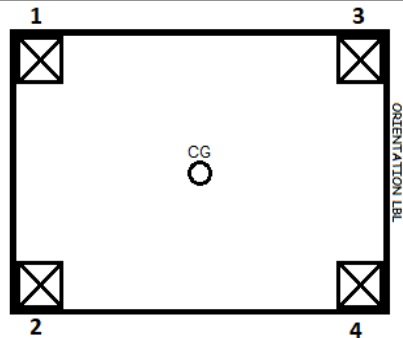


Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		1		0		0		56	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
1	1	No	-467	-421	313	358	194	312	559	476	1064
1	2	No	-201	-181	313	729	647	-312	2635	794	2970
2	1	No	-381	-343	414	358	341	312	-559	548	94
2	2	No	-114	-103	414	729	794	-312	-2635	839	-2153

Level 2 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		2		0		0		56	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-467	-421	313	358	194	420	753	476	1367
1	2	No	-201	-181	313	729	647	-420	3552	794	3778
2	1	No	-381	-343	414	358	341	420	-753	548	8
2	2	No	-114	-103	414	729	794	-420	-3552	839	-3177



Tag/Iso	2521509-02 : FHS-2				2521509-03 : FHS-2						
Iso Num	Load	Isolator	Defl	Iso Num	Load	Isolator	Defl	Iso Num	Load	Isolator	Defl
1/2	416 / 308	2-500 / 2-500	1.66 / 1.23	1/2	456 / 338	2-500 / 2-500	1.82 / 1.35				
3/4	204 / 151	2-225 / 2-225	1.81 / 1.34	3/4	212 / 157	2-225 / 2-225	1.88 / 1.40				
ORIENTATION LBL	INLET SIDE				INLET SIDE						
ISO/REST QTY	4 PER TAG, 4 TOTAL				4 PER TAG, 4 TOTAL						
CG LEN/WDTH	-12.493 / -1.65 From Center				-13.806 / -1.64 From Center						
UNIT WEIGHT	1,078.00 lbs				1,163.00 lbs						
MAN/MODEL	HARTZELL/A 41-10-27				HARTZELL/A 41-10-27						
REPRESENTATIVE	Hartzell Air Movement			DESIGNER	J						
REFERENCE/ COMMENTS	COMPONENTS SELECTED FOR VIBRATION ISOLATION ONLY. WIND/SEISMIC RESTRAINT IS TO BE BY OTHERS. CENTER OF GRAVITY IS ASSUMED TO BE THE GEOMETRIC CENTER OF THE ISOLATED EQUIPMENT UNLESS OTHERWISE SPECIFIED.										

	TITLE	DATE	DRAWN BY	KNC	DRAWING NO.	REV
	GENERAL ISOLATION LOCATIONS	06/19/2025	J	217055	C217055-01	0

FHS 2" DEFLECTION SEISMIC ISOLATOR

IP UNITS (INCHES AND POUNDS)

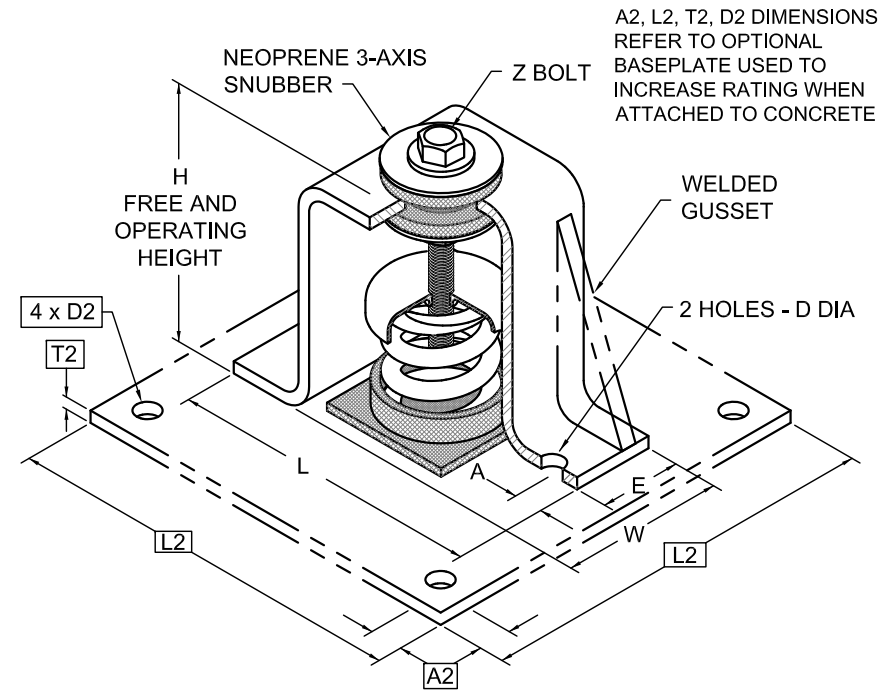
TYPE	L	W	A	D	E	H	Z
2-100/1600	10.38	5.00	0.75	0.69	2.50	9.38	0.50
2-1975	10.38	5.00	0.75	0.69	2.50	9.38	0.50

STANDARD RATINGS		SPRING COIL					
		COLOR		FREE		RATED	
TYPE	SIZE	OUTER	INNER	HT.	O.D.	LOAD	DEFL.
FHS	2-100	GRAY		6.09	3.50	100	2.00
FHS	2-135	BEIGE		6.09	3.50	135	2.00
FHS	2-185	BROWN		6.09	3.50	185	2.00
FHS	2-225	GRAY	BROWN	6.09	3.50	225	2.00
FHS	2-250	BLUE		6.09	3.50	250	2.00
FHS	2-375	BLUE	BROWN	6.09	3.50	375	2.00
FHS	2-500	GREEN		6.09	3.50	500	2.00
FHS	2-625	GREEN	BROWN	6.09	3.50	625	2.00
FHS	2-750	BLACK		6.09	3.50	750	2.00
FHS	2-875	BLACK	BROWN	6.09	3.50	875	2.00
FHS	2-995	ORANGE		6.09	3.50	995	2.00
FHS	2-1120	ORANGE	BROWN	6.09	3.50	1120	2.00
FHS	2-1400	ORANGE	GREEN	6.09	3.50	1400	2.01
FHS	2-1600	RED		6.09	3.50	1600	2.00
*FHS	2-1975	RED	GREEN	6.09	3.50	1975	1.98

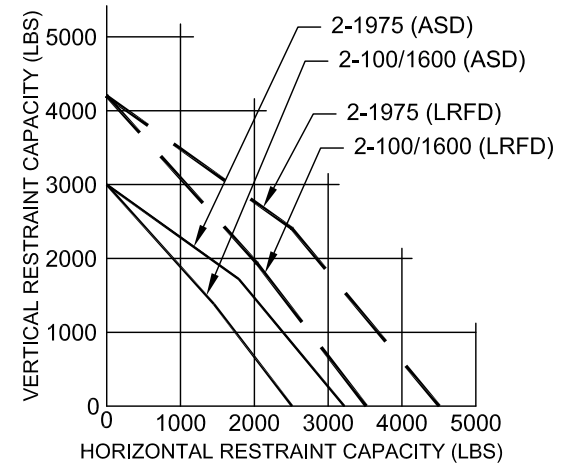
*INCLUDES WELDED GUSSET AS SHOWN

SPECIFICATIONS:

- VERTICALLY AND Laterally RESTRAINED SPRING ISOLATOR WITH CONSTANT FREE AND OPERATING HEIGHT AND REPLACEABLE NEOPRENE SNUBBING ELEMENTS.
- 0.25 NOMINAL CLEARANCE IN SNUBBING ELEMENTS.
- MINIMUM OVERLOAD OF 50%.
- MINIMUM K_x/K_y RATIO OF 1.0.
- ALL ELEMENTS SAFE AT SOLID LOADING.
- POWDER COATED COILS AND HOT DIPPED GALVANIZED BRACKET.



WELDED TO STEEL



Computed horizontal and vertical seismic load combinations must fall within the restraint capacity envelope. Contact KNC for bolted to steel data.



KINETICS NOISE CONTROL, INC
6300 IRELAN PL,
DUBLIN, OH 43017 USA
Ph: 614 889-0480, Fax: 614 889-0540
www.kineticsnoise.com

Model:

FHS-2-
100/1975

By: PWM

Date: 12/21/01

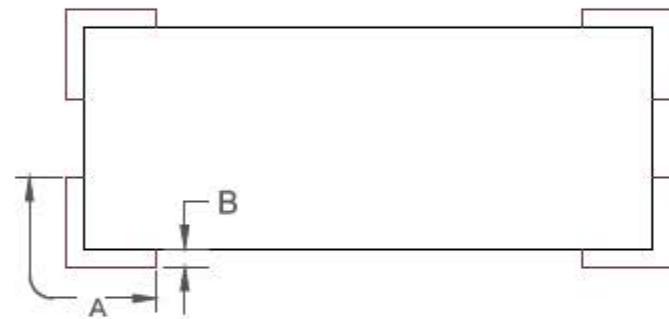
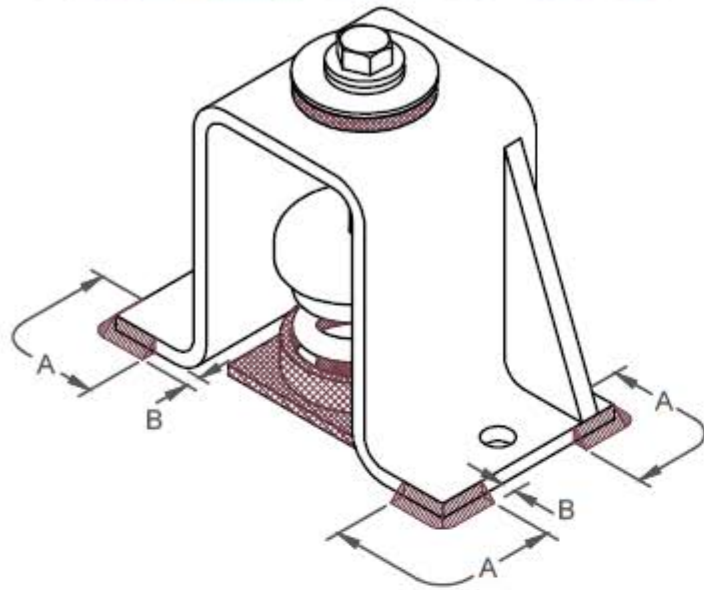
Revised: 05/26/22 / BB

Drawing No:

S-01.35-21

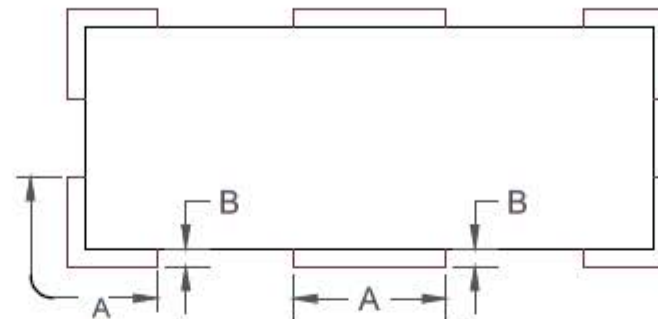
ISOLATOR WELD DATA

FHS WELD DOWN DATA



WELD 4 PLACES AS SHOWN

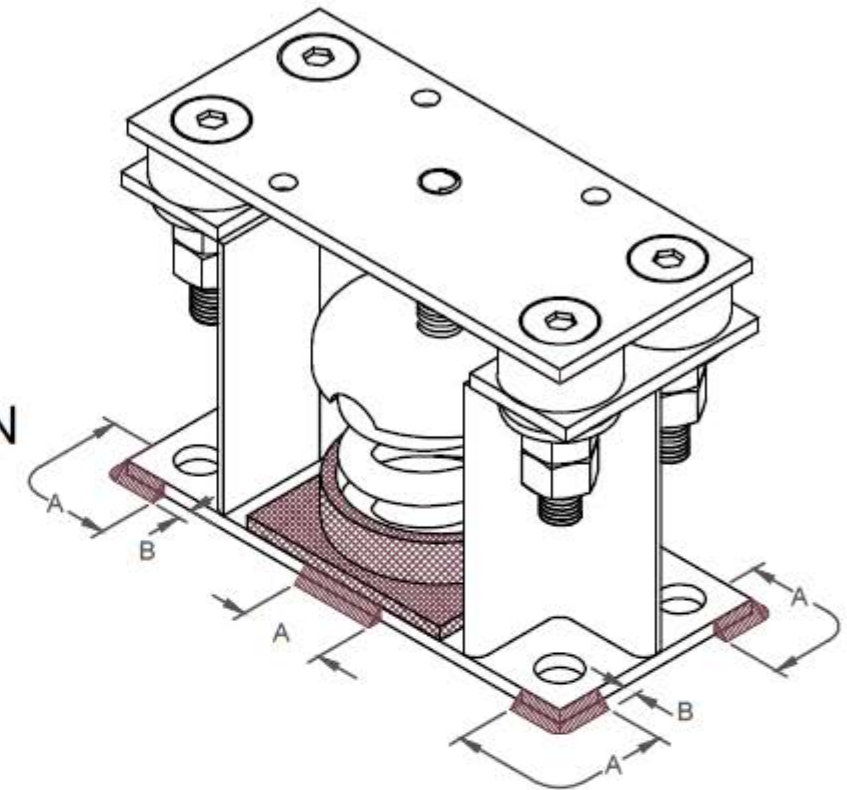
MODEL	LGTH (A)	SIZE(B)
1-35/805	3.25	5/16
1-50/3500	3.25	5/16
2-100/1975	3.25	5/16
2-2000/4500	3.25	5/16
4-100/1600	3.25	5/16
4-2250/5800	3.25	5/16



WELD 6 PLACES AS SHOWN

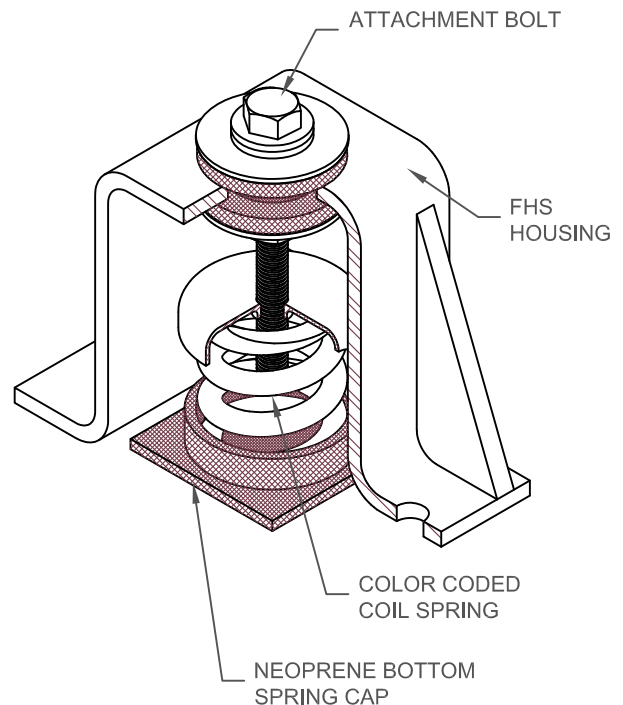
MODEL	LGTH(A)	SIZE(B)
1-35/805	2.25	1/4
1-50/3500	3.50	1/4
1-2500/7000	3.50	1/4
1-5000/14000	4.50	5/16
2-100/1975	3.50	1/4
2-2000/4500	3.50	1/4
2-4000/9000	3.50	1/4
2-8000/18000	4.50	5/16
4-100/1600	3.50	1/4
4-2250/5800	3.50	1/4
4-5500/11600	3.50	1/4
4-11000/23200	4.50	5/16

FLS, FLSS, FTS WELD DOWN DATA



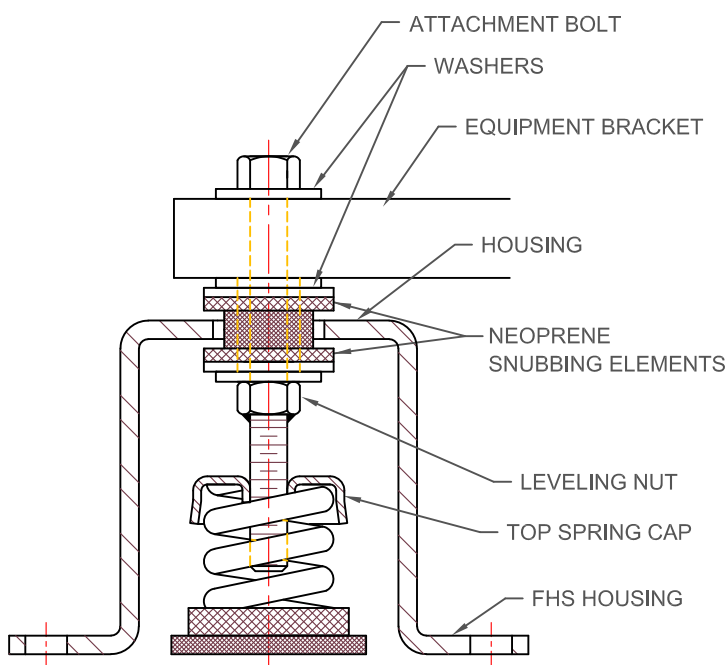
FHS INSTALLATION INSTRUCTIONS

- 1) COORDINATE THE LOCATION OF EACH ISOLATOR WITH THE APPROPRIATE SUBMITTAL DRAWING.
- 2) REMOVE THE ATTACHMENT BOLT AND ONE WASHER FROM THE TOP OF THE ISOLATOR. INSERT THEM INTO THE EQUIPMENT BRACKET.
- 3) ALIGN ISOLATORS AND CENTER THE ISOLATOR HOUSINGS BENEATH THE EQUIPMENT BOLTING HOLES. MARK THE ISOLATOR HOUSING ANCHOR BOLT LOCATIONS ON THE FLOOR.
- 4) RAISE THE EQUIPMENT AND INSTALL THE ANCHOR BOLTS INTO THE FLOOR.
- 5) LOCATE THE ISOLATORS AND LOWER THE EQUIPMENT ONTO THEM. NOTE THE HOUSING SHOULD SUPPORT THE EQUIPMENT WITH APPROXIMATELY THE DESIRED FLOOR CLEARANCE. (TAKE CARE NOT TO PUSH THE EQUIPMENT SIDWAYS).
- 6) INSTALL ANCHOR NUTS TO FIRMLY ATTACH THE FHS HOUSING TO THE FLOOR OR HOUSEKEEPING PAD.
- 7) LOOSELY BOLT EQUIPMENT TO THE ISOLATOR USING THE ATTACHMENT BOLTS. BEFORE ADJUSTING THE ISOLATORS, IT IS NECESSARY THAT THE EQUIPMENT BE BROUGHT UP TO ITS OPERATING WEIGHT. ADJUST THE ISOLATOR BY TURNING THE LEVELING NUT COUNTER CLOCKWISE ONE OR TWO TURNS AT EACH ISOLATOR,



IN SUCCESSION, UNTIL THE UPPER NEOPRENE ELEMENT BARELY LIFTS OFF OF THE TOP OF THE FHS HOUSING. WHEN THIS HAPPENS, NO LONGER ADJUST THIS ISOLATOR, BUT CONTINUE TO ADJUST THE REMAINING ISOLATORS UNTIL ALL ARE IN THIS CONDITION.

- 8) AT THIS POINT, TURN ALL THE LEVELING SCREWS THE APPROPRIATE AMOUNT TO LIFT THE EQUIPMENT 1/4". THE NUMBER OF TURNS IS A FUNCTION OF THE LEVELING BOLT SIZE. 1/2" = 3 TURNS, 5/8" = 2-3/4 TURNS, 3/4" = 2-1/2 TURNS, 7/8" = 2-1/4 TURNS, 1" = 2 TURNS.
- 9) TIGHTEN THE ATTACHMENT BOLTS, THUS BOLTING THE ISOLATORS TO THE SUPPORTED EQUIPMENT AND LOCKING THE LEVELING BOLT AGAINST TURNING.
- 10) DO NOT ATTEMPT TO MOVE THE ISOLATORS Laterally WITH THE WEIGHT OF THE EQUIPMENT ON THEM. IF IT IS NECESSARY TO MOVE THE EQUIPMENT ON THEM, REMOVE THE WEIGHT FROM THE ISOLATORS BY RAISING THE EQUIPMENT BEFORE MOVING.



APPENDIX:

KNC #217055

SUBMITTAL #1.0

CUSTOMER PROVIDED EQUIPMENT INFORMATION FOR REFERENCE ONLY

IT IS CRITICAL TO NOTE THAT THE CERTIFICATION PROVIDED BY KINETICS NOISE CONTROL ONLY COVERS THOSE HARDWARE ITEMS WHICH HAVE BEEN PROVIDED BY KINETICS NOISE CONTROL. IT IS BASED ON BUT DOES NOT EXTEND TO THE EQUIPMENT REFERENCED IN THIS APPENDIX.

SECTION 03.02 LINE ITEM TECHNICAL INFORMATION



Line No.: 2521509-02

Description: A41-0-271FA-66FGI8LT - Fiberglass Backward
Curved Centrifugal Fan

Tag: 02-EF-1330 & 02-EF-1340



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 3 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 99WE
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:

Ship To Phone:

Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
----------	----------------------	----------	---------	-------------------------------	----------------

2	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				
---	---	--	--	--	--

12/31/2025

2.00

Each

A41-0-271FA-66FGI8LT
41 - Fiberglass Backward Curved Centrifugal Fan

Tag: 02-EF-1330 & 02-EF-1340

Performance
Volume Flow Rate: 5025.0 cfm
SP: 3.5 in. w.g. / TP: 3.61 in. w.g.
RPM: 1168.0
Density: 0.075 lbs/ft³
Operating Temp: 70 °F
OPwr: 3.652 hp / SPwr: 3.652 hp

Motor
HP: L - 5 HP
RPM/Volt/Hz/Phase: T - 1750, 460, 60, 3
Enclosure: I8 - TEFC IEEE841
Frame Size: 184T
Operating Voltage: 460
Vendor: BALDOR

Motor Accessories
Hermetically Sealed Thermostat
Space Heater - 115V/120V
Hartzell Motor P/N: 3109-0053-B-(MODIFIED), Mfct P/N: ECP83665T-4

Fan
Arrangement: 10
Rotation and Discharge: Up Blast (Clockwise)
Mounting Location: Floor
Motor Position: Standard
Material: Fiberglass
Construction: ASTM D4167 97 (Includes Single Veil & Electric Grounding)
Fan Coating: Hartkoate
Prop/Wheel: FA - Backward Airfoil Centrifugal Wheel - Fiberglass
Prop/Wheel Coating: Hartkoate & Electric Grounding



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 4 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date:	5/19/2025 8:45:57 AM	Ship Method:	PPD CC
Salesman Code:	99WE	FOB:	Piqua
Partial Ship:	False	AR Terms:	Per Invoice

Ship To : WILL ADVISE
WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID	Req Date	Req Qty	Unit Price	Extended Price
	Item Name			Unit of Measure	

Standard Accessories (Included)
Water Slinger
Inlet Support, Steel - Inorganic Zinc Coating w/ Epoxy Topcoat

Additional Accessories
Bolted Access Door, Fiberglass
Drain, CPVC w/ CPVC Plug (Factory Installed)
Extended Lube Lines to Fan Bearings, 316SS
316SS Hardware to Complete Fan
316SS Shaft
Teflon Double Lip Seal in a 316SS Casing
Drilled Inlet Flange, PS15-69
Drilled Outlet Flange, PS15-69
Inlet Box, Fiberglass w/ 316SS Support Legs ; 12:00 Orientation
FAB Airfoil Backdraft Damper - Fiberglass: Max SP - 14.5: Max FPM - 4000
Damper Style: Parallel
Polycarbonate Jamb Seal for Damper
EPDM Blade Seal for Damper
FRP (Standard) Axle for Damper
Both Flanges Drilled For Damper

Seismically Restrained Spring Isolators

Inlet Support, Motor & Bearing Base, and Weather Cover to be 316 SS
Material of Construction: 316 SS,

ACCESS DOOR TO HAVE ONE PIECE NEOPRENE GASKET PUNCHED TO MATCH BOLT PATTERN

SPECIAL ARRANGEMENT 10 WEATHER COVER WITH CLEAR WINDOWS FOR BELT AND BEARING VIEWING

Inlet Box Drain, CPVC w/ CPVC Plug

2-Sets of Spare Belts (2 years worth)

ADDER FOR -- 15830 2.03 B 5 L10 100,000 hour bearings

Spare set of bearings

Total Weight (approx.): 1153 lbs. (Each) (quoted items are not included in the weight.)



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 5 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 99WE
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

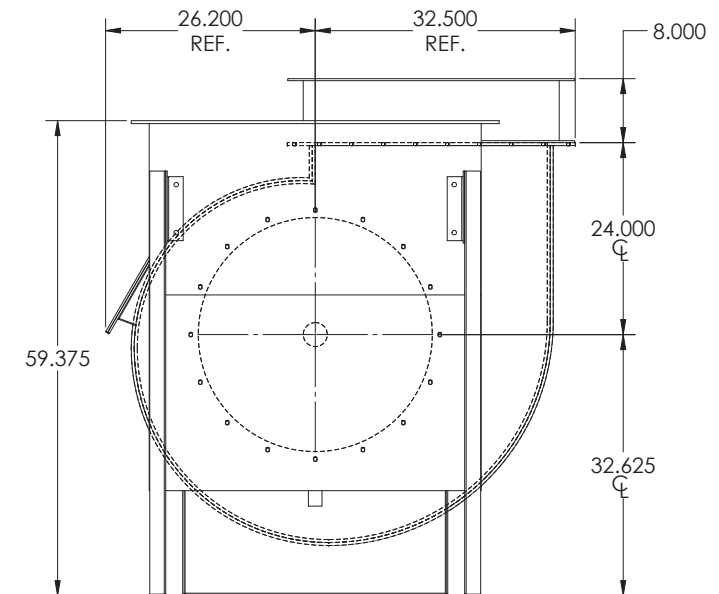
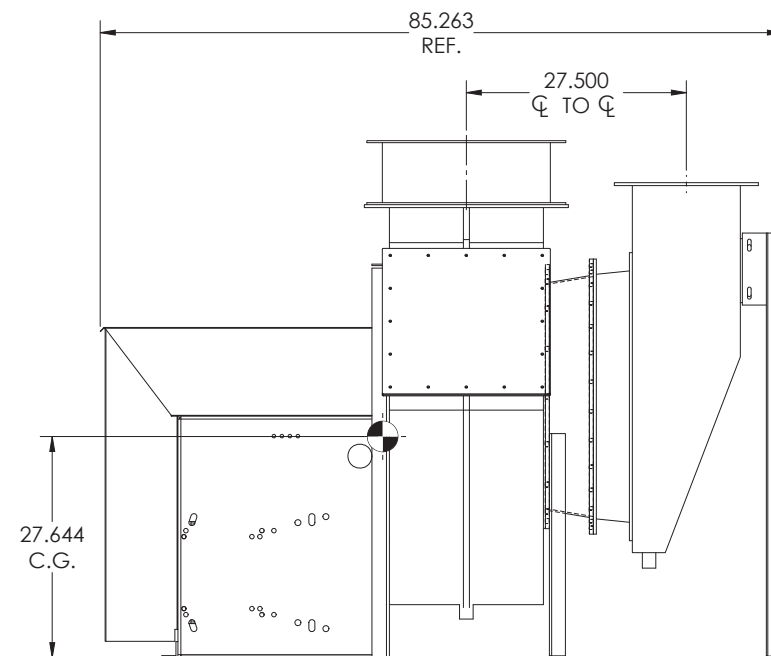
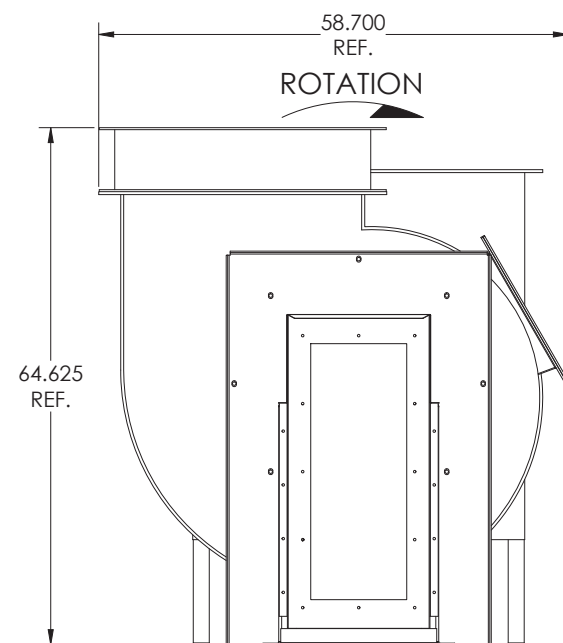
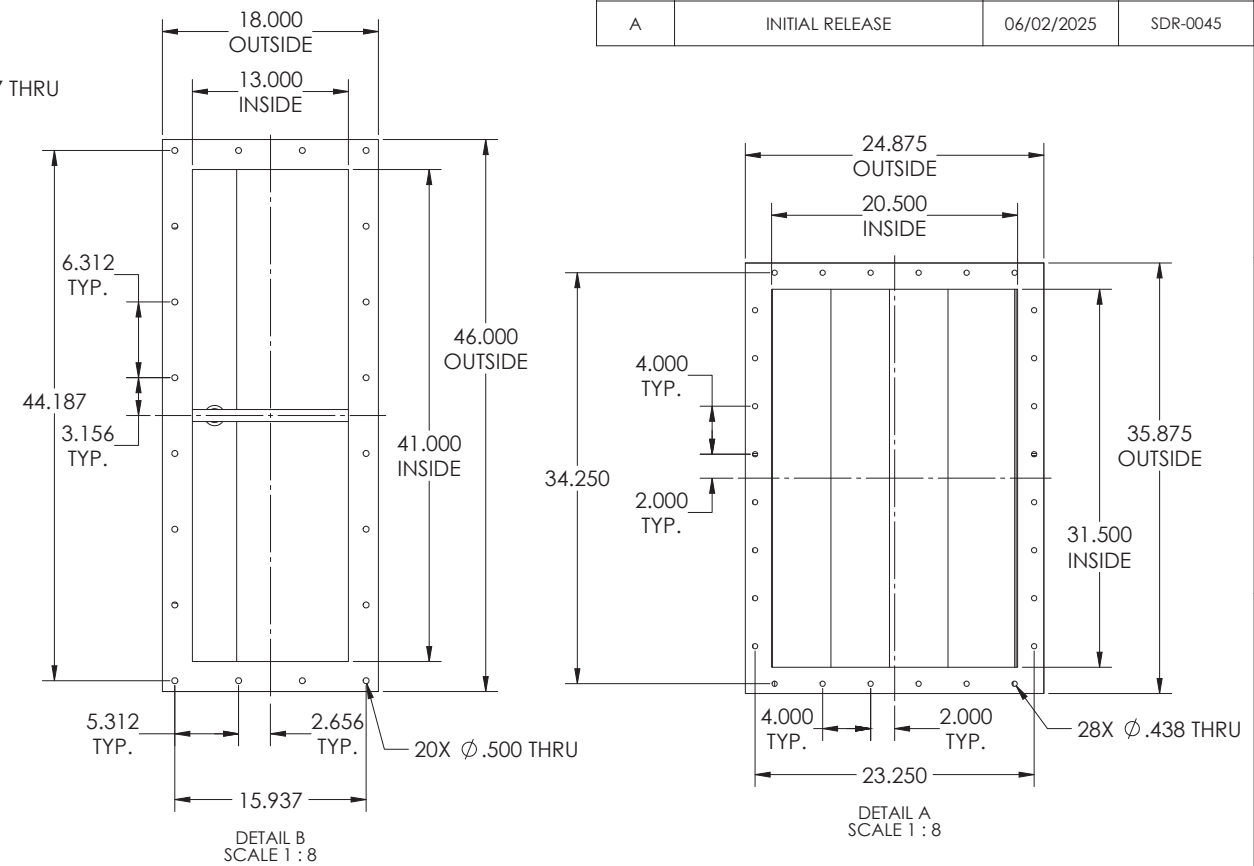
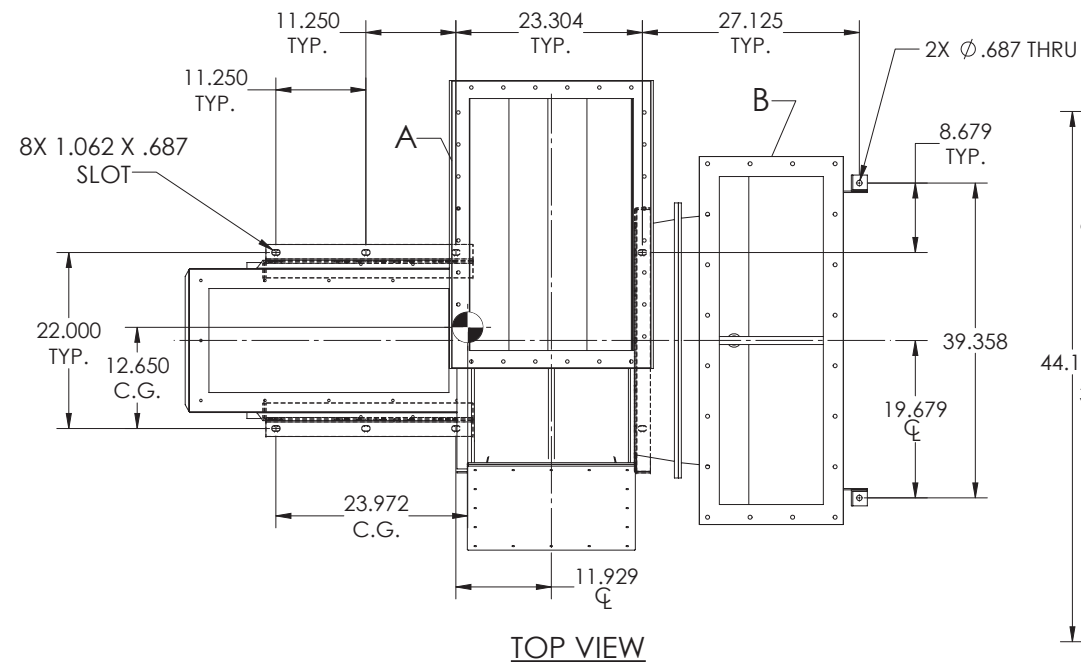
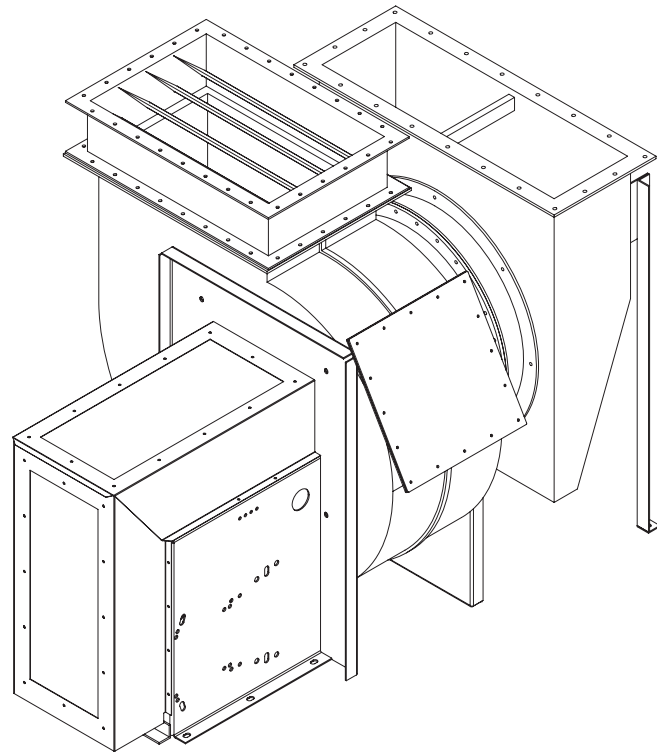
Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID	Req Date	Req Qty	Unit Price	Extended Price
	Item Name			Unit of Measure	

Blanket Qty: 0.00

Blanket Exp Date:

REV.	DESCRIPTION	DATE	CHANGE NO.
A	INITIAL RELEASE	06/02/2025	SDR-0045



CERTIFIED PRINT

BY  DATE: 06/02/2025

SALES ORDER # 2521509

CUSTOMER PO # 24077-00083

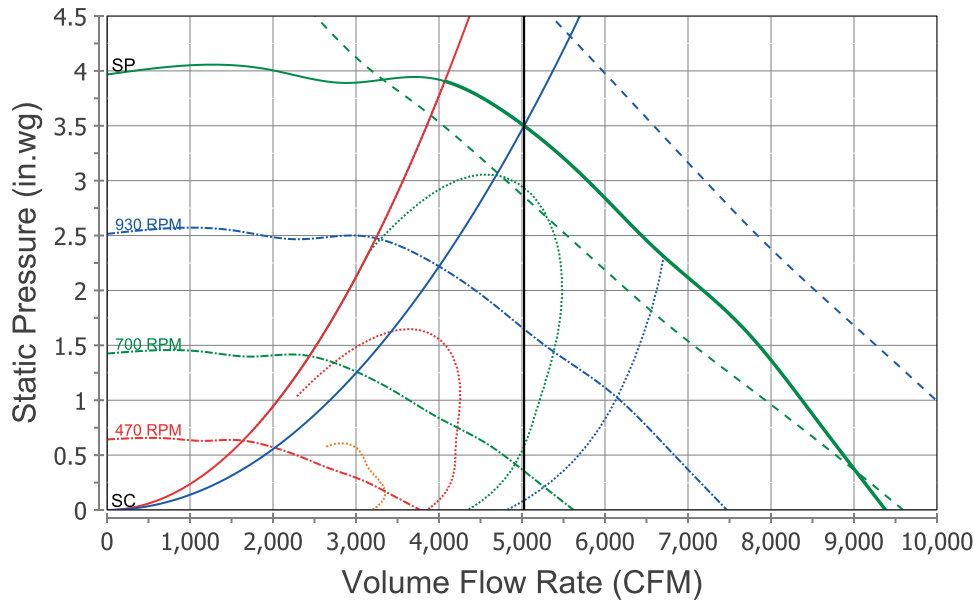
FAN TAG # 02-EF-1330 & 02-EF-1340

APPROX. WEIGHT (lb): 1078

TYPE	PRODUCT SERIES	ARRANGEMENT / DUTY / COLOR			SIZE	# OF BLADES / CLASS		BLADE / WHEEL		BLADE ANGLE / WHEEL WIDTH		MATERIAL OF CONSTRUCTION	MOTOR ENCLOSURE	MOTOR HP/RPM		ADDITIONAL INFORMATION:				
		ARRANGEMENT	DUTY	COLOR		# OF BLADES	CLASS	BLADE	WHEEL	BLADE ANGLE	WHEEL WIDTH			HP	RPM	MOTOR FRAME	BLADE OD	ROTATION	DISCHARGE	
A	41-	10			27		1		FA		66	FG	18	5	1750	184T		CW	UB	

 AIR MOVEMENT PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF HARTZELL AIR MOVEMENT. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF HARTZELL AIR MOVEMENT IS PROHIBITED. FILE LOCATION: C:\Vaulth\Custom\ORDERS\2520000\2521509\	MATERIAL INFORMATION MATERIAL: SEE BOM MATERIAL THICKNESS WEIGHT 1078 SURFACE AREA 57155 PERIMETER LENGTH	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES DIMENSIONAL TOLERANCE $\pm 1/8"$ ANGULAR $\pm$ MACH. $\pm 5^\circ$ BEND $\pm 1^\circ$ DIMENSIONS MARKED AS REC. ARE NOT SUBJECT TO TOLERANCE. NAME DATE DRAWN BY NDW 6/2/2025 CHECKED APPROVED	Hartzell Air Movement Piqua, Ohio 45356 TITLE: 41-27" 66% ARR. 10 FIBERGLASS BACKWARD CURVED CENTRIFUGAL FAN SIZE DWG. NO. D 41-2521509-02 REV A DO NOT SCALE DRAWING SHEET 1 OF 1
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Hartzell-FLOW™ v1.0.18 / February 2019
A41-0-271FA-66FGI8LT



Fan Tag#: 02-EF-1330 & 02-EF-1340

Vol Flow Rate	5025
Pressure	3.5
Density (lbs/ft³)	0.075
Oper. Temp. (°F)	70
Fan RPM	1168
Max Safe RPM	1521
Operating Power	3.652
Standard Power	3.652
Static Efficiency	0.758
Outlet Velocity (fpm)	1347
Fan Energy Index (FEI)	1.37
Fan Efficiency Grade (FEG)	FEG80

Discharge Sound Power Levels referred to 10 ⁻¹² watts							
1	2	3	4	5	6	7	8
87	85	84	85	80	79	70	61

Radiated Sound Power Levels referred to 10 ⁻¹² watts							
1	2	3	4	5	6	7	8
83	78	74	74	68	66	57	48
Radiated Sound is not AMCA Licensed							

— System Curve — Static Pressure (in.wg) — Do Not Select — 5 HP — 3 HP — FEI=1.3 — FEI=1.4 — FEI=1.5 — FEI=1.6 — AMCA Certified

Hartzell Air Movement certifies that the model shown is licensed to bear the AMCA Seal. The ratings shown are based on tests and procedures performed in accordance with AMCA publication 211 and AMCA publication 311 and comply with the requirements of the AMCA Certified Ratings Program. FEI values are calculated in accordance with ANSI/AMCA Standard 208 and are based on four-pole TEFC motors of the size shown. Power rating BHP excludes transmission losses. Performance certified is for Installation Type D: Ducted Inlet, Ducted Outlet. AMCA Licensed for Sound and Air Performance. Performance ratings do not include the effects of appurtenances (Accessories). Sound ratings are based on sound level data obtained in accordance with AMCA Standard 300. The sound power level ratings shown are in decibels, referred to 10⁻¹² watts, calculated per AMCA Standard 301. Fan Outlet Sound Testing. Values shown are for outlet Lwo sound power levels for: Installation Type D: Ducted Inlet, Ducted Outlet. Ratings include the effects of duct end correction.

Discharge Sound Pressure = 75 dBA @ 5ft

Radiated Sound Pressure = 63 dBA @ 5ft

Discharge Sound Power = 86 LwA

Radiated Sound Power = 75 LwA

The A-weighted sound pressure level (dBA) is based on Hartzell Laboratory sound power tests, and is calculated in accordance with AMCA standard 303.

The FEG, dBA, LwA and radiated values are not AMCA International Licensed.

The calculation assumes a free field condition with a directivity factor for hemi-spherical radiation (Q=2).

The installed sound pressure levels are influenced by the installation and acoustic environment, and cannot be guaranteed. Use of this estimate level along for field acceptability test is not recommended.

Although the calculation can be done for any stated distance, the free field does not start until 20 to 50 ft from the equipment in most installations.

Contact Hartzell Air Movement for more information concerning dBA values.



Fan Shaft Critical Speed and Bearing Life Analysis

Model: 411-27FA-66

Reference: 2521509-02

Actual Fan Diameter (in.): 27.625

Tip Speed: 8,447

Motor Horsepower: 5

Fan Sheave Dia. (in.): 6.2

Sheave Wt. (lbs.): 9

Shaft Speed (rpm): 1168

Shaft Diameter (in.): 2.1875

Wheel Wt. (lbs.): 81

Shaft Material: 316 SS

Tensile Yield Strength: 30000

Hub Material: 1020 HR

Tensile Yield Strength: 30000

Arrangement No.: 10

Fan Air Temp (deg. F): 70

Ambient Temp (deg. F): 70

Belt Tension Factor: 1.82

Belt Angle from 6 O'Clock (deg.): 45

Shaft Modulus of Elasticity (lbs/in^2): 2.92E+07

Wheel WR2 (lb-ft^2): 67.8

WR2 Factor: 0.5

Distance From Wheel to Drive Bearing (in.): 29.13

Distance Between Bearings (in.): 20.5625

Distance From Drive Bearing to Sheave (in.): 3.938

Wheel Overhang: 8.5675

	w/o Belt Pull	w/Belt Pull
Critical Speed	7622	5935
Safety Factor	6.526	5.081

G Shock Load: 10

Minimum Shaft Dia.: 1.331

Hub O.D.: 3.500

Hub Wall: 0.65625

Static Pressure (neg. for downblast): 3.5

Expeller Vanes?: No

Minimum Bearing L10 Life: 200,000

Shaft Orientation: Horizontal

Belt Pull: 158

Belt Pull: 158

Key Size: 0.5

Key Length: 0.066

Load on Bearing A (Wheel): 119

Load on Bearing B (Drive): 183

Total Axial Thrust: 30

Bearing A: Fixed

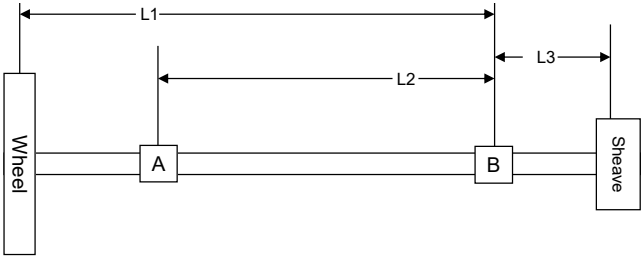
Bearing B: Fixed

Base Load

Base Load

Series	Position	Dynamic Rating	Status	Equivalent Load	Fixed or Expansion	L10 Life
Link-Belt U200	A	7,540	Good Selection	117	Fixed	>1 million
	B	7,540	Good Selection	183	Fixed	995,500

V041555/21/2025



Thrust Multiplier

0.67

Today is: Wednesday, May 21, 2025



Document:

ES 7.5.1

Rev

1Title: **Inlet Boxes for Centrifugal Fans – Series 03, 05, and 41**Owner: **Engineering**Document Type: **Engineering Standard**Creation Date: **5/2/1989**

Inlet Boxes for Centrifugal Fans – Series 03, 05, and 41

1.0 Scope

This standard defines the dimensional and performance requirements for inlet boxes used on centrifugal fans

2.0 Products Affected

- 2.1 Series 03 Backward Curved Centrifugal Fan, Single Width
- 2.2 Series 05 Industrial Exhausters
- 2.3 Series 41 Fiberglass Backward Curved Centrifugal Fans

3.0 Dimensions

Inlet dimensions are based on the fan wheel diameters as shown below. For Series 03 and Series 41 fans, they are approximately 1.5 wheel diameters wide by 0.5 wheel diameters deep. For Series 05 fans, they are approximately 1.0 wheel diameters wide by 0.5 wheel diameters deep

Series 03 / 41			Series 05		
Fan Size	Inlet Width (in.)	Inlet Depth (in.)	Fan Size	Inlet Width (in.)	Inlet Depth (in.)
12	18	6	12	12	6
15	24	8	15	16	8
18	28	10	19	20	10
22	34	12	22	22	12
24	36	12	26	26	14
27	42	14	29	30	14
30	46	16	33	34	16
33	50	16	36	36	18
36	54	18	40	40	20
40	69	20	45	46	22
44	66	22	50	50	26
49	74	24	57	58	28
54	86	28			
60	96	30			
66	105	34.25			
73	116	38			
81	129	41.5			
89	141.75	46			
98	156.5	51			

Actual Inlet Vel. - 1231 fpm
Actual Press. Drop - 0.09 in w.c.

4.0 Performance

The pressure drop through an inlet box is based on tests and is shown below. Use the inlet areas listed above to calculate the inlet velocity. These values include all system effects

Inlet Velocity (fpm)	1300	1500	2000	2500	3000	3500	4000	4500	5000
Pressure Drop (in W.G.)	0.10	0.16	0.31	0.50	0.72	0.99	1.30	1.64	1.97

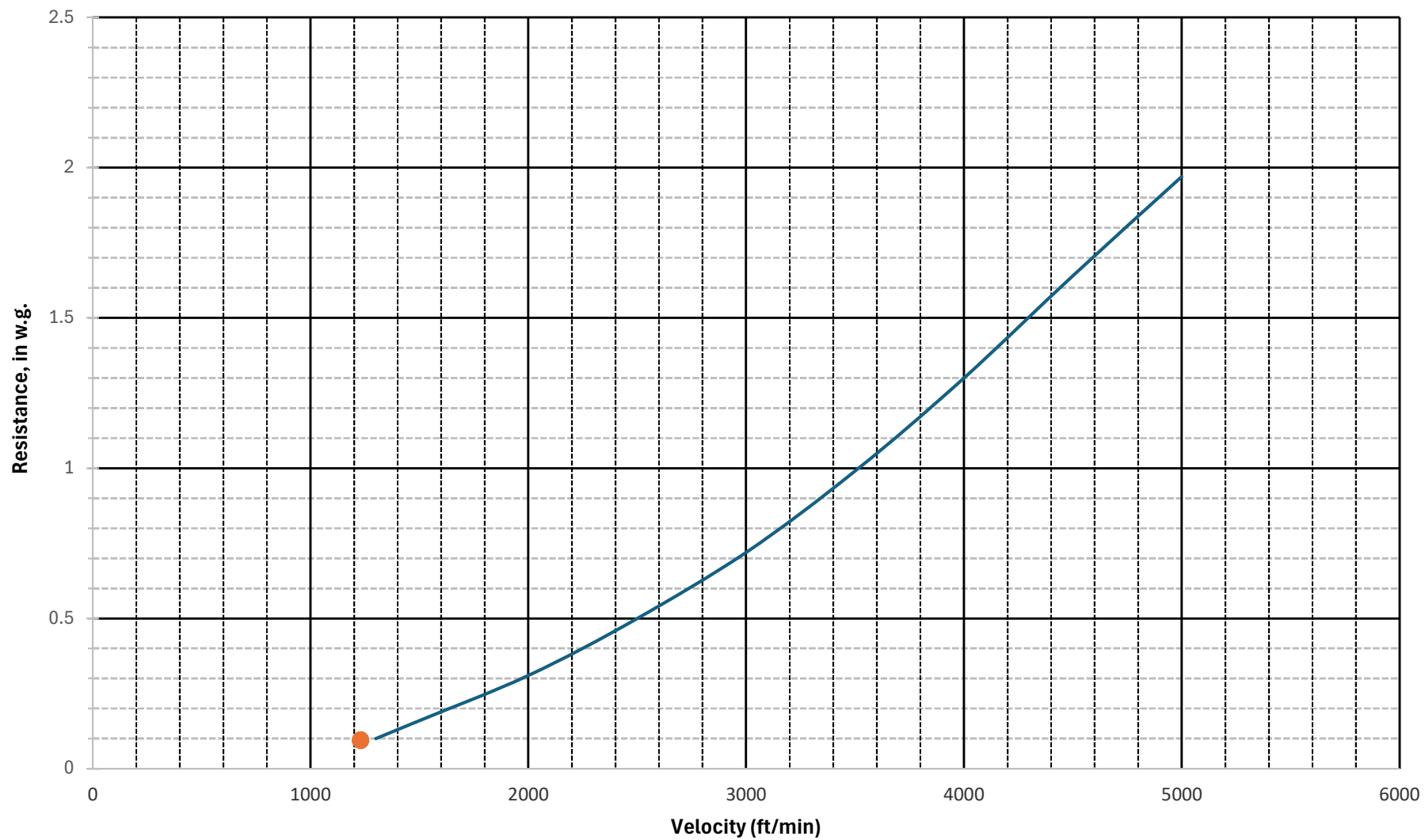
5.0 References

- 5.1 N/A

6.0 Revision History

Rev	Description	Date	Changed By
1	Remove from Drive System and Release in QMS	10/30/2024	J.Felver

Inlet Box Resistance





Customer information packet

ECP83665T-4

5HP, 1750RPM, 3PH, 60HZ, 184T, 0641M, TEFC, F1

Class - CLI GP A,B,C,D

Division - Division II

Specifications

Enclosure	TEFC
Frame	184T
Frame Material	Iron
Frequency	60.00 Hz
Haz Area Class and Group	CL I GP A,B,C,D
Haz Area Division	Division II
Motor Letter Type	Three Phase
Output @ Frequency	5.000 HP @ 60 HZ
Phase	3
Synchronous Speed @ Frequency	1800 RPM @ 60 HZ
Voltage @ Frequency	460.0 V @ 60 HZ
Agency Approvals	CSA EEV NEMA PREMIUM NEMA_PREMIUM UR CCSA US
Ambient Temperature	40 °C
Auxillary Box	No Auxillary Box
Auxillary Box Lead Termination	None
Base Indicator	Rigid
Bearing Grease Type	Polyrex EM (-20F +300F)
Blower	None
Constant Torque Speed Range	1.7
Current @ Voltage	6.450 A @ 460.0 V
Design Code	B
Drip Cover	No Drip Cover
Duty Rating	CONT
Efficiency @ 100% Load	89.5 %
Electrically Isolated Bearing	Not Electrically Isolated
Enclosure Modification	841
Feedback Device	NO FEEDBACK
Front Face Code	Standard

Part detail

Revision	AG
Type	AC
Mech. spec.	06H759
Base	
Status	PRD/A
Elec. spec.	06WGX203
Layout	06LYH759
Eff. date	07-17-2024
CD Diagram	CD0006
Poles	04
Leads	3#16
Proprietary	False
Created date	04-04-2012

Front Shaft Indicator	None
Haz Area Temp Code	T3C
Heater Indicator	No Heater
High Voltage Full Load Amps	6.5 a
Insulation Class	F
Inverter Code	Inverter Duty
KVA Code	J
Lifting Lugs	Standard Lifting Lugs
Locked Bearing Indicator	Locked Bearing
Max Speed	2700 rpm
Motor Lead Exit	Ko Box
Motor Lead Quantity/Wire Size	3 @ 16 AWG
Motor Lead Termination	Ring Terminals
Motor Standards	NEMA
Motor Type	0642M
Mounting Arrangement	F1
Number of Poles	4
Overall Length	15.93 IN
Power Factor	81
Product Family	Chem Process S/P 32-8 IEEE 841
Pulley End Bearing Type	Ball
Pulley Face Code	Standard
Pulley Shaft Indicator	Standard
Rodent Screen	None
RoHS Status	ROHS COMPLIANT
Service Factor	1.15
Shaft Diameter	1.125 IN
Shaft Extension Location	Pulley End
Shaft Ground Indicator	No Shaft Grounding
Shaft Rotation	Reversible
Shaft Slinger Indicator	Shaft Slinger
Speed	1750 rpm
Speed Code	Single Speed
Starting Method	Direct on line

Thermal Device - Bearing	None
Thermal Device - Winding	None
Vibration Sensor Indicator	No Vibration Sensor
Winding Thermal 1	None
Winding Thermal 2	None

Nameplate

NP4328											
CAT.NO.	ECP83665T-4										
SPEC.	06H759X203G2										
HP	5 TE				IP	56					
VOLTS	460										
AMPS	6.45										
R.P.M.	1750										
FRAME	184T		HZ		60		PH	3			
SER.F.	1.15	CODE	J	DES.	B	CLASS		F			
RATING	40C AMB-CONT										
SN											
DE	6206		ODE		6206						
NEMA NOM. EFF.	89.5		P.F.		81						
GUAR. MIN. EFF.	87.5		CC		010A						
T. CODE	T3C		TEMP=		160						

NP3186

SPEC.	06H759X203G2				
ABMA DE BRG	30BC02X30X				
ABMA ODE BRG	30BC02X30X				
GREASE	POLYREX EM				
MOTOR WEIGHT	127	ROTOR BARS	28	STATOR BARS	36
MAX. R.P.M.	2700	MAX. KVAR	1		
INV.TYPE	PWM				
T=	160				
CHP	60	TO	90		
CT	1.7	TO	60		
VT	-0	TO	60		
HTR-VOLTS	N/A	HTR-AMPS	N/A		
HTR-WATTS	N/A	MAX. SPACE HEATER TEMP.	N/A		

AC Induction Motor Performance Data

Record # 92406

Typical performance - not guaranteed values

Winding: 06WGX203-R061		Type: 0642M	Enclosure: TEFC
Nameplate Data		460 V, 60 Hz: Single Voltage Motor	
Rated Output (HP)	5	Full Load Torque	15 LB-FT
Volts	460	Start Configuration	direct on line
Full Load Amps	6.6	Breakdown Torque	53.9 LB-FT
R.P.M.	1750	Pull-up Torque	22.4 LB-FT
Hz	60 Phase	Locked-rotor Torque	31.8 LB-FT
NEMA Design Code	B KVA Code	Starting Current	46 A
Service Factor (S.F.)	1.15	No-load Current	3.02 A
NEMA Nom. Eff.	89.5 Power Factor	Line-line Res. @ 25°C	2.63 Ω
Rating - Duty	40C AMB-CONT	Temp. Rise @ Rated Load	53°C
S.F. Amps		Temp. Rise @ S.F. Load	65°C
		Locked-rotor Power Factor	40.5
		Rotor inertia	0.391 lb-ft ²

Load Characteristics 460 V, 60 Hz, 5 HP

% of Rated Load	25	50	75	100	125	150	S.F.
Power Factor	41	63	75	81	84	85	81
Efficiency	85.2	89.8	90.4	89.5	88.4	86.9	88.7
Speed	1789	1777	1765	1752	1737	1721	1739
Line amperes	3.35	4.14	5.21	6.45	7.91	9.47	7.52

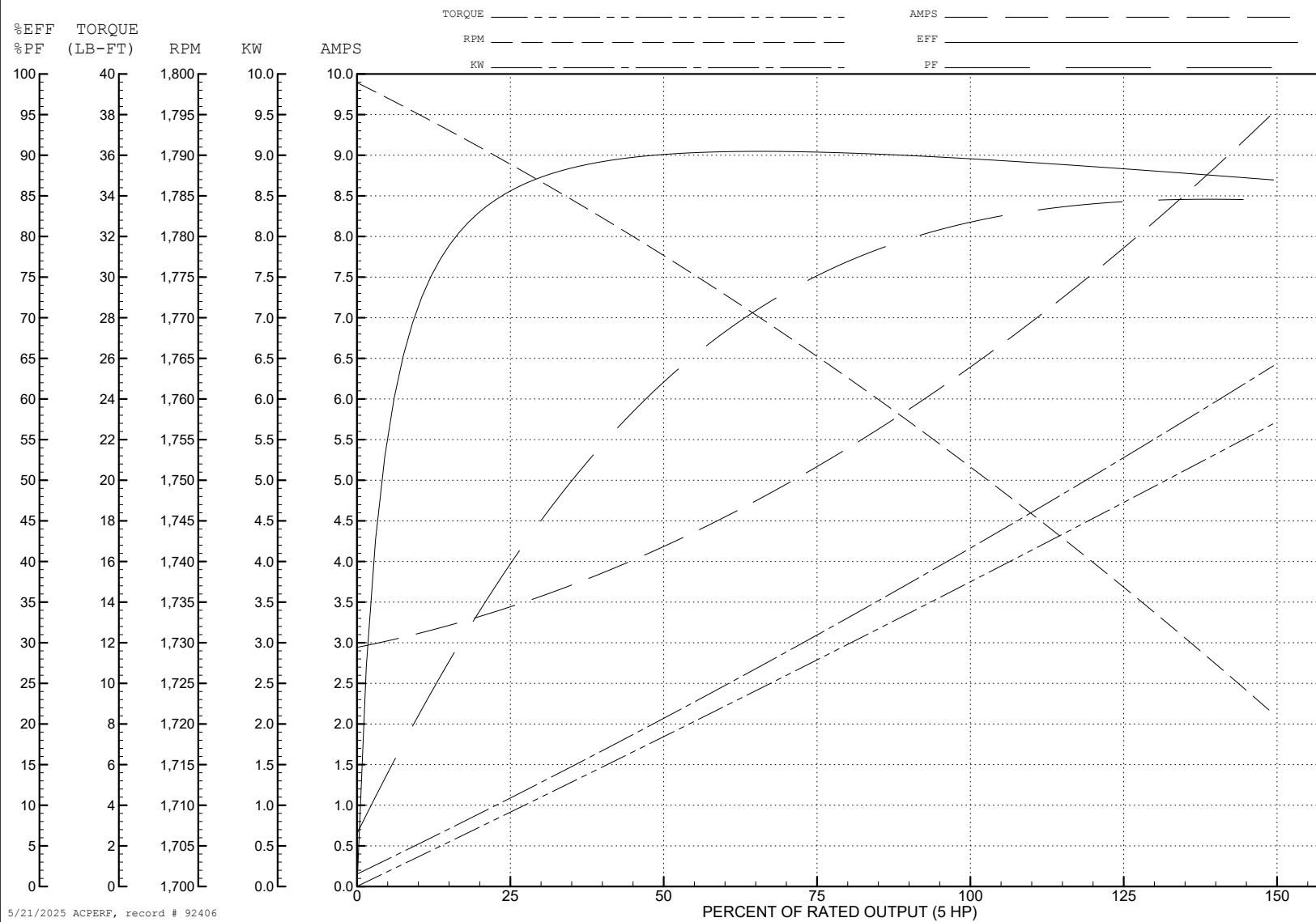
ABB Motors and Mechanical Inc.

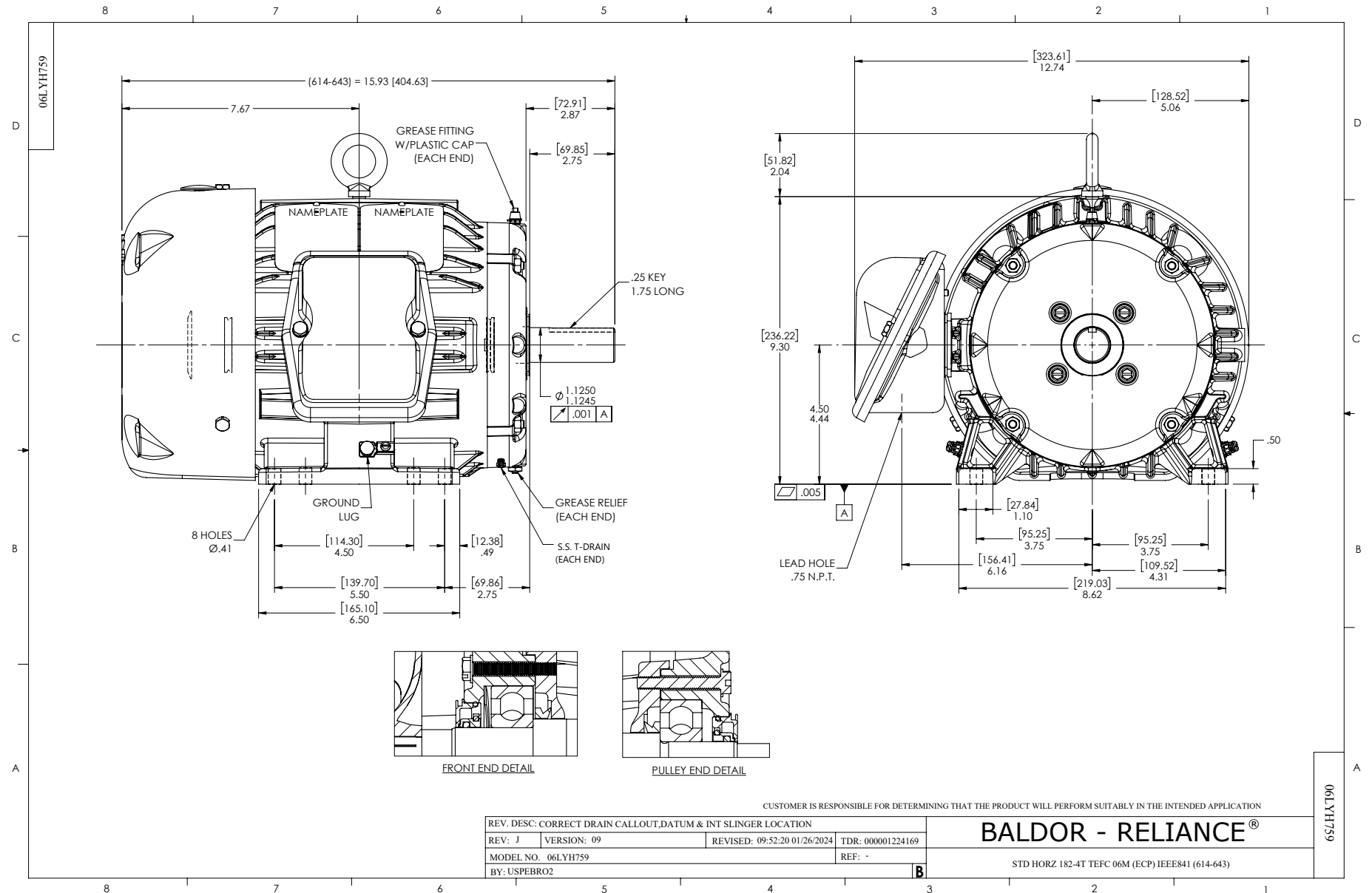
WINDING # 06WGX203

Typical performance - not guaranteed values.

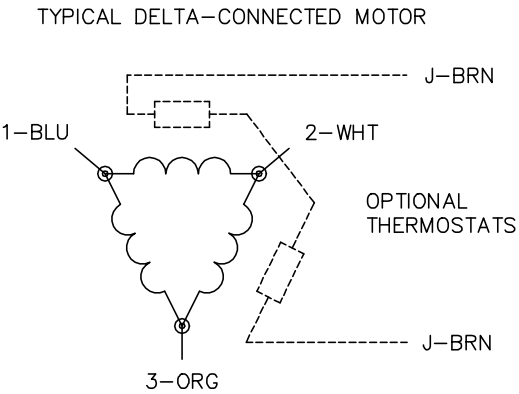
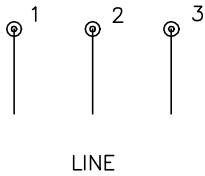
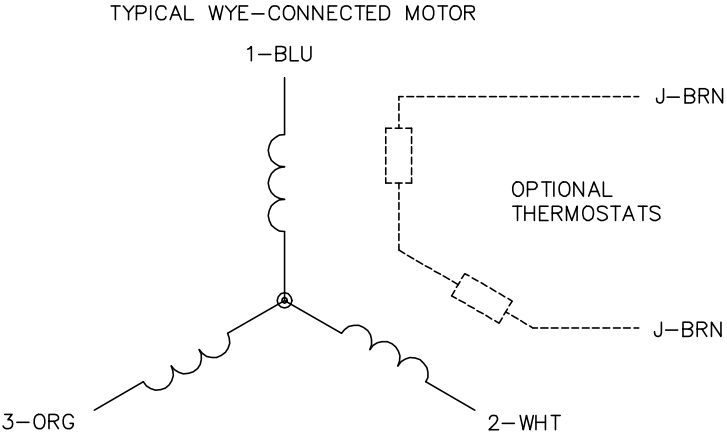
5 HP 3 PH 60 HZ 1750 RPM 460 V 0642M

TORQUES (LB-FT): PO=53.9 PU=22.4 LR=31.8 LRA=46





CD0006



- NOTES:
1. THREE LEAD MOTOR MAY BE EITHER WYE CONNECTED OR DELTA CONNECTED.
 2. INTERCHANGE ANY TWO LINE LEADS TO REVERSE ROTATION.
 3. OPTIONAL THERMOSTATS ARE PROVIDED WHEN SPECIFIED.
 4. ACTUAL NUMBER OF INTERNAL PARALLEL CIRCUITS MAY VARY.
 5. LEAD COLORS ARE OPTIONAL. LEADS MUST BE NUMBERED AS SHOWN.

CD0006

REV. DESC: ADD CLASS CONN00000007		
REV. LTR: E	VERSION: 01	TDR: 000001099922
FILE: \AAA\00005\141	REVISED: 10: 24: 49 02/19/2019	BY: ENBRIRO
MTL: —		

BALDOR - RELIANCE®

3PH, SV, 3 LEADS, WYE OR DELTA CONNECTED

SH 1 of 1

Motor Mod Express®

Modification number	Motor Mod Express®
Balance	
M1A	Dynamic balance of rotors
Bearings	
M2A	Sealed
M2B	Ball to roller 1800 RPM or below
M2C	RPM AC ball to roller
M2F	Add isolated bearings NEMA 56 through 449T (SKF Insocoat or equivalent)
Brake motors	
M3D	Change stearns brake voltage
Conduit box	
M4A	Tap or provide additional lead hole
M4B	Rotate conduit box lead opening
M4F	Replace die cast aluminum conduit box
M4G	Add conduit box for thermostat or heater leads
Cord	
M5A	Install line cord in conduit box or terminal panel
M5B	Install 6 foot, 3 conductor, type SO cord with plug
Drains	
M7A	Add condensation drain holes
M7C	Close standard condensation drain holes
M7D	Add breather drains to cast iron explosion-proof motor
Dripcovers	
M8A	Install dripcover on TEFC or ODP
Leads	
M10A	Install terminal lugs
M10B	Reconnect motor from nine leads to three
M10C	Reconnect motor to high or low voltage in conduit box
M10D	Longer leads, additional leads added
Lubrication	
M11A	Install low temperature lubricant in bearings
M11B	Add high temperature grease
Seals	
M12A	install oil seals
M12B	Install Inpro/Seal® drive end only
M12C	Install V-ring seals to motor

Motor Mod Express®

Modification number	Motor Mod Express®
Mountings	
M13A	Add rigid base
M13B	Install NEMA C-face – steel band motor construction
M13C-1	Install NEMA C-face – cast iron construction
M13C-2	Install NEMA C-face – IEEE 841 motors
M13D	Add D-flange
M13E	Remove rigid base from foot mounted steel band motors
M13F-2	Convert mounting configuration for cast iron motor
M13G	Install C-Face to explosion-proof motors
M13I	Labor only to mount grinder or buffer onto a pedestal and package for shipment
Mounting drawings	
M44A	RPM AC mounting conversions
Export packaging	
M14A	Open crate packaging
M14B	Totally enclosed crate packaging
M14C	Motor and nameplate photograph
Nameplate	
M15A	Provide original nameplate for remote mounting
M15B	Replace nameplate
M15C	Laser engrave nameplate into stainless steel motor
M15E	Mail nameplates to customer
Blowers	
M16A	Add constant velocity blower unit
M16B	Convert from a 3 phase to a single phase blower
Paint	
M17A	Paint motor Baldor-Reliance® stock paint color
M17B	Paint motor or parts customer specified color
Lifting lugs	
M18A	Add lifting lugs up to a 326T frame size
Hardware	
M20A	Exchange plated hardware with stainless steel
Space heaters	
M21A	Add space heaters to TEFC or open motors
Thermal protectors	
M23A	Thermostats
Winding thermocouples	
M24A	Iron constantan (3 per motor)
Winding thermistors	
M25A	Thermistors

Motor Mod Express®

Modification number	Motor Mod Express®
Tropicalization	
M26A	Protection of windings and mechanical parts
Weatherproofing	
M27A	Treatment of windings and mechanical parts
Shaft modifications	
M29A	Drill and tap one hole in end of shaft or mill flat on shaft extension
M29B	Convert from TEFC to TEAO
M29C	Threading
M29D	Convert "T" frame to "U" frame shaft dimension or smaller (TS)
M29F	Shorten shafts without reducing the overall diameter
Tach and encoder mounting kits	
M31D	Add encoder kit to inverter duty V*S Master or RPM AC motor
M31E	Additional mounted encoder options for IDRPM motors
Testing	
M32A	CSA C390 method 1 equivalent to IEEE 112 method B performance testing
M33A	Short motor test
M33B	Routine motor test
M33C	Complete performance test
M33D	Calibration test
M33E	Sound measurement test
Terminal blocks	
M35A	Install IEC terminal blocks in conduit box and connect leads
Starters	
	Add magnetic or manual starter to grinder or buffer
M36A	Starter – single phase
M36B	Starter – three phase
M36C	Magnetic starter – three phase
M36D	Overload protection for dust control unit
Shaft grounding	
M39A	Shaft grounding systems
M39B-1	Shaft grounding ring - bearing protection
M39C	Baldor-Reliance® shaft grounding brush assembly
M39D	Add shaft grounding ring's colloidal silver shaft coating to shaft
Check Total Indicated Runout (TIR)	
M40A	Check Total Indicated Runout (TIR)
ABB Ability™	
M43A	Install ABB Ability™ Smart Sensor for motors



Transmittal of Drawings

Kinetics Noise Control Inc.
6300 Irelan Place
Dublin, Ohio 43017
PHONE: (614) 889-0480
FAX: (614) 889-0540

Send Via:

- | | |
|---|-----------------------------------|
| ☐ Email | ☐ Next Day |
| ☐ First Class Mail | ☐ Other |
| ☐ UPS | |

To: Hartzell Air Movement

Date	06/19/2025
KNC #	217055
Submittal #	1.0
PO #	2521509
RE	Nan Inc Hilo WWTP Ph 1

Attn Dakota Iddings

For:

- | | |
|--|---------------------------------|
| ☒ Approval | ☐ Record |
|--|---------------------------------|

Please Return:

- | | |
|---|---|
| ☒ Completed Submittal Review/Approval Sheet | ☐ No Return Required |
|---|---|

Order:

- | | |
|---|--|
| ☒ Released After Approval | ☐ Released to Manufacturing |
|---|--|

Comments:

NOTES:

- 1.ASSUMED ATTACHED TO STEEL.
2. BUILDING HEIGHT NOT PROVIDED.

**THIS SHEET MUST BE RETURNED TO KINETICS NOISE CONTROL PRIOR TO
BEGINNING MANUFACTURE OF THE REFERENCED ITEMS IN THIS SUBMITTAL**

Job Title: Nan Inc Hilo WWTP Ph 1
Customer: Hartzell Air Movement
Customer P.O.: 2521509
Date Submitted: 6/19/2025

KNC Order No.: 217055
Submittal No.: 1.0
KNC Rep Name: Dakota Iddings

- ☐ Drawings Approved As Shown.
- ☐ Revise Drawings As Shown. Resubmittal Required.
- ☐ Revise Drawings As Shown. Resubmittal Not Required.
- ☐ KNC Is To Release All To Manufacture.
- ☐ KNC Is To Release Only The Tags Noted Below.
- ☐ Preliminary approval pending PE Review

Schedule of Drawing Changes:		
<u>Drawing No.</u>	<u>Change Notes</u>	<u>Reason for Change</u>

Note:
ALL VERBAL COMMENTS OR MODIFICATIONS
TO SUBMITTAL PACKAGE MUST BE
DOCUMENTED IN WRITING TO KINETICS NOISE
CONTROL PRIOR TO RELEASE TO
MANUFACTURING. K.N.C. ASSUMES NO
LIABILITY FOR NON-CONFIRMED CHANGES OR
FOR ANY CHANGES OTHER THAN THOSE
LISTED.

Revised By:

Name:

Company:

Date:

Disclaimer:

By approving this submittal you are confirming that the information provided to Kinetics is correct and complete and that all pertinent dimensions and data included herein have been checked and found to be in conformance with the requirements.

NOTES ON TAGS

Sub Num	Tag	Tag Notes
1.0	2521509-02	USED ONLY 4 ISOLATORS. USED FHS-2s TO GET ATLEAST MINIMUM 1" DEFLECTION AS REQUESTED BY CUSTOMER.
1.0	2521509-03	USED ONLY 4 ISOLATORS. USED FHS-2s TO GET ATLEAST 1" MINIMUM DEFLECTION AS REQUESTED BY CUSTOMER.
1.0	2521509-01	ASSUMED 10% FROM CENTER FOR C.G.

P.O.: 2521509
Project: Nan Inc Hilo WWTP Ph 1

Date: 6/19/2025

DWG No.	Description
-	Transmittal of Drawings
-	Review/Approval
-	Tag Notes
-	Submittal Index
-	MATERIAL REQUIRED LIST
-	SEISMIC CERTIFICATION
GSS-1	Seismic and Wind Certification Disclaimer (2 pgs)
2521509-01	Seismic Calculations
2521509-02	Seismic Calculations
2521509-03	Seismic Calculations
-	GENERAL ISOLATION LOCATIONS
S-01-35-21	FHS-2-100/1975
SS-20111000-L	Isolator Weld Data
INS-01-35-0A.PDF	MODEL FHS INSTALLATION INSTRUCTIONS
-	PE Appendix

Material Required List:

Thank you for choosing KINETICS NOISE CONTROL, the world leader in innovative products and solutions for acoustics, noise control, vibration isolation and seismic restraint systems since 1959.

WARNING

IMPORTANT NOTE REGARDING CERTIFICATIONS PROVIDED BY KINETICS NOISE CONTROL – PLEASE READ CAREFULLY
 Certifications provided by Kinetics Noise Control (KNC) pertain ONLY and SOLELY to the components provided by KNC with the following exception: The use of industry standard hardware to connect the equipment and the structure to components provided by KNC is permitted ONLY and SOLELY if said hardware is specified in the KNC Certification.
 For equipment directly connected to the structure (with no KNC provided component in between) the use of non-KNC provided hardware will invalidate the Certification.

Tag/Drawing No.	Part #	Material Required Description	Qty Each	Total Qty	Reference Drawing
2521509-01	5100410	5/8-11 GR 5 HARDWARE KIT, 3.5" FULL THREAD NUT WASHERS	6	6	
2521509-02	013510003	FHS-2-500	2	2	S-01-35-21
	013510019	FHS-2-225	2	2	S-01-35-21
	5100410	5/8-11 GR 5 HARDWARE KIT, 3.5" FULL THREAD NUT WASHERS	8	8	
2521509-03	013510003	FHS-2-500	2	2	S-01-35-21
	013510019	FHS-2-225	2	2	S-01-35-21
	5100410	5/8-11 GR 5 HARDWARE KIT, 3.5" FULL THREAD NUT WASHERS	8	8	

KINETICS RESTRAINT COMPONENT CERTIFICATION

Date: 06/19/2025
Project: Nan Inc Hilo WWTP Ph 1
Location: HONULULU, Hawaii 96819
KNC Number: 217055
Tags: 2521509-01, 2521509-02, 2521509-03

Reviewed By: James Gerloff

Registration: Hawaii

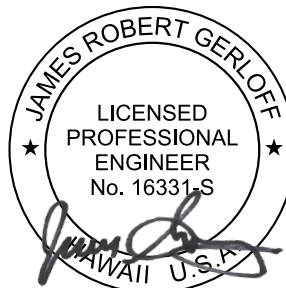
License Number: PE-16331

Expiration Date: 4/30/2026

This packet contains certifications of Kinetics Noise Control restraint products tailored to the specific load conditions indicated on the individual certification documents. Unless indicated otherwise on the attached documents, the basis of the certification is code minimum requirements. Unidentified Equipment Importance factors are assumed to be 1.5 for components in critical and life safety structures and 1.0 in other structures. Load and equipment data has been drawn from information provided to Kinetics Noise Control by our client. Where required input data was not provided, Kinetics Noise Control will attempt to extract data from publicly available mapped data, however it remains the responsibility of those with full access to the project data to verify that the loads used, as well as the equipment and installation conditions are consistent with the system as certified. Any significant deviations from the conditions identified in these documents including equipment substitutions, relocations, mounting arrangements, Importance Factors, special load conditions, etc. will void this certification.

It is critical to note that this certification covers only those hardware items which have been provided by Kinetics Noise Control. It does not extend to global or localized stresses in the building/supporting structure, the equipment, or where relevant, piping/ductwork and their suspension components.

This document and all documents referenced above have been sealed in the name of Kinetics Noise Control, Inc. by the listed Professional engineer. The seal represents only that components have been selected to meet the requirements of ASCE 7 and IBC as indicated in the project requirements. Any specialized Professional Engineering services desired for this project are beyond the scope of Kinetics Noise Control. In conjunction with these certifications, Kinetics Noise Control, Inc. guarantees only the use of that degree of care and skill ordinarily exercised under similar conditions by reputable members of our profession. No other warranty, expressed or implied, is made or intended.



06/22/2025

THIS WORK WAS PREPARED BY
ME OR UNDER MY SUPERVISION. 06/22/2025
Signature4/30/2026
Expiration Date of the License

Certification Agency:

SEISMIC AND WIND CERTIFICATION – GENERAL ASSUMPTIONS AND DISCLAIMER

All Seismic and Wind Certifications performed by Kinetics Noise Control, Inc and/or its associates, unless clearly stated otherwise in the body of that certification document, will be performed in accordance with the assumptions and disclaimers identified here in.

Loads Considered

The loads considered in this certification are limited to those forces described in the seismic portion of the code and/or that portion of the specification for the project provided to Kinetics Noise Control. If Kinetics Noise Control is not otherwise informed, the most recent version of the appropriate code will be used. Because equipment can be indoors, or shielded, wind will not be considered during the analysis unless it is specified to be included in the seismic certification request. In the absence of wind velocity and the appropriate factors, Kinetics Noise Control will use the equivalent of ~45 psf (2,155 Pa) as a wind load requirement. If this is not adequate, it is the responsibility of the Design Professional of Record to notify Kinetics Noise Control.

Extent of the Certification

This certification applies only to those components provided by Kinetics Noise Control. It does not extend to the building structure, equipment, or pipe/duct and their supports.

Equipment Data

The equipment weight, geometry and CG data used to perform the certification have been provided to Kinetics Noise Control by others, no attempt has been made by Kinetics Noise Control to verify its accuracy and it is up those providing the information to do so. Where CG data is not provided, associates of Kinetics Noise Control will attempt to make reasonable, yet conservative estimates as to the magnitude of any imbalance although it must be recognized that the direction of the imbalance is often unknown. Unless the equipment orientation is obvious from the diagram in the certification document, it should be assumed that the orientation is not known. Under these conditions, the worst case restraint, attachment and/or anchorage selection indicated for any particular location must be used for all locations.

Equipment Durability

Kinetics Noise Control and its associates make no representations as to equipment durability and its ability to survive a seismic event and remain functional.

Installation

Where detailed installation procedures are not addressed in documentation provided by Kinetics Noise Control, all seismic hardware and components must be installed in conformance with FEMA 412, 413, and 414, Installing Seismic Restraints for Mechanical Equipment, Installing Seismic Restraints for Electrical Equipment, and Installing Seismic Restraints for Pipe and Duct respectively. Free copies available from FEMA (1-800-480-2520) or through Kinetics Noise Control

SEISMIC & WIND CERTIFICATION – ASSUMPTIONS & DISCLAIMER

PAGE 1 of 2

GSS-1

RELEASED ON: 08/12/2009



Dublin, Ohio, USA • Cambridge, Ontario, Canada

Toll Free (USA Only): 800-959-1229
International: 614-889-0480
FAX: 614-889-0540
World Wide Web: www.kineticsnoise.com
E-mail: sales@kineticsnoise.com



Member

Equipment, Restraint, and Component attachment holes

For seismic restraint, it is necessary that any attachment bolts positioned in the path between the equipment to be restrained and the building structure, be a tight fit with their mating holes (The hole is to be not more than 1/8" (3 mm) in diameter larger than the attachment bolt). In the case of Kinetics Noise Control supplied restraint components, attachment safety factors are based on hardware sized per the above. In the case of directly attached equipment, the hardware and components provided by Kinetics Noise Control are the minimum required to withstand the seismic loading. If attachment holes in the equipment exceed the recommendation above, the attachment hole is to be sleeved or grouted to bring its effective diameter down to not more than 1/8" (3 mm) larger than the attachment hardware used.

Anchor Capacity and Edge Distances

All anchor allowable loads are based on ICC – ES test data and assume full anchor embedment in 3,000 psi (20,684 kPa) concrete and a minimum spacing between the anchor centerline and the edge of the slab into which it is sunk in accordance with the included anchorage data. The anchor data used is appropriate for the anchors provided by Kinetics Noise Control, unless otherwise noted. Under some conditions as noted in the calculations, "Undercut" type anchors may be required

Stamps

Stamped documents are intended to support the Engineer of Record on the project. If the project is located in an area for which Kinetics does not have a valid PE license, the documents will be signed and sealed by a Licensed Professional Engineer in the State of Ohio. This practice is intended solely to indicate that a competent individual has reviewed the document. It is not intended to imply that the licensee is legally empowered to practice in the jurisdiction of the project.

General

Kinetics Noise Control, Inc. and its associates guarantee that we will use that degree of care and skill ordinarily exercised under similar conditions by reputable members of our profession to determine restraint and/or attachment safety factors based on customer supplied input data. No other warranty, expressed or implied is made or intended.

SEISMIC & WIND CERTIFICATION – ASSUMPTIONS & DISCLAIMER

PAGE 2 of 2

GSS-1

RELEASED ON: 08/12/2009



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FAX: 614-889-0540
World Wide Web: www.kineticsnoise.com
E-mail: sales@kineticsnoise.com



Dublin, Ohio, USA • Cambridge, Ontario, Canada

Member



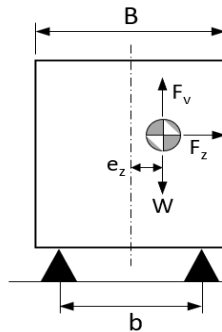
Project:	Nan Inc Hilo WWTP Ph 1	KNC#: 217055
Tag:	2521509-01	Designer: FJ
Date:	6/19/2025	Rev. 0.0

Seismic Design Basis

Building Code:	2018 IBC	Site Class:	C
Risk Category:	III	Spectral Acceleration, S_{DS} :	1.20
Component Importance Factor, I_p :	1.5	Overstrength Factor, Ω_0 :	2.0
Component Response Factor, R_p :	6	Minimum Lateral Acceleration (g's):	NA
Component Amplification Factor, a_p :	2.5	Minimum Vertical Acceleration (g's):	NA
Mounting Height, z (ft)	0	Roof Height, h (ft)	N/A

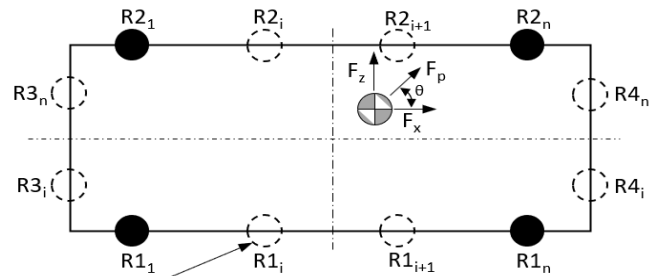
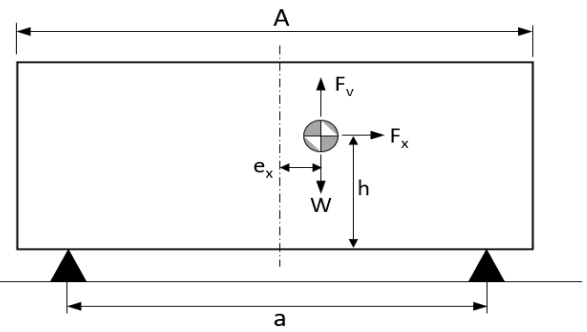
Geometry Description

Isolated System =	No
Eq Length, A (in) =	122
Eq Width, B (in) =	112
Eq Height, H (in) =	94.00
Eq Weight, W (lbs.) =	3596.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	97.84
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	52.358002
Angle of Maximum Effect, θ (deg) =	90
x-dir CG offset, e_x (in) =	12.20
z-dir CG offset, e_z (in) =	11.20
y-dir CG offset, h (in) =	55.00



$$F_x = F_p \cos(\theta)$$

$$F_z = F_p \sin(\theta)$$



Additional restraints when required

Seismic Force Calculation

$$F_p = \frac{0.4 a_p S_{DS} W}{(R_p/I_p)} \left(1 + 2 \frac{z}{h}\right)$$

$$0.3 S_{DS} I_p W \leq F_p \leq 1.6 S_{DS} I_p W$$

$$F_v = 0.2 S_{DS} W$$

$$F_p, (\text{lbs.}) = 1941.8$$

$$F_v, (\text{lbs.}) = 863.0$$

For Anchoring to Concrete

$$F'_p = \Omega_0 F_p$$

Otherwise

$$F'_p = F_p$$

Governing Load Combinations (LRFD)

$$0.9W - F_v + F'_p$$

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	6	15.09
Level 2					
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

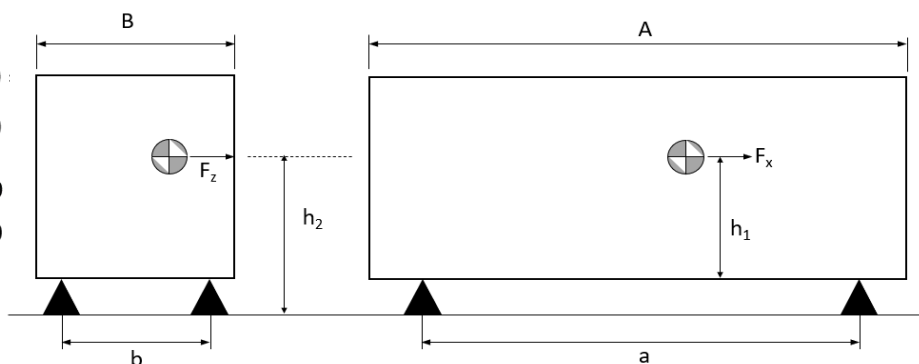
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

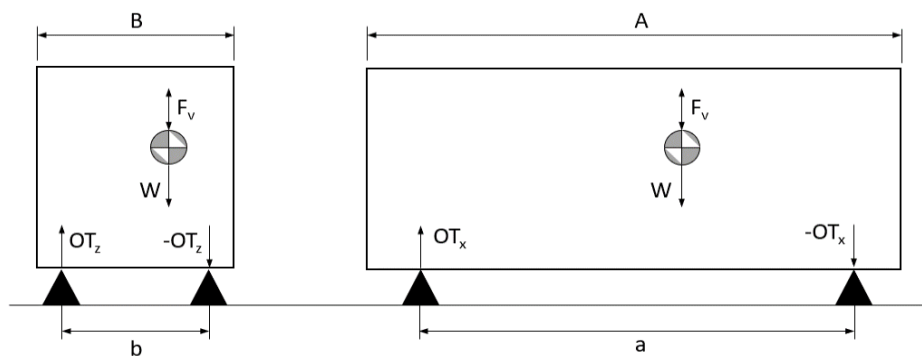
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



Note: The direction of F_v is dependent on which load combination is being applied.

$$(F_v)_{comb} = \pm F_v + OT_x + OT_z - W^*$$

Level 1 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

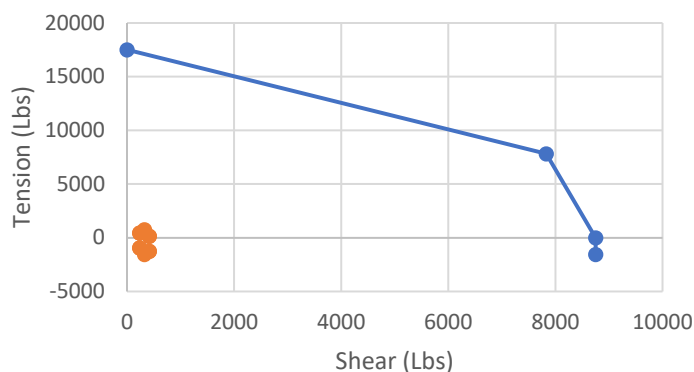
Notes:

- For non-sustained loading conditions.
- Based on maximum shear stress theory.

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$

Capacity Curve - LRFD



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).

Product Type: Gr5 Bolts

Product Size: 5/8 Dia

Product Qty: 6

Orientation: Vertical

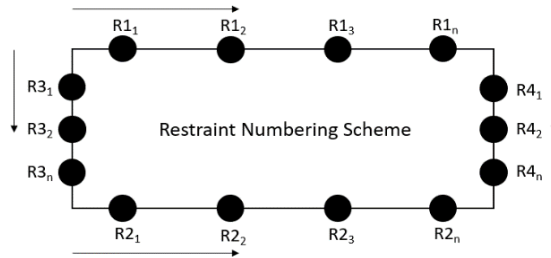
Worst Case Tension Load
(lbs.): 760

Worst Case Shear Load
(lbs.): 413

Worst Case Factor of
Safety: 15.09

	Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
	Tag:	2521509-01	Designer:	FJ
	Date:	6/19/2025	Rev:	0.0

Level Inputs							
Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	97.843	52.358	55	3596
2							
3							
4							



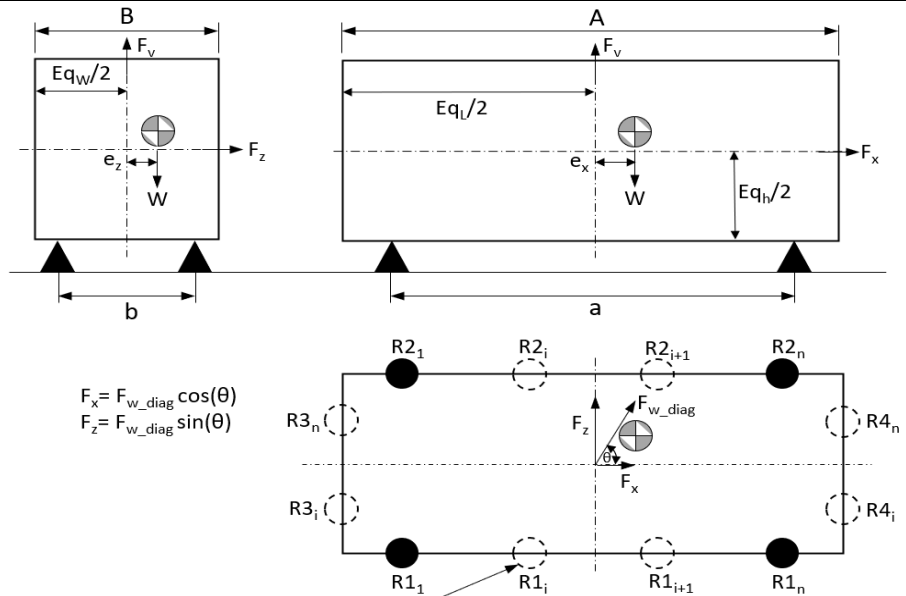
Level 1 Reaction Loads (lbs)											
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
3		0		1		0		0		90	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat -x Load	Lat -z Load	Vertical Load	OT-x Load	OT-z Load	Lat Tot Load	Combined Vertical Load
1	1	Yes	-110	-99	0	234	27	0	527	234	454
1	2	Yes	-343	-309	0	324	82	0	986	324	760
1	3	Yes	-575	-518	0	413	138	0	527	413	147
2	1	Yes	-623	-561	0	234	150	0	-527	234	-938
2	2	Yes	-856	-770	0	324	205	0	-986	324	-1551
2	3	Yes	-1088	-979	0	413	261	0	-527	413	-1245

Wind Design Basis

Building Code:	2018 IBC	Velocity Press, $P = .00256 K_d K_{zt} K_z V^2$ (psf)	49.9
Risk Category:	III	Gust Factor, G	0.85
Basic Wind Speed, V (mph):	145	Lateral Force Coefficient, C_{fh}	1.09
Wind Directionality Factor, K_d :	0.90	Vertical Force Coefficient, C_{fv}	1.09
Exposure Category:	D	GC_{fh}	0.93
Topographic Factor, K_{zt} :	1	GC_{fv}	0.93
Velocity Pressure Coefficient, K_z :	1.03	Mean Roof Height, h (ft)	N/A
Mounting height, z (ft)	0		

Geometry Description

Isolated System =	No
Eq Length, A (in) =	122.0
Eq Width, B (in) =	112.0
Eq Height, H (in) =	94.0
Eq Weight, W (lbs.) =	3596.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	97.8
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	52.4
Turning Point, b (in) =	
Angle of Diagonal Wind Force, θ (deg) =	47.4
x-dir CG offset, e_x (in) =	12.2
z-dir CG offset, e_z (in) =	11.2



Additional restraints when required

Wind Force Calculation

Front Area, (Eq Length * Eq Height)/144 (ft ²) =	79.6	x-Dir Wind Force (lbs.) =	3383.8
Side Area, (Eq Width * Eq Height)/144 (ft ²) =	73.1	z-Dir Wind Force (lbs.) =	3686.0
Top Area, (Eq Length * Eq Width)/144 (ft ²) =	94.9	Diagonal Wind Force (lbs.) =	5003.7
Lateral Wind Pressure, $P * GC_{fh}$ (psf) =	46.3	Vertical Wind Force (lbs.) =	4391.8
Vertical Wind Pressure, $P * GC_{fv}$ (psf) =	46.3		

Governing Load Combinations (LRFD)

0.9W + Fw,lat - Fw,vert

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	6	4.10
Level 2					
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

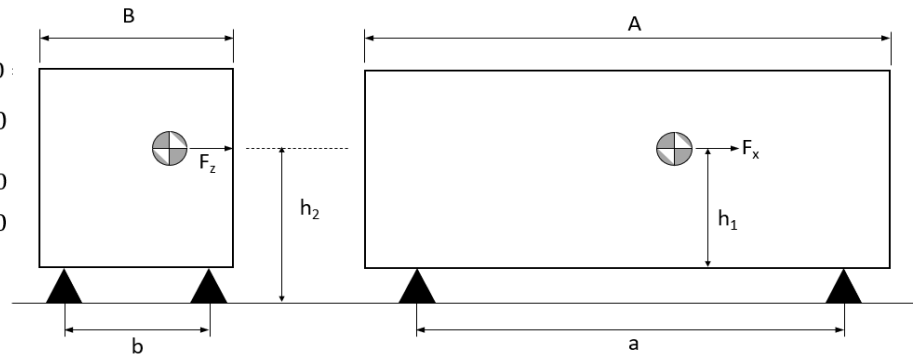
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

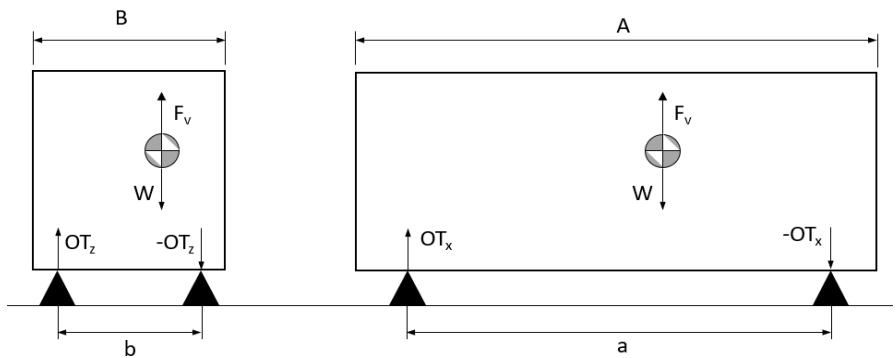
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



$$(F_v)_{comb} = F_v + OT_x + OT_z - W^*$$

Level 1 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

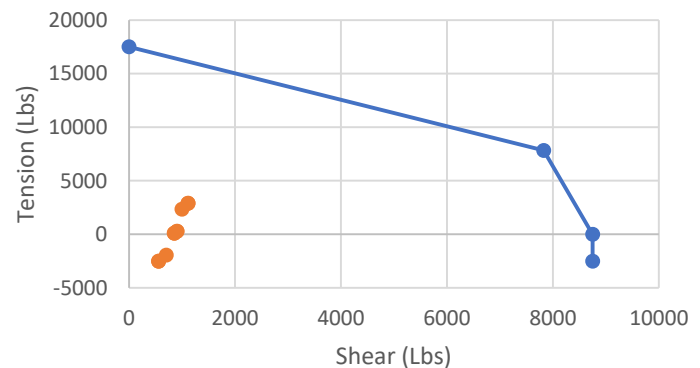
Notes:

- For non-sustained loading conditions.
- Based on maximum shear stress theory.

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$

Capacity Curve - LRFD



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).

Product Type: Gr5 Bolts

Product Size: 5/8 Dia

Product Qty: 6

Orientation: Vertical

Worst Case Tension Load
(lbs.): 2897

Worst Case Shear Load
(lbs.): 1113

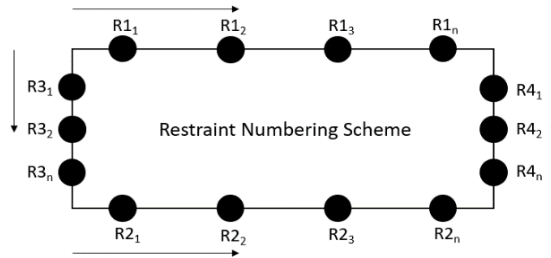
Worst Case Factor of
Safety: 4.10



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-01	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Structure	Gr5 Bolts	5/8 Dia	97.843	52.358	55.0	3596
2							
3							
4							



Level 1 Reaction Loads (lbs)

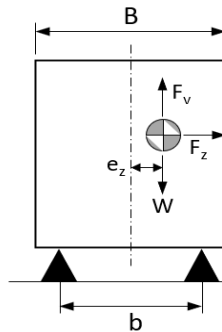
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
3		0		1		0		0		47	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
1	1	Yes	-110	-99	789	785	1329	813	855	1113	2897
1	2	Yes	-343	-309	789	614	1045	0	1599	1000	2336
1	3	Yes	-575	-518	789	444	761	-813	855	905	285
2	1	Yes	-623	-561	339	785	703	813	-855	855	100
2	2	Yes	-856	-770	339	614	419	0	-1599	702	-1951
2	3	Yes	-1088	-979	339	444	135	-813	-855	559	-2512

Seismic Design Basis

Building Code:	2018 IBC	Site Class:	C
Risk Category:	III	Spectral Acceleration, S_{DS} :	1.20
Component Importance Factor, I_p :	1.5	Overstrength Factor, Ω_0 :	2.0
Component Response Factor, R_p :	2	Minimum Lateral Acceleration (g's):	NA
Component Amplification Factor, a_p :	2.5	Minimum Vertical Acceleration (g's):	NA
Mounting Height, z (ft)	0	Roof Height, h (ft)	N/A

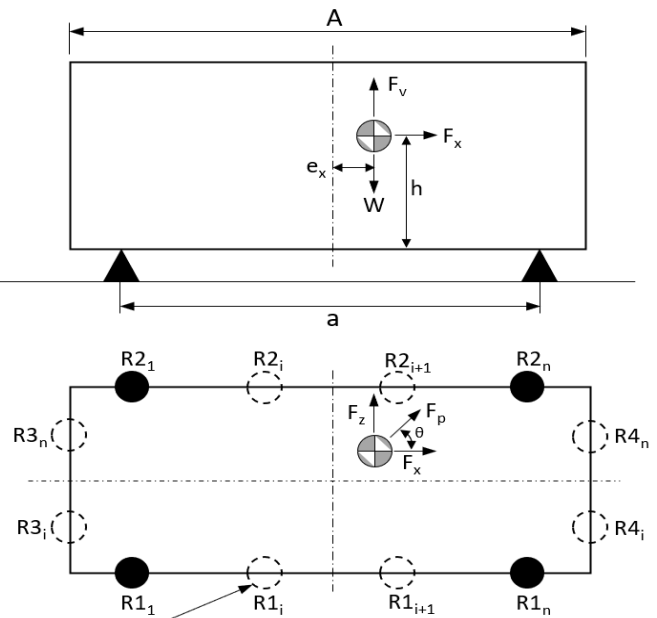
Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	85.263
Eq Width, B (in) =	58.700001
Eq Height, H (in) =	64.63
Eq Weight, W (lbs.) =	1078.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	72.93
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	22
Angle of Maximum Effect, θ (deg) =	290
x-dir CG offset, e_x (in) =	-12.49
z-dir CG offset, e_z (in) =	-1.65
y-dir CG offset, h (in) =	27.64



$$F_x = F_p \cos(\theta)$$

$$F_z = F_p \sin(\theta)$$



Additional restraints when required

Seismic Force Calculation

$$F_p = \frac{0.4 a_p S_{DS} W}{(R_p/I_p)} \left(1 + 2 \frac{z}{h}\right)$$

$$0.3 S_{DS} I_p W \leq F_p \leq 1.6 S_{DS} I_p W$$

$$F_v = 0.2 S_{DS} W$$

$$F_p, (\text{lbs.}) = 970.2$$

$$F_v, (\text{lbs.}) = 258.7$$

For Anchoring to Concrete

$$F'_p = \Omega_0 F_p$$

Otherwise

$$F'_p = F_p$$

Governing Load Combinations (LRFD)

$$0.9W - F_v + F'_p$$

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	3.93
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	26.09
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

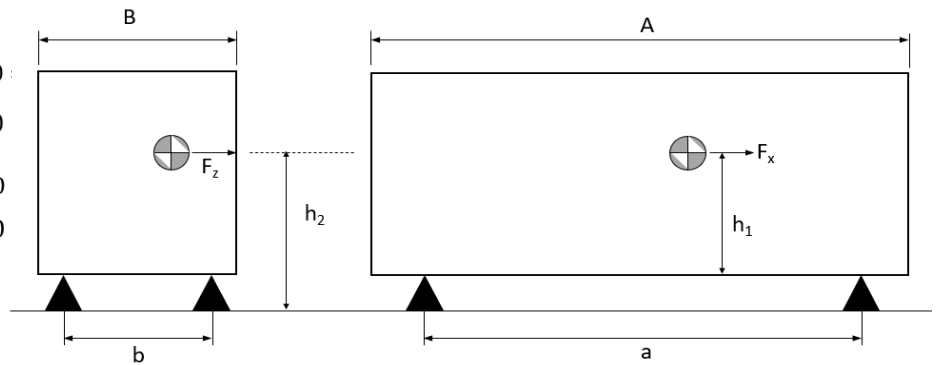
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

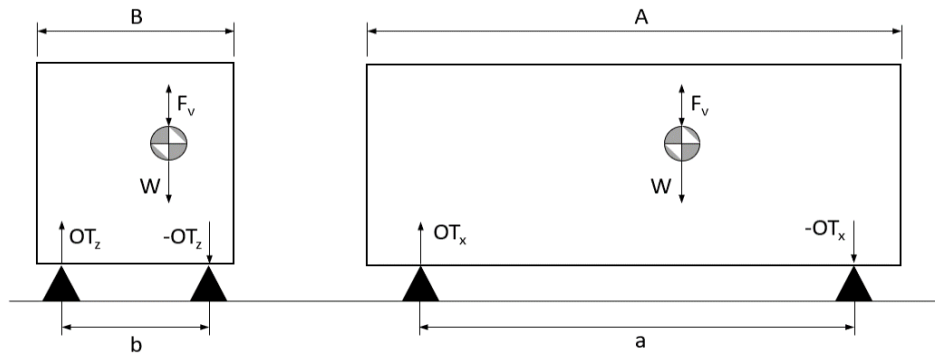
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



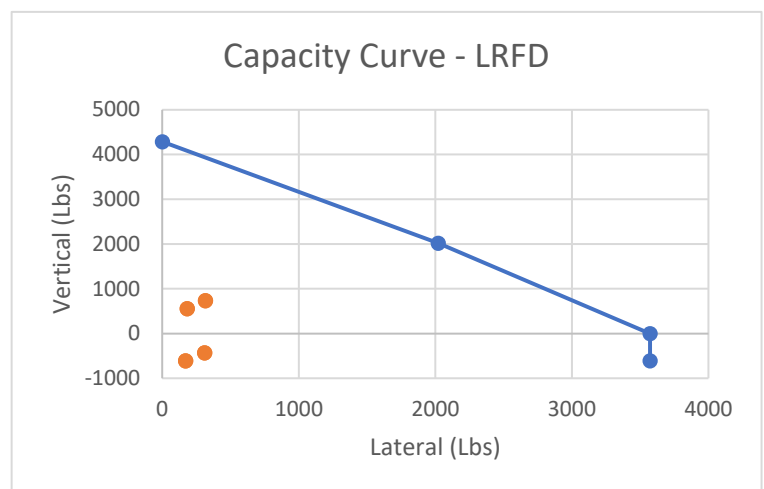
Note: The direction of F_v is dependent on which load combination is being applied.

$$(F_v)_{comb} = \pm F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	738
Worst Case Shear Load (lbs.):	315
Worst Case Factor of Safety:	3.93



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-02

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

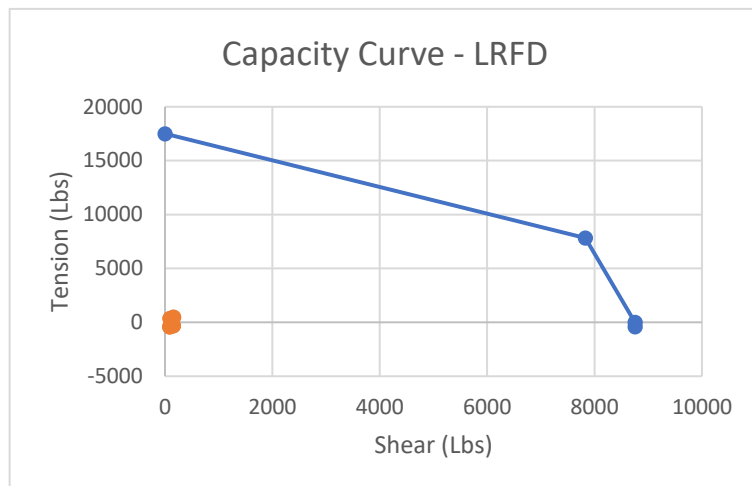
²Shear Capacity = 0.5* Tensile Capacity

Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	476
Worst Case Shear Load (lbs.):	158
Worst Case Factor of Safety:	26.09

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



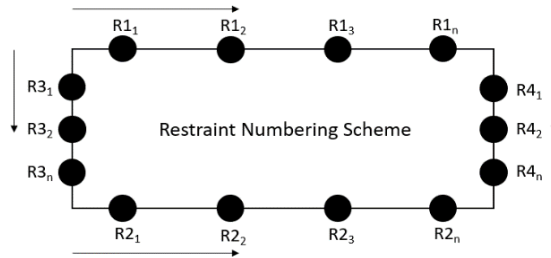
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-02	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	72.929	22	27.644	1078
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	72.929	22	36.954	1078
3							
4							



Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		1		0		0		290	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
	1	No	-426	-383	95	-301	102	63	573	315	738
	1	No	-194	-175	95	-155	47	-63	573	182	557
	2	No	-345	-310	71	-301	83	63	-573	309	-427
2	2	No	-113	-102	71	-155	27	-63	-573	171	-608

Level 2 Reaction Loads (lbs)

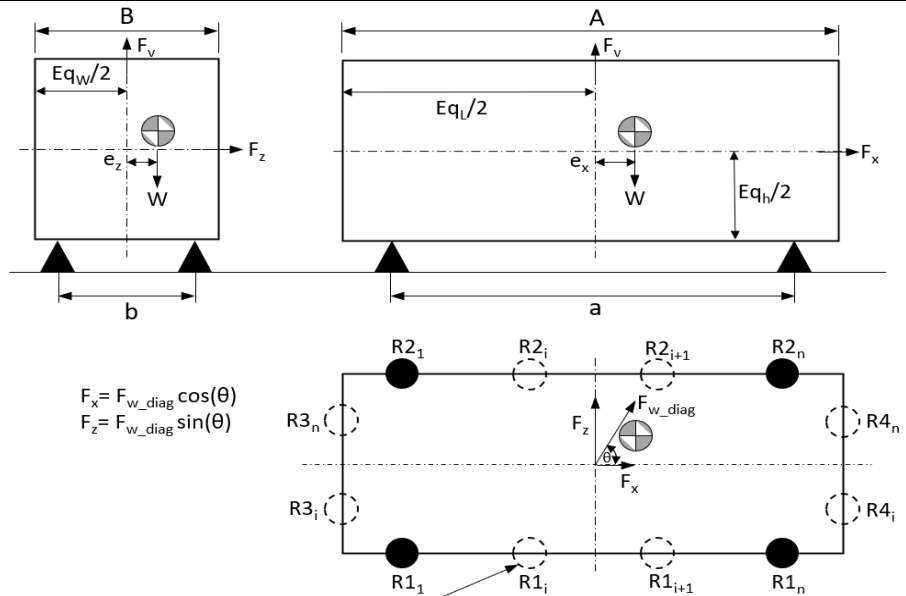
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		2		0		0		290	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-426	-383	95	-301	102	84	766	315	952
1	2	No	-194	-175	95	-155	47	-84	766	182	728
2	1	No	-345	-310	71	-301	83	84	-766	309	-599
2	2	No	-113	-102	71	-155	27	-84	-766	171	-823

Wind Design Basis

Building Code:	2018 IBC	Velocity Press, $P = .00256 K_d K_{zt} K_z V^2$ (psf)	49.9
Risk Category:	III	Gust Factor, G	0.85
Basic Wind Speed, V (mph):	145	Lateral Force Coefficient, C_{fh}	1.30
Wind Directionality Factor, K_d :	0.90	Vertical Force Coefficient, C_{fv}	1.30
Exposure Category:	D	GC_{fh}	1.11
Topographic Factor, K_{zt} :	1	GC_{fv}	1.11
Velocity Pressure Coefficient, K_z :	1.03	Mean Roof Height, h (ft)	N/A
Mounting height, z (ft)	0		

Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	85.3
Eq Width, B (in) =	58.7
Eq Height, H (in) =	64.6
Eq Weight, W (lbs.) =	1078.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	72.9
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	22.0
Angle of Diagonal Wind Force, θ (deg) =	55.5
x-dir CG offset, e_x (in) =	-12.5
z-dir CG offset, e_z (in) =	-1.6



Additional restraints when required

Wind Force Calculation

Front Area, (Eq Length * Eq Height)/144 (ft ²) =	38.3	x-Dir Wind Force (lbs.) =	1454.6
Side Area, (Eq Width * Eq Height)/144 (ft ²) =	26.3	z-Dir Wind Force (lbs.) =	2112.9
Top Area, (Eq Length * Eq Width)/144 (ft ²) =	34.8	Diagonal Wind Force (lbs.) =	2565.2
Lateral Wind Pressure, $P * GC_{fh}$ (psf) =	55.2	Vertical Wind Force (lbs.) =	1919.2
Vertical Wind Pressure, $P * GC_{fv}$ (psf) =	55.2		

Governing Load Combinations (LRFD)

0.9W + Fw,lat - Fw,vert

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	1.35
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	7.69
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

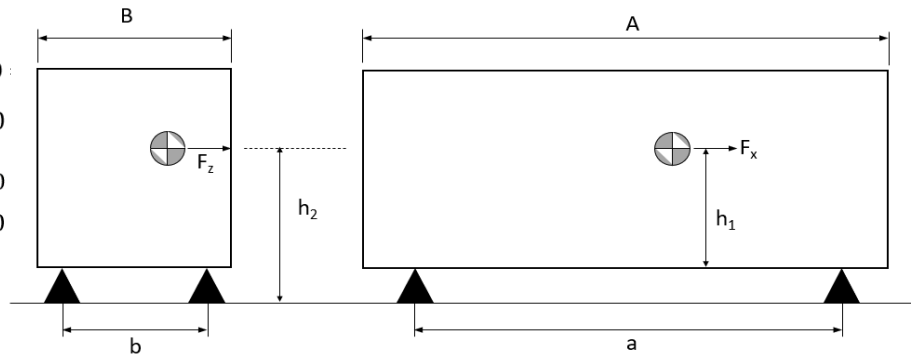
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

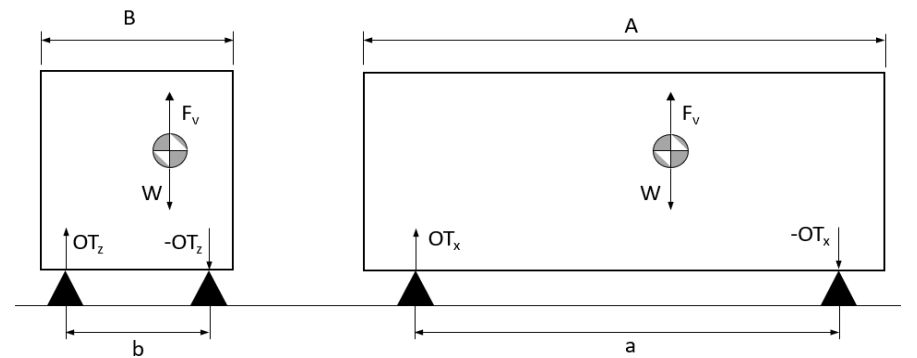
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.

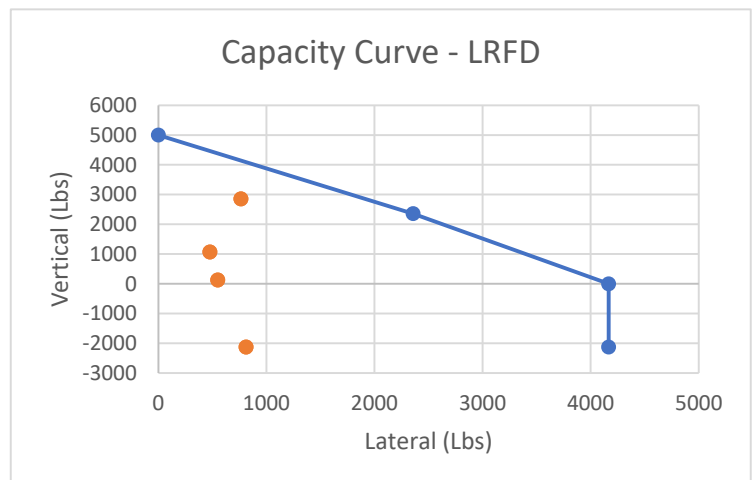


$$(F_v)_{comb} = F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	2852
Worst Case Shear Load (lbs.):	811
Worst Case Factor of Safety:	1.35



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-02

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

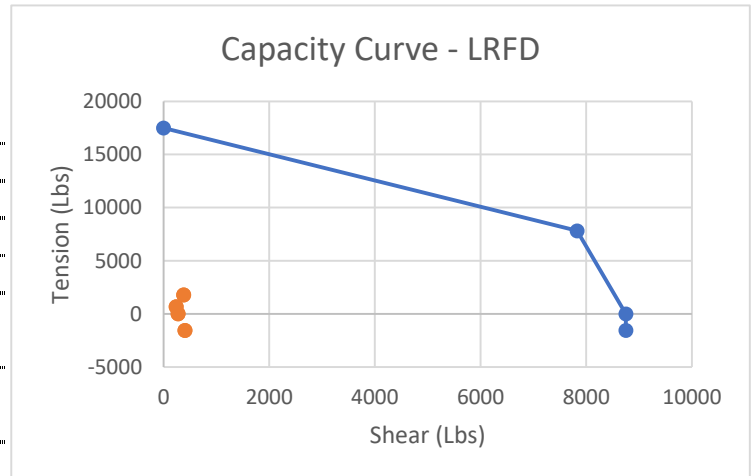
Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	1803
Worst Case Shear Load (lbs.):	405
Worst Case Factor of Safety:	7.69

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



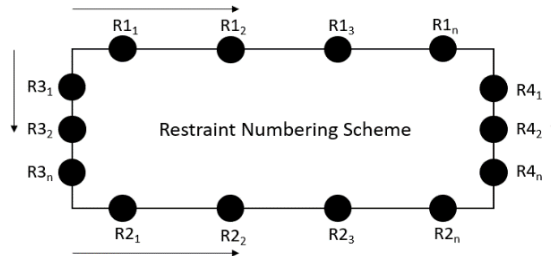
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-02	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	72.929	22	27.6	1078
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	72.929	22	37.0	1078
3							
4							



Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		1		0		0		55	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat -x Load	Lat -z Load	Vertical Load	OT-x Load	OT-z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-426	-383	313	360	202	322	543	477	1067
1	2	No	-194	-175	313	697	614	-322	2560	764	2852
2	1	No	-345	-310	415	360	346	322	-543	549	125
2	2	No	-113	-102	415	697	758	-322	-2560	811	-2125

Level 2 Reaction Loads (lbs)

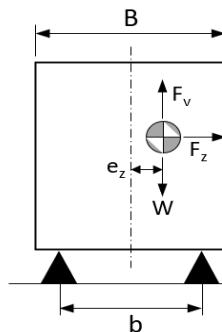
# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		2		0		0		55	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-426	-383	313	360	202	431	726	477	1359
1	2	No	-194	-175	313	697	614	-431	3422	764	3605
2	1	No	-345	-310	415	360	346	431	-726	549	51
2	2	No	-113	-102	415	697	758	-431	-3422	811	-3096

Seismic Design Basis

Building Code:	2018 IBC	Site Class:	C
Risk Category:	III	Spectral Acceleration, S_{DS} :	1.20
Component Importance Factor, I_p :	1.5	Overstrength Factor, Ω_0 :	2.0
Component Response Factor, R_p :	2	Minimum Lateral Acceleration (g's):	NA
Component Amplification Factor, a_p :	2.5	Minimum Vertical Acceleration (g's):	NA
Mounting Height, z (ft)	0	Roof Height, h (ft)	N/A

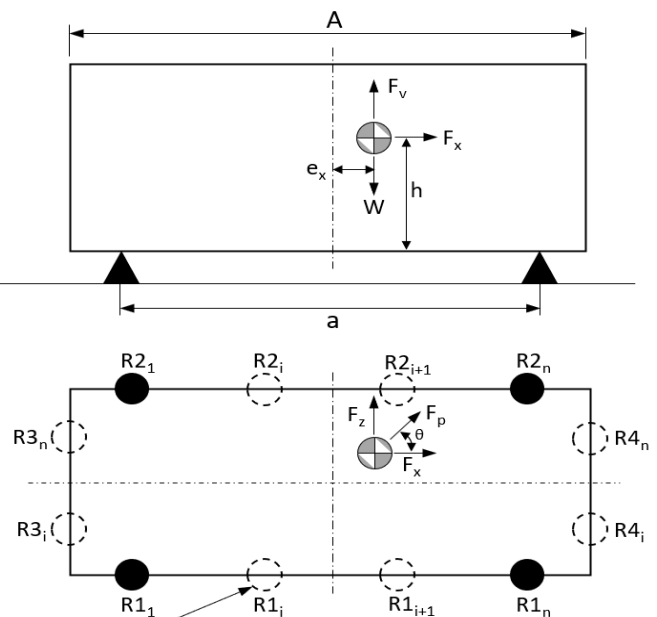
Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	87.763
Eq Width, B (in) =	58.700001
Eq Height, H (in) =	64.63
Eq Weight, W (lbs.) =	1163.00
x Distance Between Outer Restraint and Closest Over	75.43
Turning Point, a (in) =	
z Distance Between Outer Restraint and Closest Over	22
Turning Point, b (in) =	
Angle of Maximum Effect, θ (deg) =	290
x-dir CG offset, e_x (in) =	-13.81
z-dir CG offset, e_z (in) =	-1.64
y-dir CG offset, h (in) =	26.78



$$F_x = F_p \cos(\theta)$$

$$F_z = F_p \sin(\theta)$$



Additional restraints when required

Seismic Force Calculation

$$F_p = \frac{0.4 a_p S_{DS} W}{(R_p/I_p)} \left(1 + 2 \frac{z}{h}\right)$$

$$F_p \text{ (lbs.)} = 1046.7$$

$$0.3 S_{DS} I_p W \leq F_p \leq 1.6 S_{DS} I_p W$$

$$F_v \text{ (lbs.)} = 279.1$$

$$F_v = 0.2 S_{DS} W$$

For Anchoring to Concrete

$$F'_p = \Omega_0 F_p$$

Otherwise

$$F'_p = F_p$$

Governing Load Combinations (LRFD)

$$0.9W - F_v + F'_p$$

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	3.69
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	24.45
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

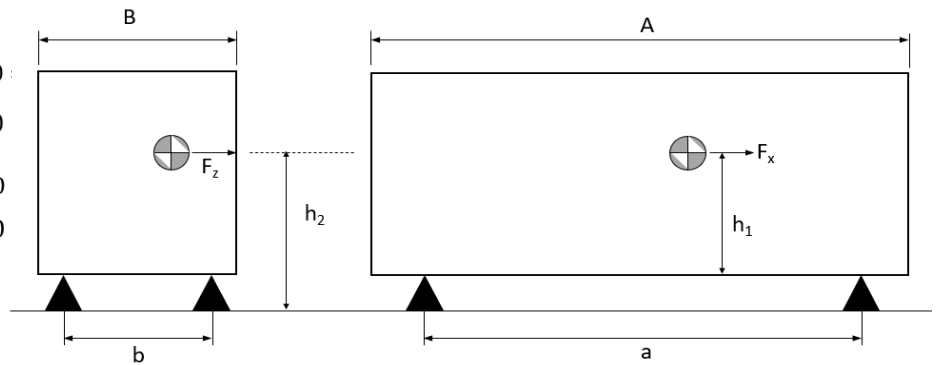
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

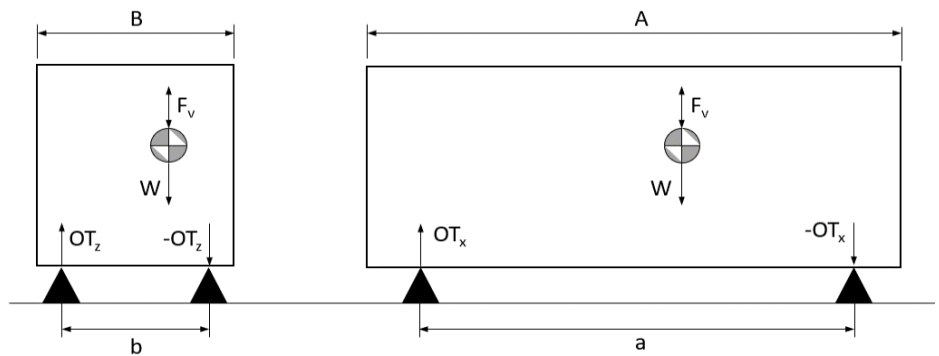
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.



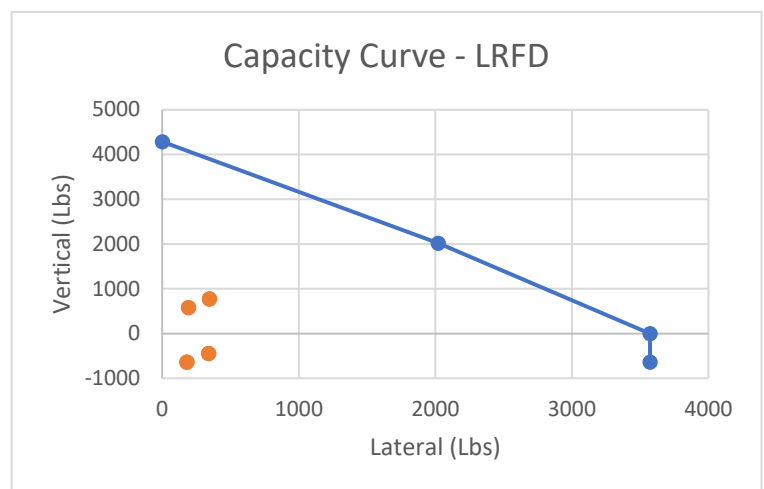
Note: The direction of F_v is dependent on which load combination is being applied.

$$(F_v)_{comb} = \pm F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	774
Worst Case Shear Load (lbs.):	345
Worst Case Factor of Safety:	3.69



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-03

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

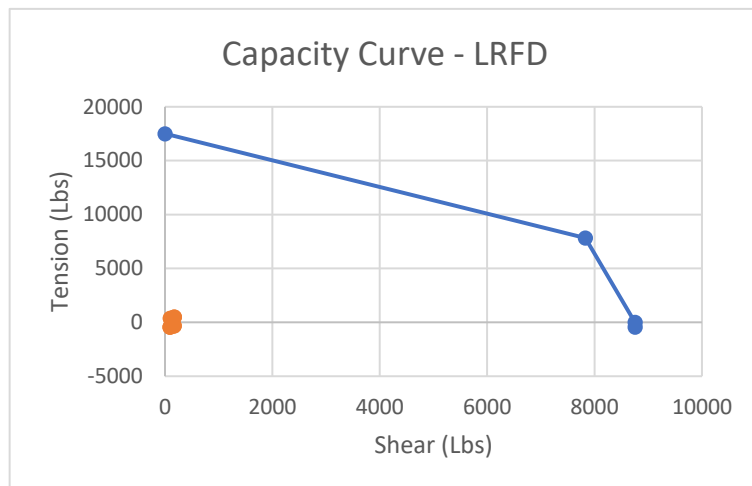
²Shear Capacity = 0.5* Tensile Capacity

Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	502
Worst Case Shear Load (lbs.):	173
Worst Case Factor of Safety:	24.45

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



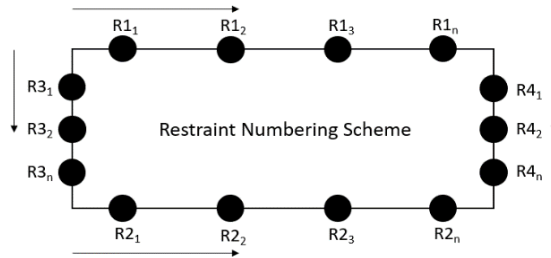
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-03	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	75.429	22	26.778	1163
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	75.429	22	36.088	1163
3							
4							



Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic	
2		0		1		0		0		290	Pass	
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined	
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load	
	1	No	-467	-421	102	-330	112	64	599	345	774	
	1	2	No	-201	-181	102	-162	48	-64	599	191	583
	2	1	No	-381	-343	77	-330	91	64	-599	339	-444
2	2	No	-114	-103	77	-162	27	-64	-599	179	-635	

Level 2 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Seismic
2		0		2		0		0		290	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-467	-421	102	-330	112	86	807	345	1005
1	2	No	-201	-181	102	-162	48	-86	807	191	769
2	1	No	-381	-343	77	-330	91	86	-807	339	-630
2	2	No	-114	-103	77	-162	27	-86	-807	179	-865



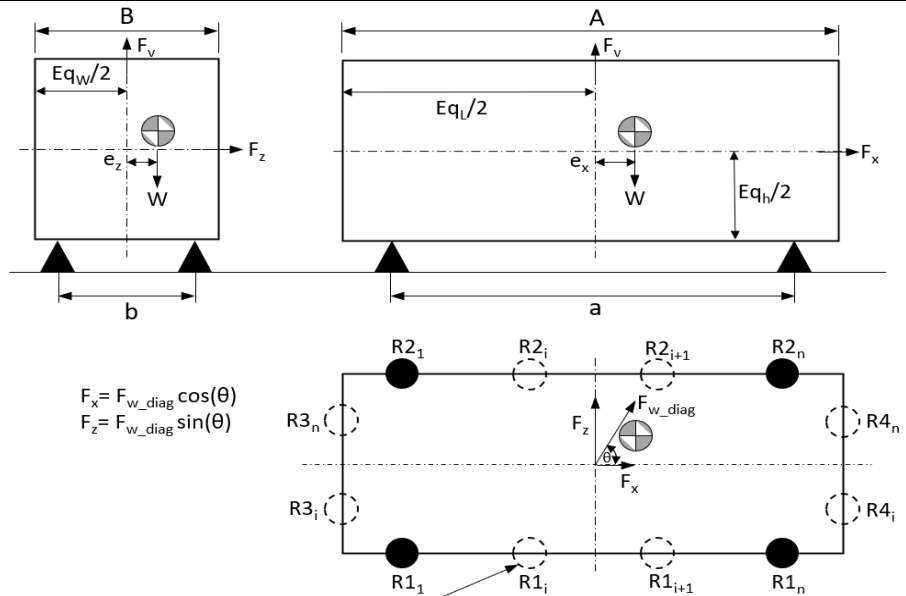
Project:	Nan Inc Hilo WWTP Ph 1	KNC: 217055
Tag:	2521509-03	Designer: FJ
Date:	6/19/2025	Rev: 0.0

Wind Design Basis

Building Code:	2018 IBC	Velocity Press, $P = .00256 K_d K_{zt} K_z V^2$ (psf)	49.9
Risk Category:	III	Gust Factor, G	0.85
Basic Wind Speed, V (mph):	145	Lateral Force Coefficient, C_{fh}	1.30
Wind Directionality Factor, K_d :	0.90	Vertical Force Coefficient, C_{fv}	1.30
Exposure Category:	D	GC_{fh}	1.11
Topographic Factor, K_{zt} :	1	GC_{fv}	1.11
Velocity Pressure Coefficient, K_z :	1.03	Mean Roof Height, h (ft)	N/A
Mounting height, z (ft)	0		

Geometry Description

Isolated System =	Yes
Eq Length, A (in) =	87.8
Eq Width, B (in) =	58.7
Eq Height, H (in) =	64.6
Eq Weight, W (lbs.) =	1163.00
x Distance Between Outer Restraint and Closest Over Turning Point, a (in) =	75.4
z Distance Between Outer Restraint and Closest Over Turning Point, b (in) =	22.0
Angle of Diagonal Wind Force, θ (deg) =	56.2
x-dir CG offset, e_x (in) =	-13.8
z-dir CG offset, e_z (in) =	-1.6



Additional restraints when required

Wind Force Calculation

Front Area, (Eq Length * Eq Height)/144 (ft ²) =	39.4	x-Dir Wind Force (lbs.) =	1454.6
Side Area, (Eq Width * Eq Height)/144 (ft ²) =	26.3	z-Dir Wind Force (lbs.) =	2174.8
Top Area, (Eq Length * Eq Width)/144 (ft ²) =	35.8	Diagonal Wind Force (lbs.) =	2616.5
Lateral Wind Pressure, $P * GC_{fh}$ (psf) =	55.2	Vertical Wind Force (lbs.) =	1975.4
Vertical Wind Pressure, $P * GC_{fv}$ (psf) =	55.2		

Governing Load Combinations (LRFD)

0.9W + Fw,lat - Fw,vert

Calculation Scope & Result Summary

	Description	Type	Size	Qty	Worst Case Safety Factor
Level 1	Equipment to Restraint	FHS	2-100/1600	4	1.30
Level 2	Restraint to Structure	Gr5 Bolts	5/8 Dia	8	7.35
Level 3					
Level 4					

Calculation Summary

Overturning & Lateral Loads

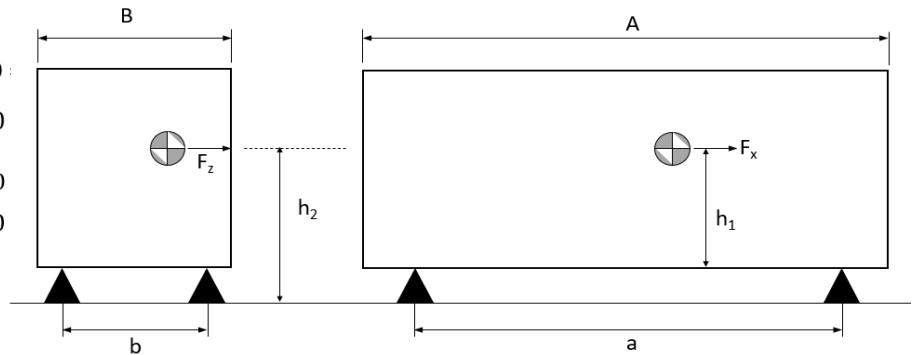
$$(OT_z)_{tot} = \frac{F_z h_i}{b} = \sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTz})_j; \text{ for } (F_{OTz})_j < 0$$

$$(OT_x)_{tot} = \frac{F_x h_i}{a} = \sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j \geq 0$$

$$= -\sum_{j=1}^n (F_{OTx})_j; \text{ for } (F_{OTx})_j < 0$$

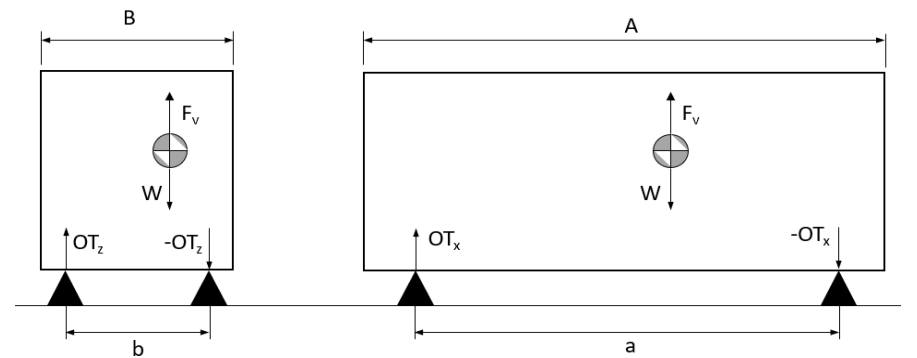
$$Lat_{tot} = (F_x^2 + F_z^2)^{\frac{1}{2}} = \sum_{j=1}^n (F_{xj}^2 + F_{zj}^2)^{\frac{1}{2}}$$



Combined Vertical Loads

Notes:

- *Weight does not apply for isolated restraints
- *Weight does not apply for isolated restraint to structure connections, where the weight bypasses the connection.
- See Load Tables for Combined Vertical Load values at each location.

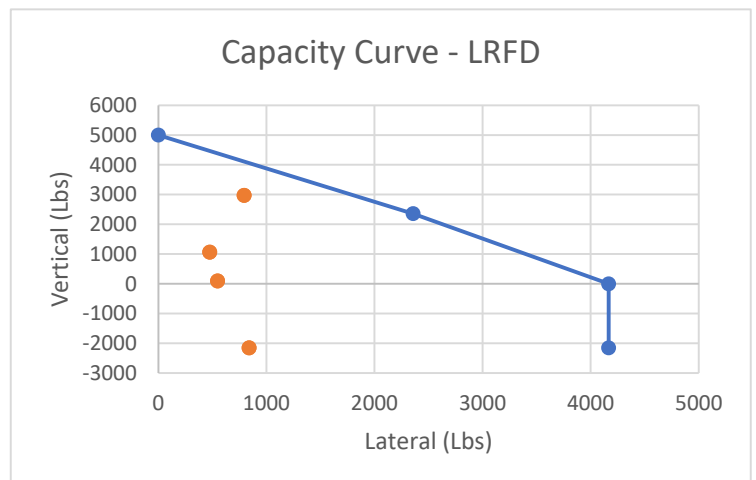


$$(F_v)_{comb} = F_v + OT_x + OT_z - W^*$$

Level 1 - Kinetics Product Calculations (Based on Testing and/or Analysis)

Vertical, Horizontal, and Combined Capacities are based on product testing and/or analysis.

Product Type:	FHS
Product Size:	2-100/1600
Product Qty:	4
Worst Case Tension Load (lbs.):	2970
Worst Case Shear Load (lbs.):	839
Worst Case Factor of Safety:	1.30



Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project: Nan Inc Hilo WWTP Ph 1

KNC#: 217055

Tag: 2521509-03

Designer: FJ

Date: 6/19/2025

Rev. 0.0

Level 2 - Bolt Calculations

¹Tensile Capacity = Proof Strength

²Shear Capacity = 0.5* Tensile Capacity

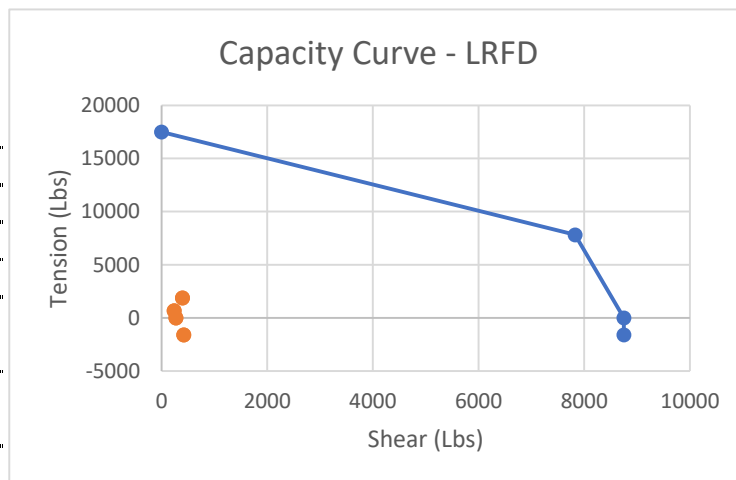
Notes:

1. For non-sustained loading conditions.
2. Based on maximum shear stress theory.

Product Type:	Gr5 Bolts
Product Size:	5/8 Dia
Product Qty:	8
Orientation:	Vertical
Worst Case Tension Load (lbs.):	1889
Worst Case Shear Load (lbs.):	419
Worst Case Factor of Safety:	7.35

Combined Capacity Definition

$$\frac{(\text{Shear Load})^2}{(\text{Shear Capacity})^2} + \frac{(\text{Tensile Load})^2}{(\text{Tensile Capacity})^2} \leq 1.0$$



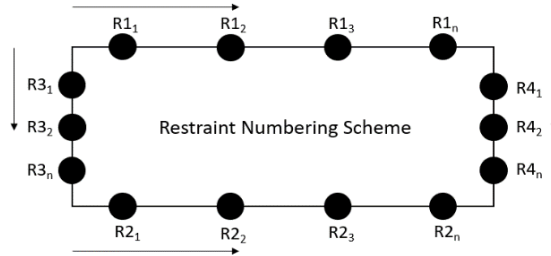
Graph shows the values of all restraints (in orange) included in the system in relation to their LRFD capacity curve (in blue).



Project:	Nan Inc Hilo WWTP Ph 1	KNC#:	217055
Tag:	2521509-03	Designer:	FJ
Date:	6/19/2025	Rev:	0.0

Level Inputs

Level	Description	Type	Size	a (in)	b (in)	h (in)	Weight (lbs)
1	Equipment_to_Isolator_Type	FHS	2-100/1600	75.429	22	26.8	1163
2	Isolator_to_Structure	Gr5 Bolts	5/8 Dia	75.429	22	36.1	1163
3							
4							

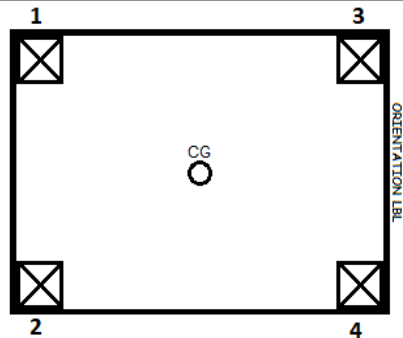


Level 1 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		1		0		0		56	Pass
Side	Load	Weight	Static	Factored	Lat -x	Lat -z	Vertical	OT-x	OT-z	Lat Tot	Combined
	Point #	Transfer	Wt	Wt	Load	Load	Load	Load	Load	Load	Vertical Load
1	1	No	-467	-421	313	358	194	312	559	476	1064
1	2	No	-201	-181	313	729	647	-312	2635	794	2970
2	1	No	-381	-343	414	358	341	312	-559	548	94
2	2	No	-114	-103	414	729	794	-312	-2635	839	-2153

Level 2 Reaction Loads (lbs)

# Load Points Sides 1 & 2		Added # Load Points Side 3 & 4		# Connections per load point		Connection eccentricity x-dir		Connection eccentricity z-dir		Worst Case θ	Pass/Fail Wind
2		0		2		0		0		56	Pass
Side	Load Point #	Weight Transfer	Static Wt	Factored Wt	Lat-x Load	Lat-z Load	Vertical Load	OT-x Load	OT-Z Load	Lat Tot Load	Combined Vertical Load
1	1	No	-467	-421	313	358	194	420	753	476	1367
1	2	No	-201	-181	313	729	647	-420	3552	794	3778
2	1	No	-381	-343	414	358	341	420	-753	548	8
2	2	No	-114	-103	414	729	794	-420	-3552	839	-3177



Tag/Iso		2521509-02 : FHS-2				2521509-03 : FHS-2							
Iso Num	Load	Isolator	Defl	Iso Num	Load	Isolator	Defl	Iso Num	Load	Isolator	Defl		
1/2	416 / 308	2-500 / 2-500	1.66 / 1.23	1/2	456 / 338	2-500 / 2-500	1.82 / 1.35						
3/4	204 / 151	2-225 / 2-225	1.81 / 1.34	3/4	212 / 157	2-225 / 2-225	1.88 / 1.40						
ORIENTATION LBL		INLET SIDE			INLET SIDE								
ISO/REST QTY		4 PER TAG, 4 TOTAL			4 PER TAG, 4 TOTAL								
CG LEN/WDTH		-12.493 / -1.65 From Center			-13.806 / -1.64 From Center								
UNIT WEIGHT		1,078.00 lbs			1,163.00 lbs								
MAN/MODEL		HARTZELL/A 41-10-27			HARTZELL/A 41-10-27								
REPRESENTATIVE		Hartzell Air Movement		DESIGNER	J								
REFERENCE/ COMMENTS		COMPONENTS SELECTED FOR VIBRATION ISOLATION ONLY. WIND/SEISMIC RESTRAINT IS TO BE BY OTHERS. CENTER OF GRAVITY IS ASSUMED TO BE THE GEOMETRIC CENTER OF THE ISOLATED EQUIPMENT UNLESS OTHERWISE SPECIFIED.											

	TITLE	DATE	DRAWN BY	KNC	DRAWING NO.	REV
	GENERAL ISOLATION LOCATIONS	06/19/2025	J	217055	C217055-01	0

FHS 2" DEFLECTION SEISMIC ISOLATOR

IP UNITS (INCHES AND POUNDS)

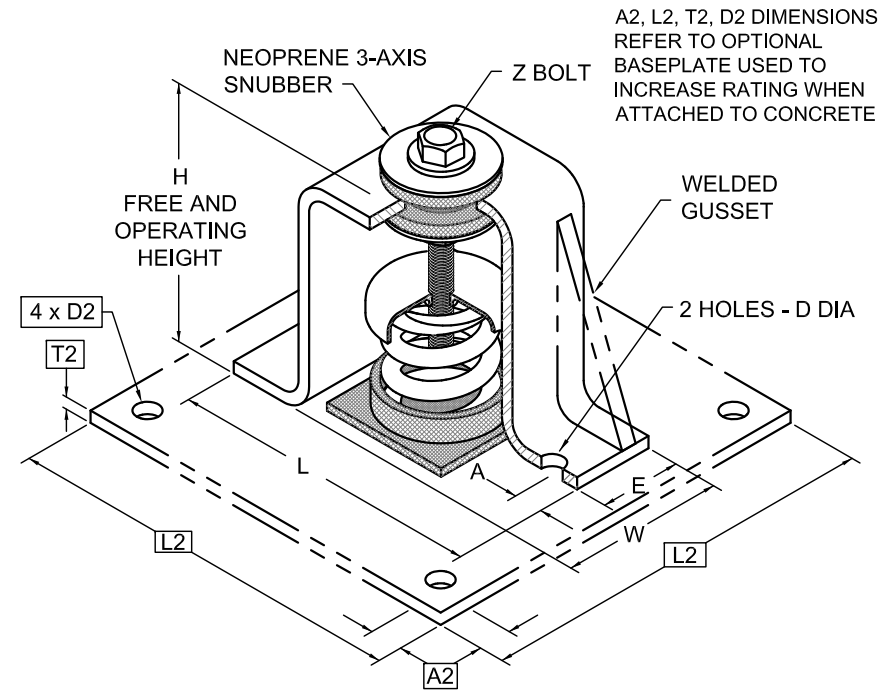
TYPE	L	W	A	D	E	H	Z
2-100/1600	10.38	5.00	0.75	0.69	2.50	9.38	0.50
2-1975	10.38	5.00	0.75	0.69	2.50	9.38	0.50

STANDARD RATINGS		SPRING COIL					
		COLOR		FREE		RATED	
TYPE	SIZE	OUTER	INNER	HT.	O.D.	LOAD	DEFL.
FHS	2-100	GRAY		6.09	3.50	100	2.00
FHS	2-135	BEIGE		6.09	3.50	135	2.00
FHS	2-185	BROWN		6.09	3.50	185	2.00
FHS	2-225	GRAY	BROWN	6.09	3.50	225	2.00
FHS	2-250	BLUE		6.09	3.50	250	2.00
FHS	2-375	BLUE	BROWN	6.09	3.50	375	2.00
FHS	2-500	GREEN		6.09	3.50	500	2.00
FHS	2-625	GREEN	BROWN	6.09	3.50	625	2.00
FHS	2-750	BLACK		6.09	3.50	750	2.00
FHS	2-875	BLACK	BROWN	6.09	3.50	875	2.00
FHS	2-995	ORANGE		6.09	3.50	995	2.00
FHS	2-1120	ORANGE	BROWN	6.09	3.50	1120	2.00
FHS	2-1400	ORANGE	GREEN	6.09	3.50	1400	2.01
FHS	2-1600	RED		6.09	3.50	1600	2.00
*FHS	2-1975	RED	GREEN	6.09	3.50	1975	1.98

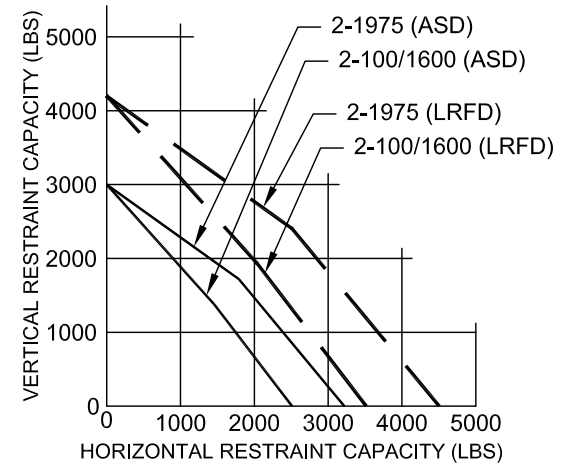
*INCLUDES WELDED GUSSET AS SHOWN

SPECIFICATIONS:

- VERTICALLY AND Laterally RESTRAINED SPRING ISOLATOR WITH CONSTANT FREE AND OPERATING HEIGHT AND REPLACEABLE NEOPRENE SNUBBING ELEMENTS.
- 0.25 NOMINAL CLEARANCE IN SNUBBING ELEMENTS.
- MINIMUM OVERLOAD OF 50%.
- MINIMUM K_x/K_y RATIO OF 1.0.
- ALL ELEMENTS SAFE AT SOLID LOADING.
- POWDER COATED COILS AND HOT DIPPED GALVANIZED BRACKET.



WELDED TO STEEL



Computed horizontal and vertical seismic load combinations must fall within the restraint capacity envelope. Contact KNC for bolted to steel data.



KINETICS NOISE CONTROL, INC
6300 IRELAN PL,
DUBLIN, OH 43017 USA
Ph: 614 889-0480, Fax: 614 889-0540
www.kineticsnoise.com

Model:

FHS-2-
100/1975

By: PWM

Date: 12/21/01

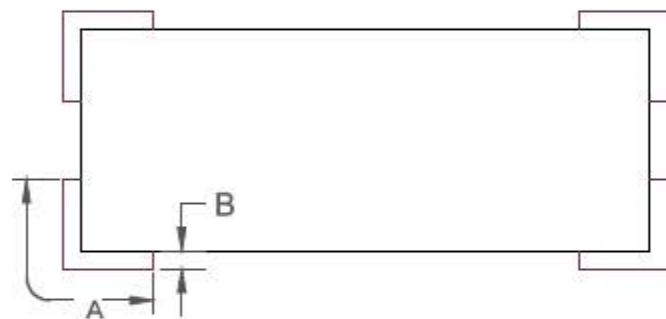
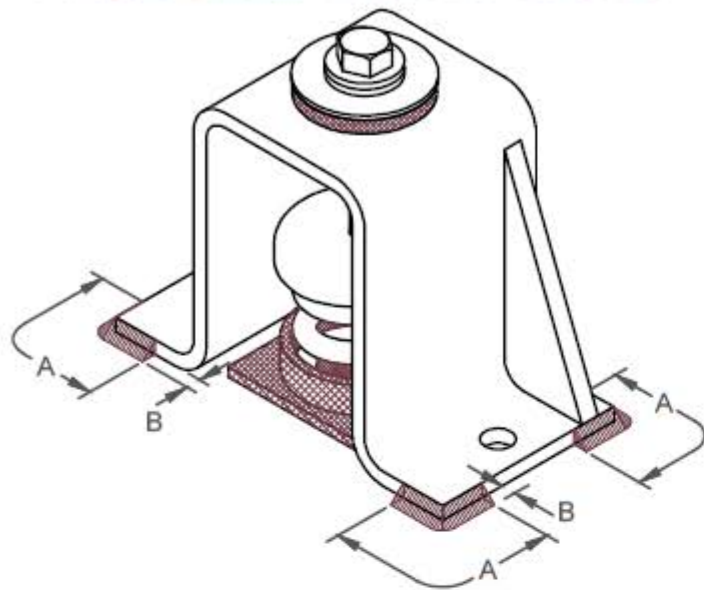
Revised: 05/26/22 / BB

Drawing No:

S-01.35-21

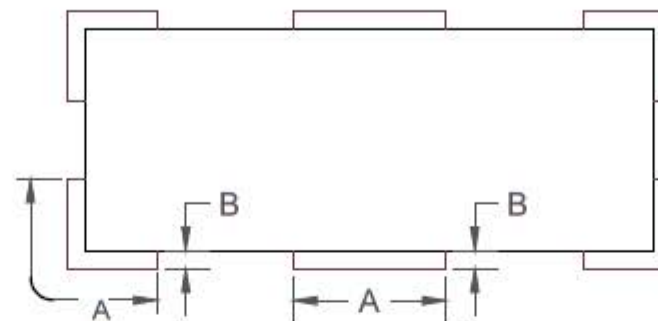
ISOLATOR WELD DATA

FHS WELD DOWN DATA



WELD 4 PLACES AS SHOWN

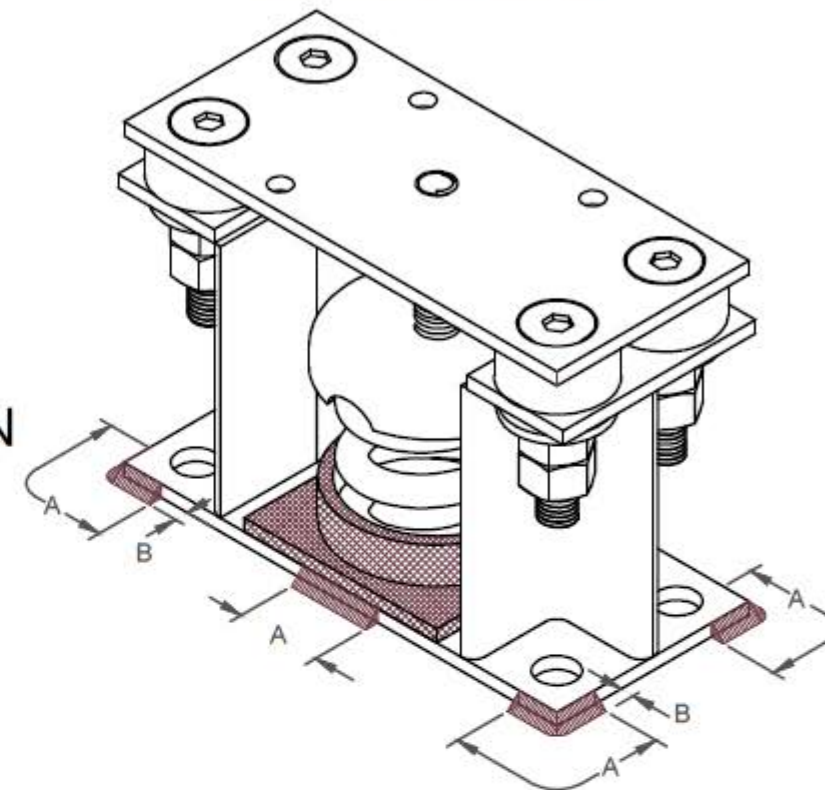
MODEL	LGTH (A)	SIZE(B)
1-35/805	3.25	5/16
1-50/3500	3.25	5/16
2-100/1975	3.25	5/16
2-2000/4500	3.25	5/16
4-100/1600	3.25	5/16
4-2250/5800	3.25	5/16



WELD 6 PLACES AS SHOWN

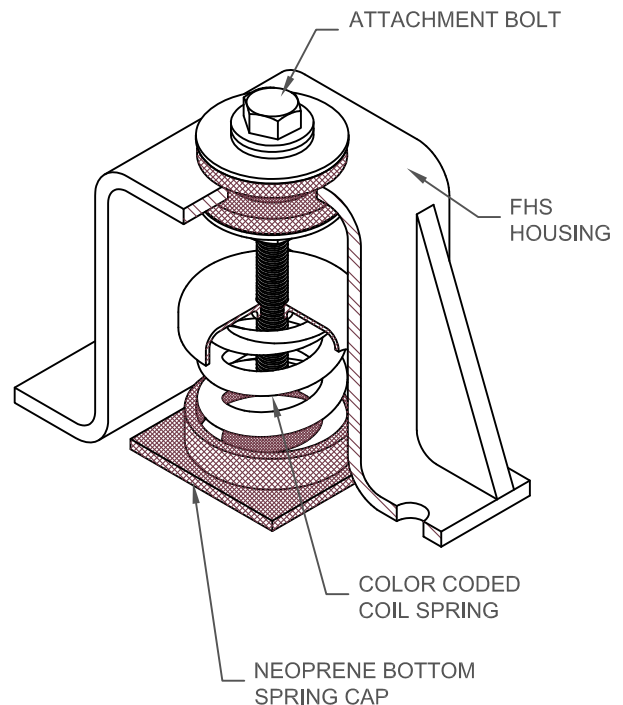
MODEL	LGTH(A)	SIZE(B)
1-35/805	2.25	1/4
1-50/3500	3.50	1/4
1-2500/7000	3.50	1/4
1-5000/14000	4.50	5/16
2-100/1975	3.50	1/4
2-2000/4500	3.50	1/4
2-4000/9000	3.50	1/4
2-8000/18000	4.50	5/16
4-100/1600	3.50	1/4
4-2250/5800	3.50	1/4
4-5500/11600	3.50	1/4
4-11000/23200	4.50	5/16

FLS, FLSS, FTS WELD DOWN DATA



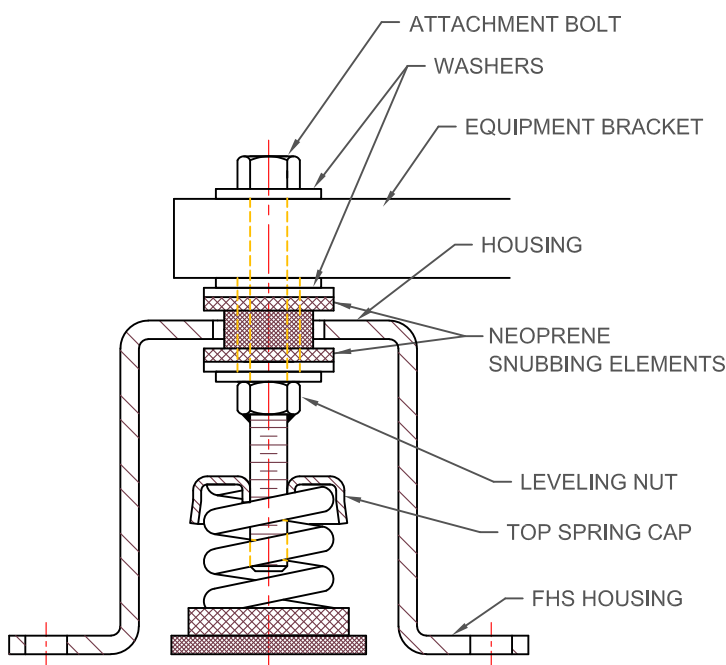
FHS INSTALLATION INSTRUCTIONS

- 1) COORDINATE THE LOCATION OF EACH ISOLATOR WITH THE APPROPRIATE SUBMITTAL DRAWING.
- 2) REMOVE THE ATTACHMENT BOLT AND ONE WASHER FROM THE TOP OF THE ISOLATOR. INSERT THEM INTO THE EQUIPMENT BRACKET.
- 3) ALIGN ISOLATORS AND CENTER THE ISOLATOR HOUSINGS BENEATH THE EQUIPMENT BOLTING HOLES. MARK THE ISOLATOR HOUSING ANCHOR BOLT LOCATIONS ON THE FLOOR.
- 4) RAISE THE EQUIPMENT AND INSTALL THE ANCHOR BOLTS INTO THE FLOOR.
- 5) LOCATE THE ISOLATORS AND LOWER THE EQUIPMENT ONTO THEM. NOTE THE HOUSING SHOULD SUPPORT THE EQUIPMENT WITH APPROXIMATELY THE DESIRED FLOOR CLEARANCE. (TAKE CARE NOT TO PUSH THE EQUIPMENT SIDWAYS).
- 6) INSTALL ANCHOR NUTS TO FIRMLY ATTACH THE FHS HOUSING TO THE FLOOR OR HOUSEKEEPING PAD.
- 7) LOOSELY BOLT EQUIPMENT TO THE ISOLATOR USING THE ATTACHMENT BOLTS. BEFORE ADJUSTING THE ISOLATORS, IT IS NECESSARY THAT THE EQUIPMENT BE BROUGHT UP TO ITS OPERATING WEIGHT. ADJUST THE ISOLATOR BY TURNING THE LEVELING NUT COUNTER CLOCKWISE ONE OR TWO TURNS AT EACH ISOLATOR,



IN SUCCESSION, UNTIL THE UPPER NEOPRENE ELEMENT BARELY LIFTS OFF OF THE TOP OF THE FHS HOUSING. WHEN THIS HAPPENS, NO LONGER ADJUST THIS ISOLATOR, BUT CONTINUE TO ADJUST THE REMAINING ISOLATORS UNTIL ALL ARE IN THIS CONDITION.

- 8) AT THIS POINT, TURN ALL THE LEVELING SCREWS THE APPROPRIATE AMOUNT TO LIFT THE EQUIPMENT 1/4". THE NUMBER OF TURNS IS A FUNCTION OF THE LEVELING BOLT SIZE. 1/2" = 3 TURNS, 5/8" = 2-3/4 TURNS, 3/4" = 2-1/2 TURNS, 7/8" = 2-1/4 TURNS, 1" = 2 TURNS.
- 9) TIGHTEN THE ATTACHMENT BOLTS, THUS BOLTING THE ISOLATORS TO THE SUPPORTED EQUIPMENT AND LOCKING THE LEVELING BOLT AGAINST TURNING.
- 10) DO NOT ATTEMPT TO MOVE THE ISOLATORS Laterally WITH THE WEIGHT OF THE EQUIPMENT ON THEM. IF IT IS NECESSARY TO MOVE THE EQUIPMENT ON THEM, REMOVE THE WEIGHT FROM THE ISOLATORS BY RAISING THE EQUIPMENT BEFORE MOVING.



APPENDIX:

KNC #217055

SUBMITTAL #1.0

CUSTOMER PROVIDED EQUIPMENT INFORMATION FOR REFERENCE ONLY

IT IS CRITICAL TO NOTE THAT THE CERTIFICATION PROVIDED BY KINETICS NOISE CONTROL ONLY COVERS THOSE HARDWARE ITEMS WHICH HAVE BEEN PROVIDED BY KINETICS NOISE CONTROL. IT IS BASED ON BUT DOES NOT EXTEND TO THE EQUIPMENT REFERENCED IN THIS APPENDIX.

SECTION 03.03 LINE ITEM TECHNICAL INFORMATION



Line No.: 2521509-03

Description: A41-0-271FA100FGI8MT - Fiberglass Backward
Curved Centrifugal Fan

Tag: 02-EF-1390 & 02-EF-1396



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 6 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE
Contact Name: WILL ADVISE
Ship To Phone: WILL ADVISE
Ship To Fax: WILL ADVISE

Line Nbr	Item ID	Req Date	Req Qty	Unit Price	Extended Price
	Item Name			Unit of Measure	

3	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				
---	---	--	--	--	--

12/31/2025

1.00

Each

A41-0-271FA100FGI8MT
41 - Fiberglass Backward Curved Centrifugal Fan

Tag: 02-EF-1390 & 02-EF-1396

Performance
Volume Flow Rate: 6465.0 cfm
SP: 3.6 in. w.g. / TP: 3.75 in. w.g.
RPM: 1150.0
Density: 0.075 lbs/ft³
Operating Temp: 70 °F
OPwr: 4.724 hp / SPwr: 4.724 hp

Motor
HP: M - 7.5 HP
RPM/Volt/Hz/Phase: T - 1750, 460, 60, 3
Enclosure: I8 - TEFC IEEE841
Frame Size: 213T
Operating Voltage: 460
Vendor: BALDOR

Motor Accessories
Hermetically Sealed Thermostat
Space Heater - 115V/120V
Hartzell Motor P/N: 3110-0040-B-(MODIFIED), Mfct P/N: ECP83770T-4

Fan
Arrangement: 10
Rotation and Discharge: Up Blast (Clockwise)
Mounting Location: Floor
Motor Position: Standard
Material: Fiberglass
Construction: ASTM D4167 97 (Includes Single Veil & Electric Grounding)
Fan Coating: Hartkoate
Prop/Wheel: FA - Backward Airfoil Centrifugal Wheel - Fiberglass
Prop/Wheel Coating: Hartkoate & Electric Grounding



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 7 of 18

DUNS NO. 05-655-1039

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636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:

Ship To Phone:

Ship To Fax:

Line Nbr	Item ID	Req Date	Req Qty	Unit Price	Extended Price
	Item Name			Unit of Measure	

Standard Accessories (Included)
Water Slinger
Inlet Support, Steel - Inorganic Zinc Coating w/ Epoxy Topcoat

Additional Accessories
Bolted Access Door, Fiberglass
Drain, CPVC w/ CPVC Plug (Factory Installed)
Extended Lube Lines to Fan Bearings, 316SS
316SS Hardware to Complete Fan
316SS Shaft
Teflon Double Lip Seal in a 316SS Casing
Drilled Inlet Flange, PS15-69
Drilled Outlet Flange, PS15-69
FAB Airfoil Backdraft Damper - Fiberglass: Max SP - 14.5: Max FPM - 4000
Damper Style: Parallel
Polycarbonate Jamb Seal for Damper
EPDM Blade Seal for Damper
FRP (Standard) Axle for Damper
Both Flanges Drilled For Damper

Seismically Restrained Spring Isolators

Inlet Support, Motor & Bearing Base, and Weather Cover to be 316 SS

ACCESS DOOR TO HAVE ONE PIECE NEOPRENE GASKET PUNCHED TO MATCH BOLT PATTERN

SPECIAL ARRANGEMENT 10 WEATHER COVER WITH CLEAR WINDOWS FOR BELT AND BEARING VIEWING

2-Sets of Spare Belts (2 years worth)

ADDER FOR ---15830 2.03 B 5 L10 100,000 hour bearings

Spare set of bearings

Total Weight (approx.): 1147 lbs. (Each) (quoted items are not included in the weight.)

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 8 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
4	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

Spare Sets of sheaves to be provided after final balancing and the belt sizes determined.
Reference Series and Size: 10-FAN-2230 FAN ONLY

Tag: Config-324164

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 9 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
5	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

Spare Sets of Belts (2 per fan) to be provided after final balancing and the belt sizes determined.
Reference Series and Size: A41-1-402FA100FGFQT3

Tag: Config-324163

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 10 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
6	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

2 sets of spare belts per fan (2 years worth)
Reference Series and Size: A41-0-271FA100FGI8MT

Tag: Config-324162

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 11 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
7	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

Spare Set of Belts (2 Years Worth)
Reference Series and Size: A41-0-271FA-66FGI8LT

Tag: Config-324270

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 12 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
8	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

Spare Set of Bearings
Reference Series and Size: A41-1-402FA100FGFQT3

Tag: Config-324272

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
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DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
9	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

Spare Set of Bearings
Reference Series and Size: A41-0-271FA100FGI8MT

Tag: Config-324268

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 14 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
10	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

2.00

Each

Spare Set of Bearings
Reference Series and Size: A41-0-271FA-66FGI8LT

Tag: Config-324273

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 15 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
11	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

6.00

Each

AMCA PERFORMANCE WITNESS TEST TO BE PERFORMED AT HARTZELL CERTIFIED LAB
Description and Dimensions: AMCA 210

Tag: Config-324165

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 16 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:

Ship To Phone:

Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price	Extended Price
				Unit of Measure	
12	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

6.00

Each

Field Service Trips five (6) total all detailed below

Description and Dimensions: Each Trip for a total of 6 Trips Per Section 15830 Fans & 15832 Odor Control Fans (all travel expenses included) and up to 2 service personnel. Trip 1 & 2 - Installation Assistance (3 days on site and 2 days in transit) X 2 Trips Trip 3 - Installation Inspection (3 days on site and 2 days in transit) X 1 Trip Trip 4 - Startup and performance testing assistance (3 days on site and 2 days in transit) X 1 Trip Trip 5 - Training and Final Acceptance Checkiout (3 days on site and 2 days in transit) X 1 Trip Trip 6 - Floater Trip included per Section 15830 (3 days on site and 2 days in transit)

Tag: Config-324166

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 17 of 18

DUNS NO. 05-655-1039

Bill To : NAN INC
636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice

Ship To : WILL ADVISE

WILL ADVISE
WILL ADVISE

Contact Name:
Ship To Phone:
Ship To Fax:

Line Nbr	Item ID Item Name	Req Date	Req Qty	Unit Price Unit of Measure	Extended Price
13	HFA/PBA-FRP HOLD FOR APPROVAL - PROGRESS BILLING ADVANCE				

12/31/2025

1.00

Each

Seismic Calcs Per Line (1 of each line item)

Tag: Config-324611

Blanket Qty: 0.00

Blanket Exp Date:



Hartzell Air Movement
1025 S Roosevelt Ave.
Piqua, Ohio 45356
PH: (937) 773-7411
FAX: (937) 773-8894

Sales Order Acknowledgement

Sales Order ID: 2521509
Customer Purchase Order ID: 24077-00083
Page 18 of 18

DUNS NO. 05-655-1039

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636 LAUMAKA STREET
HONOLULU, HI 96819
USA

Order Date: 5/19/2025 8:45:57 AM
Salesman Code: 03ES
Partial Ship: False

Ship Method: PPD CC
FOB: Piqua
AR Terms: Per Invoice



All fan products
eligible for the
5-Year Warranty

Website: www.hartzellairmovement.com/five-year-warranty

Total:
Sales Tax:
Estimated Ship Charges:
Estimated Tax Freight:
Sales Order Total:

Special Instructions: **HFA-PBA 05.19.25 JW**
CONTACT JYUN-CHENG JHUO JCHURO@NANHAWII.COM
PREPAID_ADD
SHIP VIA: Best Way
FREIGHT CONTACT: Jyun-Cheng Jhuo PHONE: 765.607.8397
AGENT ORDER #:
ATTN: Nan Inc.
MARK: 24077-00083
DWG: Yes- Certified - 0COPY FOR Hold For Approval
FREIGHT COMMENTS:
JW

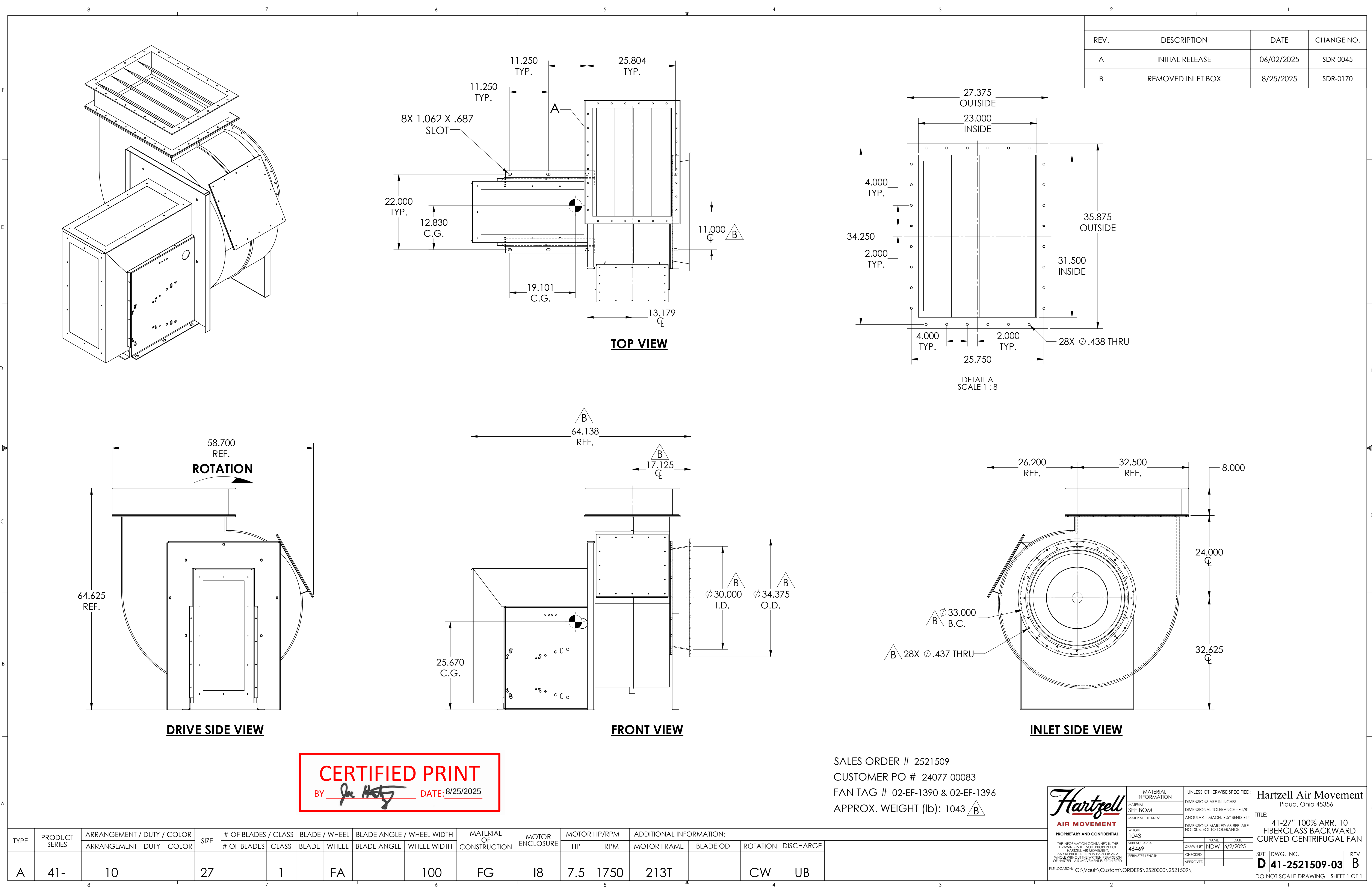
Change Nbr: 10
Change Date:

IMPORTANT: THE GOODS DESCRIBED ABOVE ARE SOLD AND SHIPPED ACCORDING TO THE CONDITIONS OF SALE SET FORTH ON THE LAST PAGE.

-----IMPORTANT PLEASE READ-----

The safe application and use of equipment supplied by Hartzell Air Movement. is the responsibility of the installer, user, owner and employer. To evaluate the safe application of this equipment, the following should be considered: the location of the installation, accessibility of employees and other persons to the equipment, any adjacent equipment, applicable building and safety codes, and requirements of OSHA. Since the application and use of its equipment can vary greatly, Hartzell Air Movement offers various product types, optional safety accessories, and sound performance data per laboratory tests. An industry publication: "Recommended Safety Practices for Users and Installers of Industrial and Commercial Fans" is available from Hartzell upon request.

We hereby certify that these goods were produced in compliance with all applicable requirements of section 6, 7 and 12 of the Fair Labor Standards Act, as amended, and of regulations and orders of the United States Department of Labor issues under section 14, there of.



REV.	DESCRIPTION	DATE	CHANGE NO.
A	INITIAL RELEASE	06/02/2025	SDR-0045
B	REMOVED INLET BOX	8/25/2025	SDR-0170

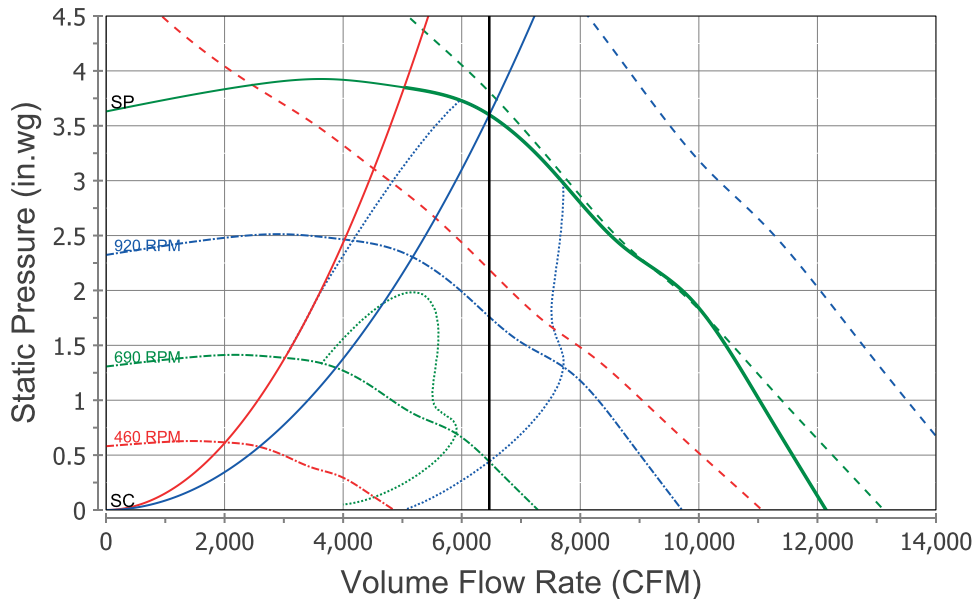
CERTIFIED PRINT
BY *[Signature]* DATE: 8/25/2025

SALES ORDER # 2521509
CUSTOMER PO # 24077-00083
FAN TAG # 02-EF-1390 & 02-EF-1396
APPROX. WEIGHT (lb): 1043 B

TYPE	PRODUCT SERIES	ARRANGEMENT / DUTY / COLOR			SIZE	# OF BLADES / CLASS		BLADE / WHEEL		BLADE ANGLE / WHEEL WIDTH		MATERIAL OF CONSTRUCTION	MOTOR ENCLOSURE	MOTOR HP/RPM		ADDITIONAL INFORMATION:			
		ARRANGEMENT	DUTY	COLOR		# OF BLADES	CLASS	BLADE	WHEEL	BLADE ANGLE	WHEEL WIDTH			HP	RPM	MOTOR FRAME	BLADE OD	ROTATION	DISCHARGE
A	41-	10			27		1		FA		100	FG	I8	7.5	1750	213T		CW	UB

 AIR MOVEMENT PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF HARTZELL AIR MOVEMENT. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF HARTZELL AIR MOVEMENT IS PROHIBITED. FILE LOCATION: C:\Vault\Custom\ORDERS\2520000\2521509\ <td> MATERIAL INFORMATION MATERIAL SEE BOM MATERIAL THICKNESS </td> <td colspan="3"> UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES DIMENSIONAL TOLERANCE = ±1/8" ANGULAR = MACH. ± 5° BEND ± 1° DIMENSIONS MARKED AS REF. ARE NOT SUBJECT TO TOLERANCE. </td> <td colspan="3"> Hartzell Air Movement Piqua, Ohio 45356 </td>	MATERIAL INFORMATION MATERIAL SEE BOM MATERIAL THICKNESS	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES DIMENSIONAL TOLERANCE = ±1/8" ANGULAR = MACH. ± 5° BEND ± 1° DIMENSIONS MARKED AS REF. ARE NOT SUBJECT TO TOLERANCE.			Hartzell Air Movement Piqua, Ohio 45356		
	WEIGHT 1043	SURFACE AREA 46469	TITLE: 41-27" 100% ARR. 10 FIBERGLASS BACKWARD CURVED CENTRIFUGAL FAN				
	PERIMETER LENGTH	CHECKED	DRAWN BY NDW	DATE 6/2/2025	SIZE D 41-2521509-03	REV B	
		APPROVED	DO NOT SCALE DRAWING			SHEET 1 OF 1	

Hartzell-FLOW™ v1.0.18 / February 2019
A41-0-271FA100FGI8MT

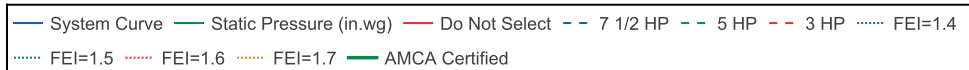


Fan Tag#: 02-EF-1390 & 02-EF-1396

Vol Flow Rate	6465
Pressure	3.6
Density (lbs/ft³)	0.075
Oper. Temp. (°F)	70
Fan RPM	1150
Max Safe RPM	1521
Operating Power	4.724
Standard Power	4.724
Static Efficiency	0.775
Outlet Velocity (fpm)	1532
Fan Energy Index (FEI)	1.42
Fan Efficiency Grade (FEG)	FEG85

Discharge Sound Power Levels referred to 10 ⁻¹² watts							
1	2	3	4	5	6	7	8
90	89	89	86	81	76	70	62

Radiated Sound Power Levels referred to 10 ⁻¹² watts							
1	2	3	4	5	6	7	8
86	82	79	75	69	63	57	49
Radiated Sound is not AMCA Licensed							



Hartzell Air Movement certifies that the model shown is licensed to bear the AMCA Seal. The ratings shown are based on tests and procedures performed in accordance with AMCA publication 211 and AMCA publication 311 and comply with the requirements of the AMCA Certified Ratings Program. FEI values are calculated in accordance with ANSI/AMCA Standard 208 and are based on four-pole TEFC motors of the size shown. Power rating BHP excludes transmission losses. Performance certified is for Installation Type D: Ducted Inlet, Ducted Outlet. AMCA Licensed for Sound and Air Performance. Performance ratings do not include the effects of appurtenances (Accessories). Sound ratings are based on sound level data obtained in accordance with AMCA Standard 300. The sound power level ratings shown are in decibels, referred to 10⁻¹² watts, calculated per AMCA Standard 301. Fan Outlet Sound Testing. Values shown are for outlet two sound power levels for: Installation Type D: Ducted Inlet, Ducted Outlet. Ratings include the effects of duct end correction.

Discharge Sound Pressure = 76 dBA @ 5ft

Radiated Sound Pressure = 65 dBA @ 5ft

Discharge Sound Power = 87 LwA

Radiated Sound Power = 76 LwA

The A-weighted sound pressure level (dBA) is based on Hartzell Laboratory sound power tests, and is calculated in accordance with AMCA standard 303.

The FEG, dBA, LwA and radiated values are not AMCA International Licensed.

The calculation assumes a free field condition with a directivity factor for hemi-spherical radiation (Q=2).

The installed sound pressure levels are influenced by the installation and acoustic environment, and cannot be guaranteed. Use of this estimate level along for field acceptability test is not recommended.

Although the calculation can be done for any stated distance, the free field does not start until 20 to 50 ft from the equipment in most installations.

Contact Hartzell Air Movement for more information concerning dBA values.



Fan Shaft Critical Speed and Bearing Life Analysis

Model: 411-27FA100

Reference: 2521509-03

Actual Fan Diameter (in.): 27.625

Tip Speed: 8,317

Motor Horsepower: 7.5

Fan Sheave Dia. (in.): 6.4

Sheave Wt. (lbs.): 9

Shaft Speed (rpm): 1150

Shaft Diameter (in.): 2.1875

Wheel Wt. (lbs.): 91.6

Shaft Material: 316 SS

Tensile Yield Strength: 30000

Hub Material: 1020 HR

Tensile Yield Strength: 30000

Arrangement No.: 10

Fan Air Temp (deg. F): 70

Ambient Temp (deg. F): 70

Belt Tension Factor: 1.68

Belt Angle from 6 O'Clock (deg.): 45

Shaft Modulus of Elasticity (lbs/in^2): 2.92E+07

Wheel WR2 (lb-ft^2): 78.2

WR2 Factor: 0.5

Distance From Wheel to Drive Bearing (in.): 30.137

Distance Between Bearings (in.): 20.5625

Distance From Drive Bearing to Sheave (in.): 4

Wheel Overhang: 9.5745

	w/o Belt Pull	w/Belt Pull
Critical Speed	5875	4760
Safety Factor	5.109	4.139

G Shock Load: 10

Minimum Shaft Dia.: 1.439

Hub O.D.: 3.500

Hub Wall: 0.65625

Static Pressure (neg. for downblast): 3.6

Expeller Vanes?: No

Minimum Bearing L10 Life: 200,000

Shaft Orientation: Horizontal

Belt Pull: 216

Belt Pull: 216

Key Size: 0.5

Key Length: 0.100

Load on Bearing A (Wheel): 134

Load on Bearing B (Drive): 246

Total Axial Thrust: 31

Bearing A: Fixed

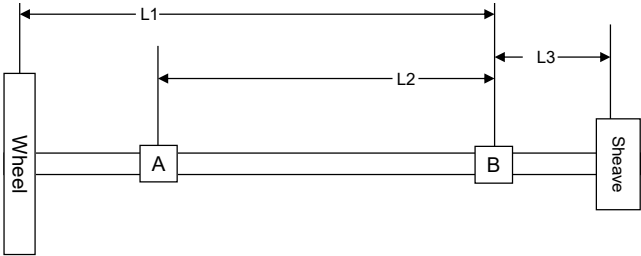
Bearing B: Fixed

Base Load

Base Load

Series	Position	Dynamic Rating	Status	Equivalent Load	Fixed or Expansion	L10 Life
Link-Belt U300	A	12,400	Good Selection	134	Fixed	>1 million
	B	12,400	Good Selection	246	Fixed	>1 million

V041555/21/2025



Thrust Multiplier

0.67

Today is: Wednesday, May 21, 2025



Document:

ES 7.5.1

Rev

1Title: **Inlet Boxes for Centrifugal Fans – Series 03, 05, and 41**Owner: **Engineering**Document Type: **Engineering Standard**Creation Date: **5/2/1989**

Inlet Boxes for Centrifugal Fans – Series 03, 05, and 41

1.0 Scope

This standard defines the dimensional and performance requirements for inlet boxes used on centrifugal fans

2.0 Products Affected

- 2.1 Series 03 Backward Curved Centrifugal Fan, Single Width
- 2.2 Series 05 Industrial Exhausters
- 2.3 Series 41 Fiberglass Backward Curved Centrifugal Fans

3.0 Dimensions

Inlet dimensions are based on the fan wheel diameters as shown below. For Series 03 and Series 41 fans, they are approximately 1.5 wheel diameters wide by 0.5 wheel diameters deep. For Series 05 fans, they are approximately 1.0 wheel diameters wide by 0.5 wheel diameters deep

Series 03 / 41			Series 05		
Fan Size	Inlet Width (in.)	Inlet Depth (in.)	Fan Size	Inlet Width (in.)	Inlet Depth (in.)
12	18	6	12	12	6
15	24	8	15	16	8
18	28	10	19	20	10
22	34	12	22	22	12
24	36	12	26	26	14
27	42	14	29	30	14
30	46	16	33	34	16
33	50	16	36	36	18
36	54	18	40	40	20
40	69	20	45	46	22
44	66	22	50	50	26
49	74	24	57	58	28
54	86	28			
60	96	30			
66	105	34.25			
73	116	38			
81	129	41.5			
89	141.75	46			
98	156.5	51			

Actual Inlet Vel. - 1583 fpm
Actual Press. Drop - 0.18 in w.c.

4.0 Performance

The pressure drop through an inlet box is based on tests and is shown below. Use the inlet areas listed above to calculate the inlet velocity. These values include all system effects

Inlet Velocity (fpm)	1300	1500	2000	2500	3000	3500	4000	4500	5000
Pressure Drop (in W.G.)	0.10	0.16	0.31	0.50	0.72	0.99	1.30	1.64	1.97

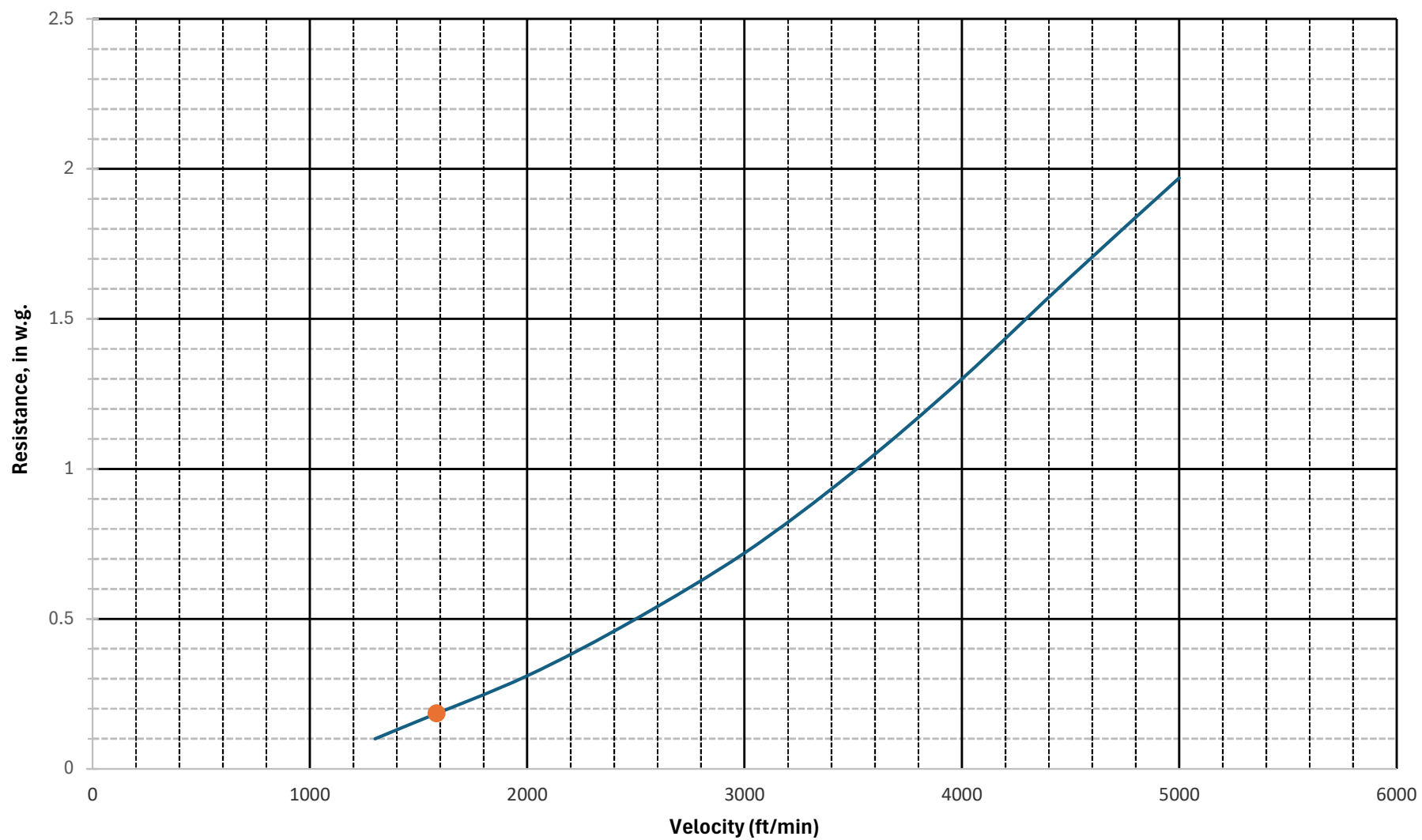
5.0 References

- 5.1 N/A

6.0 Revision History

Rev	Description	Date	Changed By
1	Remove from Drive System and Release in QMS	10/30/2024	J.Felver

Inlet Box Resistance





Customer information packet

ECP83770T-4

7.5HP, 1765RPM, 3PH, 60HZ, 213T, 0740M, TEFC, F

Class - CLI GP A,B,C,D

Division - Division II

Specifications

Enclosure	TEFC
Frame	213T
Frame Material	Iron
Frequency	60.00 Hz
Haz Area Class and Group	CL I GP A,B,C,D
Haz Area Division	Division II
Motor Letter Type	Three Phase
Output @ Frequency	7.500 HP @ 60 HZ
Phase	3
Synchronous Speed @ Frequency	1800 RPM @ 60 HZ
Voltage @ Frequency	460.0 V @ 60 HZ
Agency Approvals	CCSA US CSA CSA EEV NEMA PREMIUM NEMA_PREMIUM UR
Ambient Temperature	40 °C
Auxillary Box	No Auxillary Box
Auxillary Box Lead Termination	None
Base Indicator	Rigid
Bearing Grease Type	Polyrex EM (-20F +300F)
Blower	None
Constant Torque Speed Range	1.2
Current @ Voltage	9.500 A @ 460.0 V
Design Code	B
Drip Cover	No Drip Cover
Duty Rating	CONT
Efficiency @ 100% Load	91.7 %
Electrically Isolated Bearing	Not Electrically Isolated
Enclosure Modification	841
Feedback Device	NO FEEDBACK

Part detail

Revision	BD
Type	AC
Mech. spec.	07K629
Base	
Status	PRD/A
Elec. spec.	07WGX814
Layout	07LYK629
Eff. date	05-12-2025
CD Diagram	CD0006
Poles	04
Leads	3#14
Proprietary	False
Created date	01-24-2006

Front Face Code	Standard
Front Shaft Indicator	None
Haz Area Temp Code	T3C
Heater Indicator	No Heater
High Voltage Full Load Amps	9.5 a
Insulation Class	F
Inverter Code	Inverter Duty
KVA Code	H
Lifting Lugs	Standard Lifting Lugs
Locked Bearing Indicator	Locked Bearing
Max Speed	2700 rpm
Motor Lead Exit	Ko Box
Motor Lead Quantity/Wire Size	3 @ 14 AWG
Motor Lead Termination	Flying Leads
Motor Standards	NEMA
Motor Type	0740M
Mounting Arrangement	F1
Number of Poles	4
Overall Length	19.50 IN
Power Factor	82
Product Family	Chem Process S/P 32-8 IEEE 841
Pulley End Bearing Type	Ball
Pulley Face Code	Standard
Pulley Shaft Indicator	Standard
Rodent Screen	None
RoHS Status	ROHS COMPLIANT
Service Factor	1.15
Shaft Diameter	1.375 IN
Shaft Extension Location	Pulley End
Shaft Ground Indicator	No Shaft Grounding
Shaft Rotation	Reversible
Shaft Slinger Indicator	Shaft Slinger
Speed	1765 rpm
Speed Code	Single Speed

Starting Method	Direct on line
Thermal Device - Bearing	None
Thermal Device - Winding	None
Vibration Sensor Indicator	No Vibration Sensor
Winding Thermal 1	None
Winding Thermal 2	None

Nameplate

NP4328									
CAT.NO.	ECP83770T-4								
SPEC.	07K629X814								
HP	7.5 TE				IP	56			
VOLTS	460								
AMPS	9.5								
R.P.M.	1765								
FRAME	213T		HZ		60		PH	3	
SER.F.	1.15	CODE	H	DES.	B	CLASS		F	
RATING	40C AMB-CONT								
SN									
DE	6307		ODE		6307				
NEMA NOM. EFF.	91.7		P.F.		82				
GUAR. MIN. EFF.	90.2		CC		010A				
T. CODE	T3C		TEMP=		160				

NP3186

SPEC.	07K629X814				
ABMA DE BRG	35BC03X30X				
ABMA ODE BRG	35BC03X30X				
GREASE	POLYREX EM				
MOTOR WEIGHT	199	ROTOR BARS	28	STATOR BARS	36
MAX. R.P.M.	2700	MAX. KVAR	1.3		
INV.TYPE	PWM				
T=	160				
CHP	60	TO	90		
CT	1.2	TO	60		
VT	-0	TO	60		
HTR-VOLTS	N/A	HTR-AMPS	N/A		
HTR-WATTS		MAX. SPACE HEATER TEMP.	N/A		

AC Induction Motor Performance Data

Record # 92344

Typical performance - not guaranteed values

Winding: 07WGX814-R056				Type: 0740M		Enclosure: TEFC		
Nameplate Data				460 V, 60 Hz: Single Voltage Motor				
Rated Output (HP)		7.5		Full Load Torque		22.32 LB-FT		
Volts		460		Start Configuration		direct on line		
Full Load Amps		9.5		Breakdown Torque		66.8 LB-FT		
R.P.M.		1765		Pull-up Torque		30 LB-FT		
Hz	60	Phase	3	Locked-rotor Torque		39.7 LB-FT		
NEMA Design Code		B		KVA Code	H		Starting Current	63.5 A
Service Factor (S.F.)		1.15		No-load Current		3.71 A		
NEMA Nom. Eff.		91.7	Power Factor	81		Line-line Res. @ 25°C		1.485 Ω
Rating - Duty		40C AMB-CONT		Temp. Rise @ Rated Load		54°C		
S.F. Amps				Temp. Rise @ S.F. Load		67°C		
				Locked-rotor Power Factor		36.2		

Load Characteristics 460 V, 60 Hz, 7.5 HP

% of Rated Load	25	50	75	100	125	150	S.F.
Power Factor	46	68	78	82	84	85	83
Efficiency	88.6	92.1	92.4	91.7	90.7	89.3	91.1
Speed	1789.9	1782.7	1774.5	1765.4	1755.6	1743.7	1760
Line amperes	4.3	5.61	7.32	9.31	11.5	13.9	10.6

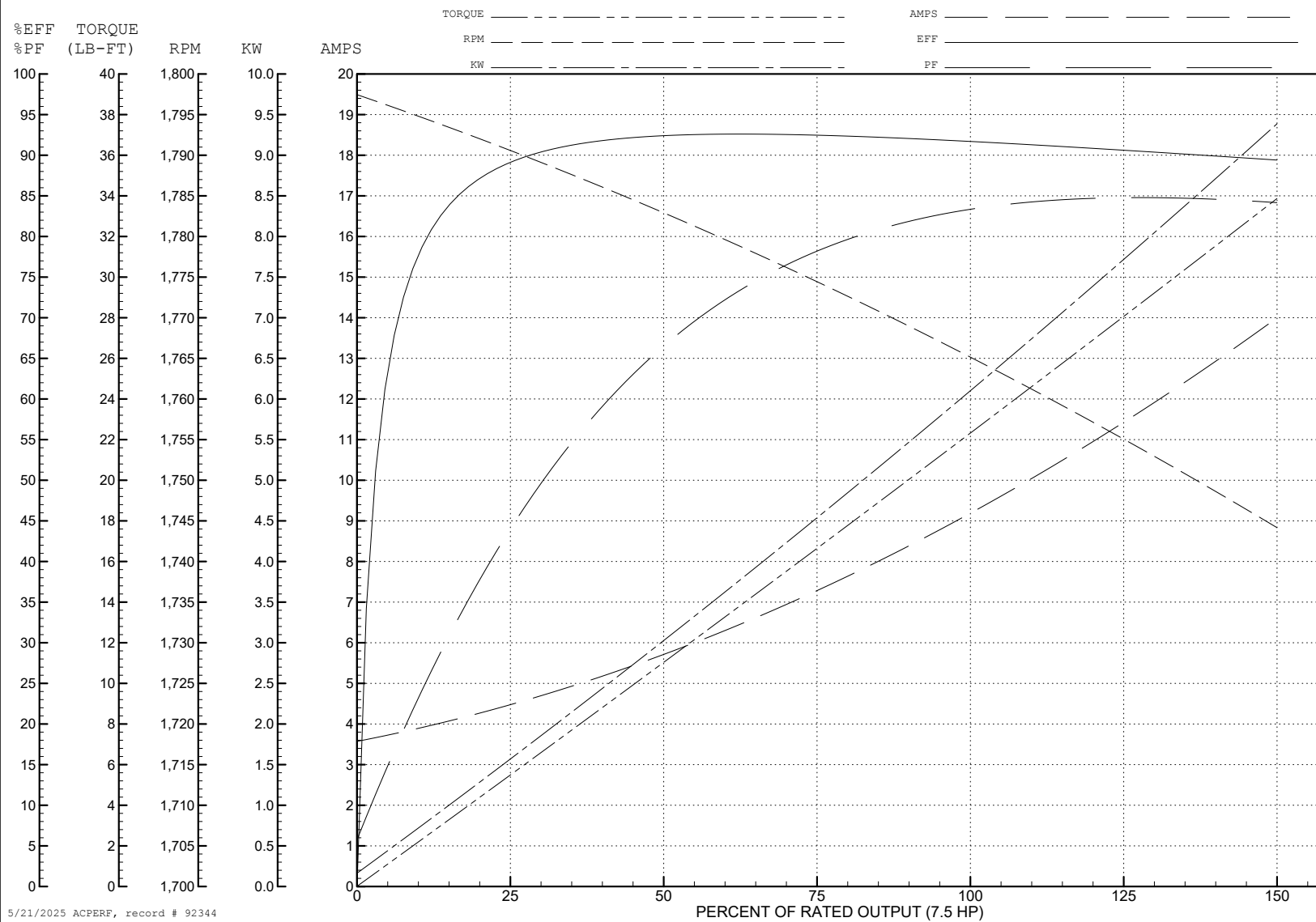
ABB Motors and Mechanical Inc.

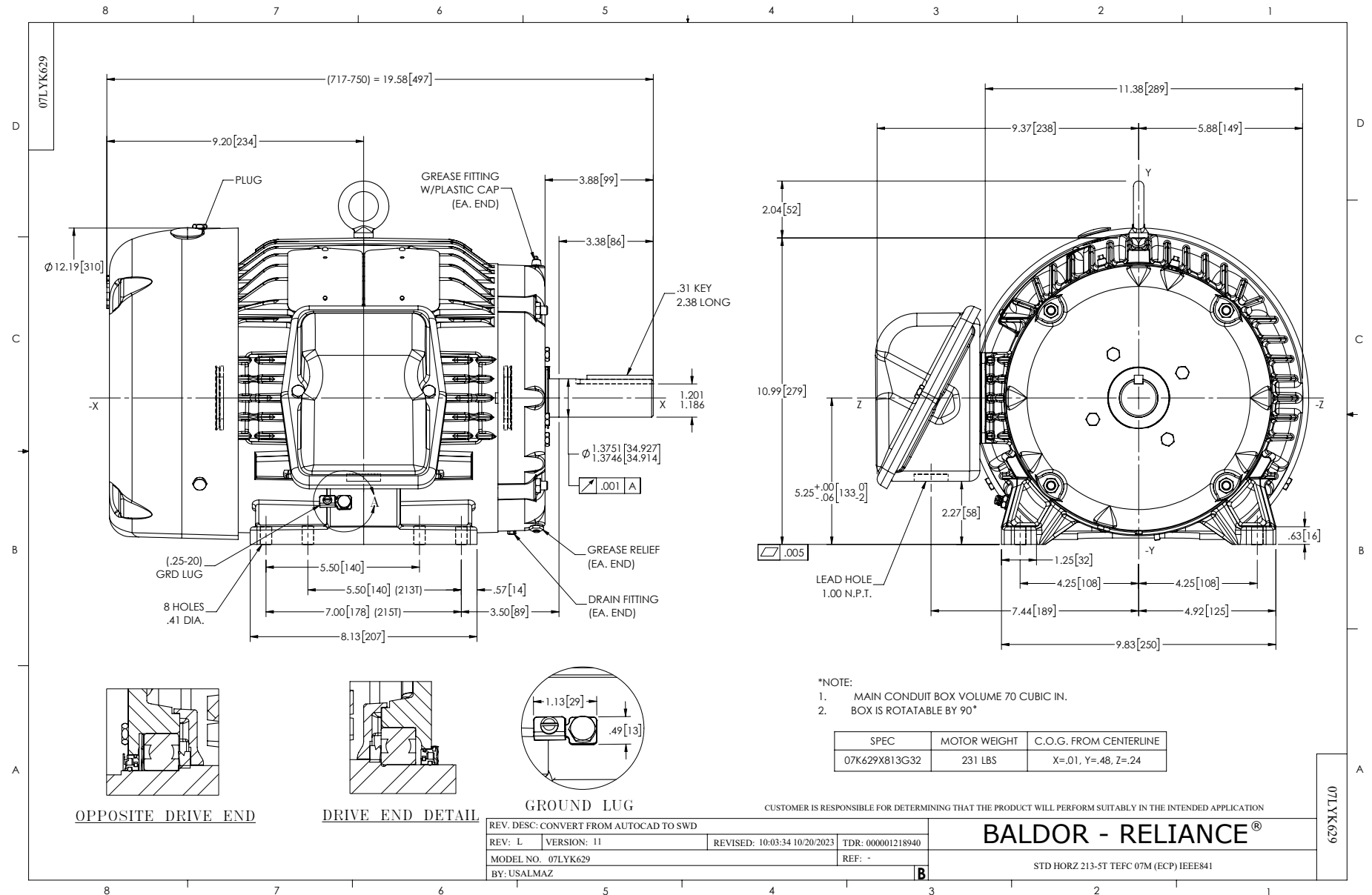
WINDING # 07WGX814

Typical performance - not guaranteed values.

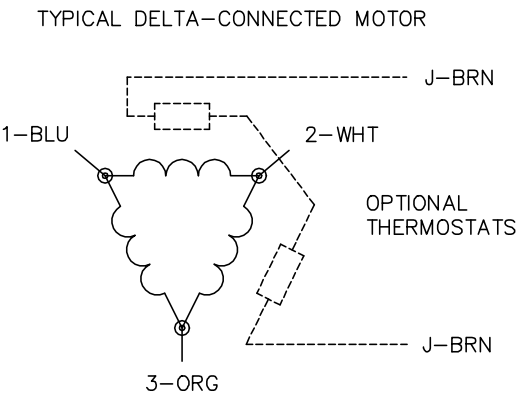
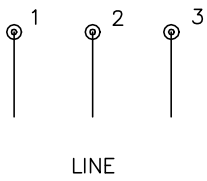
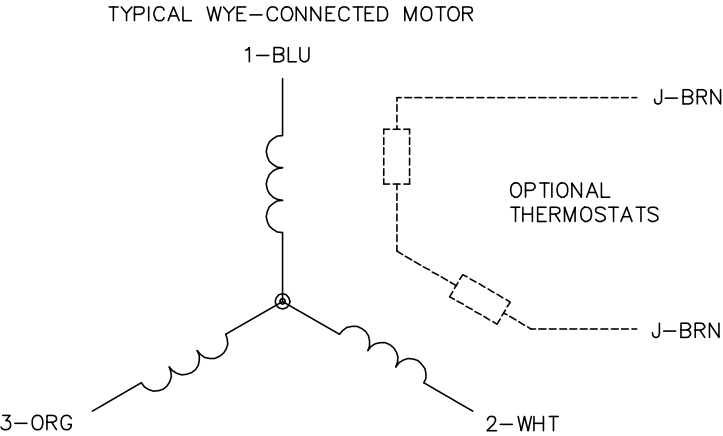
7.5 HP 3 PH 60 HZ 1765 RPM 460 V 0740M

TORQUES (LB-FT): PO=66.8 PU=30 LR=39.7 LRA=63.5





CD0006



- NOTES:
- 1. THREE LEAD MOTOR MAY BE EITHER WYE CONNECTED OR DELTA CONNECTED.
 - 2. INTERCHANGE ANY TWO LINE LEADS TO REVERSE ROTATION.
 - 3. OPTIONAL THERMOSTATS ARE PROVIDED WHEN SPECIFIED.
 - 4. ACTUAL NUMBER OF INTERNAL PARALLEL CIRCUITS MAY VARY.
 - 5. LEAD COLORS ARE OPTIONAL. LEADS MUST BE NUMBERED AS SHOWN.

CD0006

REV. DESC: ADD CLASS CONN00000007		
REV. LTR: E	VERSION: 01	TDR: 000001099922
FILE: \AAA\00005\141	REVISED: 10: 24: 49 02/19/2019	BY: ENBRIRO
MTL: —		

BALDOR - RELIANCE®

3PH, SV, 3 LEADS, WYE OR DELTA CONNECTED

SH 1 of 1

Motor Mod Express®

Modification number	Motor Mod Express®
Balance	
M1A	Dynamic balance of rotors
Bearings	
M2A	Sealed
M2B	Ball to roller 1800 RPM or below
M2C	RPM AC ball to roller
M2F	Add isolated bearings NEMA 56 through 449T (SKF Insocoat or equivalent)
Brake motors	
M3D	Change stearns brake voltage
Conduit box	
M4A	Tap or provide additional lead hole
M4B	Rotate conduit box lead opening
M4F	Replace die cast aluminum conduit box
M4G	Add conduit box for thermostat or heater leads
Cord	
M5A	Install line cord in conduit box or terminal panel
M5B	Install 6 foot, 3 conductor, type SO cord with plug
Drains	
M7A	Add condensation drain holes
M7C	Close standard condensation drain holes
M7D	Add breather drains to cast iron explosion-proof motor
Dripcovers	
M8A	Install dripcover on TEFC or ODP
Leads	
M10A	Install terminal lugs
M10B	Reconnect motor from nine leads to three
M10C	Reconnect motor to high or low voltage in conduit box
M10D	Longer leads, additional leads added
Lubrication	
M11A	Install low temperature lubricant in bearings
M11B	Add high temperature grease
Seals	
M12A	install oil seals
M12B	Install Inpro/Seal® drive end only
M12C	Install V-ring seals to motor

Motor Mod Express®

Modification number	Motor Mod Express®
Mountings	
M13A	Add rigid base
M13B	Install NEMA C-face – steel band motor construction
M13C-1	Install NEMA C-face – cast iron construction
M13C-2	Install NEMA C-face – IEEE 841 motors
M13D	Add D-flange
M13E	Remove rigid base from foot mounted steel band motors
M13F-2	Convert mounting configuration for cast iron motor
M13G	Install C-Face to explosion-proof motors
M13I	Labor only to mount grinder or buffer onto a pedestal and package for shipment
Mounting drawings	
M44A	RPM AC mounting conversions
Export packaging	
M14A	Open crate packaging
M14B	Totally enclosed crate packaging
M14C	Motor and nameplate photograph
Nameplate	
M15A	Provide original nameplate for remote mounting
M15B	Replace nameplate
M15C	Laser engrave nameplate into stainless steel motor
M15E	Mail nameplates to customer
Blowers	
M16A	Add constant velocity blower unit
M16B	Convert from a 3 phase to a single phase blower
Paint	
M17A	Paint motor Baldor-Reliance® stock paint color
M17B	Paint motor or parts customer specified color
Lifting lugs	
M18A	Add lifting lugs up to a 326T frame size
Hardware	
M20A	Exchange plated hardware with stainless steel
Space heaters	
M21A	Add space heaters to TEFC or open motors
Thermal protectors	
M23A	Thermostats
Winding thermocouples	
M24A	Iron constantan (3 per motor)
Winding thermistors	
M25A	Thermistors

Motor Mod Express®

Modification number	Motor Mod Express®
Tropicalization	
M26A	Protection of windings and mechanical parts
Weatherproofing	
M27A	Treatment of windings and mechanical parts
Shaft modifications	
M29A	Drill and tap one hole in end of shaft or mill flat on shaft extension
M29B	Convert from TEFC to TEAO
M29C	Threading
M29D	Convert "T" frame to "U" frame shaft dimension or smaller (TS)
M29F	Shorten shafts without reducing the overall diameter
Tach and encoder mounting kits	
M31D	Add encoder kit to inverter duty V*S Master or RPM AC motor
M31E	Additional mounted encoder options for IDRPM motors
Testing	
M32A	CSA C390 method 1 equivalent to IEEE 112 method B performance testing
M33A	Short motor test
M33B	Routine motor test
M33C	Complete performance test
M33D	Calibration test
M33E	Sound measurement test
Terminal blocks	
M35A	Install IEC terminal blocks in conduit box and connect leads
Starters	
	Add magnetic or manual starter to grinder or buffer
M36A	Starter – single phase
M36B	Starter – three phase
M36C	Magnetic starter – three phase
M36D	Overload protection for dust control unit
Shaft grounding	
M39A	Shaft grounding systems
M39B-1	Shaft grounding ring - bearing protection
M39C	Baldor-Reliance® shaft grounding brush assembly
M39D	Add shaft grounding ring's colloidal silver shaft coating to shaft
Check Total Indicated Runout (TIR)	
M40A	Check Total Indicated Runout (TIR)
ABB Ability™	
M43A	Install ABB Ability™ Smart Sensor for motors

SECTION 04 INSPECTION & TESTING PLAN



No.	Activity Description	Location	References to Applicable Procedures	Acceptance Criteria	Verifying Document(s)	-----Inspection Codes -----			
						Hartzell Air Movement		Client Name	
						Definition Key	Approval Type	Definition Key	Approval Type
1	Order Receipt created by Customer Service from the order entered into the iERP system per customer request	Hartzell Air Movement	WI-CSR.014 Entering Orders into iERP	Customer Review of Sales Order Acknowledgement	Sales Order Acknowledgement	R	On Approval, SOA sent to Customer	R	Notification to Manufacturer within 1 working day if not acceptable
2	Receiving Inspection inspection on receipt to Purchase Order & any non-conformances are to be tagged for correction before entering into inventory	Hartzell Air Movement	WI-INV.027 - Receiving Inspection	No outstanding non-conformances	Purchase Order / Item Drawing or Specification (internal only)	R	If not approved, will be flagged for Non-Conformance and rectified before proceeding		
3	In Process Inspection each workcenter reviews work to Job Packet & tags any non-conformances to be corrected before progressing forward to next workcenter	Hartzell Air Movement	P-8.6A - In-Process Inspection P8.7 - Control of Nonconforming Material	No outstanding non-conformances	Job Packet / Item Drawing or Specification (internal only)	R	If not approved, will be flagged for Non-Conformance and rectified before proceeding		
4	Impeller Balance dynamic balancing of impeller to ensure or bring within set tolerances	Hartzell Air Movement	ES 3.6.4 - Single Plan Dynamic Balancing ES 3.6.5 - Double Plane Dynamic Balancing	Within Tolerance per ES 3.6.7 - Fan Balance Engineering Standard	Balancer Report	R	On Approval, Balancer Report is attached per Fan & sent with paperwork	R	Review on Receipt with Product
5	Assembly Inspection review of job packet to actual product and inspected per form for consistency	Hartzell Air Movement	WI-MFG.063 - Assembly Flow of Fans F-MFG.002 - Assembly / Inspection Worksheet	Fan Complete per Job Packet / drawings / specifications	Assembly / Inspection Worksheet	A	Approval to be recorded on Assembly / Inspection Worksheet per Fan & sent with paperwork	R	Review on Receipt with Product
6	Functional Fan Test startup, run and measure vibration of fan	Hartzell Air Movement	P-8.6 - Release of Products and Services	Within Tolerance per ES 3.6.7 - Fan Balance Engineering Standard	Test Report	R	On Approval, Test Report is attached per Fan & sent with paperwork	R	Review on Receipt with Product
7	Final Quality Control Inspection review of job packet to actual product and inspected per form for consistency	Hartzell Air Movement	P-8.6 - Release of Products and Services F-MFG.002 - Assembly / Inspection Worksheet	No discrepancies found	Assembly / Inspection Worksheet	A	Approval to be recorded on Assembly / Inspection Worksheet per Fan & sent with paperwork	R	Review on Receipt with Product
Inspection codes:									
R = REVIEW - Appropriate records of activity shall be provided for review. Where practical, review of records shall take place progressively throughout the manufacture and test process. W = WITNESS - Work may proceed on schedule should the representative fail to show for the activity. Notification of Witness points shall be in writing within the agreed upon time limit. H = HOLD - Work shall not proceed beyond this point without written confirmation of waiver. Notification of Hold points shall be in writing within the agreed upon time limit. M = MONITOR - Access shall be provided to allow surveillance on site. Monitoring activities are not restricted to specifically identified monitor points. A = APPROVE - Approve documents or procedures									

SECTION 05 GENERAL STORAGE INFORMATION



GENERAL FAN STORAGE PROCEDURES:

If fans are stored for any length of time, they should be stored in a clean, dry location to prevent rust and corrosion. Outdoor storage is not recommended. When outdoor storage is necessary, they should be protected from the elements. Cover the fan inlet and outlet, grease the bearings, and keep motors dry and clean.

EXTENDED STORAGE:

Fans are to be stored in their original containers or equivalent protection and should be kept in a clean, dry, protected warehouse where exercised control over temperature, dust, dew point, shock, and vibration is reasonably maintained.

A) Temperatures: Between 50 degrees F and 120 degrees F.

B) Maximum relative humidity: 60%

C) Shock or vibration: 2 mils maximum to prevent bearings from brinelling.

Exceeding this limit will require vibration-dampening material under the units.

Motor bearings (and fan bearings on belt drive units) are to be greased at the time of going into extended storage. Motor shafts (and fan shaft on belt drive units) are to be manually rotated every month and additional grease added, purging some grease from the bearing cavity every six (6) months. ***Grease in bearings is to be purged at time of removal from storage, making sure that an ample supply of fresh grease is in each grease cavity. Grease used must be compatible with that already in motor and fan bearings.***

All motors with space heaters are to have the heaters connected if storage conditions exceed 60% relative humidity and/or if temperatures are below 50 degrees F.

Motor windings should be meggered at the time the equipment is put in storage. At the time of removal from storage, the resistance reading must not have dropped more than 50% from the initial reading. Contact Hartzell Air Movement Warranty and Service Department if the motor resistance is less than 50% of the initial reading.

NOTE: Motors in storage may absorb moisture in their windings resulting in a significant loss of insulation resistance. THE APPLICATION OF POWER TO A MOTOR WITH INSUFFICIENT INSULATION RESISTANCE MAY RESULT IN DAMAGE TO THE MOTOR OR OTHER EQUIPMENT.

Storage records complying with the above requirements should be maintained.

If an extended warranty is desired, contact your local sales representative for charges and details.

**Hartzell Air Movement does not recommend outdoor storage.
When outdoor storage is unavoidable, the fans should still be protected from the elements.**

Hartzell recommends the steps be taken below for outdoor storage

1. All motors must be meggered prior to going into storage and the readings recorded in each fans data packet.
2. Verify that the knockout on the motor conduit box is facing down - and a threaded plastic plug added in the knockout – small holes drilled in plastic plug for space heater wires.
3. Fans must be stored at least 8" off the ground, such as on rail road ties

4. VERY IMPORTANT: Motor space heaters must be powered for warranty purposes.
5. The fans must be stored in a temporary lean-to type structure that will keep the elements directly off the fan but allow some air circulation. Tarping or covering the fan is not recommended as it can trap moisture that can find its way into the fan & motor bearings.
6. Fans (blade & motor) needs to be rotated by hand on a monthly basis / At least 7 to 8 rotations per fan
7. Fan Bearing Cavity Needs to Be Full of Grease (not motor bearings) Fans should have grease added to the fan bearing cavity prior to going into storage – when adding grease, the shaft needs to be rotating and the excess grease will purge from the outer race of the bearings
8. every six (6) months / 2 to 3 shots of fresh grease in each fan bearing while the fan blade is being rotated

For Warranty purposes - Storage Records complying with the above requirements should be maintained and must be provided to Hartzell prior to a warranty claim.

Once the fans are removed from storage and ready for installation, the motors should be meggered and the fan bearings completely re-greased as indicated in the storage procedure.

Note: If fans are stored longer than 12 months - After motors are meggered please contact Hartzell, for start-up instructions on the motors.

SECTION 06 INSTALLATION, OPERATION & MAINTENANCE MANUAL



THE HARTZELL DIFFERENCE

Building the highest quality fans in America for generations!

QUALITY AT EVERY TURN.

Hartzell Air Movement exceeds the standard in the air movement industry, committed to delivering top-quality, reliable products. And our relationship with our customers, and our knowledgeable, inventive, flexible and hardworking employees are the reasons we've continued and thrived, right here in the USA, for six generations.

Customers choose Hartzell for our:

- Durability and high performance
- Low cost of ownership
- Leadership in performance testing and certification
- Advanced engineering and manufacturing processes
- Innovative design and manufacturing
- Trusted brand name

Experienced Hartzell team to assist you from design to shipping

ISO 9001:2015 Certification
AMCA Accredited Laboratory
Complete Fans and Blowers
Centrifugal Wheels
Airfoil Propellers



For more information, contact your local Hartzell Sales Representative. info@hartzell.com

HARTZELL ADVANTAGES

A LIFETIME OF VALUE

- ☐ Industrial fans are field proven, 100% tested, with virtually no incident of return
- ☐ Energy efficient designs that provide a lower total cost of ownership

LEADERSHIP IN PERFORMANCE

- ☐ Products certified by AMCA that meet DOE recommended efficiencies
- ☐ Highly efficient industrial fan designs that are the quietest in the market
- ☐ Airfoil shape blades with industry leading measured efficiencies

ADVANCED ENGINEERING & MANUFACTURING

- ☐ Proven manufacturing techniques on state of the art equipment in ISO 9001:2015 registered facilities
- ☐ Engineering team utilizes the latest design tools in 3D CAD and other modeling software

CREATIVE DESIGN & MANUFACTURING TEAM

- ☐ Offering the greatest number of industrial fan design choices
- ☐ Building products to meet your rigorous application; no limiting catalog

TRUSTED BRAND NAME

- ☐ Extensive talent pool bringing over 145 years of knowledge and experience to you

5-YEAR WARRANTY

- ☐ The industry's first and only manufacturer that stands behind their products for a full 5-years

HARTZELL WARRANTY

LIMITED WARRANTIES, LIMITATION OF LIABILITY, AND LIMITATION OF LIABILITY FOR BREACH OF WARRANTY

NO IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS

Hartzell does not warrant that said goods are of merchantable quality or that they are fit for any particular purpose. There is no implied warranty of merchantability and there is no implied warranty of fitness.

The details of the Hartzell warranty can be found at
<http://www.hartzellairmovement.com/warranty-service>



Performance Guarantee

You have high expectations. So does Hartzell Air Movement. We know you demand the most reliable and durable industrial air movement products available, so we're holding ourselves to a higher standard. We're so sure that our products will out-perform industry standards, we're backing that promise with the industry's first five-year warranty. Call your Hartzell Air Movement representative for assistance.

Register for your 5year warranty at
<https://www.hartzellairmovement.com/warranty-and-parts/five-year-warranty>



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www.hartzellairmovement.com

Safety Accessories, Application, and Use Warning

The safe installation and operation of equipment supplied by Hartzell Air Movement is the responsibility of the system designer, installer, maintainer, and user. Since the application and use of its equipment can vary greatly, Hartzell Air Movement offers various product types, optional safety accessories, and sound performance data per laboratory tests.

Hartzell Air Movement sells its equipment with and without safety accessories, and accordingly, it can supply such safety accessories only upon receipt of an order. The need for safety accessories will frequently depend upon the type of system, fan location and operating procedures being employed.

The proper protective safety accessories to meet company standards, local codes, and the requirements of the Occupational Safety and Health Act must be determined by the user since safety requirements may vary depending on the location and application of the equipment. If applicable local conditions, standard, codes or OSHA rules require the addition of the safety accessories, the user should specify and obtain the required safety accessories from Hartzell Air Movement and should not allow the operation of the equipment without them.

Owners, employers, users, and installers should read **"RECOMMENDED SAFETY PRACTICES FOR USERS AND INSTALLERS OF INDUSTRIAL AND COMMERCIAL FANS"** published by the Air Movement Control Association, International, 30 West University Drive, Arlington Heights, Illinois 60004. A copy of this publication is enclosed with each fan shipped from Hartzell Air Movement, and is available upon request at Hartzell's office in Piqua (937-773-8494).

Please contact Hartzell Air Movement or your local Hartzell representative for more information on product types, safety accessories, and sound performance estimates.

Remember, the selection of safety accessories and the safe installation, application and operation of equipment supplied by Hartzell Air Movement is your responsibility. This warning supersedes all previous editions.



INSTALLATION, OPERATION & MAINTENANCE MANUAL

Introduction

The purpose of this manual is to aid in the proper installation and operation of fans manufactured by Hartzell Air Movement. These instructions are intended to supplement good general practices and are not intended to cover detailed instruction procedures, because of the wide variety and types of fans manufactured by Hartzell Air Movement.

The safe installation and operation of fans is the responsibility of the system designer, installer, maintainer, and user. From the initial system design through the life of the equipment, safety should be a foremost consideration. Some areas, which require some special attention, include system design, layout and construction, fan performance specifications, foundation and installation details,

storage procedures, start-up and commissioning procedures, operation, maintenance, and repair.

Handling and installation should always be performed by experienced and trained personnel who are aware of the hazards associated with rotating equipment. Failure to comply with these practices may result in death or serious bodily injury.

Contact your local Hartzell representative for further assistance.

Shipment and Receiving

All equipment shipped from Hartzell Air Movement is prepared for shipment in accordance with the requirements of the commercial carrier and/or any special considerations required by the nature of the product.

The Bill of Lading or Express Receipt is an acknowledgment by the Transportation Company of the receipt in GOOD CONDITION, meeting the above requirements for the shipment covered by our invoice.

Our responsibility for the shipment has now ceased. We will not be responsible for loss or damage when you give the Transportation Company a clear receipt. (Standard conditions of sale are F.O.B. factory, unless other terms have been quoted and purchased).

Thoroughly inspect all shipments as soon as received.

Keep a record of all equipment received, including inspection details and date of receipt, because of the possibility of partial shipments.

If any of the items called for in the Bill of Lading or Express Receipt are short or damaged, do not accept them until the Freight or Express Agent makes a Damage or Short Shipment Notification on your freight bill or express receipt.

If any concealed loss or damage is discovered, **NOTIFY YOUR FREIGHT OR EXPRESS AGENT AT ONCE** and request an inspection. This is absolutely necessary. Unless you do this, the transportation companies will not entertain any claim for loss or damage. If the agent will not make an inspection, then you should make an Affidavit to the effect that you notified the agent on that particular date and the agent failed to show up. This, with your aforementioned documentation, will properly support your claim.

We will assist you in every possible manner in collecting claims for loss or damage, however, this does not make us responsible for collection of claims or replacement of material.

Handling

Handle your equipment with care. Some fans are provided with lifting lugs or holes for easy handling. Others must be handled using nylon straps or well-padded chains and cables, which protect the fan's coating and housing. Spreader bars should be used when lifting large parts.

Axial Fans

Axial fans should be lifted by using straps around the fan housing only.

DO NOT LIFT AXIAL FANS BY THE MOTOR, MOTOR BASE, PROP OR FLANGES.

Centrifugal Fans

Centrifugal fans are best lifted using one strap under the fan's scroll and another strap around the bearing base. **DO NOT LIFT CENTRIFUGAL FANS BY THE FAN SHAFT, WHEEL, FLANGES OR INLET SUPPORT.**

Roof Ventilators

Roof ventilators should be lifted by using straps around the fan housing or base only. Spreader bars should also be used to avoid damage to stack caps or hoods. **DO NOT LIFT ROOF VENTILATORS BY THE STACK CAP OR HOOD.**

Storage

If fans are stored for any length of time, they should be stored in a clean, dry location to prevent rust and corrosion. Outdoor storage is not recommended. When outdoor storage is necessary, they should be protected from the elements. Cover the fan inlet and outlet, grease the bearings, and keep motors dry and clean.

Extended Storage

Fans are to be stored in their original containers or equivalent protection and should be kept in a clean, dry, protected warehouse where exercised control over temperature, dust, dew point, shock, and vibration is reasonably maintained.

CONDITIONS

- ☐ Temperatures: Between 50° F. and 120° F. | Between 10° C. and 49° C.
- ☐ Maximum relative humidity: 60%.
- ☐ Shock or vibration: 2 mils maximum to prevent bearings from brinelling.
Exceeding this limit will require vibration-dampening material under the units.

Motor bearings (and fan bearings on belt drive units) are to be greased at the time of going into extended storage. Motor shafts (and fan shaft on belt drive units) are to be manually rotated every month and additional grease added, purging some grease from the bearing cavity every six (6) months. *Grease in bearings is to be purged at time of removal from storage, making sure that an ample supply of fresh grease is in each grease cavity. Grease used must be compatible with that already in motor and fan bearings (See Page 24).*

All motors with space heaters are to have the heaters connected if storage conditions exceed 60% relative humidity and/or if temperatures are below 50 degrees F.

Motor windings should be meggered at the time the equipment is put in storage. At the time of removal from storage, the resistance reading must not have dropped more than 50% from the initial reading. Contact Hartzell Air Movement Warranty and Service Department if the motor resistance is less than 50% of the initial reading.

NOTE: Motors in storage may absorb moisture in their windings resulting in a significant loss of insulation resistance. THE APPLICATION OF POWER TO A MOTOR WITH INSUFFICIENT INSULATION RESISTANCE MAY RESULT IN DAMAGE TO THE MOTOR OR OTHER EQUIPMENT.

Storage records complying with the above requirements should be maintained.

If an extended warranty is desired, contact your local sales representative for charges and details.

Installation

Centrifugal Fans should always be mounted to a flat, level, and rigid structure. The fan base should be shimmed and leveled. Gaps between the foundation and fan base should be grouted. This will ensure permanent alignment and a smooth-running, vibration-free fan, as well as minimize maintenance costs. Failure to properly install the fan base can contribute to excessive vibration.

Poured concrete foundations are recommended, wherever practical, for floor mounted fans. If vibration isolators are required, they should be installed between the fan and the foundation. Tighten all mounting bolts securely with lock washers and lock nuts.

Fans mounted off ground level should be rigidly mounted to a structural platform and be placed as near as possible to or over, a solid wall or column. Supports for suspended fans must be cross-braced for live load support to prevent side sway.

For roof mounted fans, place the fan curb panel on the roof curb, level, and then anchor the unit to the curb using lag screws, neoprene washers, and flat washers. It is recommended that stacks be independently mounted to the roof and use guy wires to prevent side sway.

AVOID SUPPORTING A STACK DIRECTLY ON THE FAN FLANGE.

In both axial and centrifugal fans, inlet and outlet ducts should be independently supported, and never supported by the fan flanges. Flexible duct connections are recommended. These connections will also minimize noise. The independent mounting of stacks and ducts to the fan flanges will assure that the fan will not be subjected to external forces which may twist or deform the fan housing. This also will ensure that the impeller will not strike the housing or cause misalignment of the sheaves and bearings.

It is recommended that access doors be placed in ductwork just ahead of the fan inlet and just downstream from the fan outlet for ease of inspection and maintenance. Access doors in a duct system should never be opened with the fan running, to avoid possible equipment damage and personal injury.

No turns in ductwork should be closer than 2.5 fan diameters away from the inlet or outlet of a fan. Walls or flat surfaces should also not be closer than one fan diameter from the inlet side.

Branch entries into the main duct should be spaced such that they do not enter directly opposite each other. A maximum of 45 degree angle between main branch and entering branch should be used as a guideline. Avoid sudden changes in duct size. Use a 15 degree included angle, or less, in reducing a duct to a fan, and not more than 30 degrees included angle on the discharge. Elbow turns should be kept to a centerline radius of at least one and one-half diameters of the duct.

Electrical connections for the fan motor must be connected by a qualified electrician, conforming to the National Electrical code and local codes and practices. When making electrical connections, the motor must be connected following electrical characteristics as indicated by the motor nameplate, and adhere to the wiring diagram on the motor nameplate or in the motor terminal box. ***Motors may fail immediately if improperly connected.*** It is also recommended that an overload device to protect the motor be installed between the current supply and the motor. Recommended tolerances for overload devices should be plus 10% of the motor full load amperage rating, including the allowance for the motor service factor.

NOTE: It is common for motors to draw several times full load amperage during startup, for approximately fifteen seconds. Larger sized fans may require twenty seconds or longer.

Startup

- ☐ Lock out the primary and all secondary power sources.
- ☐ A complete inspection should be made of all ductwork and the fan interior. Make certain there is no foreign material, which can be drawn into or blown through the fan or ductwork. Appropriate protective measures and safety practices should be observed when entering or working within these areas. These measures may include the use of goggles, respirators, or other personal protective devices.
- ☐ Make sure the foundation or mounting arrangement and the duct connections are adequately designed and installed per drawings and in accordance with recognized acceptable engineering practices.
- ☐ Check and tighten all bolts, fasteners, and set screws as necessary. **NOTE: Forces encountered during shipment, handling and rigging can disturb factory settings.**
- ☐ Check the fan assembly and bearings for proper grounding to prevent static electrical discharge. **NOTE: This is especially important when using FRP fans.**
- ☐ Ensure power and drive components such as motor starter, variable frequency drive, or hydraulic power unit are properly sized, matched, and connected to the fan.
- ☐ Check bearings for recommended lubricant and lubrication amount.
- ☐ Clearance should be checked between the impeller and fan housing. Spin the impeller to determine whether it rotates freely, without hitting anything, and is not grossly out of balance. **NOTE: Prior to shipment, all fans have been thoroughly inspected and have passed stringent operation and balance tests.**
- ☐ Inspect the impeller for proper rotation for the fan design. Arrows to show direction of rotation and airflow are attached to the fan housings.
- ☐ Check the belt drive for proper sheave selection and installation. Make sure the sheaves are not reversed (excessive speeds could develop).
- ☐ Check alignment of drives and all other components.
- ☐ Properly secure all safety guards.
- ☐ Assure that all appropriate warnings have been put in place.
- ☐ Secure all access doors to the fan and ductwork.

Confirm all inlet and outlet dampers connected to the fan system are open before powering up the fan.

- ☐ Restore power and momentarily energize the fan to check the direction of rotation. Listen as the fan coasts to a stop for any unusual noise, identify the source, and take corrective action as necessary.

- ☐ Switch on the electrical supply and allow the fan to reach full speed. **Check carefully for:**
(1) Excessive vibration; **(2)** Unusual noise; **(3)** Proper amperage, voltage, or power values.

If any problem is indicated, SWITCH OFF IMMEDIATELY. **NOTE: It is normal for belt drive fans to squeal briefly during startup.** If no problems are found, while the fan is running, add grease to the fan bearings until the grease purges from the outer seal.

- ☐ Lock out power supply. Secure the fan impeller if there is a potential for windmilling. Check carefully for cause of the trouble, correct as necessary, and repeat check list procedure. (Ref. Page 27 Trouble Shooting chart)

NOTE: *The fan should not need balancing, as it was balanced at the factory to be within stringent vibration levels before shipment. However, there are several things that may cause vibration, such as rough handling in shipment and erection, weak foundations, and alignments. It is recommended that the vibration levels be checked with a vibration analyzer to verify that the vibration is within levels recommended later in this manual. (See Page 23, Table 1).*

- ☐ Even if the fan appears to be operating satisfactorily, shut it down after a brief period, lock out the power supply, and re-check startup procedures, as the initial start-up may have loosened the bolts, fasteners, and set screws.
- ☐ The fan may now be put into operation, however during the first eight hour of operation, it should be closely observed and checked for excessive vibration and noise. At this time checks should be made of motor input current and motor bearing temperatures to ensure that they do not exceed manufacturer's recommendations.
- ☐ After eight hours of operation, the fan should be shut down and the power locked out. Recheck startup procedures and adjust, as necessary.
- ☐ After twenty-four hours of satisfactory operation, the fan should be shut down (locked out) and the drive belt tension should be readjusted to recommended tension (See Page 20).

Maintenance

Periodic inspection of all the fan parts is the key to long life and trouble-free fan operation. Frequency of inspection must be determined by the user and is dependent upon the severity of the application. Prepare a maintenance schedule and make sure it is strictly adhered to.

**NEVER SERVICE OR ADJUST ROTATING EQUIPMENT WHILE IT IS IN OPERATION.
LOCK OUT THE POWER SOURCE BEFORE PERFORMING MAINTENANCE.**

REGULAR FAN MAINTENANCE:

- ☐ Check the fan impeller for any buildup of foreign material or wear from abrasion. Both can cause excessive vibration which will lead to damage of the impeller and other fan components. Replace the impeller if excessive wear is noticed. Carefully clean the impeller of any foreign material.
- ☐ Check V-belt drives for proper alignment, tension, and excessive wear.
- ☐ Lubricate the fan bearings per the Hartzell Lubrication schedule.
- ☐ Lubricate the motor bearings per the motor manufacturers recommendations for grease type and intervals.
- ☐ Consult the trouble shooting guide for excessive vibration or noise, insufficient performance, or when the fan does not operate (See Page 27 Trouble Shooting Chart).
- ☐ Tighten all bolts and setscrews.

For arrangement 8 fans only: Lubricate the coupling per the coupling manufacturers recommendations for grease type and intervals. Also, check for proper alignment.

Bearings and Lubrication

All Hartzell belt drive fan bearings are heavy duty, self-aligning ball or roller type, selected for on the fan size, motor horsepower, and performance, and are re-lubricable for continuous service.

Selection of the correct bearing grease and greasing intervals depends on several factors. Extreme high or low temperatures, dirty or damp surroundings, and excessive vibration exceeding the “fair” levels in Table 1 are all things that will require more frequent greasing or special greases. (Refer to Page 24 for recommended greases and greasing intervals.)

The motor bearings and the fan bearings on the belt drive fans should be greased at regular intervals. Motor manufacturers' greasing instructions and recommendations should be followed closely. Avoid the use of a pressure greasing system which tends to fill the bearing chamber completely. Do not over-grease. Limit the use of hand guns to those with a rating of 40 psi or less. Rotate bearings during lubrication where good safety practice permits.

NOTE: *On motors with non-regreasable sealed bearings, no lubrication is required for life of the bearings.*

The most frequent causes of bearing failure are not greasing often enough, using incompatible greases or excessive belt tension. Excessive vibration, especially if the bearing is not rotating, will also cause bearings to fail. Bearings must also be protected from water and moisture to avoid internal corrosion.

NOTE: *It is typical for some anti-friction bearings to exhibit a running surface temperature in excess of 140 deg. F. This surface will be too hot to touch, but is not a cause for alarm.*

Bearing Replacement

Fan bearings on belt drive fans should not need to be replaced for many years if the above recommendations are strictly adhered to. However, use the following procedure when bearing replacement is necessary.

PROCEDURE:

- ☐ Lock out power source.
- ☐ Gain access to the fan bearings. On ducted axial flow fans, the fan probably will have to be removed from the duct system. Remove the bearing cover, if any.
- ☐ Loosen the belts by shifting the motor.
- ☐ Remove the impeller and disconnect the lube tubes.
- ☐ Remove the shaft and bearing assembly. Note the position of the bearing shims, if applicable.
- ☐ Measure the location of the bearing to the impeller end of the shaft and the bearing spacing.
- ☐ Loosen all bearing/shaft setscrews or other locking device.
- ☐ Remove bearings (may have to be pressed off the shaft).
- ☐ Polish the shaft with fine emery paper (240 Grit or finer) and file the setscrew dimples smooth.
- ☐ Install new bearings on the shaft, making sure that the collars are together, (i.e., facing each other on the shaft) and set screws are in line with each other. Lightly seat one setscrew on each bearing to hold in the approximate position from above.
- ☐ Mount the shaft/bearing assembly in the fan, on shims, with bolts. Do not tighten yet - snug up only. Loosen the setscrews.
- ☐ Center the shaft in the housing (both ends) as closely as possible and shim where necessary. The fan impeller may need to be temporarily installed to get the clearances equal.
- ☐ Tighten the bearing mounting bolts.
- ☐ With a soft-faced mallet, tap the shaft between the bearings while turning the shaft by hand to align the bearing races. The shaft must turn freely. Tighten all bearing setscrews. Spin the shaft again, making sure it turns freely.
- ☐ Reinstall the lube tubes.
- ☐ While rotating the shaft, purge bearings with fresh grease (Ref. Page 24).
- ☐ Install bearing cover, impeller, and belts, and adjust the motor to get proper belt tension. Also, make sure that the sheaves are properly aligned. (See V-belt drives , Pages 17-19).

Carefully following this procedure will ensure trouble-free service.

V-Belt Drives

V-belts on Hartzell belt drive fans are oil, heat, and static resistant type, and oversized for continuous duty. With proper installation and maintenance, years of operating efficiency can be added to the life span of the V-belt drive.

V-belt drives must be completely guarded before applying power to the fan.

A noisy V-belt indicates the need for attention. V-belt noise can be caused by the slapping of the belts against the drive guard or other obstruction. Check for an improperly installed guard, loose belts, buildup of foreign material in the sheave grooves, or excessive vibration. The cause of excessive vibration should be determined and corrected. **NOTE: It is normal for belts to squeal briefly at startup.**

Check belt tension often. Ideal tension is the tension at which the belt will not slip under peak load conditions. Over-tensioning shortens belt and bearing life. Keep belts free from foreign material which may cause slipping. The use of belt dressing is not recommended (See Page 20 for recommended tensioning).

Inspect sheaves often. Keep all sheaves grooves smooth and uniform. Burrs and rough spots along the sheave rim can damage belts. Dust, oil, and other foreign matter can lead to pitting and rust, and should be avoided as much as possible. Badly worn grooves or a shiny groove bottom indicates that the sheave, belt, or both are badly worn. Replace any belts and worn sheaves.

Check the sheave alignment. Sheaves that are not aligned properly cause excessive belt wear and sheave wear.

V-Belt Drive Replacement

Whether you are just installing new belts or a completely new drive, worn bearings, bent shafts, or other components that might cause future problems should be replaced at this time. If installing belts only, check existing sheaves carefully for worn grooves or other damage.

Although alignment is not as critical in V-belt drives as in drive types, proper alignment is essential to long belt and sheave life. **NOTE:** *Loose belts or misalignment can cause fan vibration.*

First, make sure that drive shafts are parallel. ***The most common causes of drive misalignment are nonparallel shafts and improperly located sheaves.*** Where shafts are not parallel, belts on one side are drawn tighter and pull more than their share of the load. As a result, these belts wear out faster, requiring the entire set to be replaced before it has given maximum service. If misalignment is in the sheave, belts will enter and leave the grooves at an angle, causing excessive belt and sheave wear. (See Page 19, Figure 1).

Shaft alignment can be checked by measuring the distance between the shafts at three or more locations. If the distances are equal, then the shafts will be parallel.

To check the location of the sheaves on the shaft, a straightedge or a piece of string can be used. If the sheaves are properly lined up, the string will touch them at the points indicated by the arrows (See Page 19, Figure 2). Rotating each sheave a half revolution will determine whether the sheave is wobbly or the shaft is bent. Correct any misalignment.

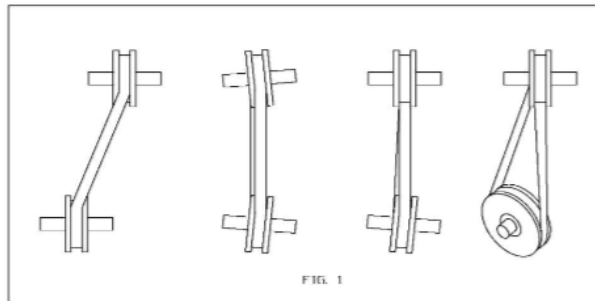
Always use matched belts and never mix new and used belts on a drive. Always replace belts with the right type of V-belt. Install belts correctly. When belts are forced into the sheave with a screwdriver or other wedge, the outer fabric is often ruptured and cords broken. It is well worth the time to move the driver unit closer so the V-belts can be slipped easily into the sheave groove without damage. Shorten the center distance between the driven and driver sheave so the belts can be put on without the use of force. While the belts are still loose on the drive, rotate the drive until all the slack is on one side. Then increase the center distance until the belts are snug. The drive is now ready for tensioning.

Tension the belts as indicated on Page 20. **NOTE:** *Never "roll" or "pry" the belts into the sheave grooves.* This can damage the belt cords and lead to belt turnover, short life, or actual breakage. Moreover, it is both difficult and unsafe to install belts this way. Keep take-up rails, motor base, or other means of center distance adjustment free of dirt, rust, and grit. Lubricate adjusting screws and slide rails from time to time.

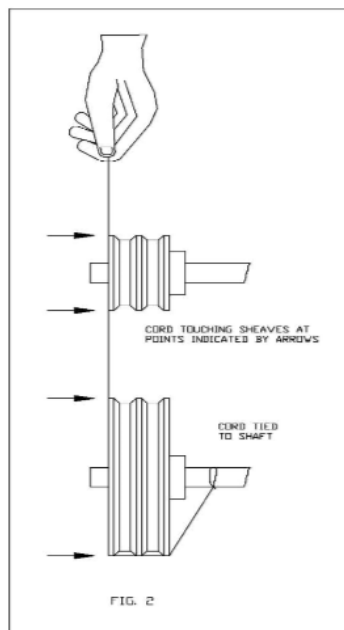
Tighten all sheave bolts and setscrews before reconnecting the power. **NOTE:** *All sheave bolts, setscrews, and belts should be checked and tightened if necessary after two days of initial operation.*

IMPORTANT:

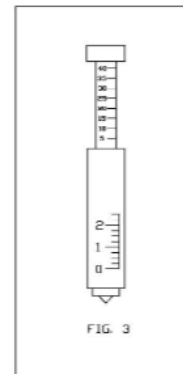
DO NOT INCREASE THE FAN SPEED BEFORE FIRST CONTACTING THE HARTZELL FACTORY. ARBITRARILY SPEEDING UP THE FAN CAN CAUSE MOTOR OVERLOAD, FAILURE, AND POTENTIAL IMPELLER FAILURE.



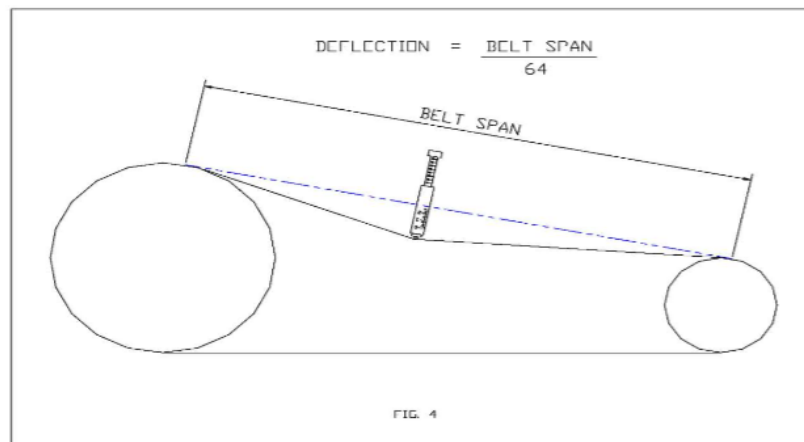
Sheave and Belt Alignment



Sheave and Belt Alignment



Belt tensioning tool



How to check for proper belt tension

Installing, Tensioning, and Checking V-Drives

GENERAL DRIVE TENSIONING GUIDELINES:

- ☐ Ideal tension at which the belt will not slip under peak load conditions.
- ☐ Over tensioning shortens belt and bearing life.
- ☐ Keep belts free from foreign material which may cause slipping.
- ☐ Make periodic V-Drive inspection, tension when slipping.
The use of belt dressing is not recommended.
- ☐ Before installing a new set of V-Belts, check the condition of the sheaves. Dirty or rusty sheaves impair the drives efficiency and abrade the belts, resulting in premature failure. Also, worn sheaves can shorten the belt life as much as 50%.
- ☐ Do not use a new or used belt as replacement for a unit of a set. If a belt breaks, a new set of matched belts is necessary. Always replace belts with the same kind as originally provided
- ☐ After properly tensioning the belts, double-check to be sure the sheave grooves are correctly aligned, and that all shafting is parallel.

INSTALLATION AND CHECKING METHODS

☐ VISUAL METHOD

1. When installing belts, reduce the center distance so that the belts may be placed in the sheave grooves without forcing. Arrange the belts so that the top and bottom spans have about the same amount of sag. Apply tension to the belts by increasing the center distance until belts are snug and have a live springing action when struck with the hand.
2. Operate the drive a few minutes to seat the belts in the sheave grooves. Observe the operation of the drive under it's highest load conditions (usually starting). A slight bowing of the slack side of the drive indicated adequate tension. If the slack inside remains taut during the peak load, the drive is too tight
3. New drive tension should be checked several times during the first 24 hours of operation, by observing the slack side span.

INSTALLATION AND CHECKING METHODS CONT.

□ TENSION GAGE METHOD

When a tension gage is available and the center of the belt span is accessible, the following method may be used. To determine the pounds force required to properly tension a drive with a belt tensioner, proceed as follows:

1. Measure the belt span as shown and calculate the deflection inches using the given equation. Set the large O-Ring for calculated inches of deflection.
2. Set small O-ring at 0 and press down the belt tensioner at center of belt span as shown.
 - On a single belt drive, compress the belt tensioner until the large O-ring is even with bottom of a straight edge placed across the outside edge of the two sheaves.
 - On multiple belt drive, compress the belt tensioner until large O-ring is even with the top of the next belt. Average readings from all belts is the value to use in the tables below.
3. Remove tension gage and observe the new position of the small O-ring. Set at the number of deflection pounds for the set number of inches.
4. Compare this reading, or the average of several readings in the case of multiple belts, to the new/used values in the tables below for the proper belt cross section. If readings do not fall in this range, readjust the belt tension as described and repeat measurement.

Example:

1. Belt span = 64 in. and small sheave is 8 in. pitch diameter with Type B notched belts at 3000 RPM.
2. $64 \text{ in.} / 64 = 1''$ required deflection.
3. Set large O-ring at 1" on gage inch scale.
4. Set small O-ring at zero on plunger.
5. Press down on belts with gage until large O-ring is even with the next belt, or a straight edge, whichever the case may be. With multiple belts, several readings are needed to get an average.
6. Use the pound force reading or average of several readings required for 1" deflecting in the tables below.
7. The "B" belt table for 8" pitch diameter small sheave should have a deflection force between 7.3 lbs and 10.9lbs.
8. Increase or decrease the tension on belts until the deflection force is between 7.3 lbs and 10.9 lbs

Installing, Tensioning, and Checking V-Drives (REF. Pg 19)

Figure 3

CROSS SECTION	SMALLEST SHEAVE DIA. (in.)	RPM RANGE	BELT DEFLECTION FORCE			
			STD. BELTS		NOTCHED BELTS	
			USED	NEW	USED	NEW
3V , 3VX	2.0 - 2.4	1000-2500 2501-4000	-	-	3.3 2.9	4.9 4.3
	2.65 - 3.65	1000-2500 2501-4000	3.6 3.0	5.1 4.4	4.2 3.8	6.2 5.6
	4.12 - 6.90	1000-2500 2501-4000	4.9 4.4	7.3 6.6	5.3 4.9	7.9 7.3
5V , 5VX	4.4 - 6.7	500-1749 1750-3000 3001-4000	- - -	- - -	10.2 8.8 5.6	15.2 13.2 8.5
	7.1 - 10.9	500-1740 1741-3000	12.7 11.2	18.9 16.7	14.8 13.7	22.1 20.1
	11.8 - 16.0	500-1740 1741-3000	15.5 14.6	23.4 21.8	17.1 16.8	25.5 25.0
8V	12.5 - 17.0	200-850 851-1500	33.0 26.8	49.3 39.9	- -	- -
	18.0 - 22.4	200-850 851-1500	39.6 35.3	59.2 52.7	- -	- -

Figure 4

CROSS SECTION	SMALLEST SHEAVE DIA. (in.)	RPM RANGE	BELT DEFLECTION FORCE			
			STD. BELTS		NOTCHED BELTS	
			USED	NEW	USED	NEW
A. AX	3 - 3.6	1000-2500 2501-4000	3.7 2.8	5.5 4.2	4.1 3.4	6.1 5.0
	3.8 - 4.8	1000-2500 2501-4000	4.5 3.8	6.8 5.7	5.0 4.3	7.4 6.4
	5.0 - 7.0	1000-2500 2501-4000	5.4 7.4	8.0 7.0	5.7 5.1	9.4 7.6
B. BX	3.4 - 4.2	860-2500 2501-4000	- -	- -	4.9 4.2	7.2 6.2
	4.4 - 5.6	860-2500 2501-4000	5.3 4.5	7.9 6.7	7.1 7.1	10.5 9.1
	5.8 - 8.6	860-2500 2501-4000	6.3 6.0	9.4 8.9	8.5 7.3	12.6 10.9
C. CX	7.0 - 9.0	500-1740 1741-3000	11.5 9.4	17.0 13.8	14.7 11.9	21.8 17.5
	9.5 - 16.0	500-1740 1741-3000	14.1 12.5	21.0 18.5	15.9 14.6	23.5 21.6
D	12.0 - 16.0	200-850 851-1500	24.9 21.2	37.0 31.3	- -	- -
	18.0 - 20.0	200-850 851-1500	30.4 25.6	45.2 38.0	- -	- -

Motors

The fundamental principle of electrical maintenance is **KEEP THE MOTOR CLEAN AND DRY**. This requires periodic inspection of the motor, the frequency of which depends upon the type of motor and the service.

Periodic checks of voltage, frequency, and current is recommended. Such checks assure the correct frequency and voltage applied to the motor, and yield an indication of the fan load. Comparison of this data with previous data will give an indication of the fan performance. Any serious deviations should be investigated and corrected.

Fractional motors usually have pre-lubricated sealed bearings with no grease fittings and are lubricated for life.

Lubricate integral horsepower motors per the motor manufacturer's recommendations. Lubrication frequency depends on the motor horsepower, speed, and service. Use compatible greases (See Page 24). Do not over grease.

If the motor is totally enclosed fan cooled (TEFC), non-ventilated (TENV), or air over (TEAO), it is recommended that the condensation drain plugs be removed. This is not necessary with motors equipped with automatic drains, which should be left in place as received.

Vibration

Excessive fan vibration can be caused by many things. **ALL POSSIBLE SOURCES OF THE EXCESSIVE VIBRATION MUST BE CHECKED OUT AND CORRECTIVE ACTION TAKEN IMMEDIATELY TO CORRECT THE PROBLEM.** See the fan trouble-shooting chart (Table 2) for possible causes of excessive fan vibration.

A vibration analyzer will be of great assistance in determining the amount of vibration. The following values give an indication of the fan vibration condition. Vibration readings should be taken on the fan bearings (if possible, on belt drive fans), or motor (on direct drive fans), or on the fan housing of ducted axial type fans. Horizontal, vertical, and axial readings should be taken. The following guidelines are based on AMCA Std. 205, Balance Grade BV-3.

Table 1		
Vibration displacement in peak velocity, inches/sec (mm/s) - Filter out		
Condition	Rigidly Mounted	Flexibly Mounted
Start Up	0.25 (6.4)	0.38 (8.8)
Alarm	0.40 (10.2)	0.65 (16.5)
Shut-Down	0.50 (12.7)	0.70 (17.8)

NOTE: *The fan should not be operated if the vibration levels are at or above the Shut-Down range.*

Grease Lubrication Schedule

Lubrication guidelines for horizontal shaft, grease lubricated fan, blower or other high speed rotating equipment. Lubrication guidelines for vertical shaft -- see note below.

Bearing Lubrications: Linkbelt

Lubrication Schedule (Months)* - Spherical Roller Bearing - Solid Pillow Blocks									
Shaft DIA	Speed (RPM)								
	500	1000	1500	2000	2500	3000	3500	4000	4500
1" Thru 1 ⁷ / ₁₆ " (25 - 35)	6	4	4	2	1	1	1	1	1/2
1 ¹¹ / ₁₆ " thru 2 ³ / ₁₆ " (40 - 55)	4	2	1 ¹ / ₂	1	1/2	1/2	1/2	1/2	1/2
2 ⁷ / ₁₆ " thru 3 ⁷ / ₁₆ " (60 - 85)	3	1 ¹ / ₂	1	1/2	1/2	1/2	1/2		
3 ¹⁵ / ₁₆ " thru 4 ¹⁵ / ₁₆ " (90 - 125)	2 ¹ / ₂	1	1/2	1/4					

*Suggested lubrication interval under ideal continuous operating conditions. Lubricate while running, if safety permits, until some purging occurs at seals. Adjust lubrication frequency depending on conditions of purged grease. Use one-half of listed interval or vertical shaft applications or for 24 hour operation. House of operation, temperature and surrounding conditions will affect the lubrication frequency required.

1. Lubricate with a high quality NLGI No. 2 lithium-base grease (SEE BELOW)
2. Lubricate bearings prior to extended shutdown or storage and rotate shaft monthly to aid corrosion protection
3. Any lubrication requirements noted on the general assembly drawing supersede requirements found here.

Lubrication Schedule (Months)* - Ball Bearing Pillow Blocks									
Shaft DIA	Speed (RPM)								
	500	1000	1500	2000	2500	3000	3500	4000	4500
1/2" Thru 1 ¹¹ / ₁₆ " (13 - 45)	6	6	5	3	3	2	2	2	1
1 ¹⁵ / ₁₆ " thru 2 ⁷ / ₁₆ " (50 - 60)	6	5	4	2	2	1	1	1	1
2 ¹¹ / ₁₆ " thru 2 ¹⁵ / ₁₆ " (65 - 75)	5	4	3	2	1	1	1		
3 ⁷ / ₁₆ " thru 3 ¹⁵ / ₁₆ " (80 - 100)	4	3	2	1	1				

*Suggested lubrication interval under ideal continuous operating conditions. Lubricate while running, if safety permits, until some purging occurs at seals. Adjust lubrication frequency depending on conditions of purged grease. Use one-half of listed interval or vertical shaft applications or for 24 hour operation. House of operation, temperature and surrounding conditions will affect the lubrication frequency required.

1. Lubricate with a high quality NGLI No. 2 lithium-based grease
2. Lubricate bearings prior to extended shutdown or storage and rotate shaft monthly to avoid corrosion protection. Any lubrication requirements noted on the general assembly drawing supersede requirements found here.

Roller Bearing Units Series P-LB6800

Relubrication Interval				Operating Speed (RPM)			
Shaft Size Range		Amt. of Grease		Max RPM			
Inches	mm	Cu. In.	Cu. cm.	6 mos	4 mos	2 mos	1 mo
1 7/16 - 1 1/2	40	0.3	4.9	2400	3600	5000	5500
1 11/16 - 1 3/4	45	0.3	5.0	2200	3300	4500	5000
1 15/16 - 2	50	0.4	6.6	2000	3000	4000	4500
2 3/16 - 2 1/4	60	0.8	12.7	1700	2500	3400	3800
2 7/16 - 2 3/4	65	0.8	12.3	1450	2200	3000	3400
2 11/16 - 2 3/4	70	0.9	14.3	1350	2000	2800	3200
2 15/16 - 3	75	1.2	19.7	1300	1900	2600	3000
3 3/16 - 3 1/4	80	1.7	27.4	1200	1800	2400	2700
3 7/16 - 3 1/2	90	2.3	37.7	1100	1650	2200	2300
3 11/16 - 4	100	3.1	50	1000	1500	1950	2100
4 3/16 - 4 1/4	110	4.2	70	900	1350	1850	1900
4 7/16 - 4 1/2	115	5.5	90.1	840	1250	1700	1800
4 15/16 - 5	125	6.4	105	780	1150	1600	1700
Clean and repack intervals				5 yrs	3 yrs	2 yrs	1yr

*Suggested lubrication interval under ideal continuous operating conditions. Remove bearing cap and observe condition of used grease after lubricating. Adjust lubrication frequency as needed. Use one-half of listed interval for vertical shaft applications for 24 hour operation. Hours of operation, temperature and surrounding conditions will affect the lubrication frequency required. Clean and repack bearings annually. Remove old grease, pack bearing full and fill housing reservoir on both sides of bearings to bottom of shaft.

1. Lubricate with a high quality NLGI No. 2 lithium-base grease.
2. Lubricate bearings prior to extended shutdown or storage and rotate shaft monthly to aid corrosion protection.
3. Purge or remove old grease when changing lubrication brands or types
4. Any lubrication requirements noted on the general assembly drawing supersede requirements found here.

Static Oil Lubrication:

1. Use only high quality mineral oil with a VG grade indicated on the customer submittal drawing.
2. Static oil level should be at the center of the lower-most roller (do not overfill).
3. Complete lubrication change should be made annually.

Note:

1. These are general recommendations only; specific manufacturer's recommendations may vary slightly.
2. Assumes clean environment, -20°F. to 120°F.
 - A. Ambient temperature greater than 120°F. Will shorten bearing life.
 - B. Under extremely dirty conditions, lubricate more frequently.
3. Assumes horizontal mounting configuration. For vertically mounted applications, lubricate twice as frequency.
4. Lubrication intervals are based on 12 hr. per day operations at a maximum temperature of 160 F hosing temperature.
 - A. For 24 hour per day operation, the intervals shall be cut in half.
5. Any type of warranty claim must be accompanied with Maintenance Log and Maintenance Schedule.
6. During storage or long term shutdowns Fan shaft bearings are to be manually rotated every month and additional grease added, purging some grease from the bearing cavity every six (6) months. Grease in bearings is to be purged at the time of removal from storage, making sure that an ample supply of fresh grease is in each grease cavity. Grease used must be compatible with that already in fan bearing.
7. Refer to Assembly/Inspection worksheet for type grease used when assembled.

Sealmaster - High Temp: Mobilith SHC 220, Lithium Complex, NGLI #2, ISO 220 synthetic oil viscosity
Standard Application: Lithium Complex, NGLI #2 ISO 110-120 oil viscosity

Rexnord - High Temp: Exxon/Mobil SHC 100 which is also a NLGI 2 grade grease made with lithium complex base and synthetic oil up to 300F
Standard Application: Exxon/Mobil Ronex MP NLGI 2 grade lithium complex -40F to +225F.

For all other Bearing manufacturers, see Assembly and Inspection Sheet or contact Hartzell

Grease Lubrication Schedule Cont.

Lubricate with a premium quality NLGI 2 or 3 multi-purpose ball bearing grease having corrosion inhibitors, anti-oxidant additives and mechanical stability for high speed operation. The grease should also have a minimum base oil viscosity of 500 SUS at 100° F and be suitable to operate continuously at 225° F. Do not use a heavy, long fibered grease.

The bearings on this fan shaft have been greased at the factory for the following application:

- ☐ General Purpose (Shell Oil-Shell Gadas S2 V220 2)
- ☐ High Temperature (Shell Oil-Aeroshell #22 (-85° F. to 400° F. | -65° C. to 204° C))
- ☐ Low Temperature (Shell Oil-Aeroshell #22 (-85° F. to 400° F. | -65° C. to 204° C))
- ☐ Extreme Moisture (Shell Oil-Shell Gadas S2 V 220 2)
- ☐ Other: _____

(Replaces Shell Alvania Grease EP2)

These units are pre-lubricated (except Series P-LB6800) with a multi-purpose Lithium soap grease. When re-lubricating with greases of different types (especially synthetic soaps or oils), complete flushing is recommended.

The presence of dirt, moisture or chemical fumes around the bearings requires more frequent lubrication. Ambient temperatures below 20° F or above 200° F require special lubricants. Consult the machinery manufacturer for recommendations.

When vibration exceeds 0.15 in./sec., frequent lubrication is required and a need for equipment trim balancing may be indicated.

Fill bearings with lubricant prior to extended shutdown or storage. Rotate the shaft several revolutions each month during idle periods.

Note: *It is possible to dislodge the seals on these units if grease is added too fast. It is preferable to use a hand gun or a regulated automatic system.*

Vertical Shafts:

For vertical applications more frequent re-lubrication is required. The above re-lubrication interval (and clean and re-pack interval for the Series P-LB 6800 bearings,) should be reduced by half.

Trouble-Shooting

Table 2	
FAN TROUBLE-SHOOTING CHART	
PROBLEM	POSSIBLE CAUSES
Excessive Vibration	- Accumulation of material on impeller - Worn or corroded impeller - Bent shaft - Impeller or sheaves loose on shaft - Motor out of balance - Impeller out of balance - Sheaves eccentric or out of balance - Bearing or drive misalignment - Mismatched belts - Belts too loose or too tight - Loose or worn bearings - Loose bearing bolts - Loose fan mounting bolts - Weak or resonant foundation - Foundation not flat and level - Structures not cross braced - System pulsation (DANGER) - Fan operation in a stall - Blades at different angles on adjustable pitch fans
Horsepower Too High	- Fan speed higher than design - Air density higher than design - Impeller rotating in the wrong direction - Angle set too high in adjustable pitch fans
Airflow Too Low	- Impeller rotating in the wrong direction - Fan speed lower than design - Actual system is more restrictive (more resistance to flow than expected) - Dampers or registers closed - Leaks or obstructions in duct work - Filters or coils are dirty or clogged - Inlet or outlet screens clogged - Restricted fan inlet or outlet - No straight duct at fan outlet - Shape elbows near fan inlet or outlet - Improperly designed turning vanes
Airflow Too Much	- Actual system is less restrictive (less resistance to flow) than expected) - Fan speed higher than design - Filter not in place - Registers, grilles and/or dampers not installed
Fan Does Not Operate	- Blown fuses - Broken belts - Loose pulleys - Electricity turned off or not wired properly - Wrong voltage - Motor too small and overload protector has broken circuit

Table 2 (continued)

FAN TROUBLE-SHOOTING CHART

PROBLEM	POSSIBLE CAUSES
Excessive Noise	- Accumulation of material on impeller - Worn or corroded impeller - Bent shaft - Impeller or sheaves loose on shaft - Impeller hitting housing - Impeller out of balance - Bearing or drive misalignment - Mismatched belts - Belts or drive misalignment - Belts hitting guard - Loose or worn bearings - Belts worn - Belts oily or dirty - Defective or bad bearings - Bearings need lubrication - Loose bearing bolts - Loose fan - System pulsation or surge - Electrical noises - Noise from high velocity air system - Duct work too small - Registers or grilles too small for application - Fan in stall condition (DANGER) - Rattle of components in high velocity air stream - Leaks in duct work - Vibrating duct work - Vibrating parts not isolated from building

It is recommended that the users and installers of this equipment familiarize themselves with AMCA publication #201, "Fans and Systems, and publication #202 "Trouble-Shooting", which are published by Air Movement and Control Association, International, 20 West University Drive, Arlington Heights, Illinois 60004.

Spare Parts:

Spare parts are not needed for the first year of operation. If spare parts are desired, then it is suggested that a spare motor and impeller be ordered for direct drive fans. For belt drive fans, in addition to the motor and impeller, it is suggested that a spare set of bearings, shaft, sheaves, and belts be ordered.

When ordering spare parts, specify the parts desired, the fan model number, and the fan serial number. Contact your local sales representative for price and delivery.



Performance Guarantee

You have high expectations. So does Hartzell Air Movement. We know you demand the most reliable and durable industrial air movement products available, so we're holding ourselves to a higher standard. We're so sure that our products will out-perform industry standards, we're backing that promise with the industry's first five-year warranty. Call your Hartzell Air Movement representative for assistance.

More than 50 Hartzell representative offices can provide specific performance and installation data to meet your requirements. Call your Hartzell Air Movement representative for assistance. Visit www.hartzellairmovement.com or call 800.336.3267 for the name of your representative.

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www.hartzellairmovement.com
800.336.3267

The safe application and use of equipment supplied by Hartzell Air Movement, Inc. is the responsibility of the installer, user, owner and employer. To evaluate the safe application of this equipment, the following should be considered: the location of the installation, accessibility of employees and other persons to the equipment, any adjacent equipment, applicable building and safety codes, and requirements of OSHA. Since the application and use of its equipment can vary greatly, Hartzell Air Movement, Inc. offers various product types, optional safety accessories, and sound performance data per laboratory tests. An industry publication: "Recommended Safety Practices for Users and Installers of Industrial and Commercial Fans" is available from Hartzell upon request. Document: QMS-0175 Rev: 2

Sales Order ID: 2521509

| 910 S. Downing St., Piqua, OH 45356 | 800.336.3267 | info@hartzell.com | www.hartzell.com |

Submittal Rev 3

SECTION 07 TERMS & WARRANTY



TERMS AND CONDITIONS OF SALE OF HARTZELL FAN, INC. dba "HARTZELL AIR MOVEMENT"

These Terms and Conditions of Sale should be read carefully. Sale of any goods or services described or referred to herein is subject to these Terms and Conditions of Sale. Any order for or any statement of intent to purchase any goods or services, or any direction to proceed with engineering, procurement, manufacture or shipment, constitutes assent to these Terms and Conditions of Sale.

REJECTION OF DIFFERENT TERMS - This document is not an acceptance of any prior written or oral offer. If any such prior written or oral offer has been made, it is hereby rejected. These Terms and Conditions of Sale supersede any additional or different written or oral terms previously or subsequently communicated. Acceptance is expressly limited to these Terms and Conditions of Sale. Hartzell Air Movement hereby gives notification of its objection to and rejection of any proposed terms different from those contained herein whether communicated previously or subsequently and whether written or oral. This document is the complete and exclusive statement of the agreement between Hartzell Air Movement and Buyer.

DELIVERY - Shipping dates are approximate and are based upon prompt receipt of all necessary information.

Hartzell Air Movement is not liable for delays in delivery or in performance or failure to manufacture or deliver, due to (1) causes beyond its reasonable control, including errors in manufacture or (2) acts of God, acts of the Buyer, acts of civil or military authority, priorities, fires, strikes, or other labor disturbances, floods, epidemics, war, riot, delays in transportation or car shortage, or (3) inability on account of causes beyond its reasonable control to obtain necessary labor, materials, components or manufacturing facilities. In the event of any such delay, the date of delivery or of performances shall be extended for a period equal to the time lost by reason of delay.

START-UP SERVICE - On certain equipment start-up service is available from Hartzell Air Movement at per diem rates plus lodging and traveling expenses. Such start-up service is not included unless specifically quoted. There will be an additional charge for time used exceeding any number of days quoted.

ELECTRICAL EQUIPMENT - Equipment includes only the electrical components referred to in the proposal. Charges arising from any local, state, provincial or other regulations necessitating changes to electrical equipment will be paid by Buyer unless agreed otherwise in writing by Hartzell Air Movement.

CANCELLATION/TERMINATION/MODIFICATION - There can be no cancellation, termination or modification by Buyer without Hartzell Air Movement's prior written consent.

CLAIMS - Claims of shortages or errors or other basis for rejection must be made within five (5) days after delivery.

PRICING - Prices quoted are firm provided release to manufacture and shipment is completed by Hartzell Air Movement within three (3) months from the date of order entry. If shipment is delayed for reasons beyond control of Hartzell Air Movement and is not made within three (3) months from date of order entry, prices are subject to change at Hartzell Air Movement's option.

PAYMENT TERMS

TAXES - Any tax or other governmental charge now or hereafter levied upon the reproduction, sale, use or shipment of goods ordered or sold will be charged to and paid for by Buyer. Such taxes are not included in Hartzell Air Movement's price unless expressly so provided.

TERMS OF PAYMENT - Terms of payment are Net 30 days from the date of invoice. If shipment is delayed by the Buyer, date of readiness for shipment shall be deemed to be date of invoice for payment purposes. If, in Hartzell Air Movement's judgment, the Buyer's financial condition at any time does not justify normal payment terms, Hartzell Air Movement may require full or partial payment as a condition to commencing or continuing manufacture, or in advance of shipment, or, if shipment has been made, recover equipment from the carrier.

FREIGHT - All freight will be shipped Ex-Works Piqua, OH.

GOODS MANUFACTURED BY OTHERS - Hartzell Air Movement has no responsibility whatever with respect to goods sold but not manufactured by Hartzell Air Movement and Buyer's sole recourse is against the manufacturer of said goods. Hartzell Air Movement will assign to Buyer any pertinent warranty rights received by Hartzell Air Movement from manufacturer.

PATENTS - Hartzell Air Movement shall defend any suit or proceeding brought against the Buyer insofar as based on a claim that any goods sold by Hartzell Air Movement, or any part thereof constitutes an infringement of any patent of the United States, if notified promptly in writing and given to authority, information and assistance for the defense of same, and Hartzell Air Movement shall pay all damages and costs awarded

therein against the Buyer. In case said Hartzell Air Movement goods, or any part thereof, is in such suit held to constitute infringement and the use of said goods is enjoined, Hartzell Air Movement shall, at its own expense and at its option, either procure for the Buyer the right to continue using said goods; or replace same with non-infringing equipment; or modify it so they become non-infringing; or remove said goods and refund purchase price and the transportation and installation costs thereof. The foregoing states the entire liability of Hartzell Air Movement for patent infringement.

The preceding paragraph shall not apply to any goods or any part thereof not manufactured by Hartzell Air Movement or to any goods or any part thereof, manufactured to Buyer's design, nor for any use to which any such goods may be put as a part of any system, mechanism or process covered by patent rights of others. As to such goods or any part thereof, Hartzell Air Movement assumes no liability whatsoever for patent infringement. If any claim is asserted against Hartzell Air Movement that products manufactured by Hartzell Air Movement in compliance with Buyer's specifications or design infringes on any United States patent or copyright, then the buyer shall defend and indemnify Hartzell Air Movement for any damages and costs.

INTELLECTUAL PROPERTY - The design, performance information, construction detail of Hartzell Air Movement goods or any part thereof, is proprietary, and remains the valuable property of Hartzell Air Movement. Buyer agrees not to copy or duplicate the goods or any part thereof, or information related thereto provided without express written authorization from Hartzell Air Movement.

LIMITED WARRANTIES - Hartzell Air Movement warrants to Buyer that any goods to be delivered hereunder will be produced in compliance with the requirements of the Fair Labor Standards Act of 1938 as amended.

Hartzell Air Movement also warrants to Buyer its goods to be free from defects in workmanship and material under normal use and service for two (2) years after tender of delivery by Hartzell Air Movement. Subject to the following limitations and conditions, the warranty will be extended to five (5) years after tender of delivery, if Buyer registers the product on-line at www.hartzellairmovement.com/five-year-warranty within ninety (90) days after tender of delivery. If the five (5) year warranty applies, applicability of the warranty to any claim arising after two (2) years after tender of delivery is conditioned upon Buyer providing to Hartzell Air Movement maintenance records (prepared contemporaneously with the performance of the maintenance) demonstrating that the product has been maintained (from the time of tender of delivery to the time the warranty claim arises) in accordance with Hartzell Air Movement's Installation, Operation and Maintenance Manual.

Goods manufactured by a third party ("Third Party Product") may constitute, contain, be contained in, incorporated into, attached to or packaged together with, the goods manufactured by Hartzell Air Movement, including but not limited to, motors. Third Party Products are not covered by foregoing warranty. For the avoidance of doubt, SELLER MAKES NO REPRESENTATIONS OR WARRANTIES WITH RESPECT TO ANY THIRD-PARTY PRODUCT, INCLUDING ANY (a) WARRANTY OF MERCHANTABILITY; (b) WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE; (c) WARRANTY OF TITLE; OR (d) WARRANTY AGAINST INFRINGEMENT OF INTELLECTUAL PROPERTY RIGHTS OF A THIRD PARTY; WHETHER EXPRESS OR IMPLIED BY LAW, COURSE OF DEALING, COURSE OF PERFORMANCE, USAGE OF TRADE, OR OTHERWISE.

The only warranty applicable to a Third Party Product is the warranty, if any, of the manufacturer of the Third Party Product. Hartzell Air Movement will provide to Buyer information regarding submission of a warranty claim to the manufacturer of a Third Party Product, but Hartzell Air Movement does not represent or assure that the warranty, if any, of the manufacturer of the Third Party Product.

Any extension of any applicable warranty period beyond that indicated above is effective only if the extension is granted by Hartzell Air Movement in writing. No warranty extends to future performance of goods, and any claim for breach of warranty or otherwise accrues upon tender of delivery.

Replacement of belts and other parts as a result of ordinary wear and tear is not covered by this warranty and is the responsibility of the Buyer.

The foregoing constitute Hartzell Air Movement's sole and exclusive warranties and are in lieu of all other warranties, whether written, oral, express, implied or statutory.

LIMITATION OF LIABILITY FOR BREACH OF WARRANTY - Hartzell Air Movement's obligation for any breach of warranty is limited to repairing or replacing, at its option, without cost to Buyer at its factory any goods which shall, within such a warranty period, be returned to it with transportation charges prepaid, and

which its examination shall disclose to its satisfaction to have been defective. Any request for repair or replacement should be directed to Hartzell Air Movement, Inc., P.O. Box 919, Piqua, Ohio 45356. Hartzell Air Movement will not pay for any repairs made outside its factory without its prior written consent. Hartzell Air Movement has no obligation to repair or replace any Hartzell Air Movement goods which have failed as a result of faulty installation or abuse, or incorrect electrical connections or alterations made by others, or use under abnormal operating conditions or misapplication of the goods.

LIMITATION OF LIABILITY - To the extent the above limitation of liability for breach of warranty is not applicable, the liability of Hartzell Air Movement on any claim of any kind, including negligence, for any loss or damage arising out of or connected with, or resulting from the sale and purchase of the goods or services covered by these Terms and Conditions of Sale or from the performance or breach of any contract pertaining to such sale or purchase or from the design manufacture, sale, delivery, resale, installation, technical direction installation, inspection repair, operation or use of any goods or services covered by these Terms and Conditions shall, in no case exceed the price allocable to the goods or services which gave rise to the claim and shall terminate one year after tender of delivery of said goods or services, plus six months allowance for shipment to approved stocking dealers and distributors.

In no event whether as a result of breach of contract, or warranty or alleged negligence, defects, incorrect advice or other causes, shall Hartzell Air Movement be liable for special or consequential damages, including, but not limited to, loss of profits or revenue, loss of use of the equipment or any associated equipment, cost of substitute equipment, facilities or services, down time costs, or claims of customers of the Buyer for such damages. Hartzell Air Movement neither assumes nor authorizes any person to assume for it any other liability in connection with the sale of its goods or services.

NO IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS – HARTZELL AIR MOVEMENT DOES NOT WARRANT THAT SAID GOODS ARE OF MERCHANTABLE QUALITY OR THAT THEY ARE FIT FOR ANY PARTICULAR PURPOSE. THERE IS NO IMPLIED WARRANTY OF MERCHANTABILITY AND THERE IS NO IMPLIED WARRANTY OF FITNESS.

REGULATORY LAWS AND/OR STANDARD - Hartzell Air Movement makes no promise or representation that its product will conform to any state or local laws, ordinances, regulations, codes or standards, except as particularly specified and agreed upon in writing. Hartzell Air Movement's prices do not include the cost of any related inspection permits or inspection fees.

EXCLUSIVE FORUM SELECTION CLAUSE - Any claims relating to these Terms and Conditions of Sale of the goods or services thereunder shall be brought (if subject matter jurisdiction exists) solely in the United States District Court for the Southern District of Ohio (Western Division). In the absence of subject matter jurisdiction in that Court, any such claims shall be brought solely in any Common Pleas Court of the State of Ohio within the area encompassed by the United States District Court for the Southern District of Ohio (Western Division).

NO DELEGATION - Buyer may not perform any duties under these Terms and Conditions of Sale through a delegate.

NO ASSIGNMENT - The rights of the Buyer hereunder cannot be assigned.

NO MODIFICATION, RESCISSION OR WAIVER - These Terms and Conditions of Sale are not subject to modification, rescission or waiver except by a writing signed by an officer of Hartzell Air Movement.

NO THIRD PARTY BENEFICIARY - These Terms and Conditions of Sale are for the benefit of Hartzell Air Movement and the Buyer and not for any other person.

GENERAL - All proposals are made and all orders are accepted by Hartzell Air Movement with the reference to the laws of the State of Ohio and the rights and duties of all persons and the construction and effect all provisions thereof shall be governed by the construed according to the laws of the State of Ohio.

The invalidity of any clause or provision of these Terms and Conditions of Sale shall not affect the validity or enforceability of any other clause or provision contained herein.

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SECTION 08 APPENDIX

